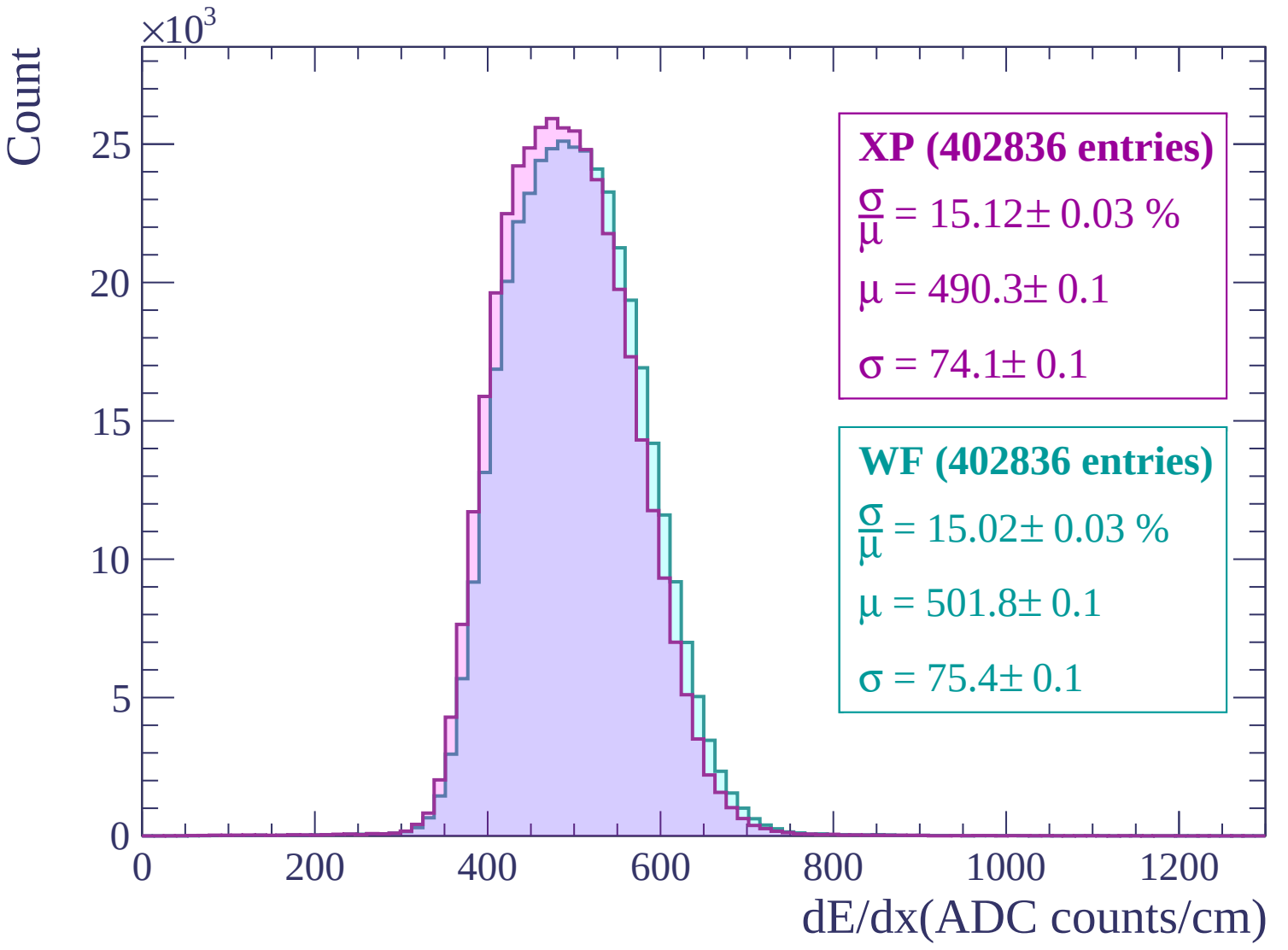
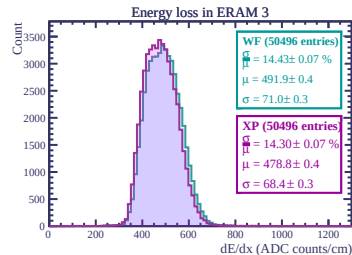
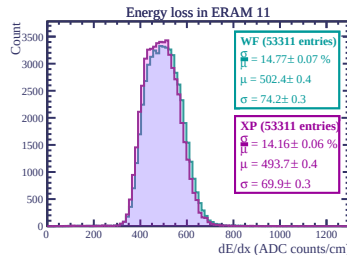
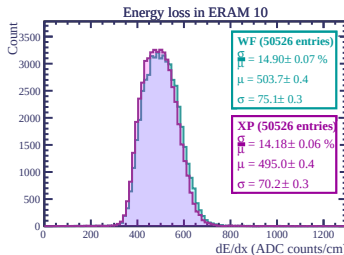
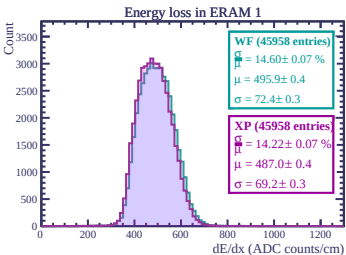
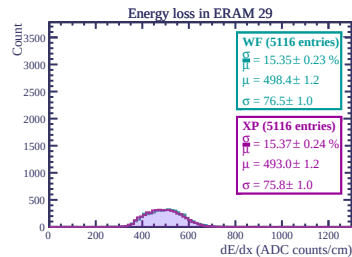
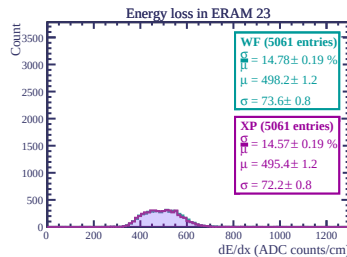
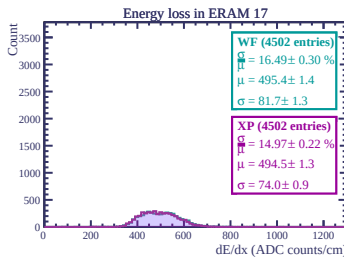
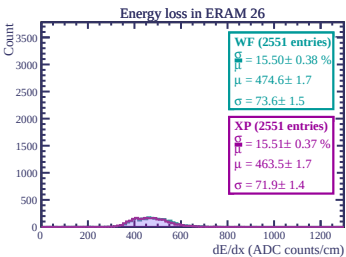
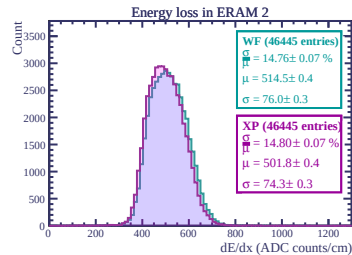
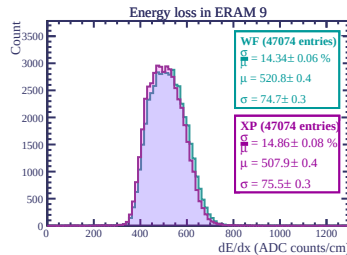
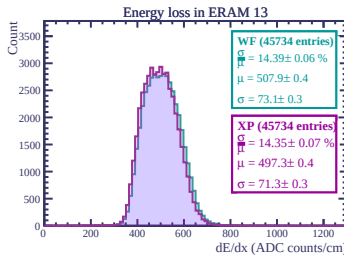
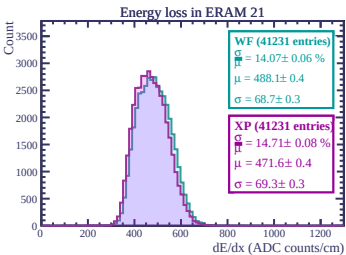
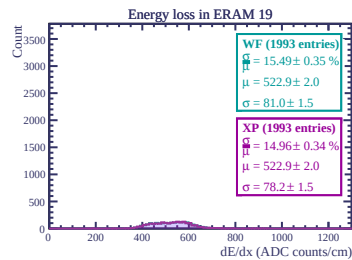
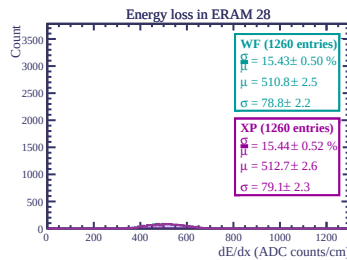
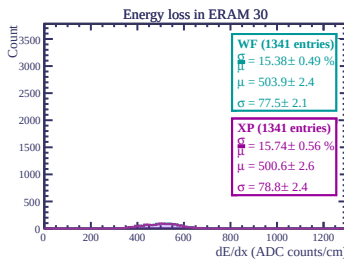
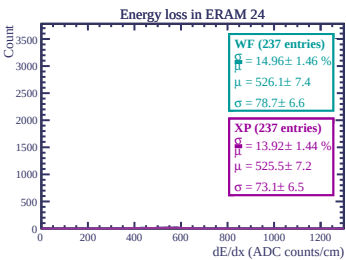
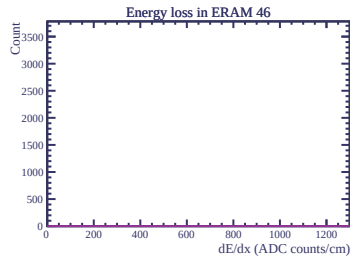
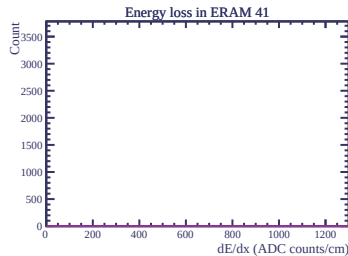
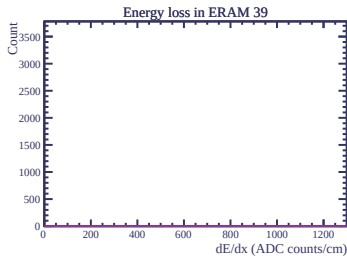
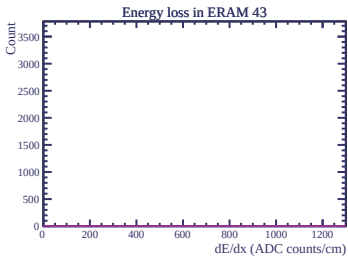
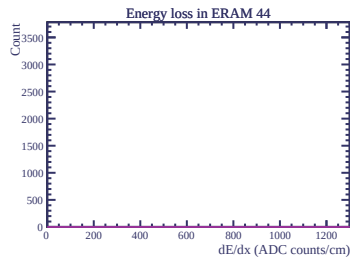
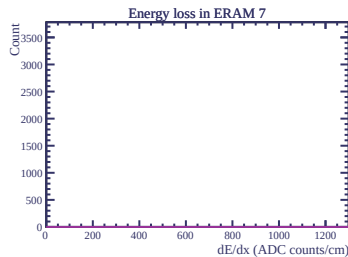
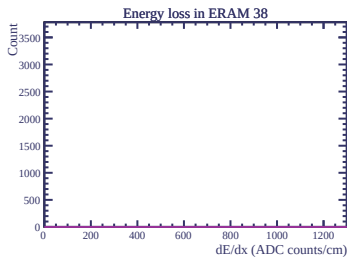
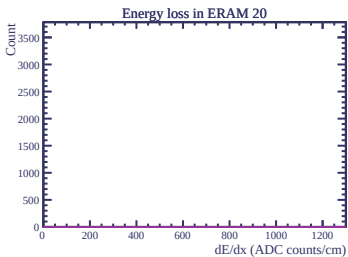
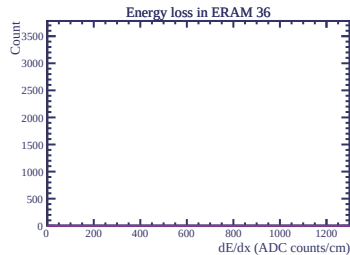
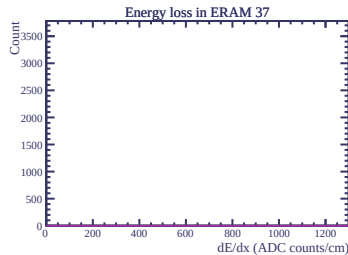
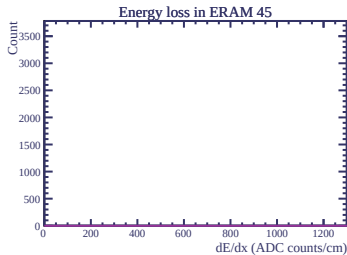
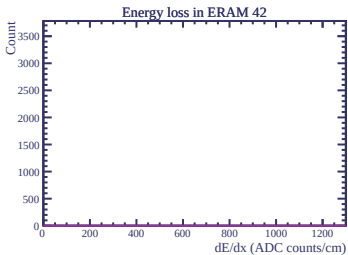
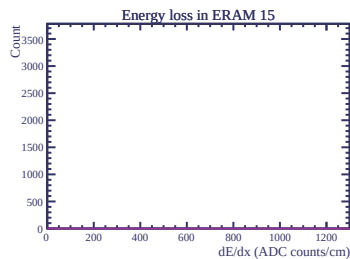
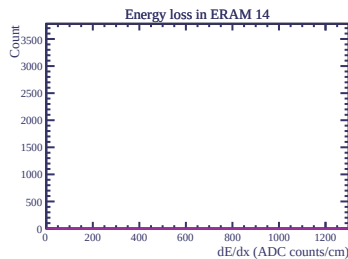
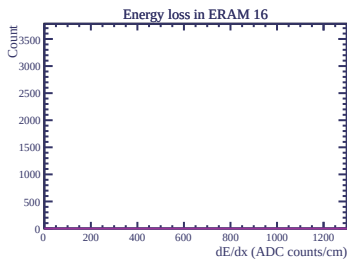
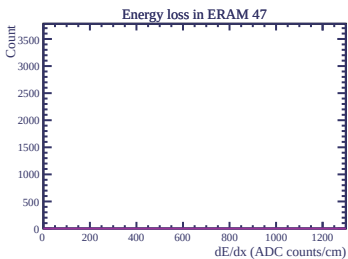
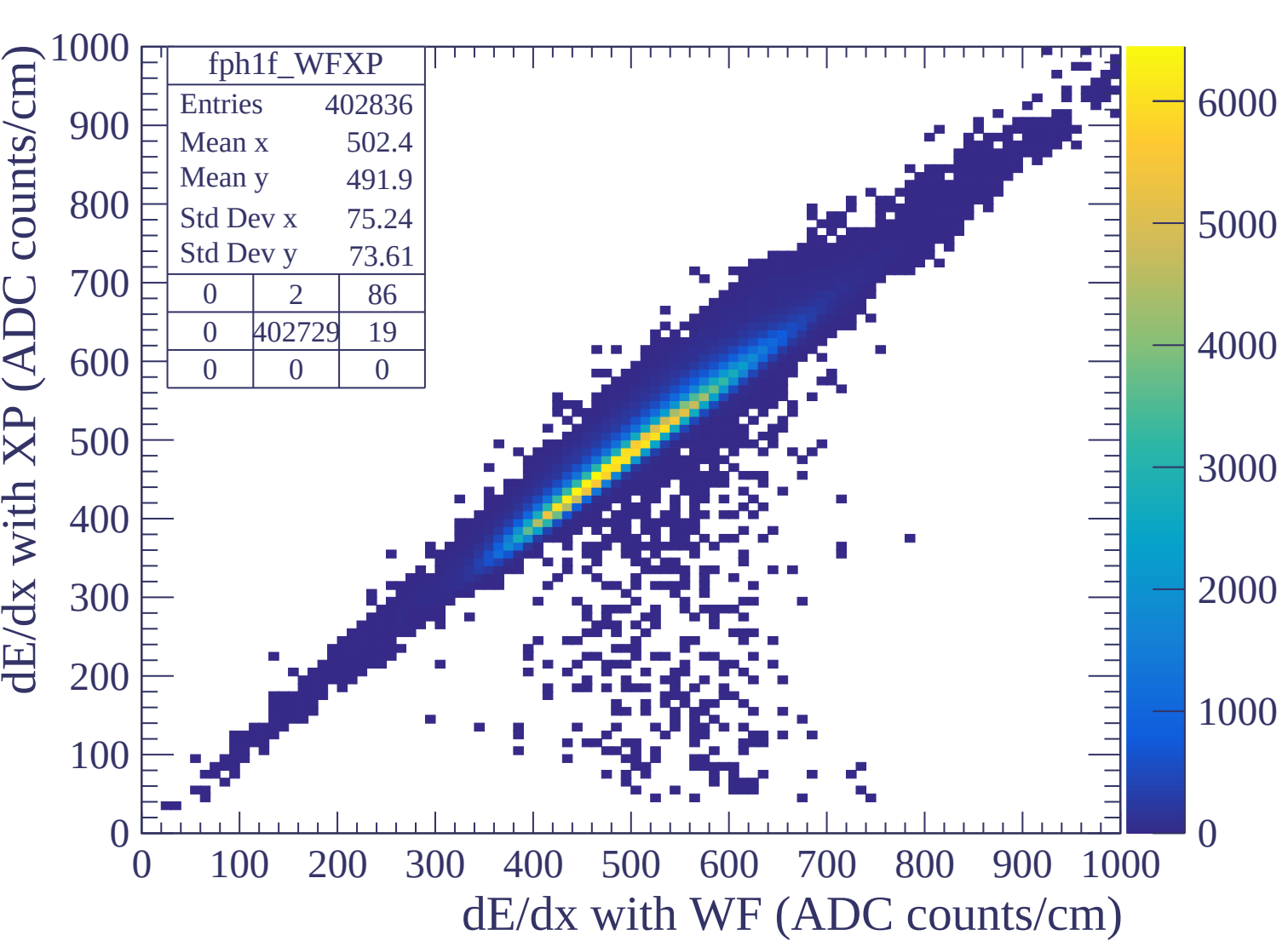
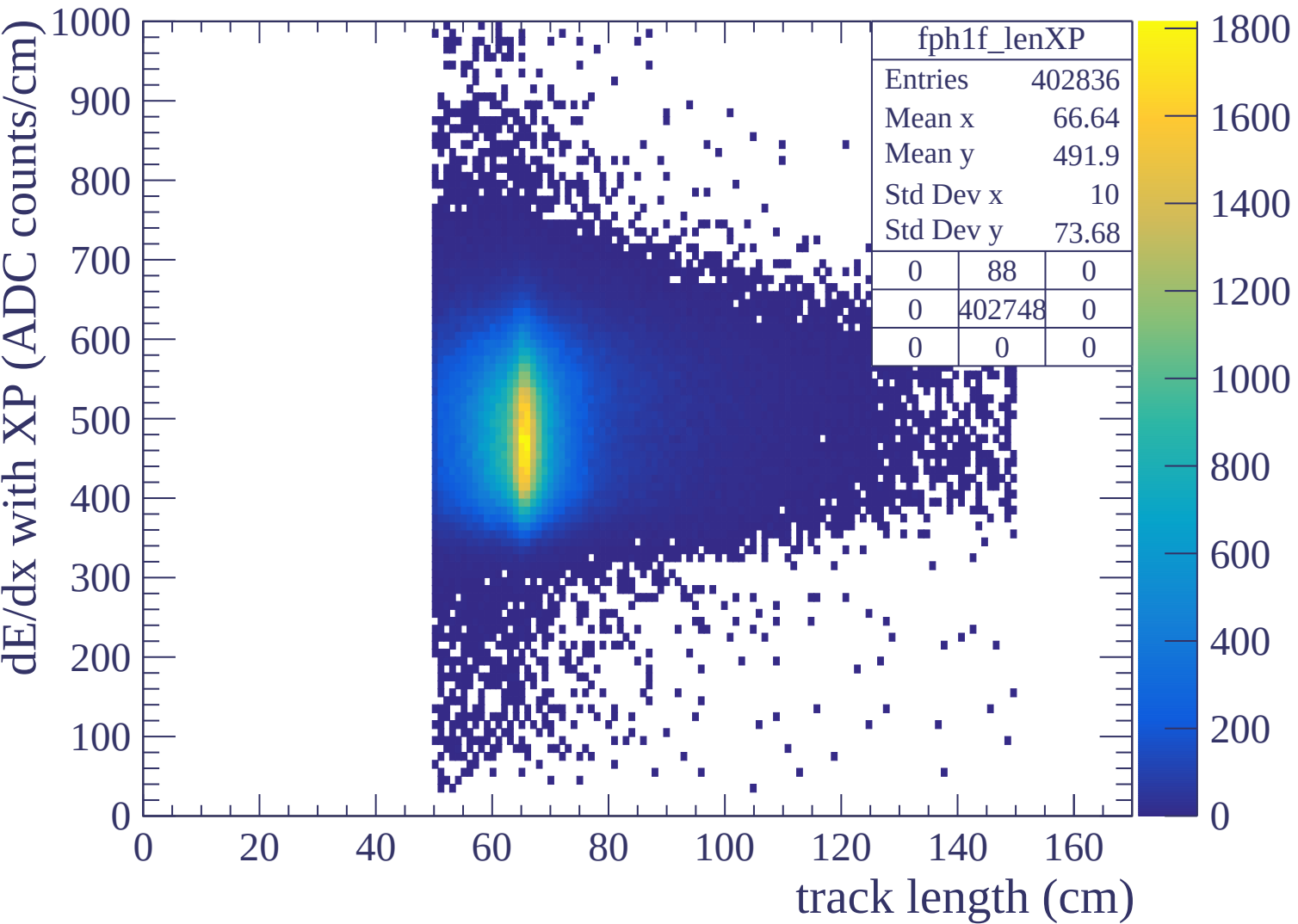


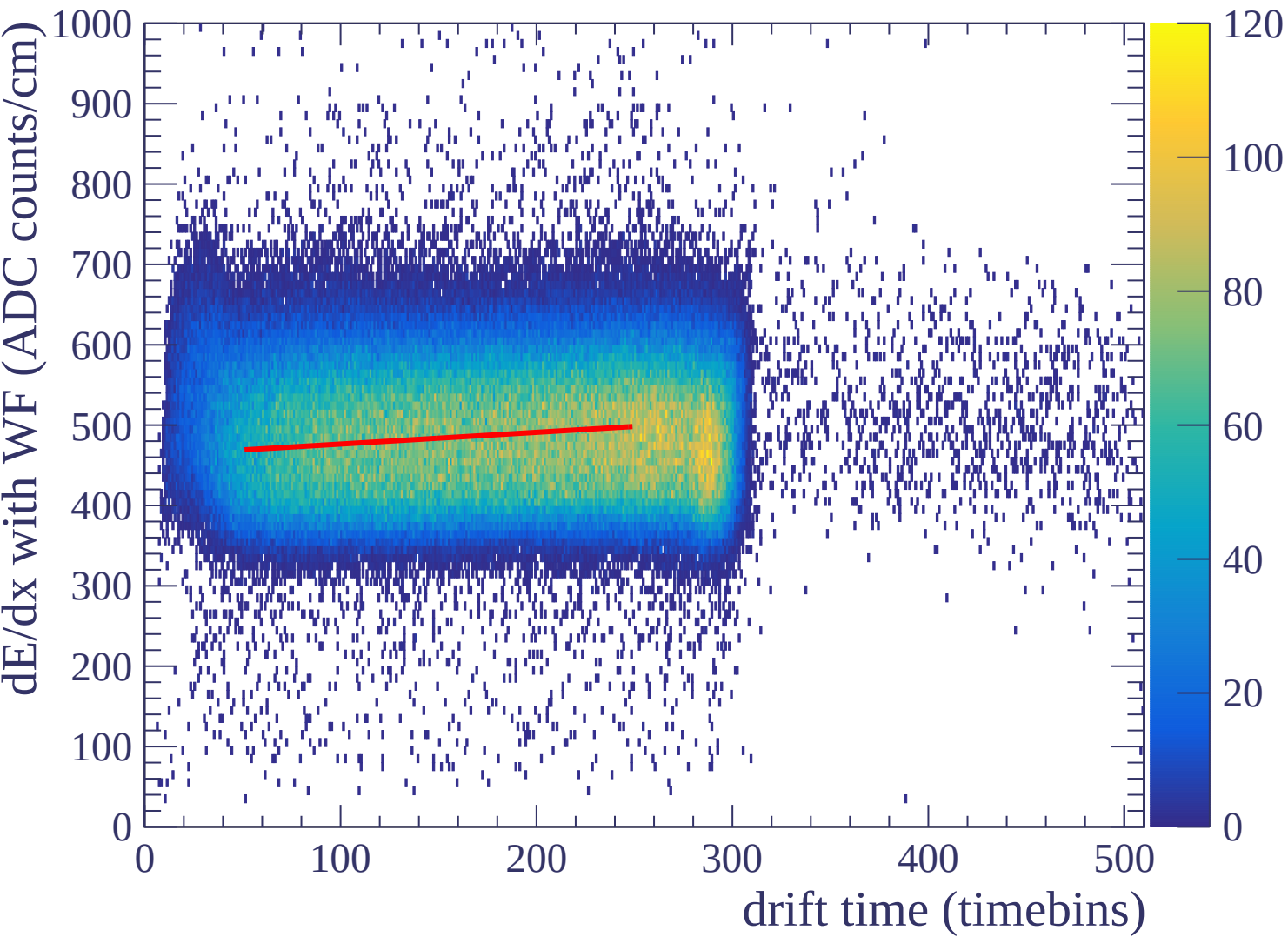


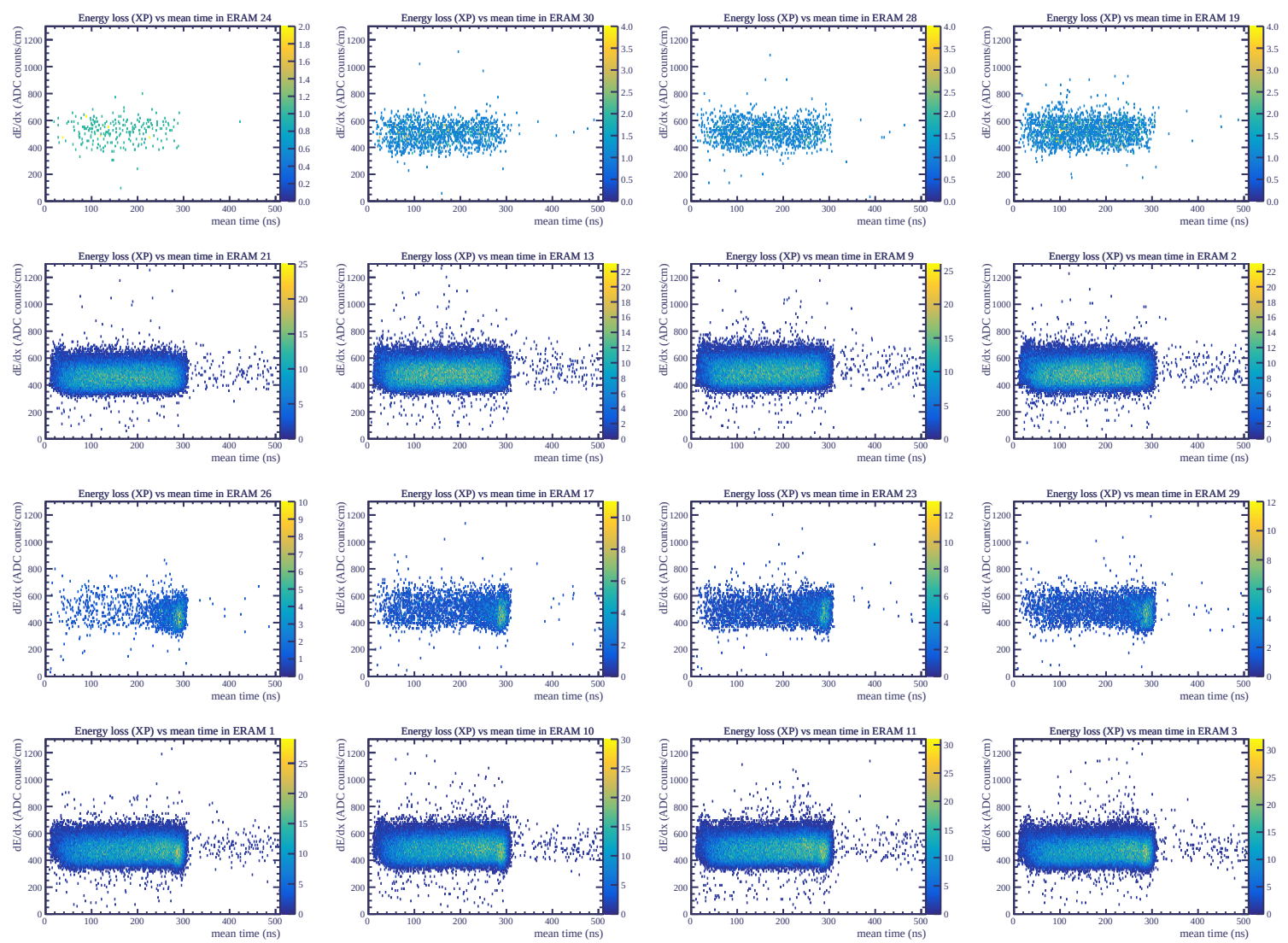


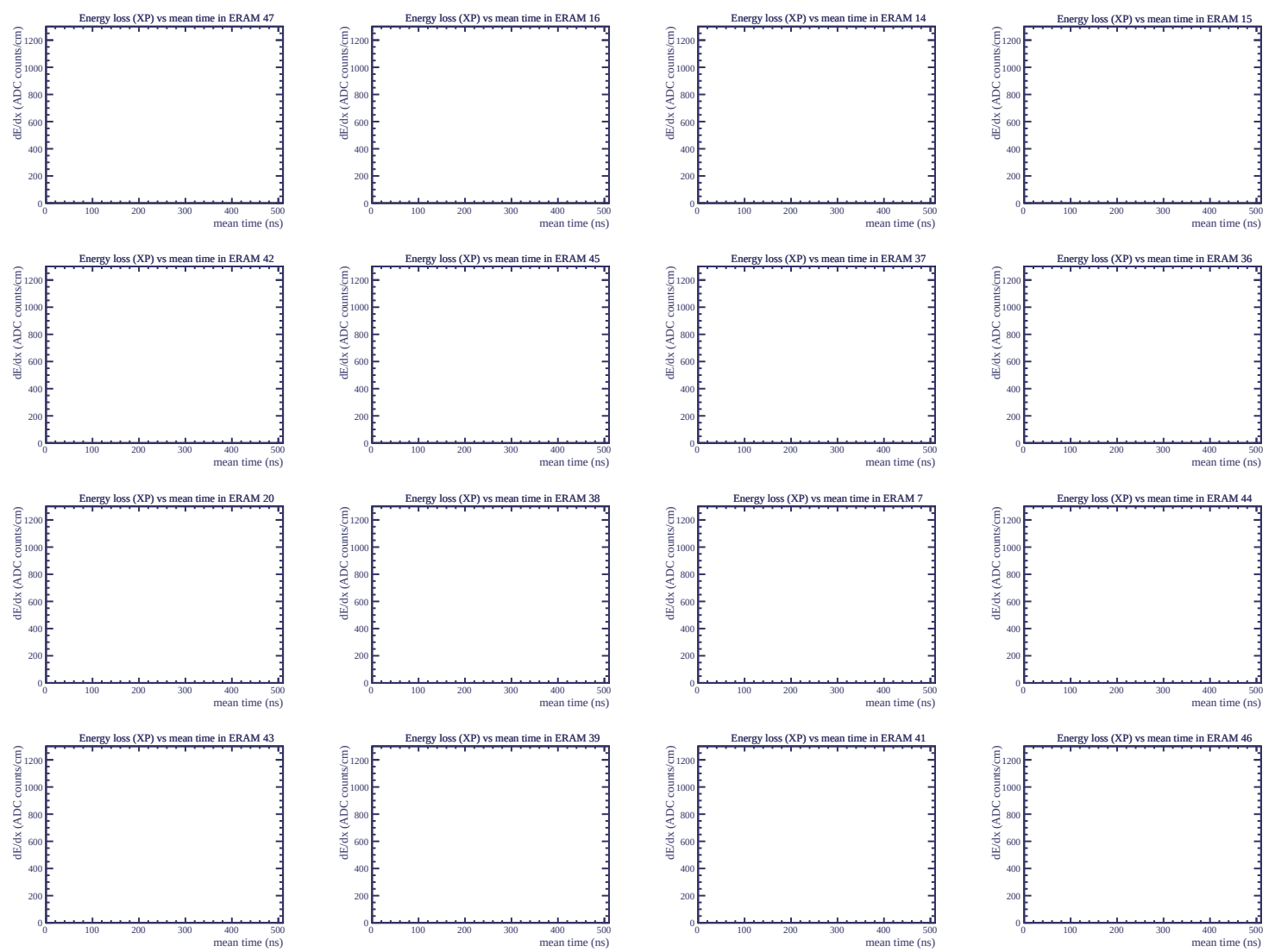




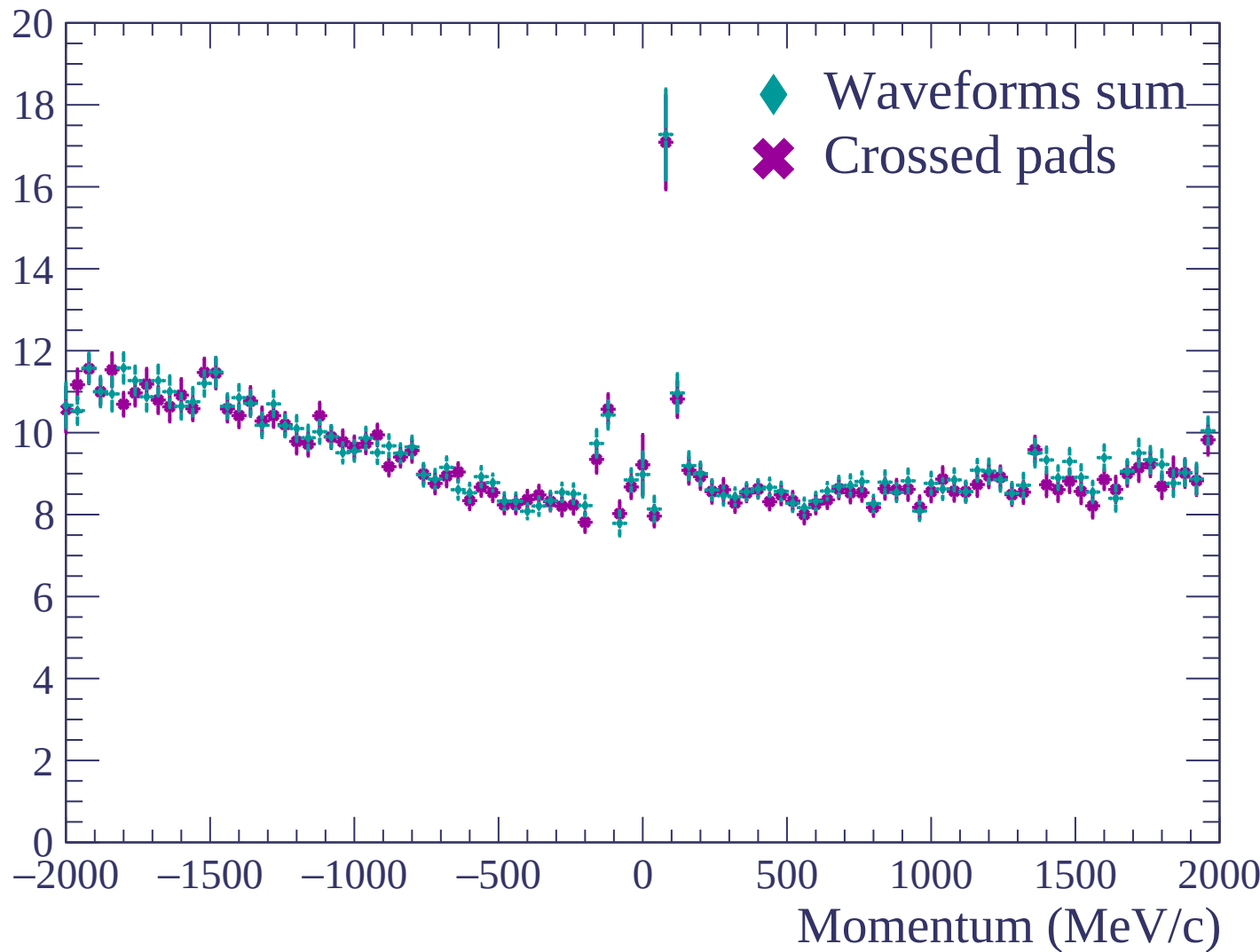


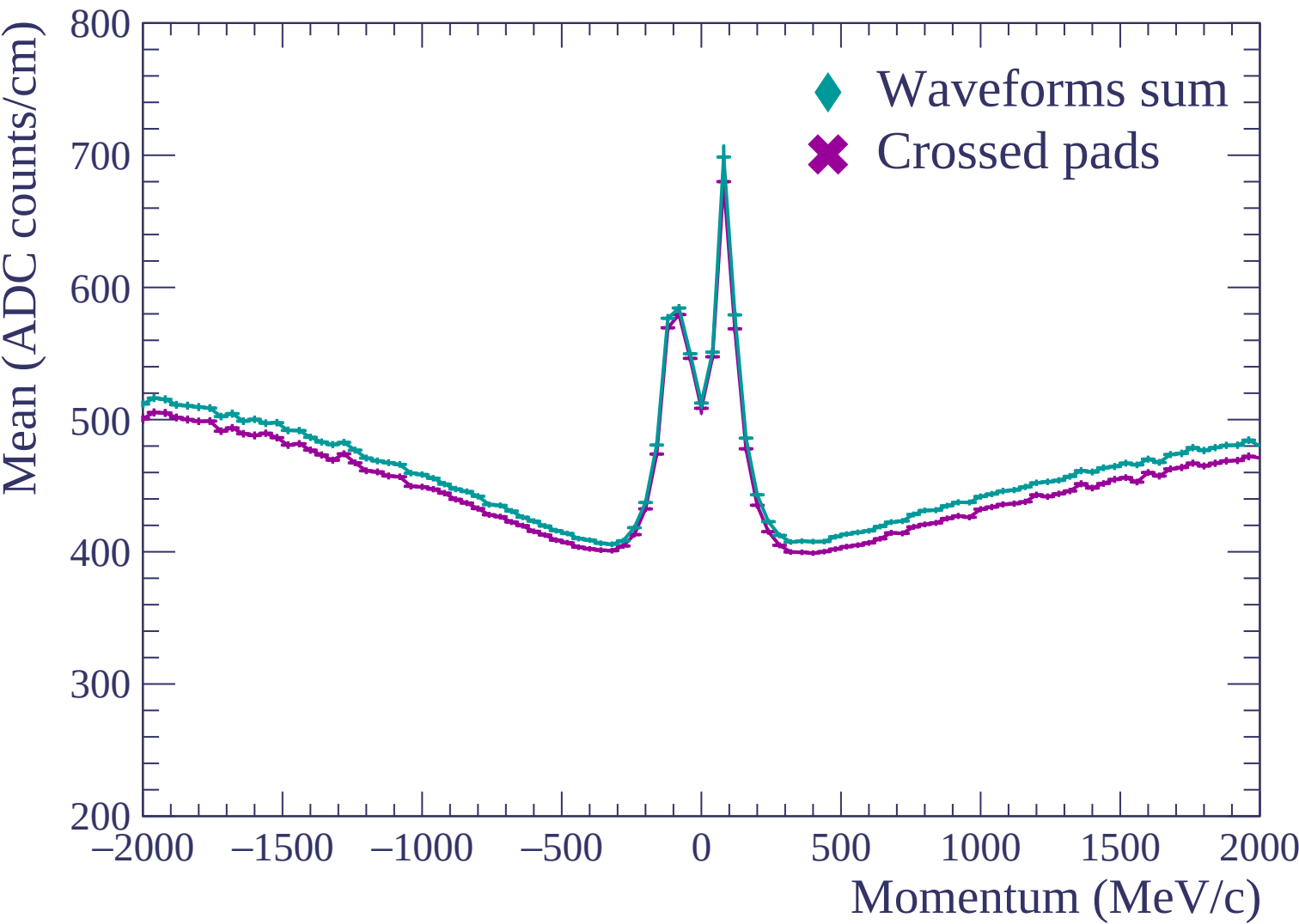


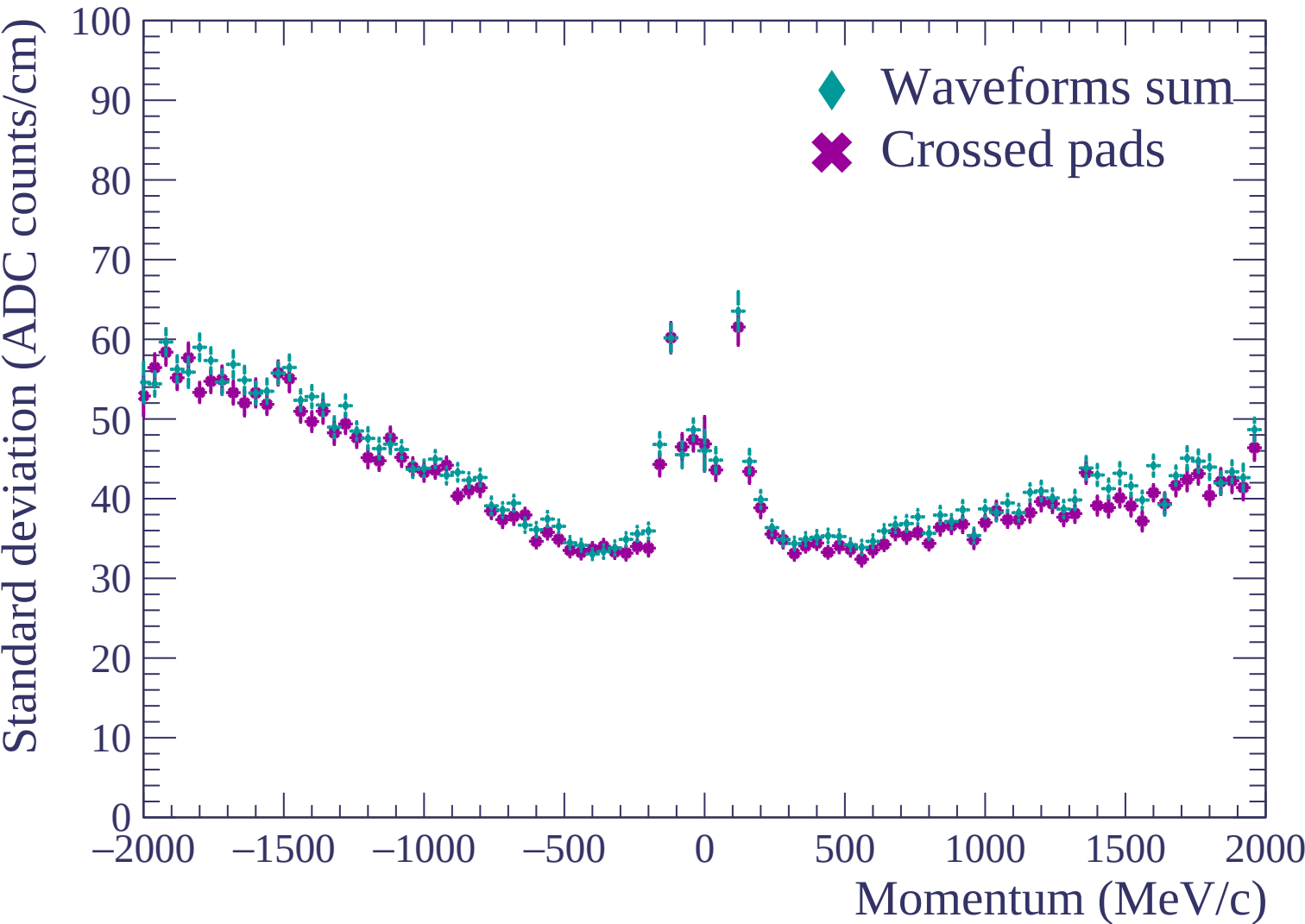


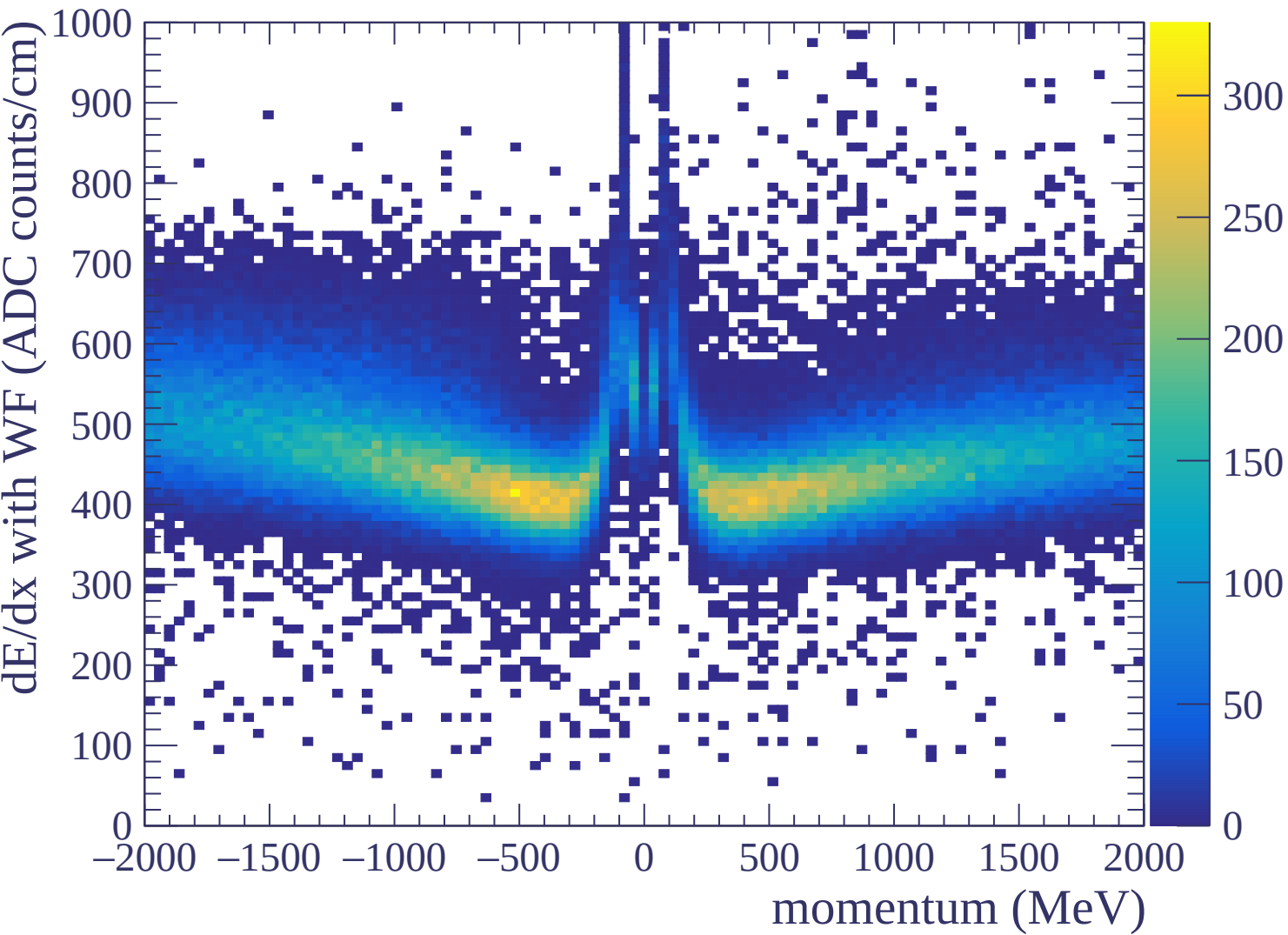


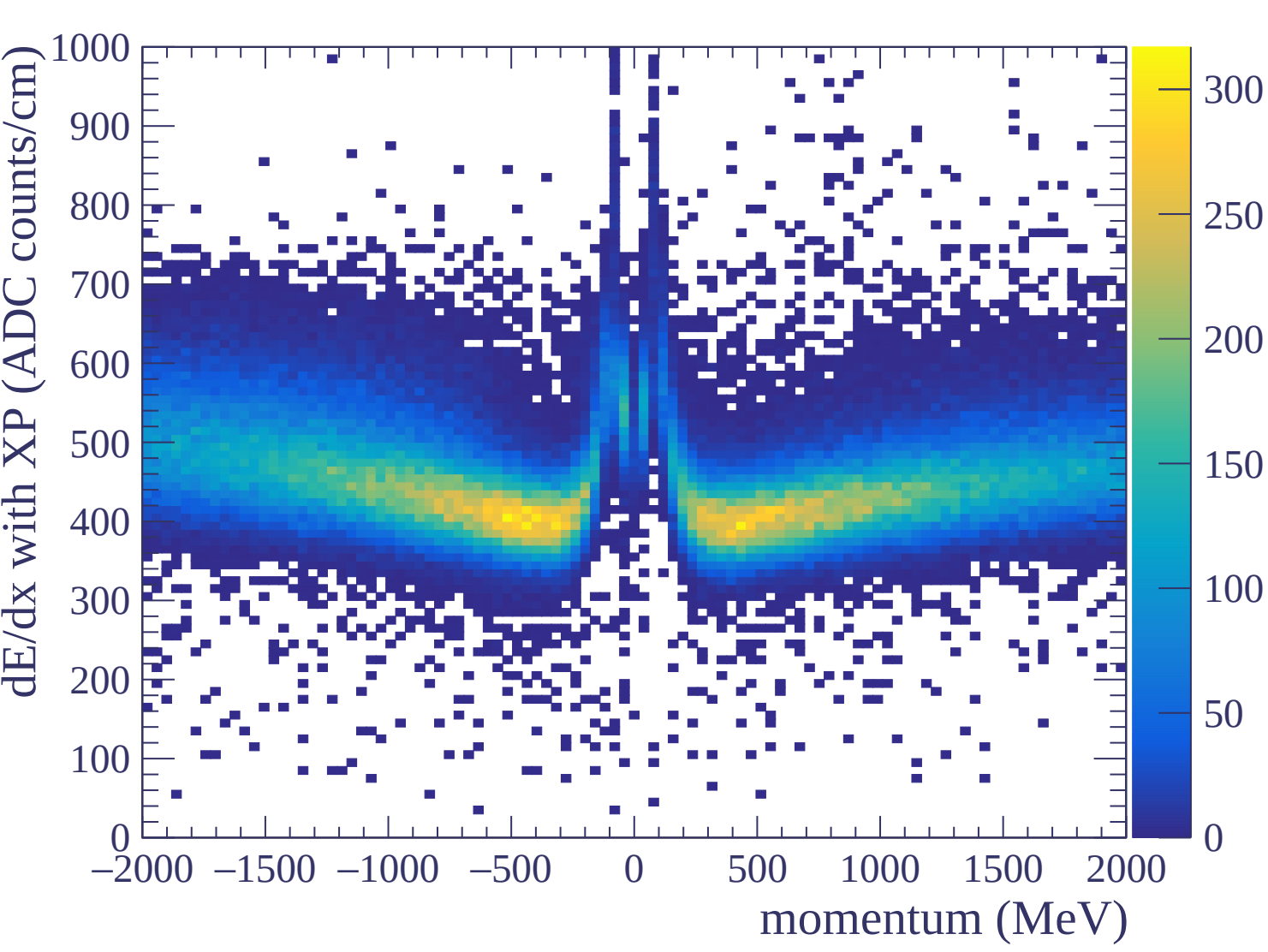
Resolution (%)

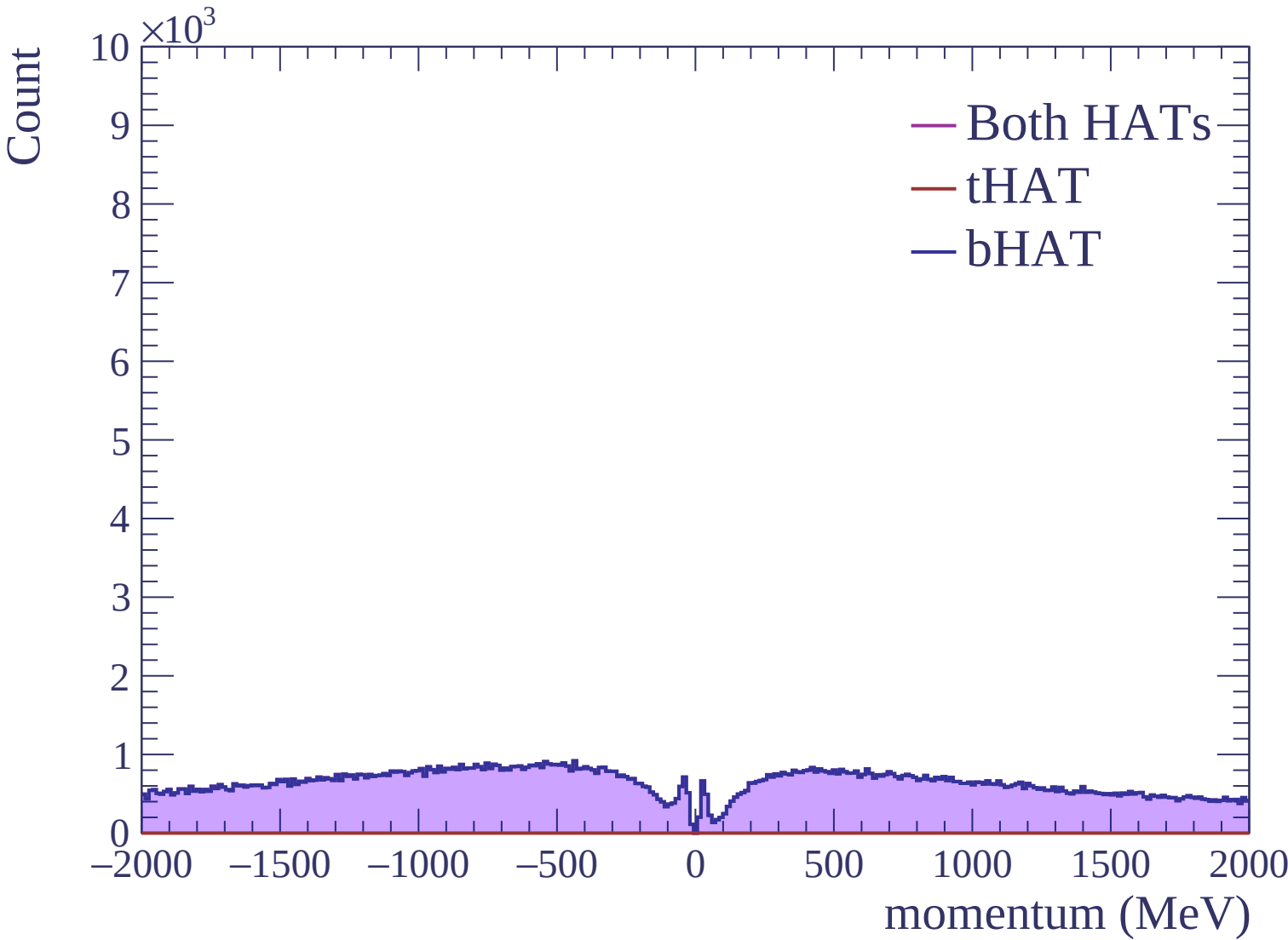






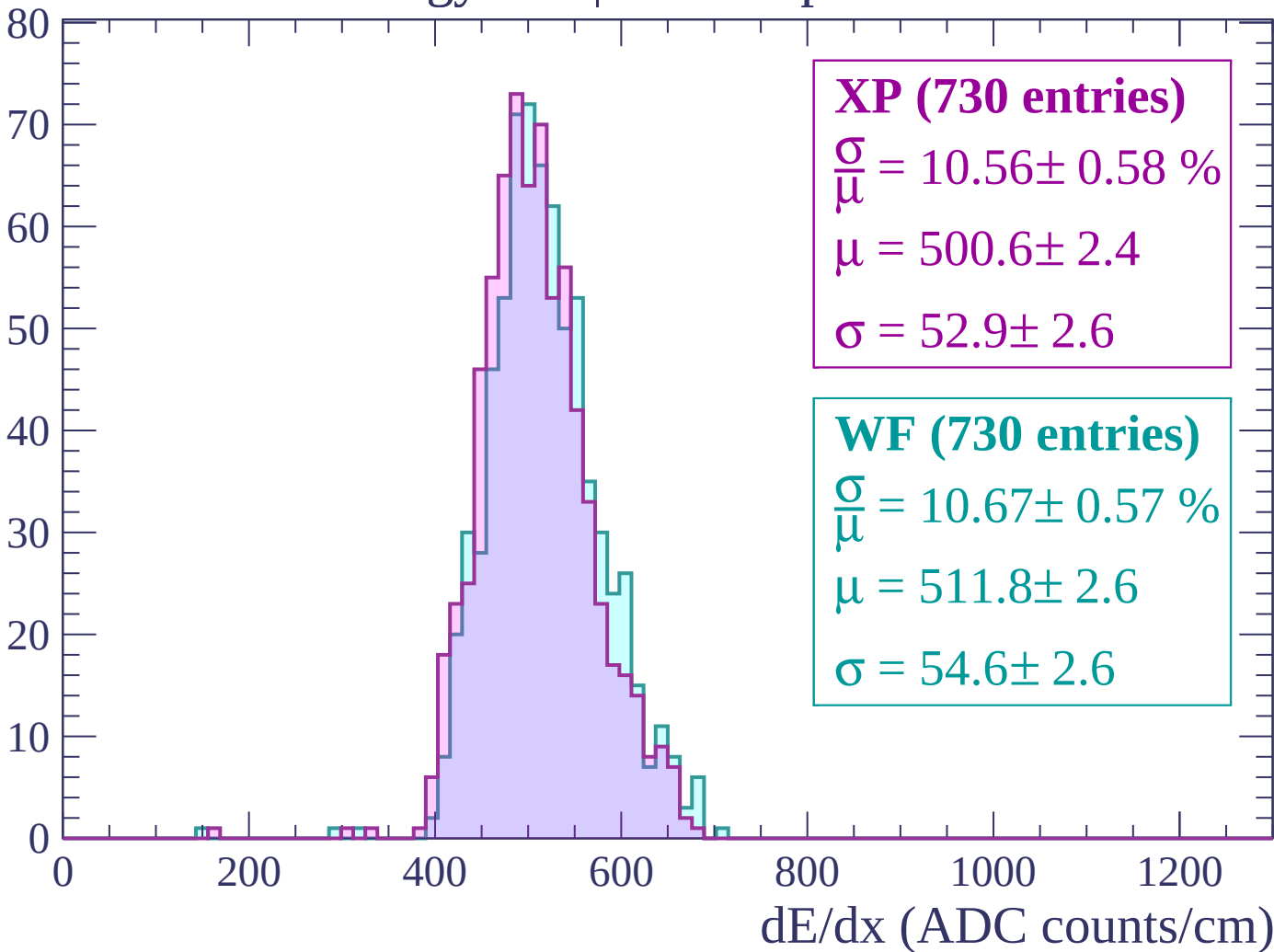




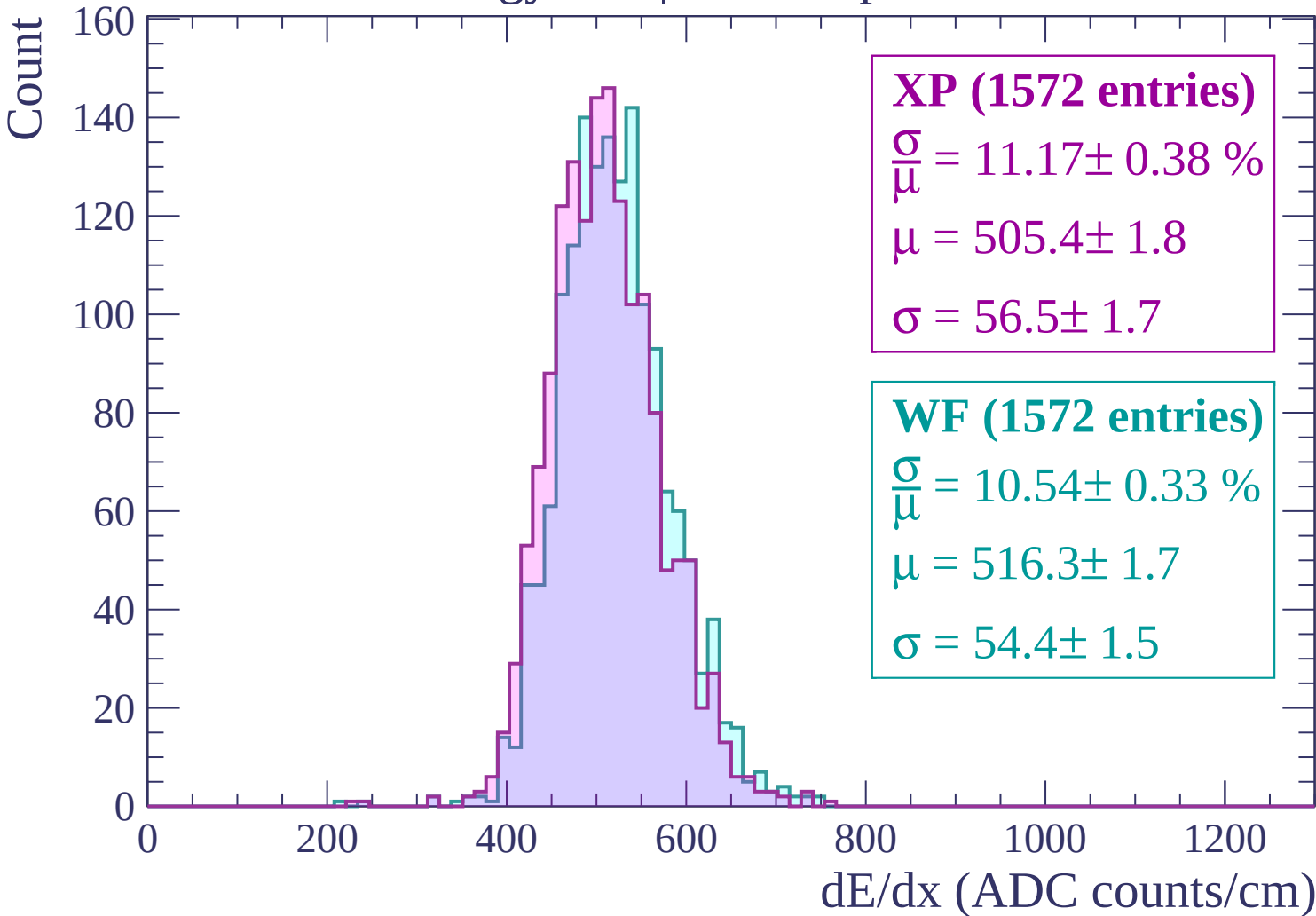


Energy loss | $-2000 < p < -1960$

Count

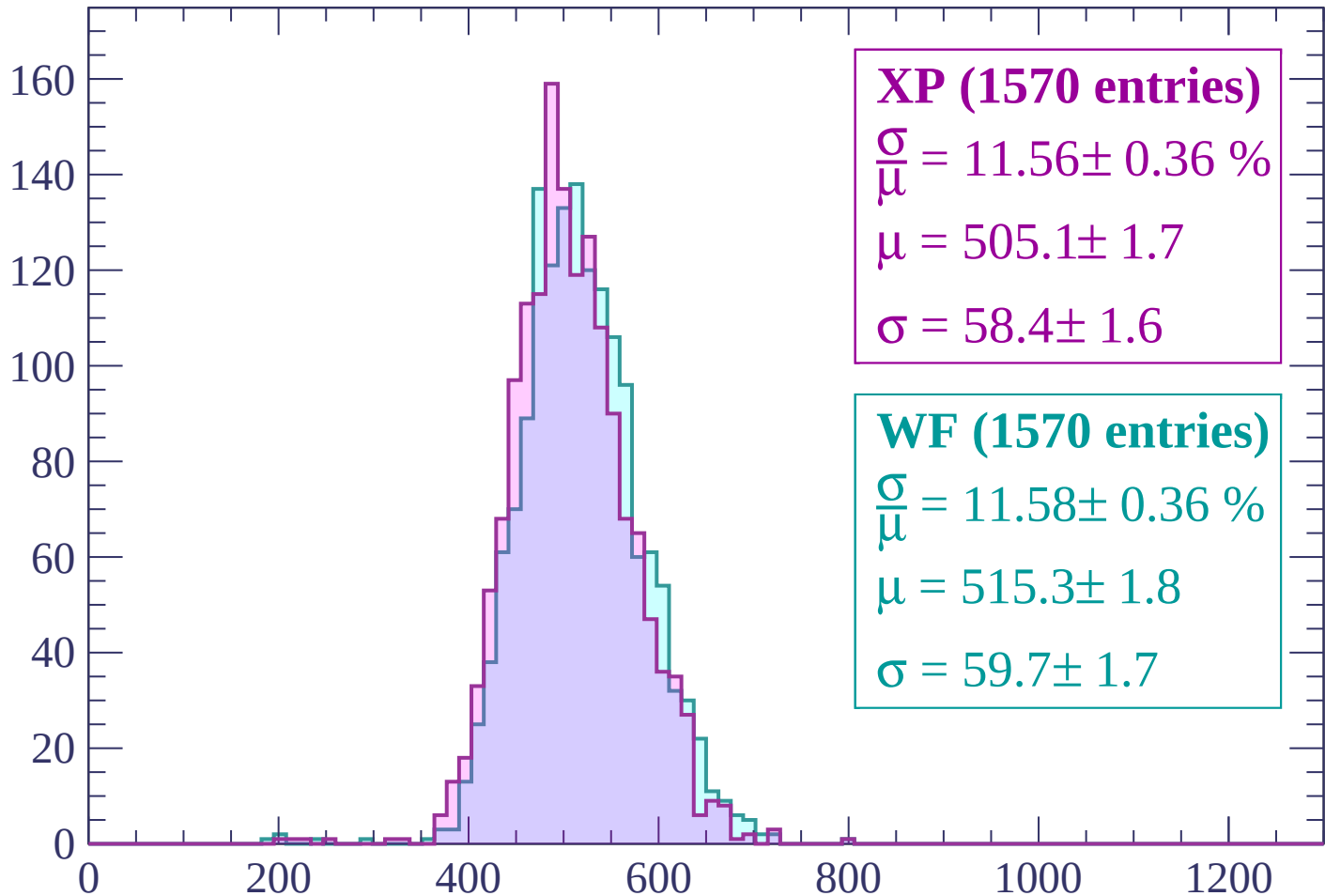


Energy loss | $-1960 < p < -1920$



Energy loss | -1920 < p < -1880

Count



XP (1570 entries)

$\frac{\sigma}{\mu} = 11.56 \pm 0.36 \%$

$\mu = 505.1 \pm 1.7$

$\sigma = 58.4 \pm 1.6$

WF (1570 entries)

$\frac{\sigma}{\mu} = 11.58 \pm 0.36 \%$

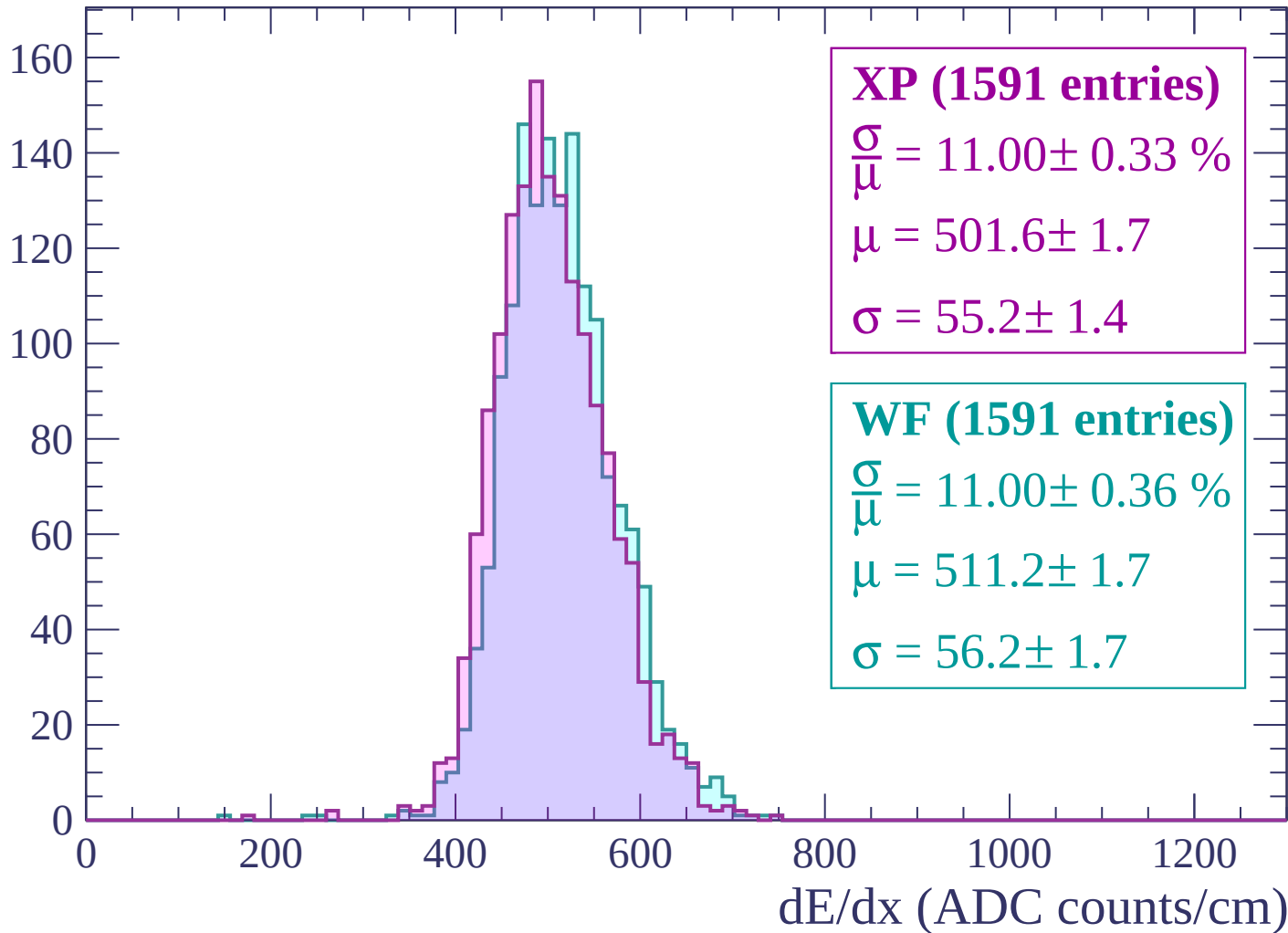
$\mu = 515.3 \pm 1.8$

$\sigma = 59.7 \pm 1.7$

dE/dx (ADC counts/cm)

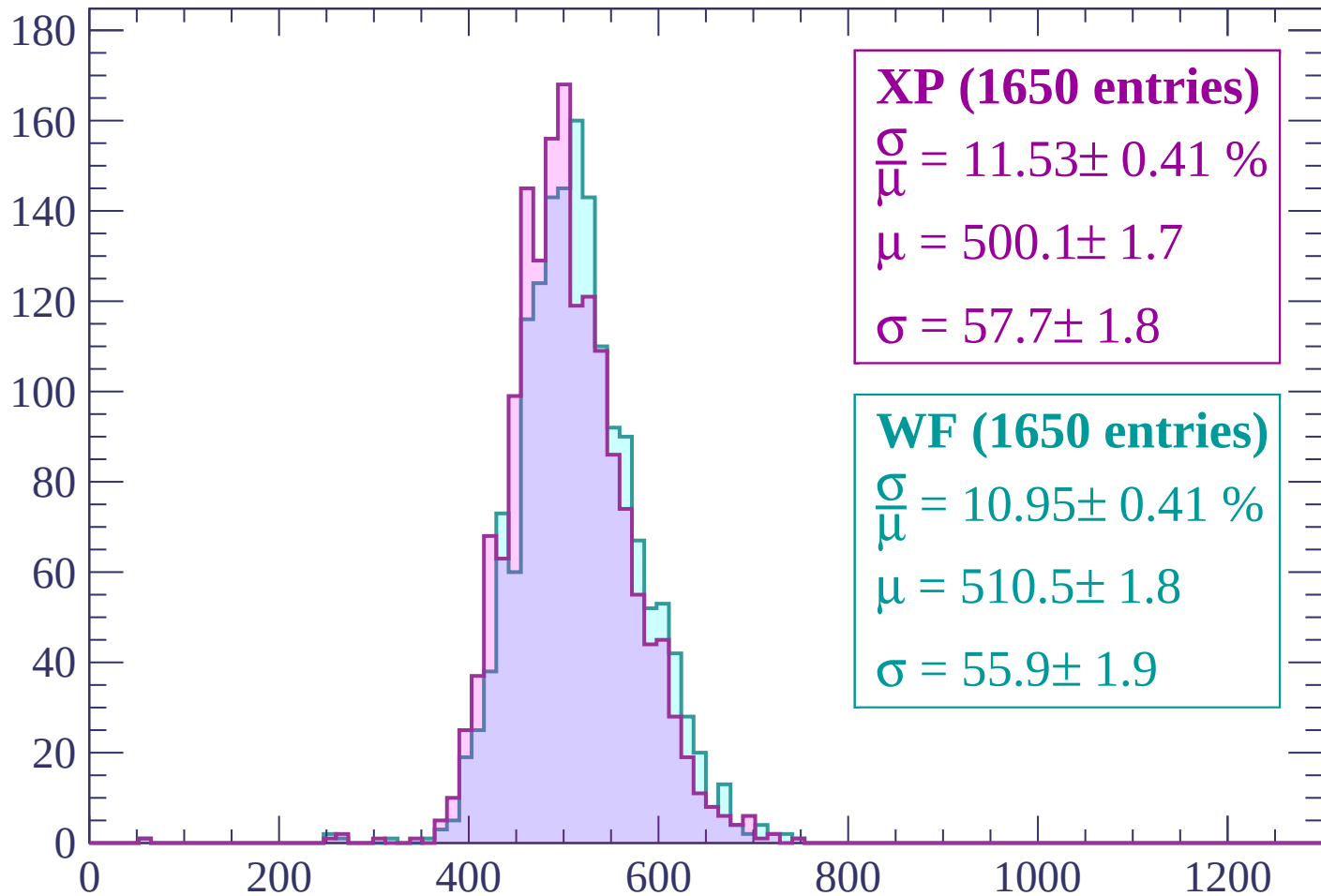
Energy loss | $-1880 < p < -1840$

Count



Energy loss | -1840 < p < -1800

Count



XP (1650 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.41 \%$$

$$\mu = 500.1 \pm 1.7$$

$$\sigma = 57.7 \pm 1.8$$

WF (1650 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.41 \%$$

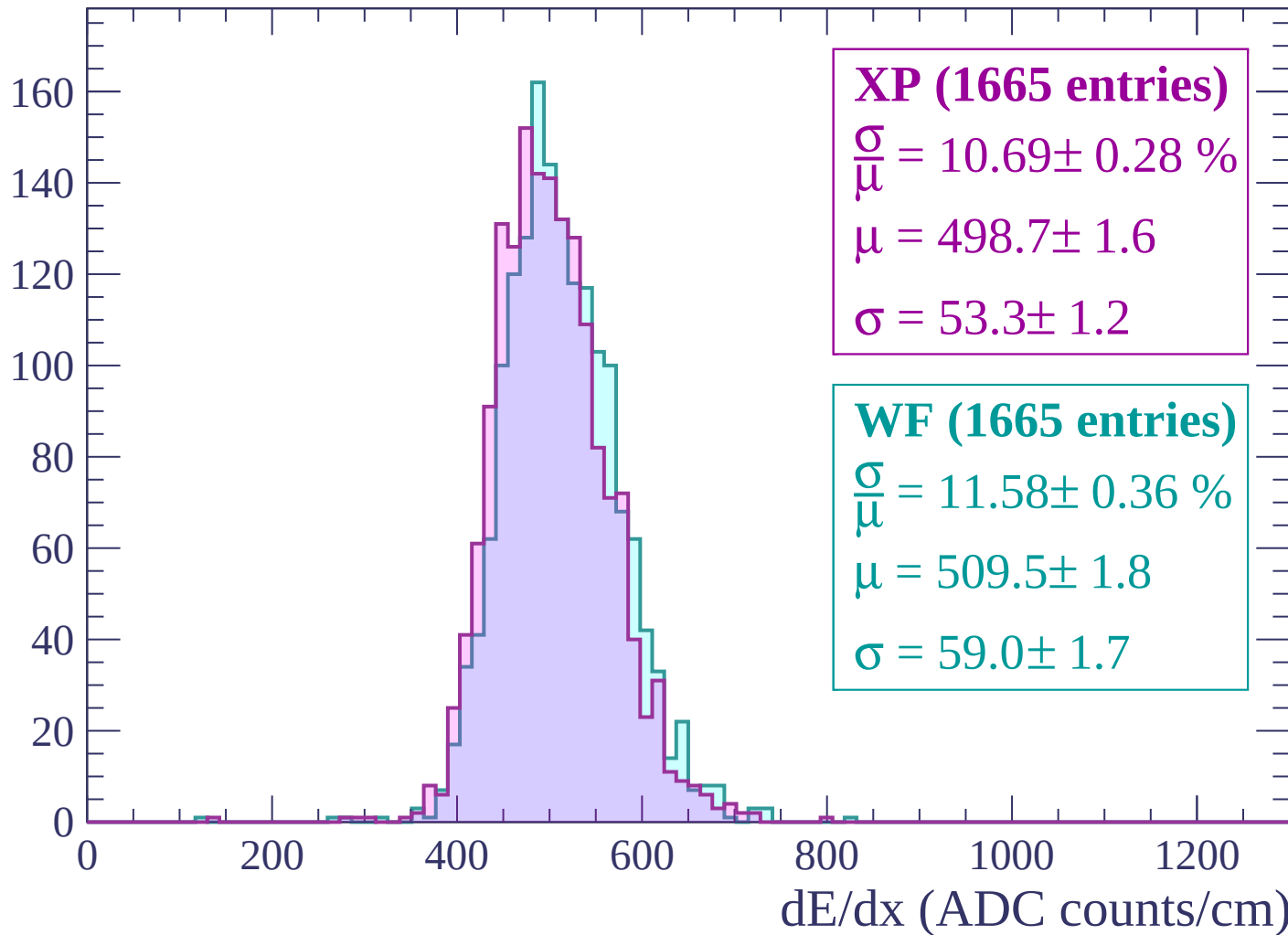
$$\mu = 510.5 \pm 1.8$$

$$\sigma = 55.9 \pm 1.9$$

dE/dx (ADC counts/cm)

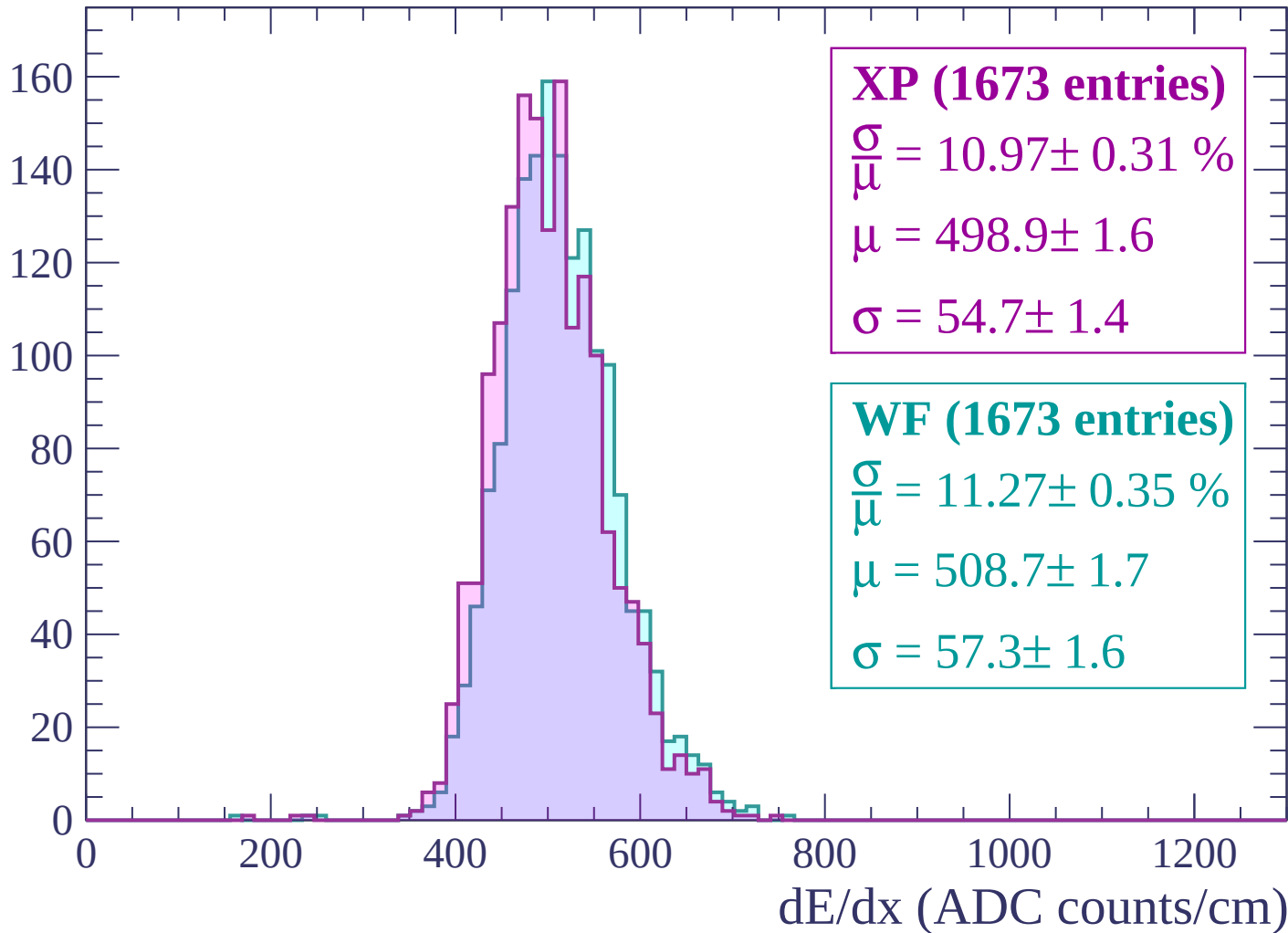
Energy loss | $-1800 < p < -1760$

Count



Energy loss | $-1760 < p < -1720$

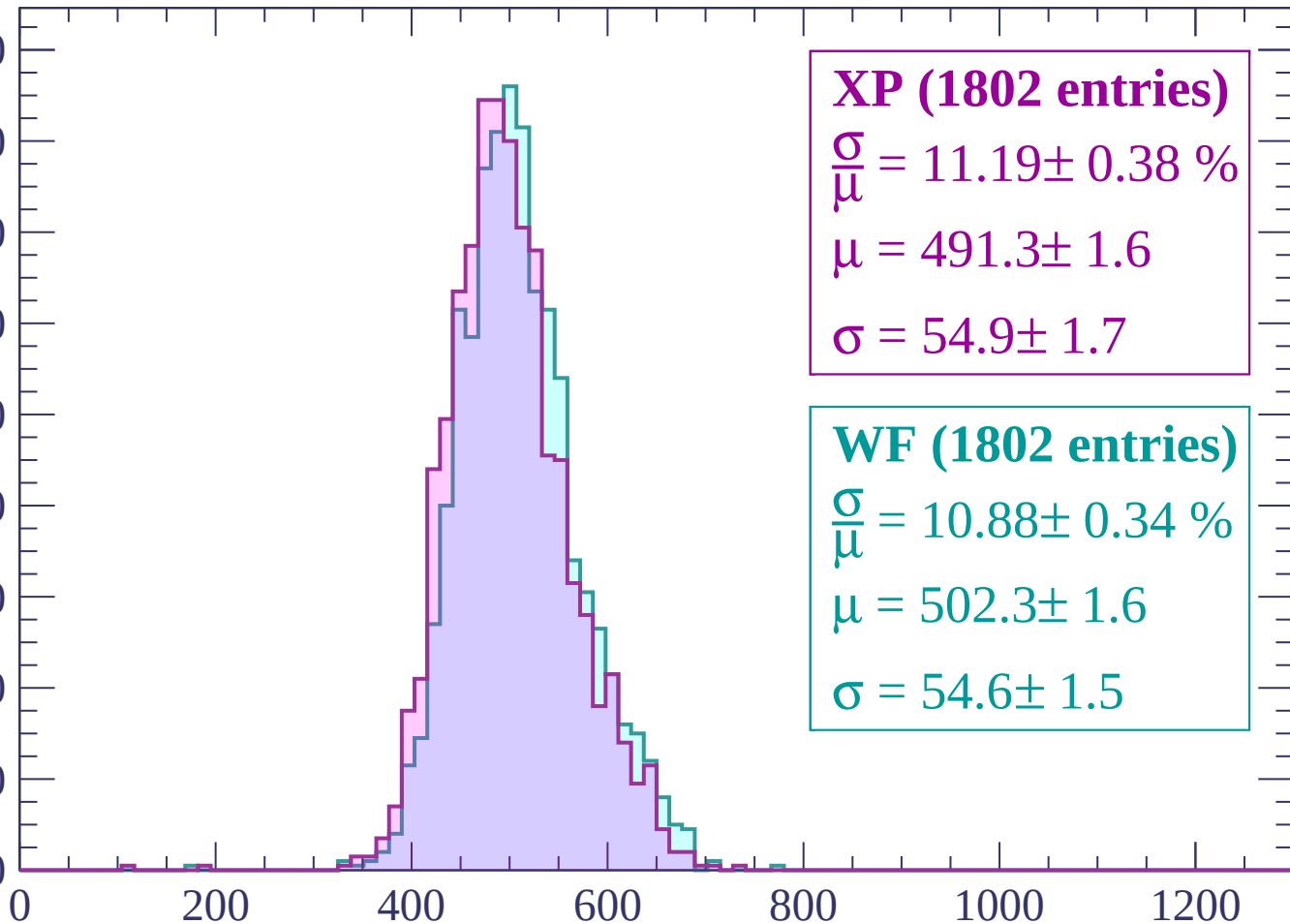
Count



Energy loss | -1720 < p < -1680

Count

180
160
140
120
100
80
60
40
20
0



XP (1802 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.38 \%$$

$$\mu = 491.3 \pm 1.6$$

$$\sigma = 54.9 \pm 1.7$$

WF (1802 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.34 \%$$

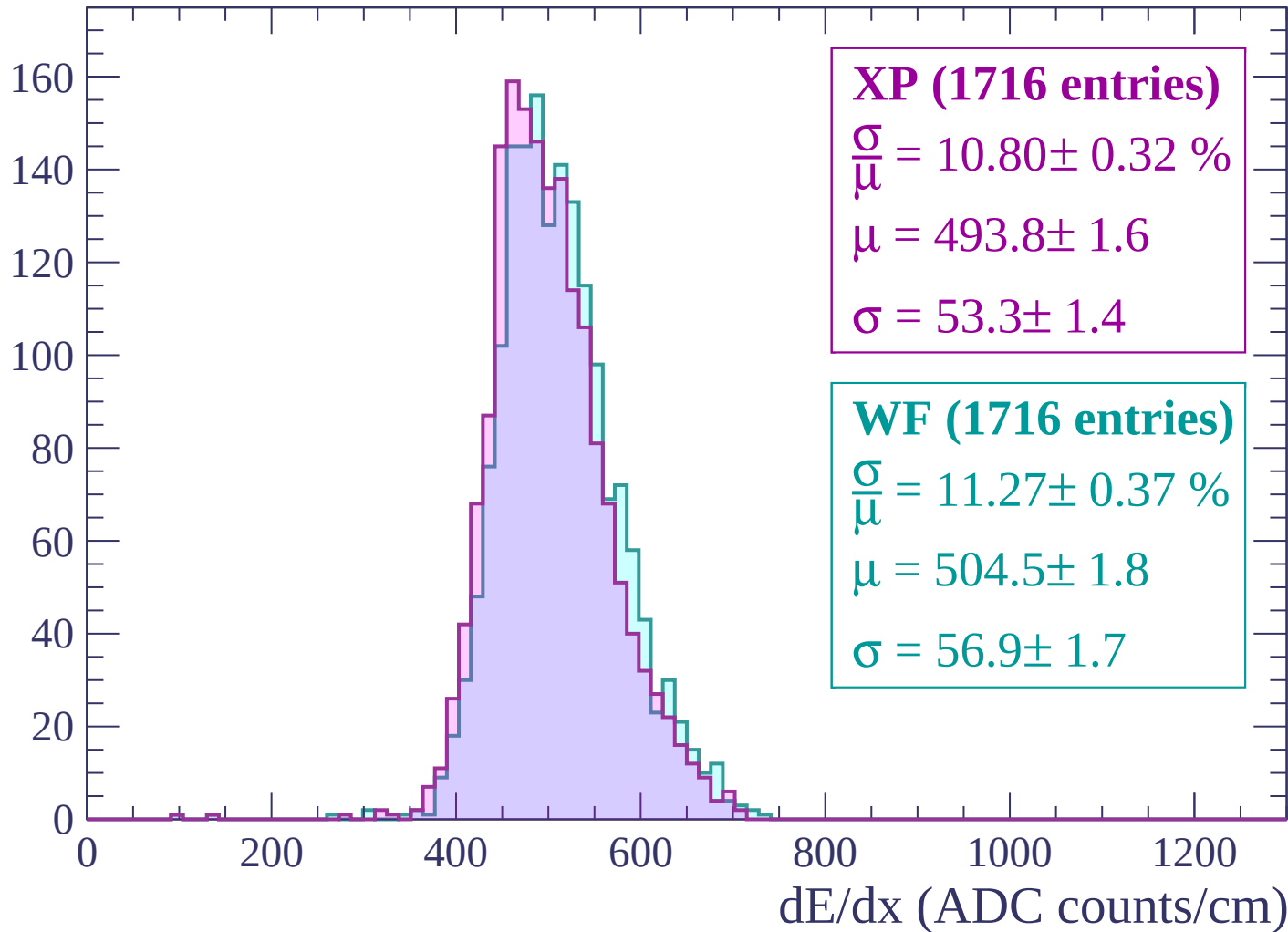
$$\mu = 502.3 \pm 1.6$$

$$\sigma = 54.6 \pm 1.5$$

dE/dx (ADC counts/cm)

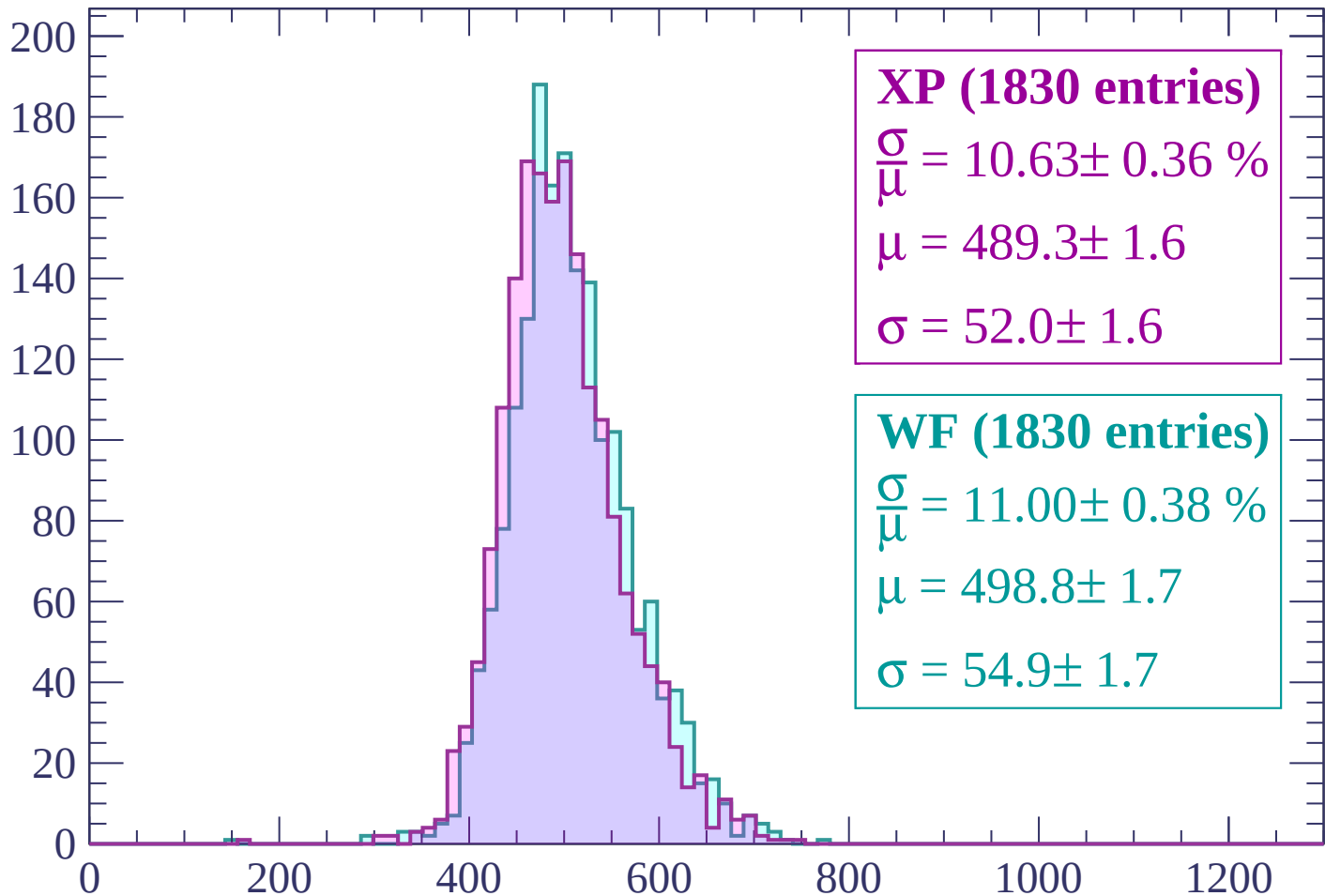
Energy loss | -1680 < p < -1640

Count



Energy loss | -1640 < p < -1600

Count



XP (1830 entries)

$$\frac{\sigma}{\mu} = 10.63 \pm 0.36 \%$$

$$\mu = 489.3 \pm 1.6$$

$$\sigma = 52.0 \pm 1.6$$

WF (1830 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.38 \%$$

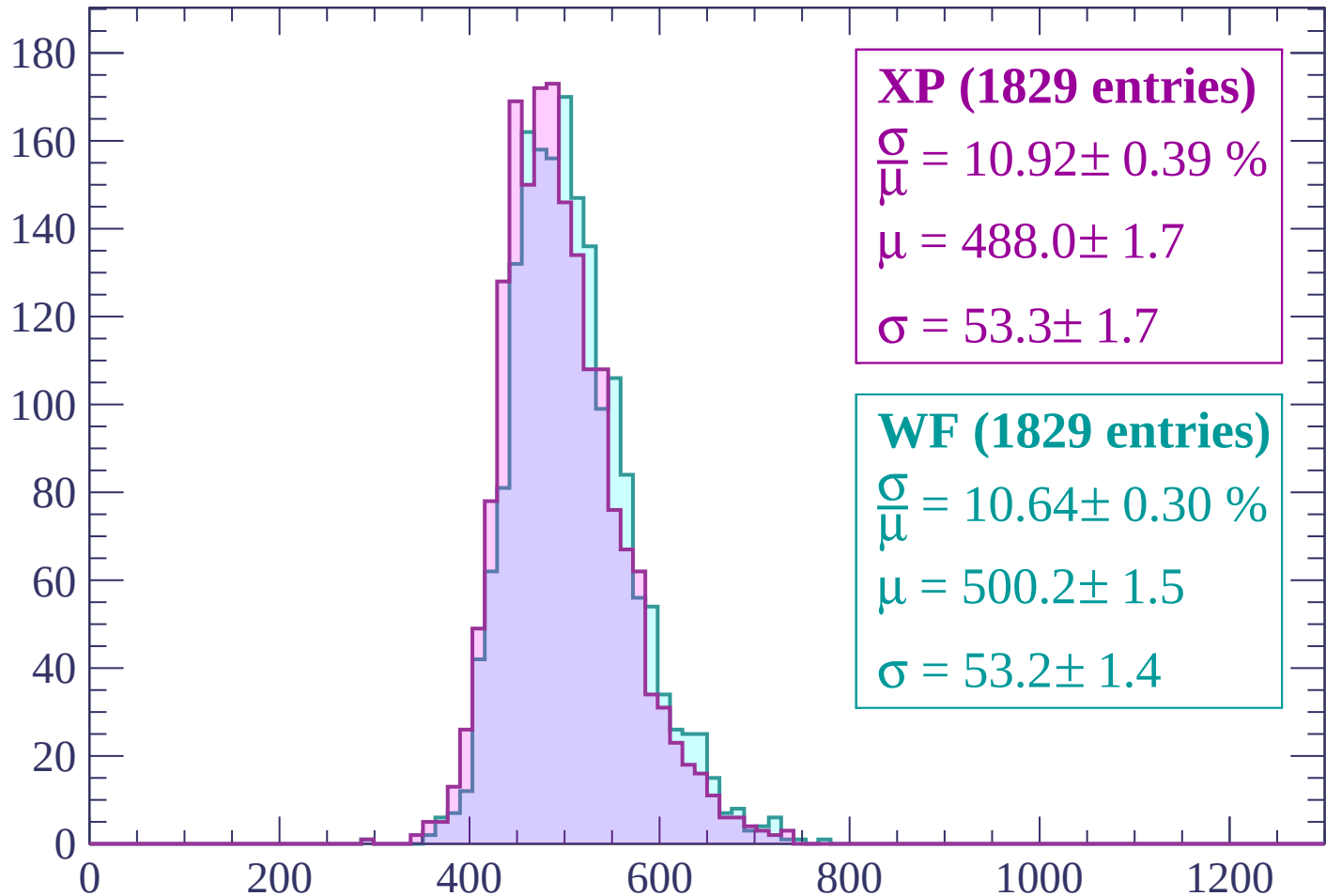
$$\mu = 498.8 \pm 1.7$$

$$\sigma = 54.9 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

Count



XP (1829 entries)

$$\frac{\sigma}{\mu} = 10.92 \pm 0.39 \%$$

$$\mu = 488.0 \pm 1.7$$

$$\sigma = 53.3 \pm 1.7$$

WF (1829 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.30 \%$$

$$\mu = 500.2 \pm 1.5$$

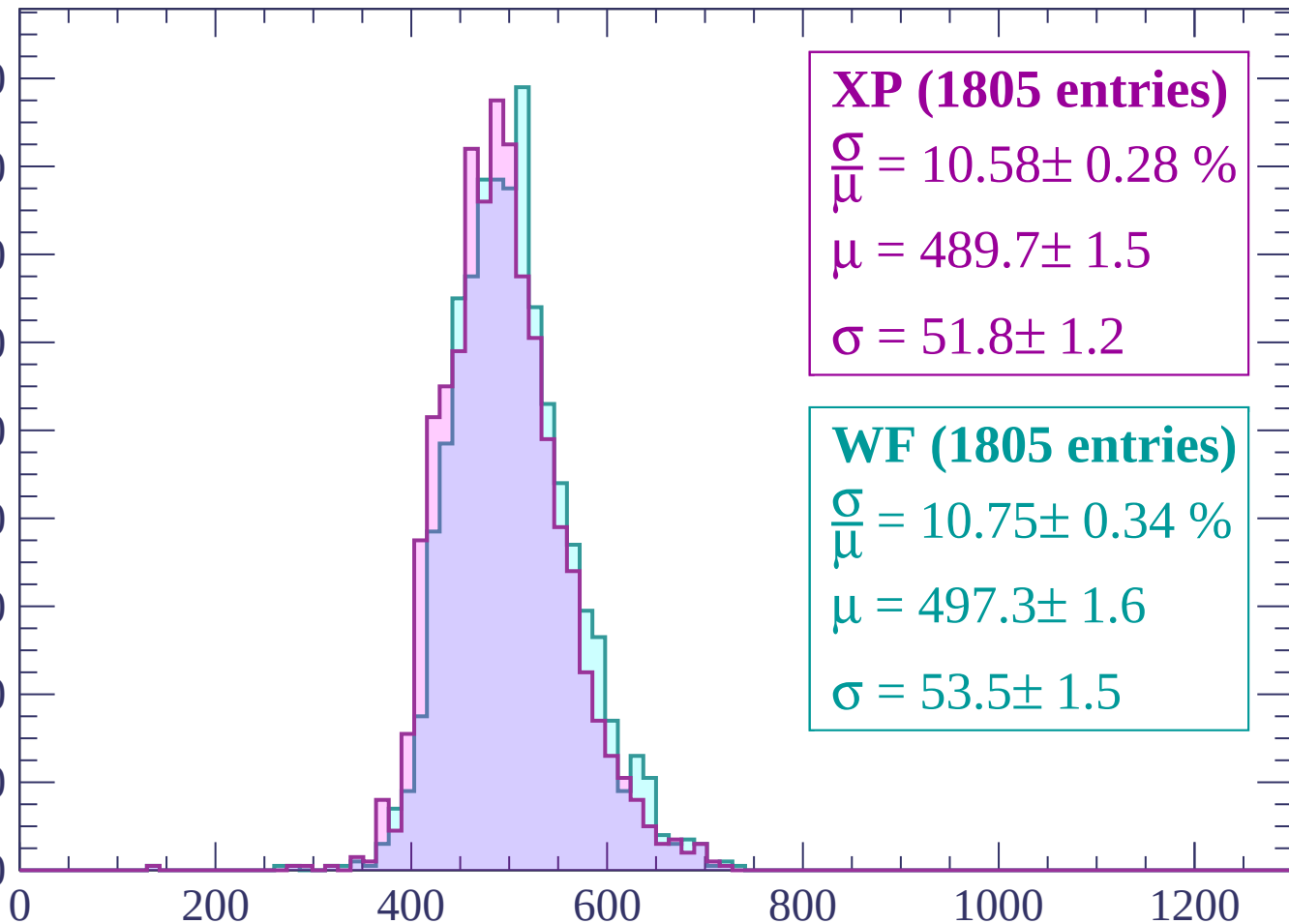
$$\sigma = 53.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1520

Count

180
160
140
120
100
80
60
40
20
0



XP (1805 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.28 \%$

$\mu = 489.7 \pm 1.5$

$\sigma = 51.8 \pm 1.2$

WF (1805 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.34 \%$

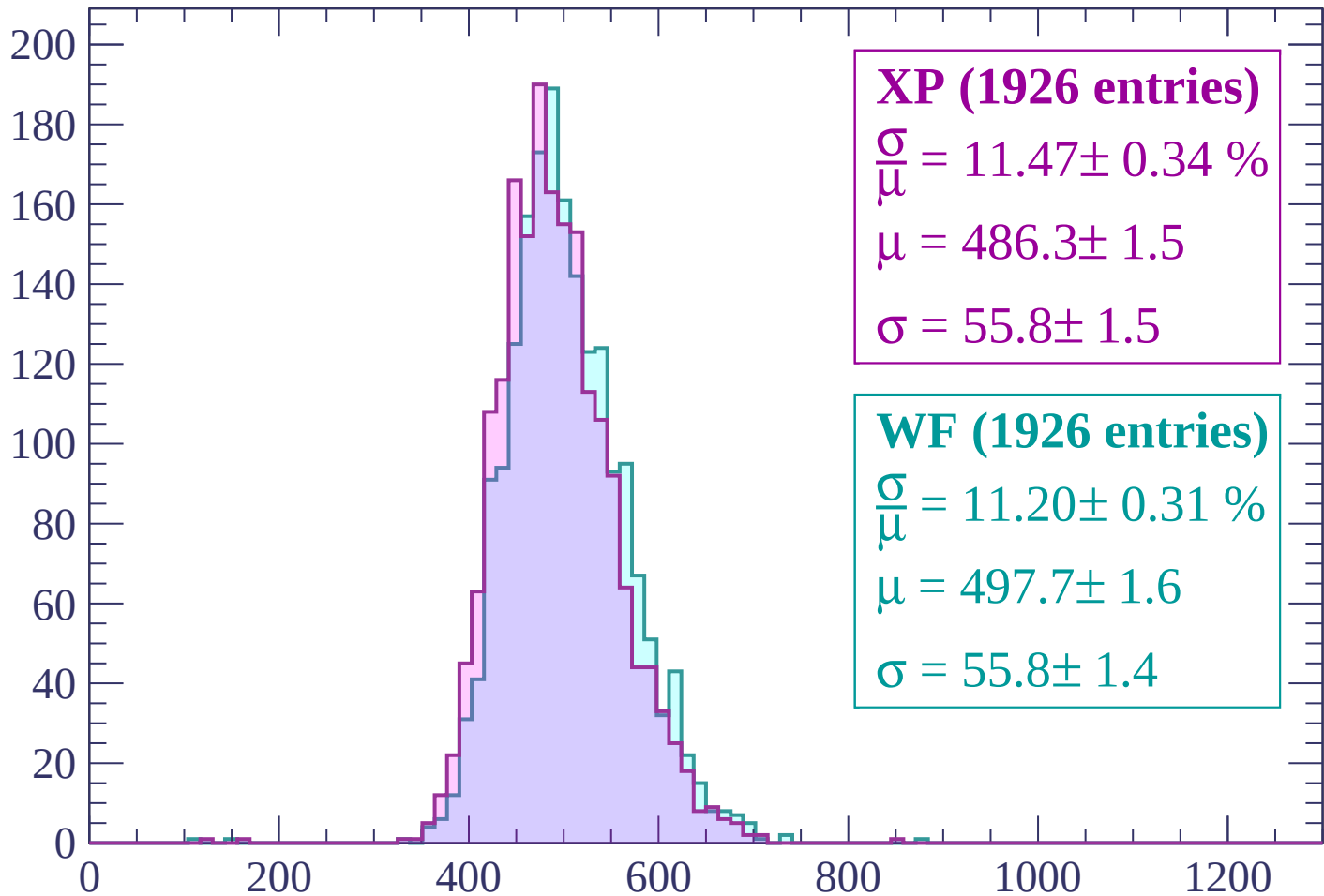
$\mu = 497.3 \pm 1.6$

$\sigma = 53.5 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | -1520 < p < -1480

Count



XP (1926 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.34 \%$$

$$\mu = 486.3 \pm 1.5$$

$$\sigma = 55.8 \pm 1.5$$

WF (1926 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.31 \%$$

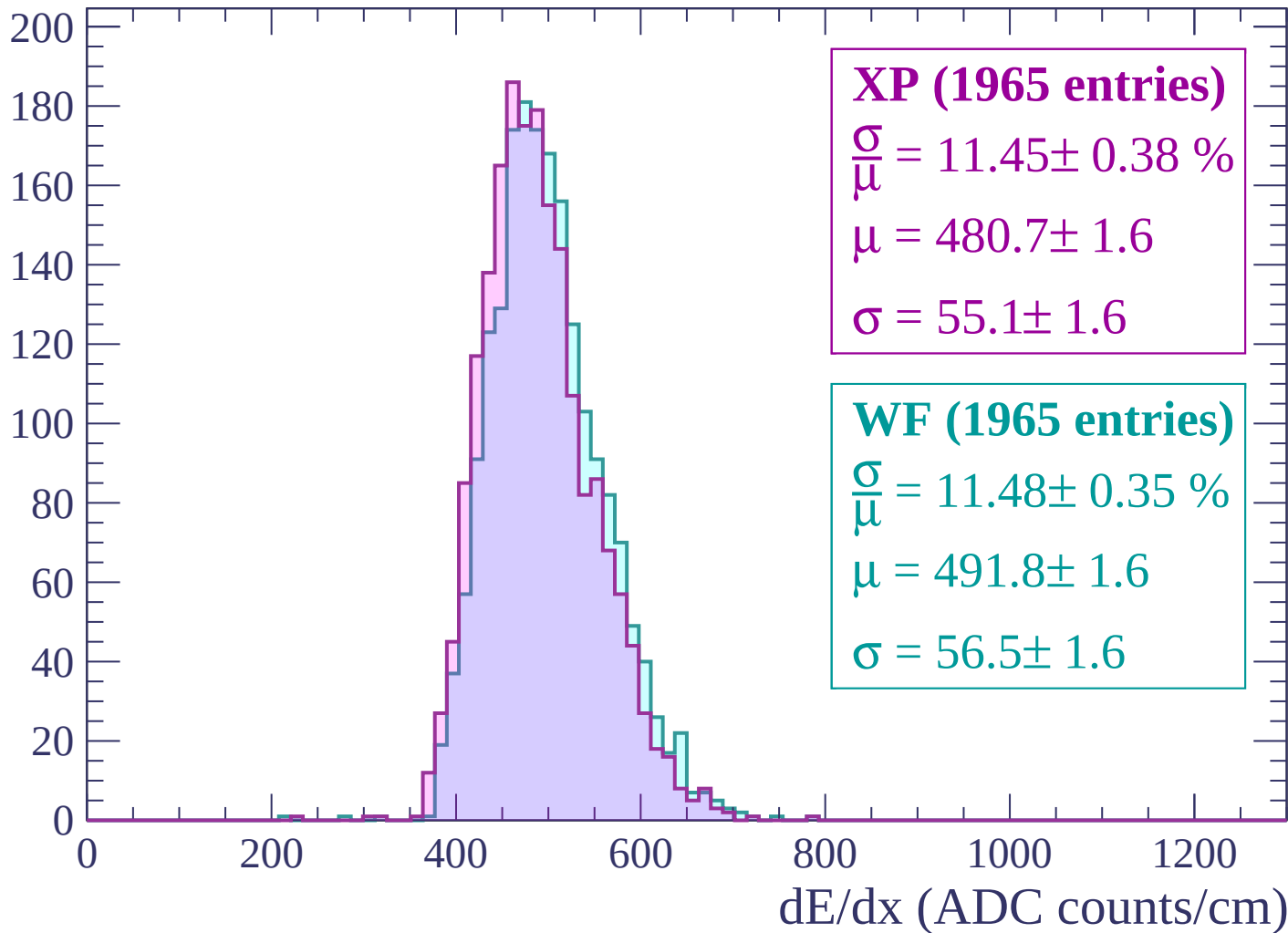
$$\mu = 497.7 \pm 1.6$$

$$\sigma = 55.8 \pm 1.4$$

dE/dx (ADC counts/cm)

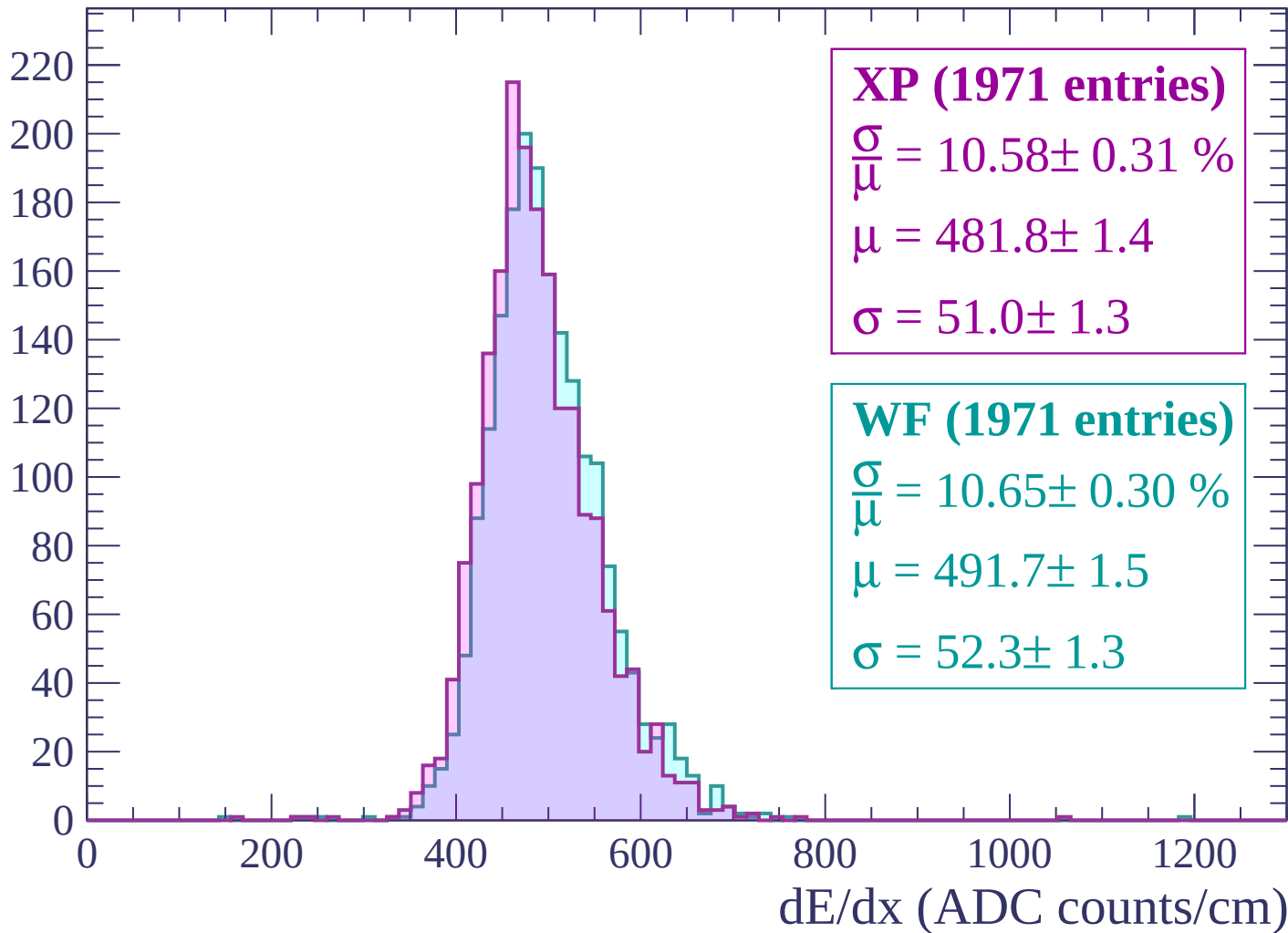
Energy loss | $-1480 < p < -1440$

Count



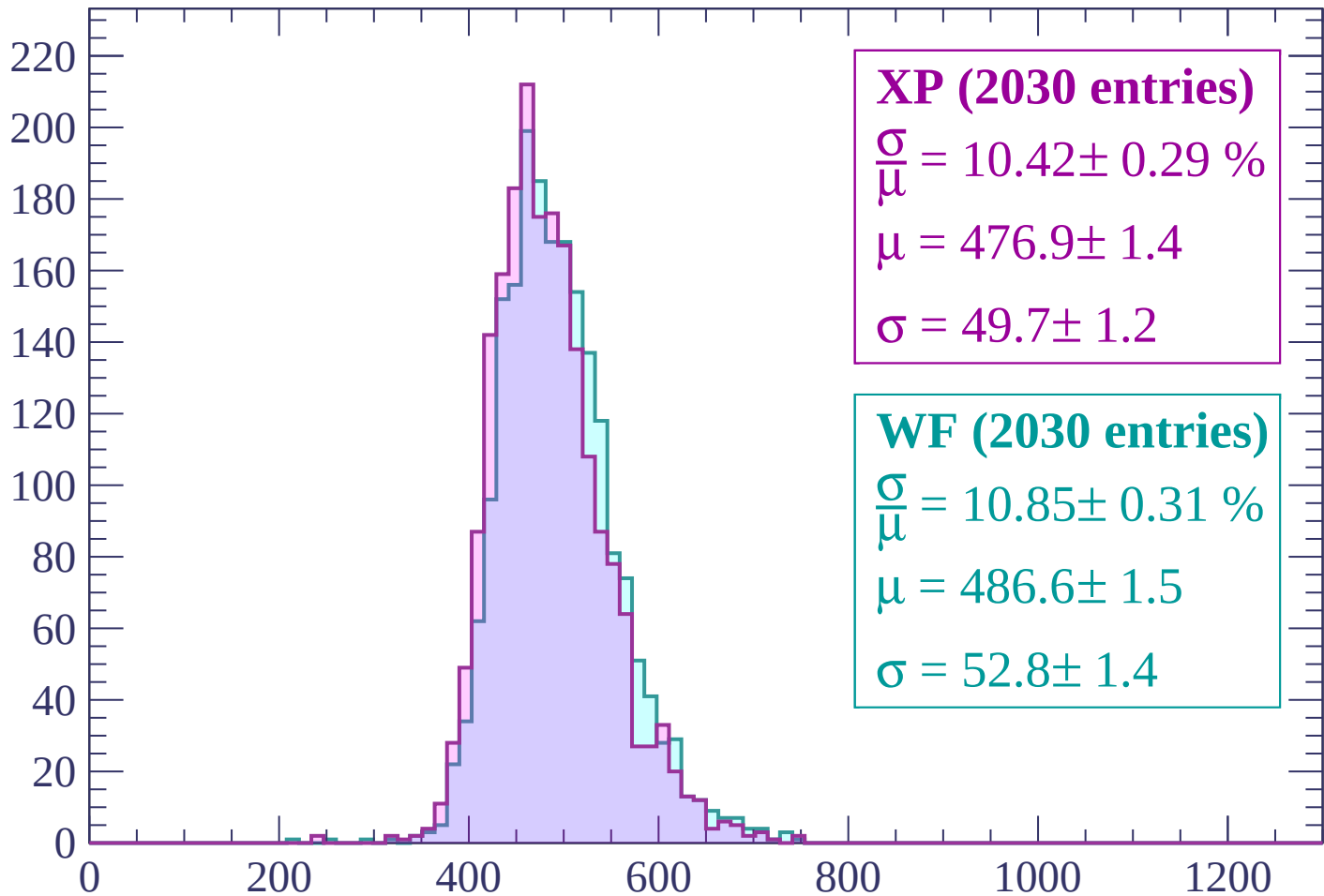
Energy loss | -1440 < p < -1400

Count



Energy loss | $-1400 < p < -1360$

Count



XP (2030 entries)

$$\frac{\sigma}{\mu} = 10.42 \pm 0.29 \%$$

$$\mu = 476.9 \pm 1.4$$

$$\sigma = 49.7 \pm 1.2$$

WF (2030 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.31 \%$$

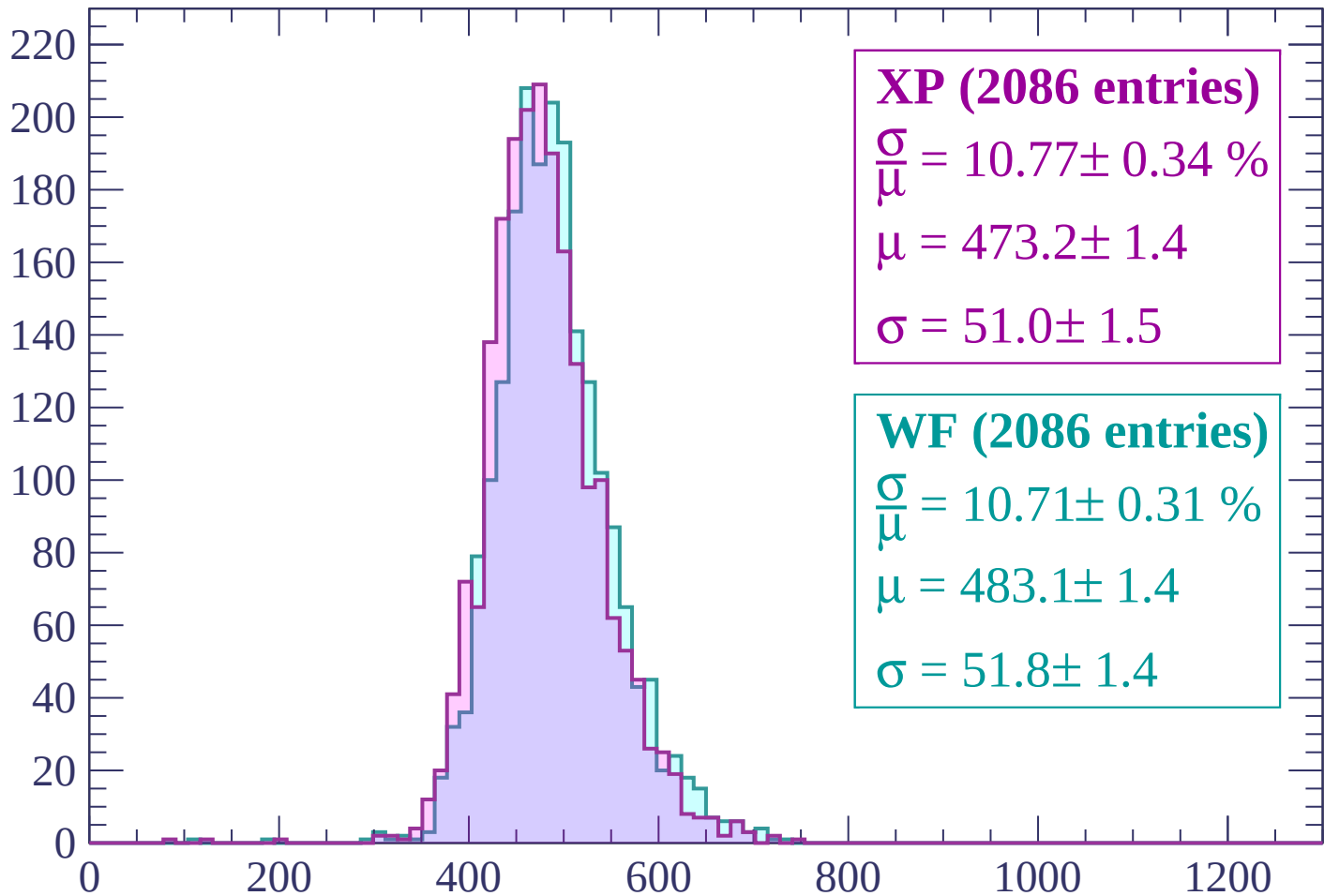
$$\mu = 486.6 \pm 1.5$$

$$\sigma = 52.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1360 < p < -1320

Count



XP (2086 entries)

$$\frac{\sigma}{\mu} = 10.77 \pm 0.34 \%$$

$$\mu = 473.2 \pm 1.4$$

$$\sigma = 51.0 \pm 1.5$$

WF (2086 entries)

$$\frac{\sigma}{\mu} = 10.71 \pm 0.31 \%$$

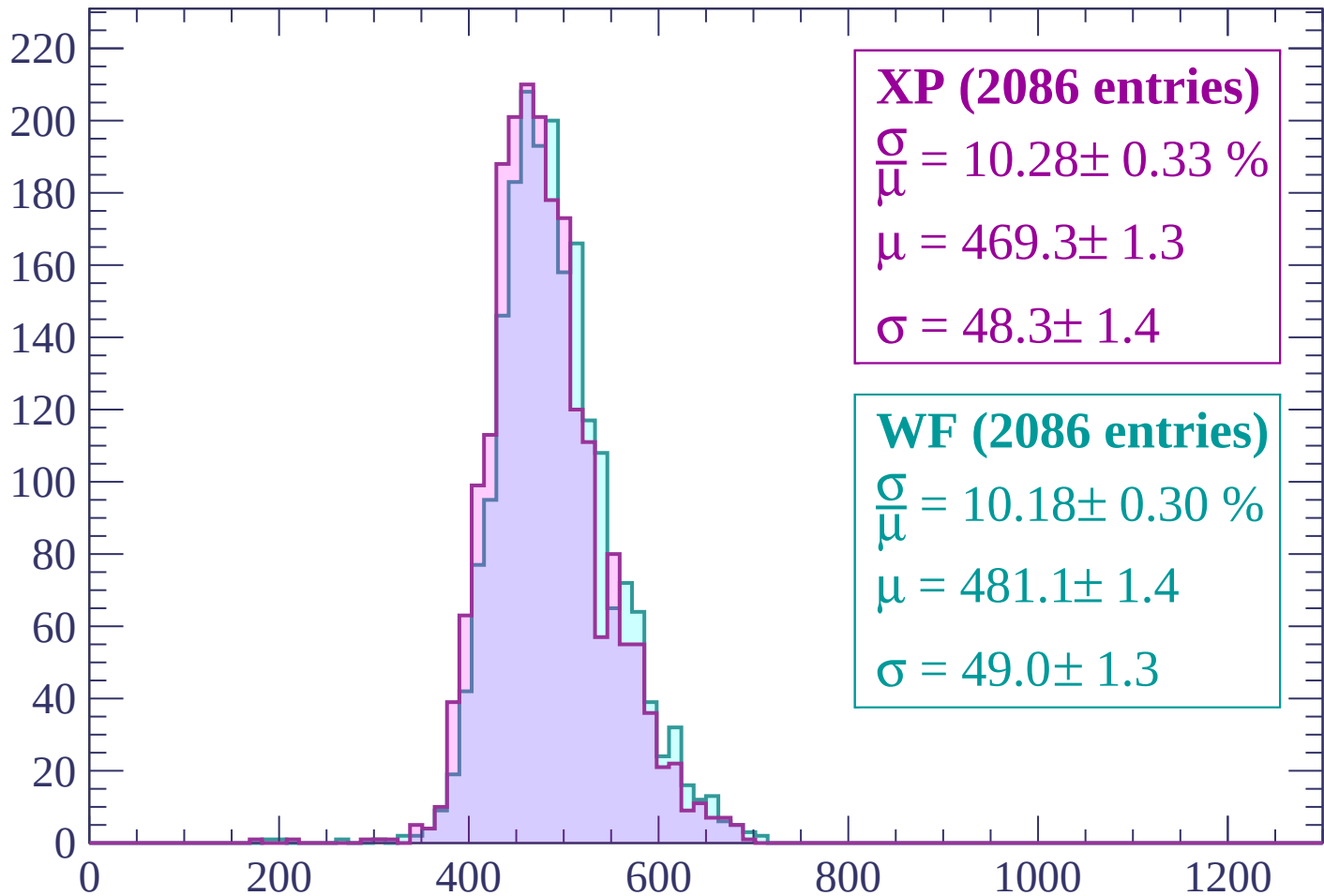
$$\mu = 483.1 \pm 1.4$$

$$\sigma = 51.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280

Count



XP (2086 entries)

$$\frac{\sigma}{\mu} = 10.28 \pm 0.33 \%$$

$$\mu = 469.3 \pm 1.3$$

$$\sigma = 48.3 \pm 1.4$$

WF (2086 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.30 \%$$

$$\mu = 481.1 \pm 1.4$$

$$\sigma = 49.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1280 < p < -1240

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (2184 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.28 \%$

$\mu = 474.1 \pm 1.3$

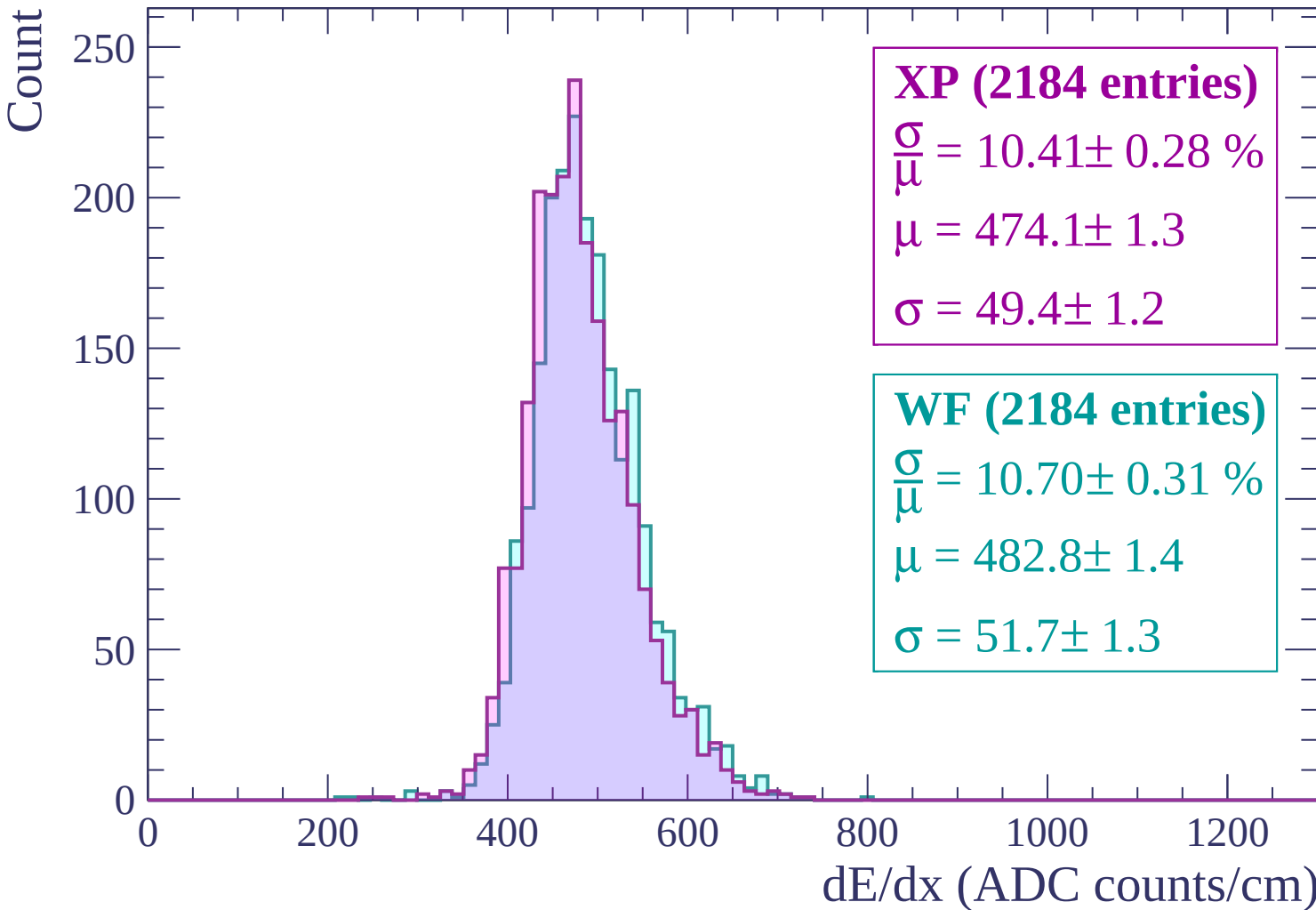
$\sigma = 49.4 \pm 1.2$

WF (2184 entries)

$\frac{\sigma}{\mu} = 10.70 \pm 0.31 \%$

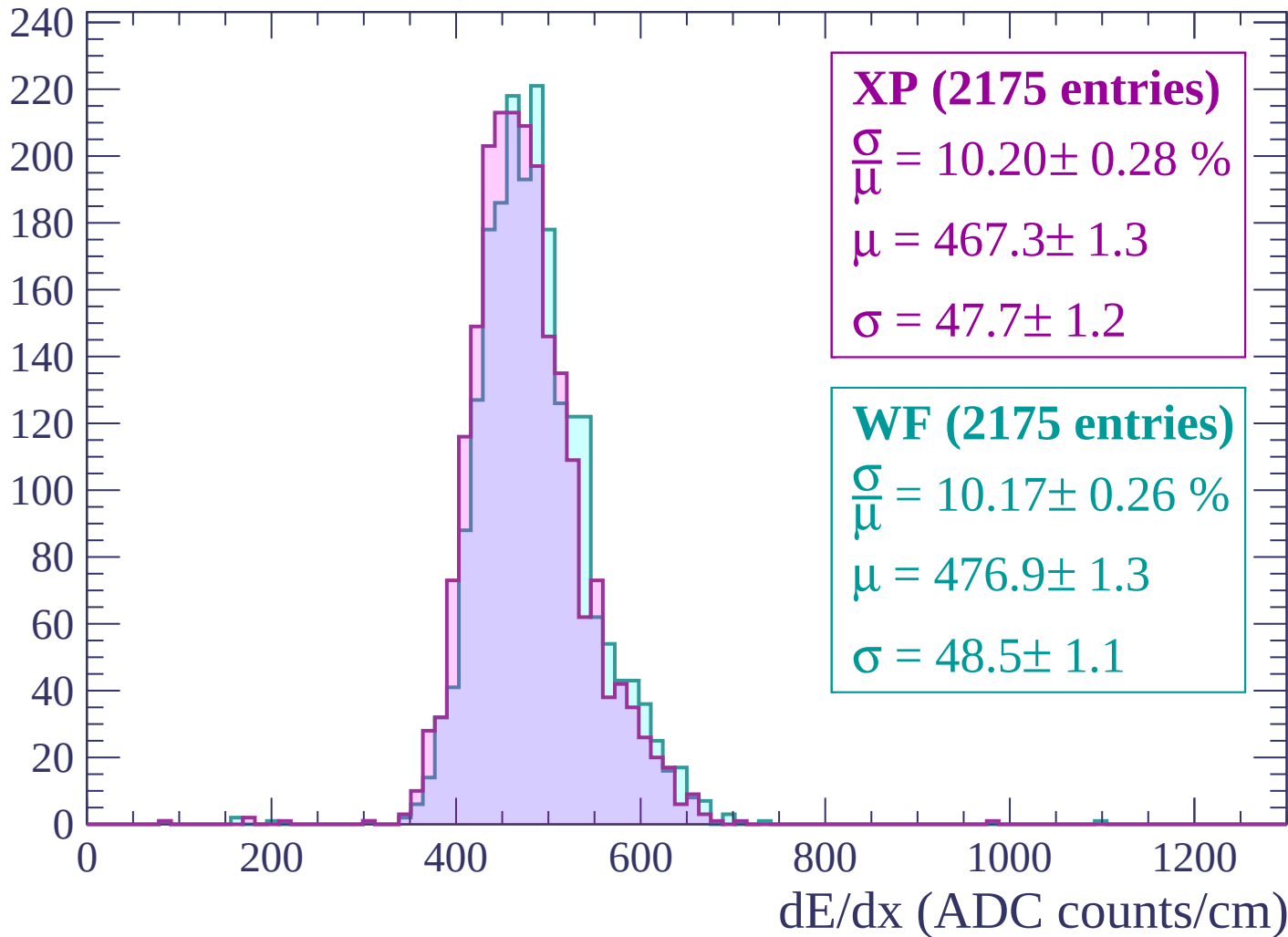
$\mu = 482.8 \pm 1.4$

$\sigma = 51.7 \pm 1.3$



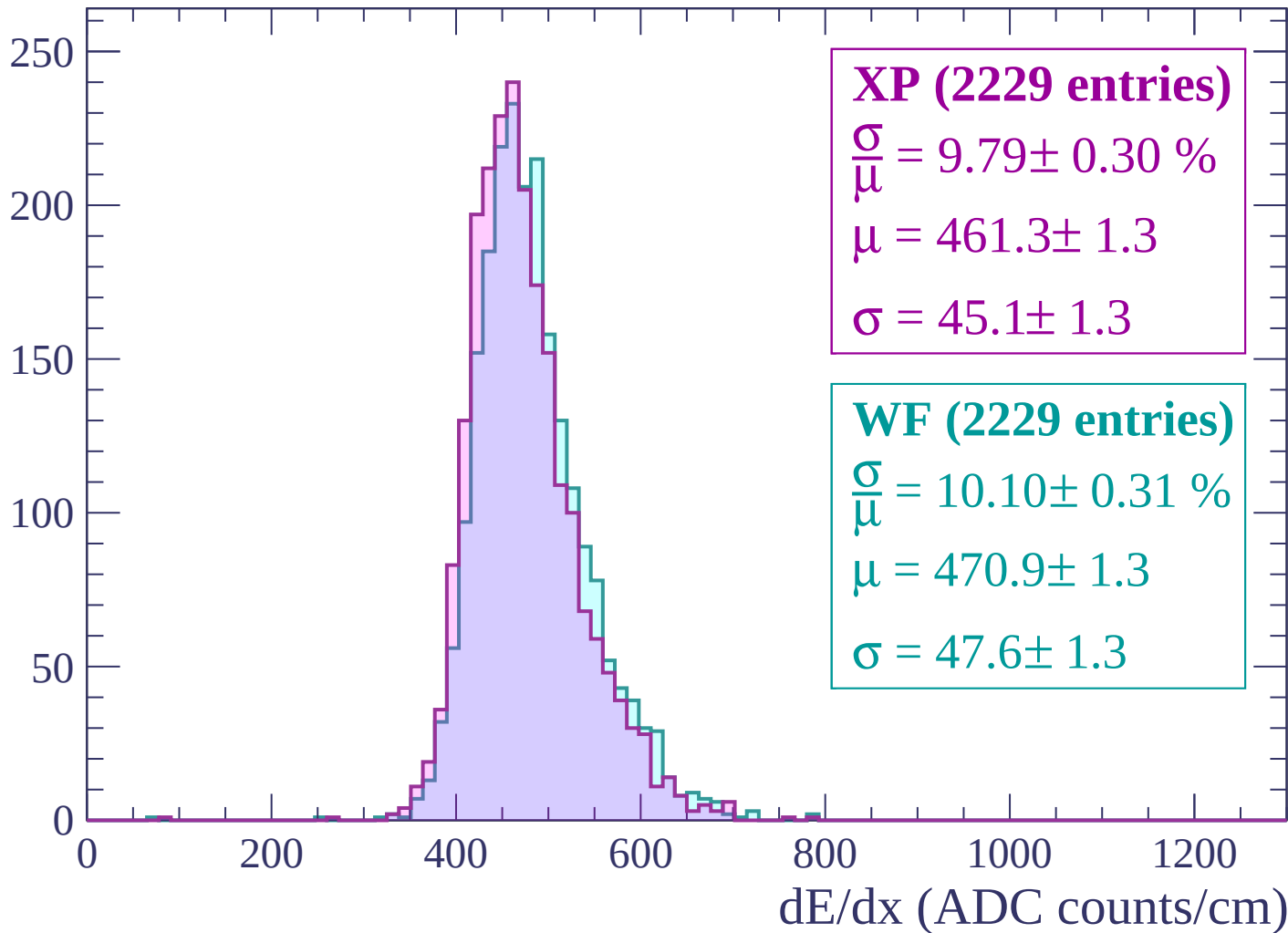
Energy loss | -1240 < p < -1200

Count



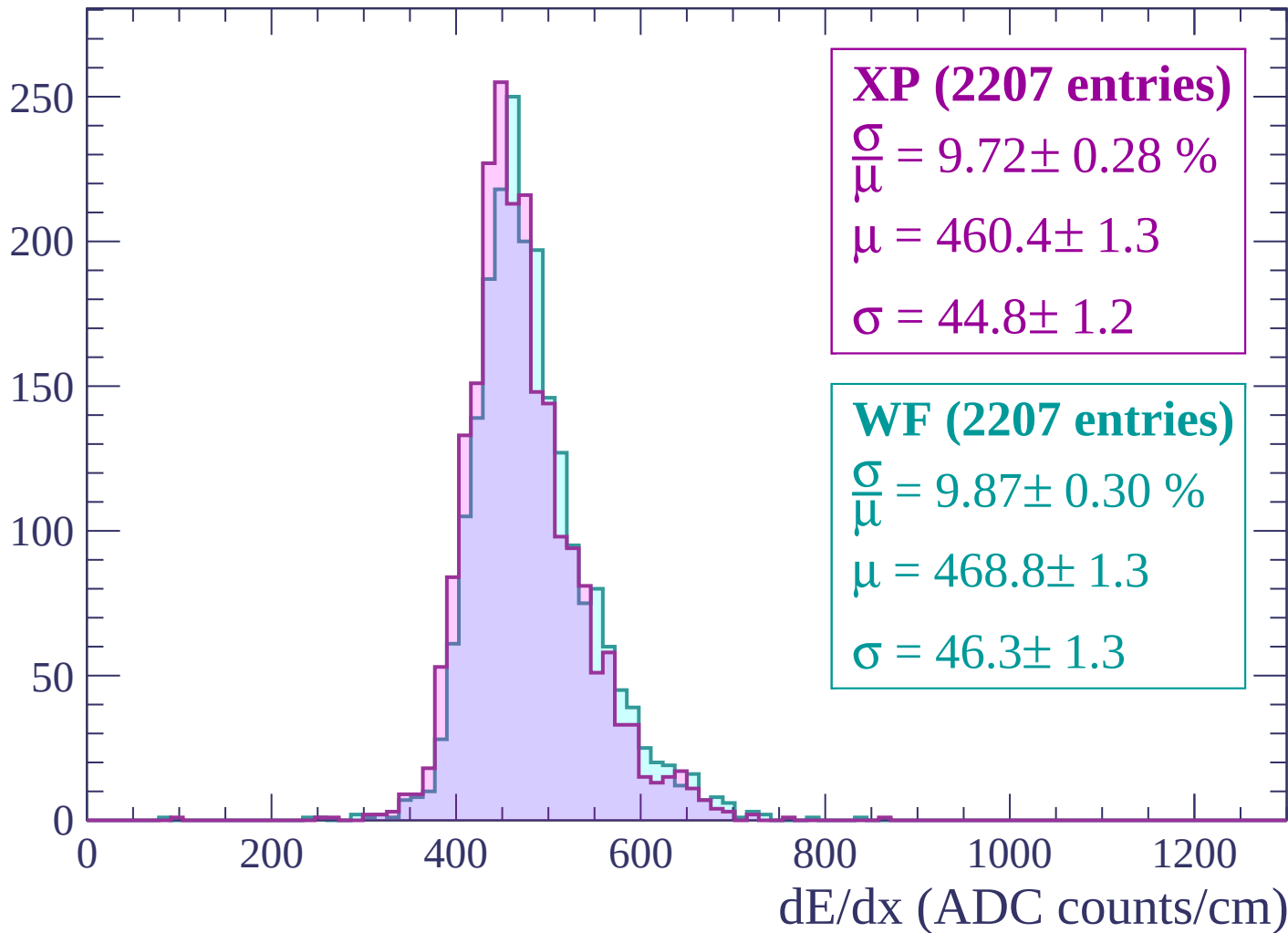
Energy loss | -1200 < p < -1160

Count



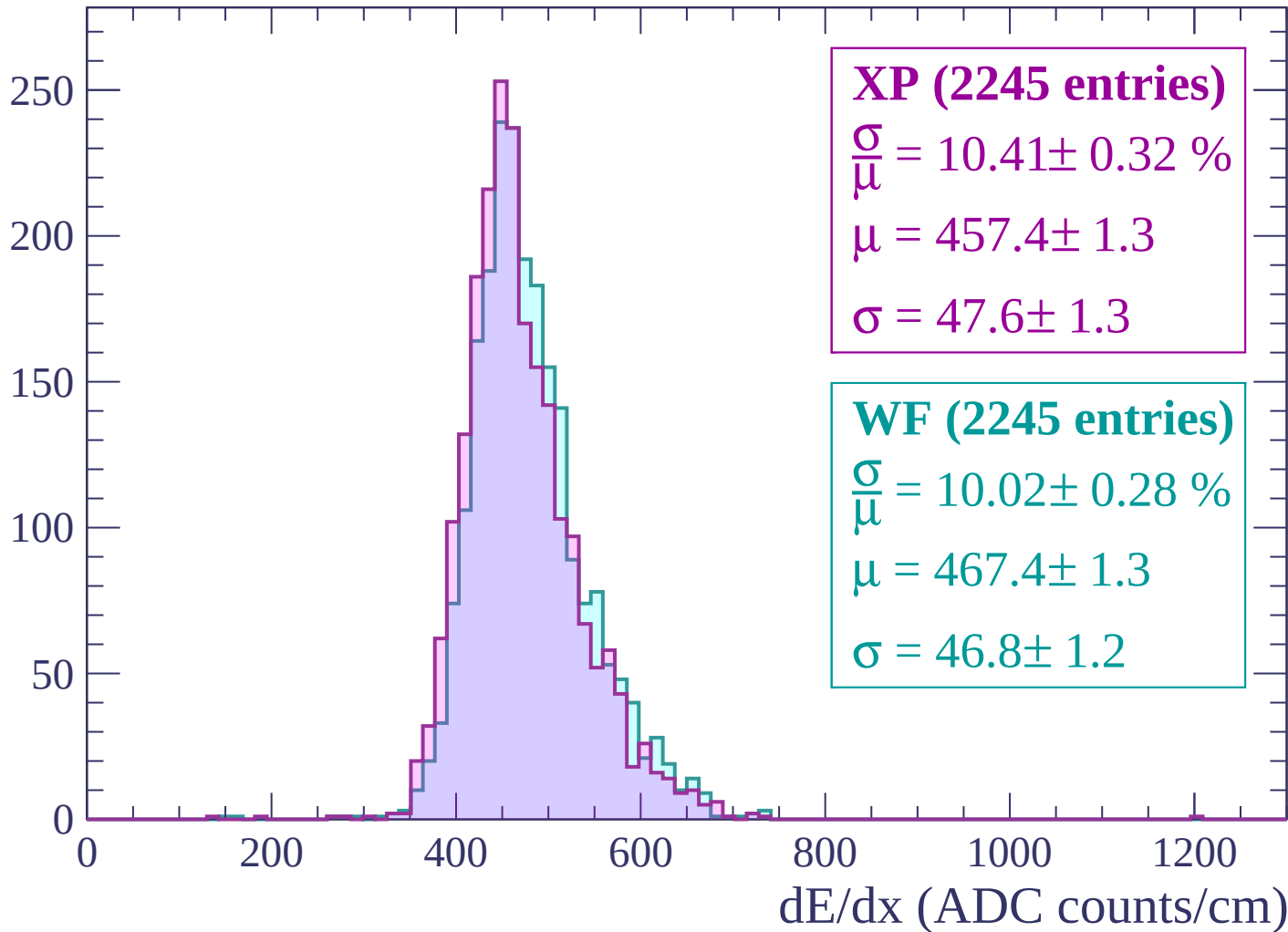
Energy loss | -1160 < p < -1120

Count



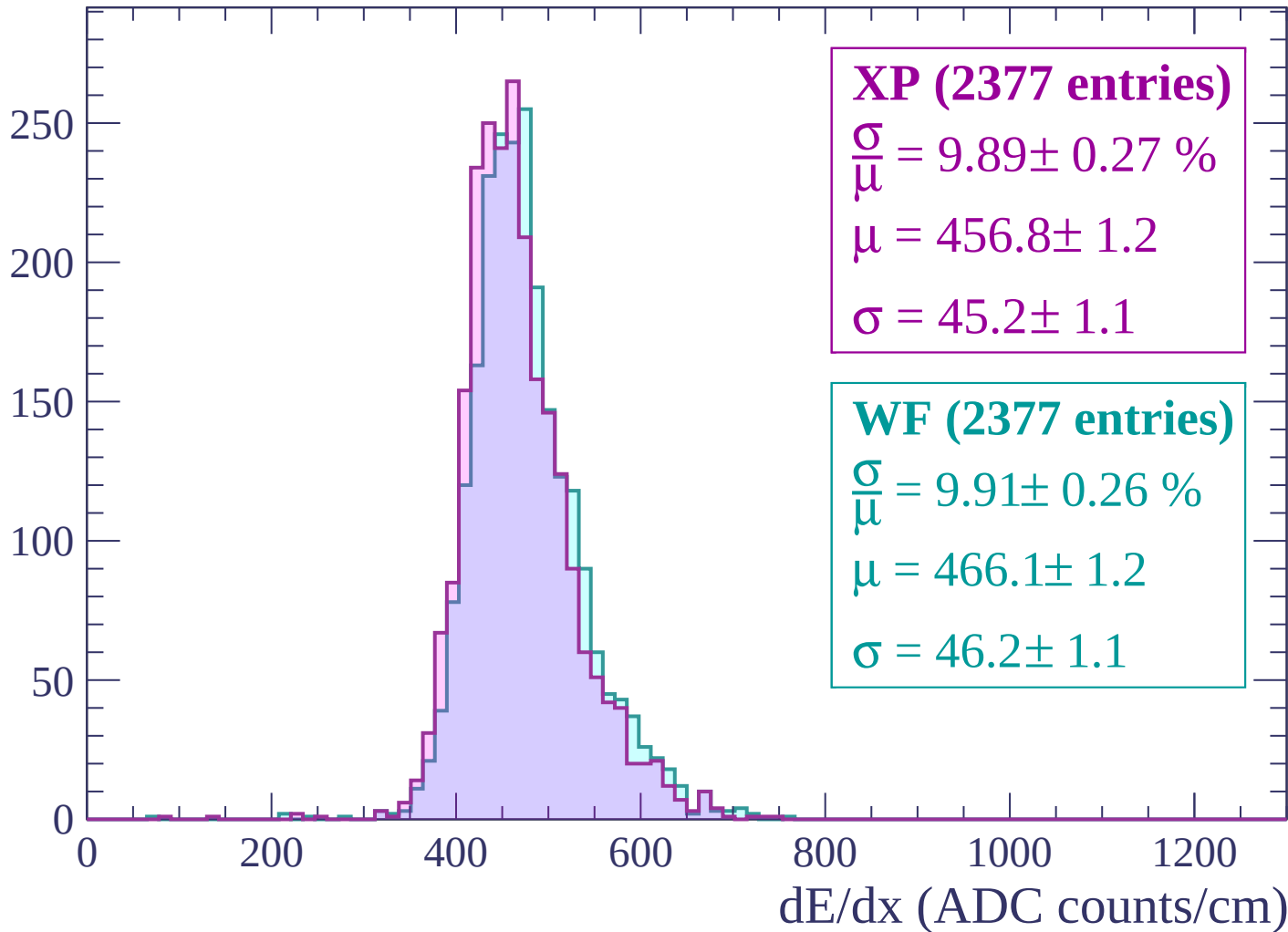
Energy loss | -1120 < p < -1080

Count



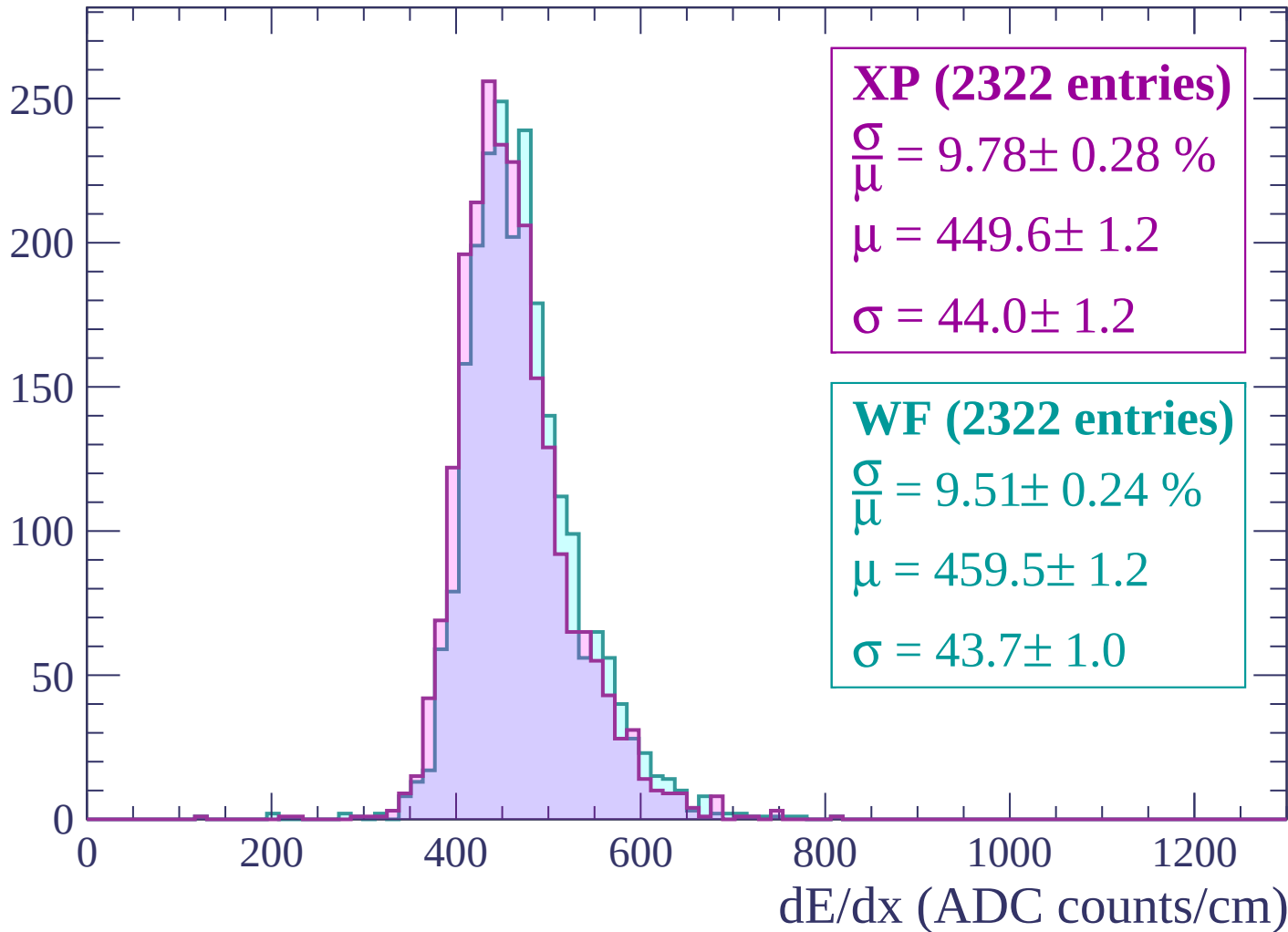
Energy loss | $-1080 < p < -1040$

Count



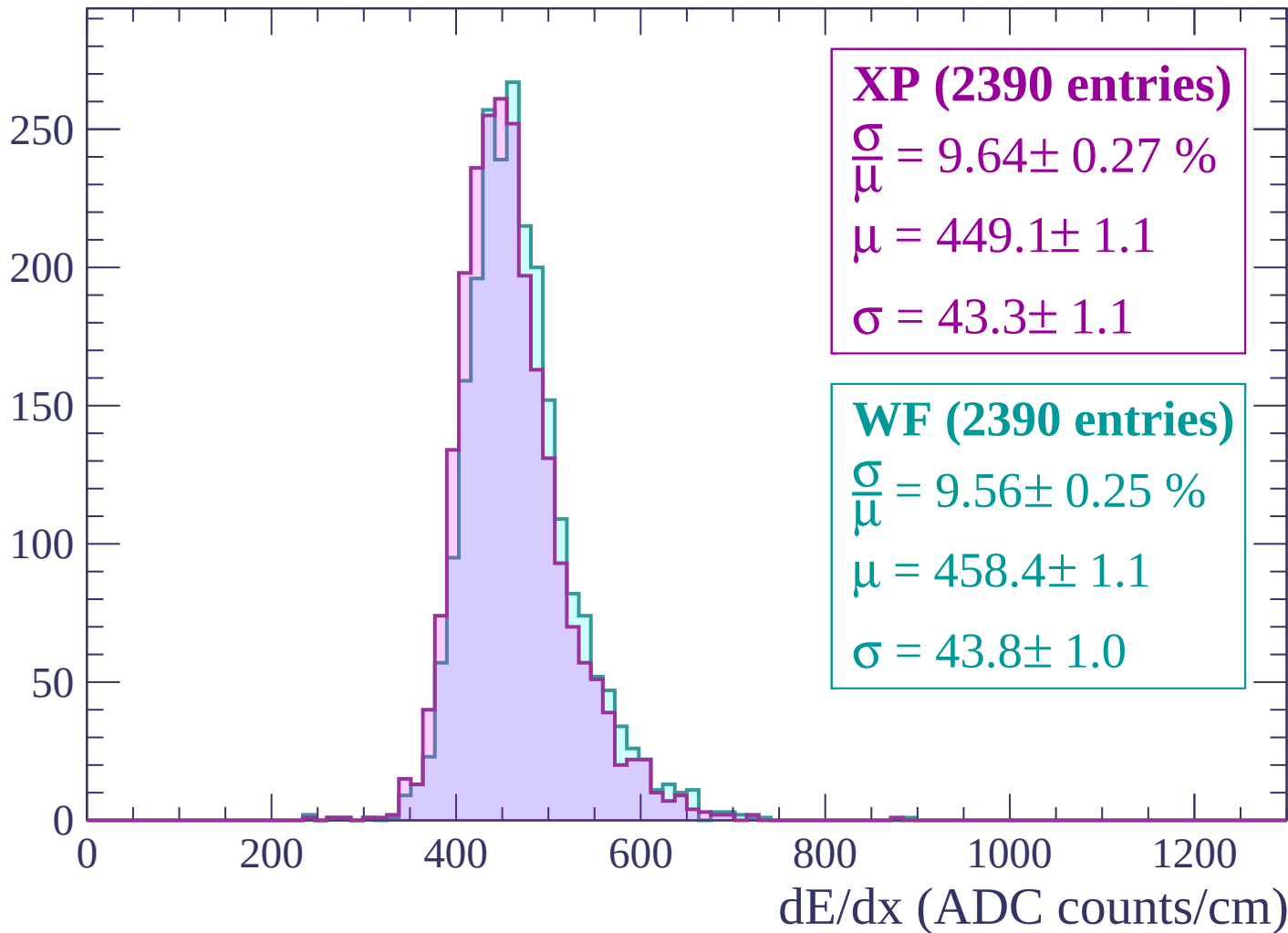
Energy loss | $-1040 < p < -1000$

Count



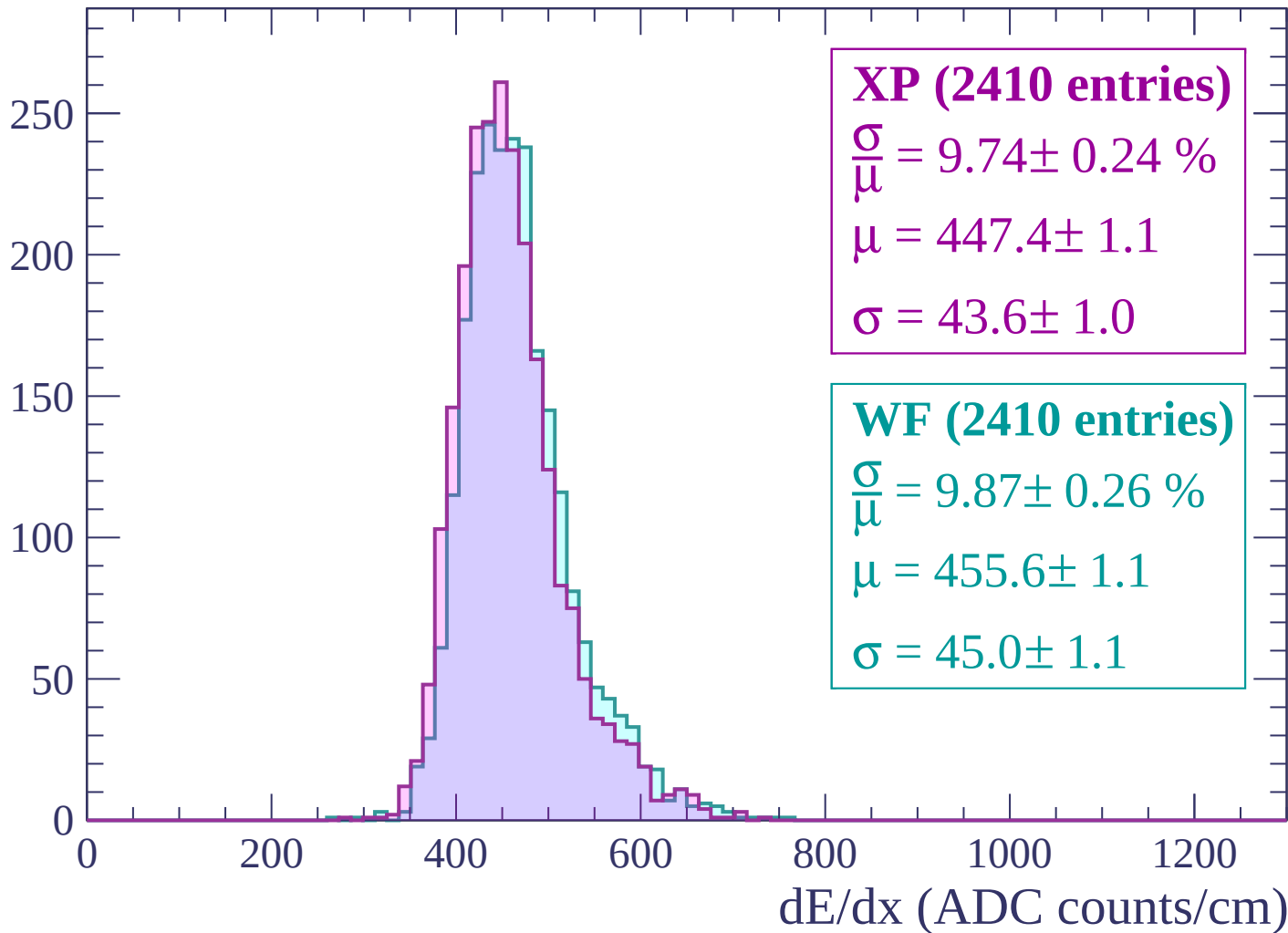
Energy loss | $-1000 < p < -960$

Count



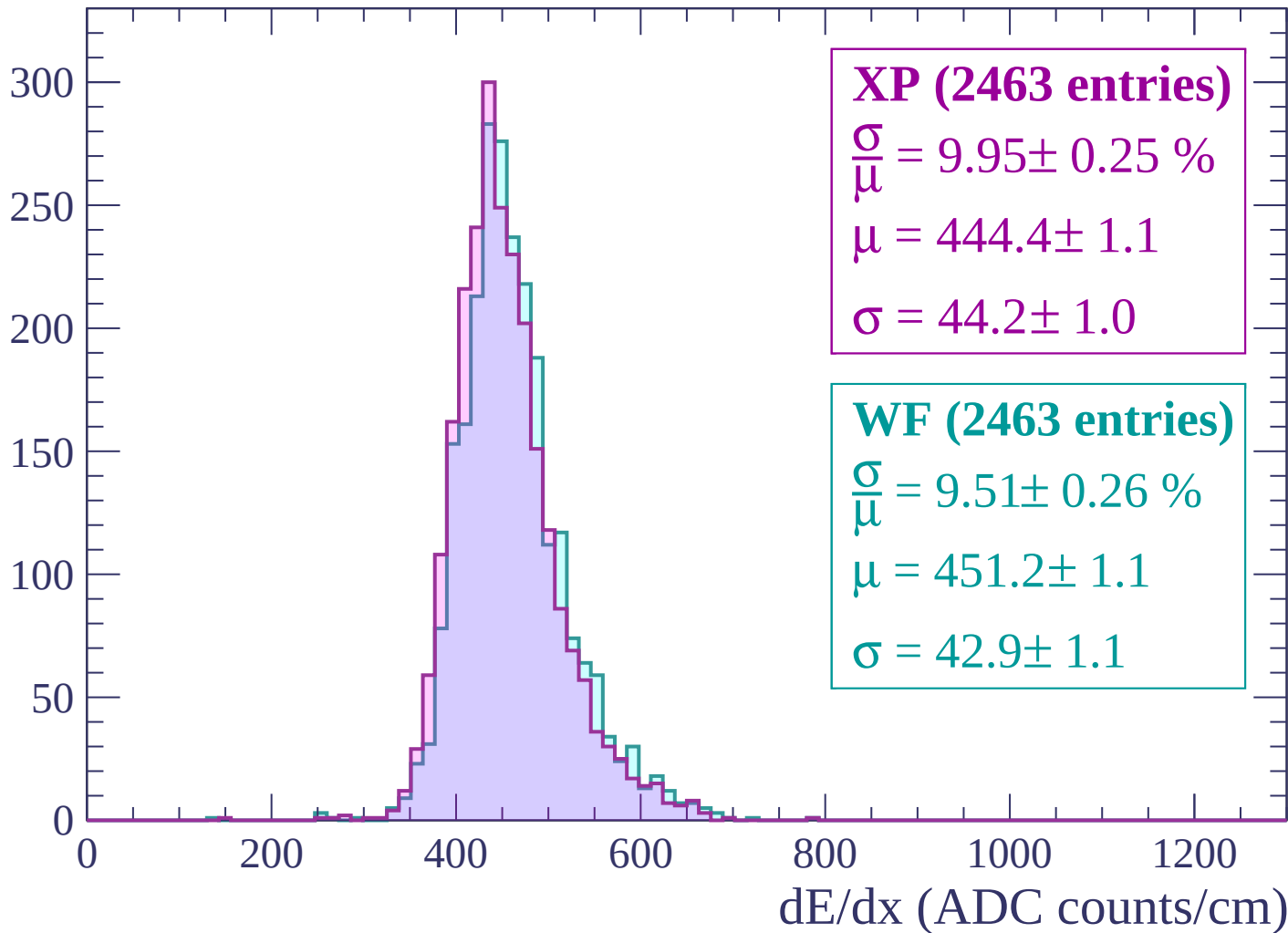
Energy loss | $-960 < p < -920$

Count



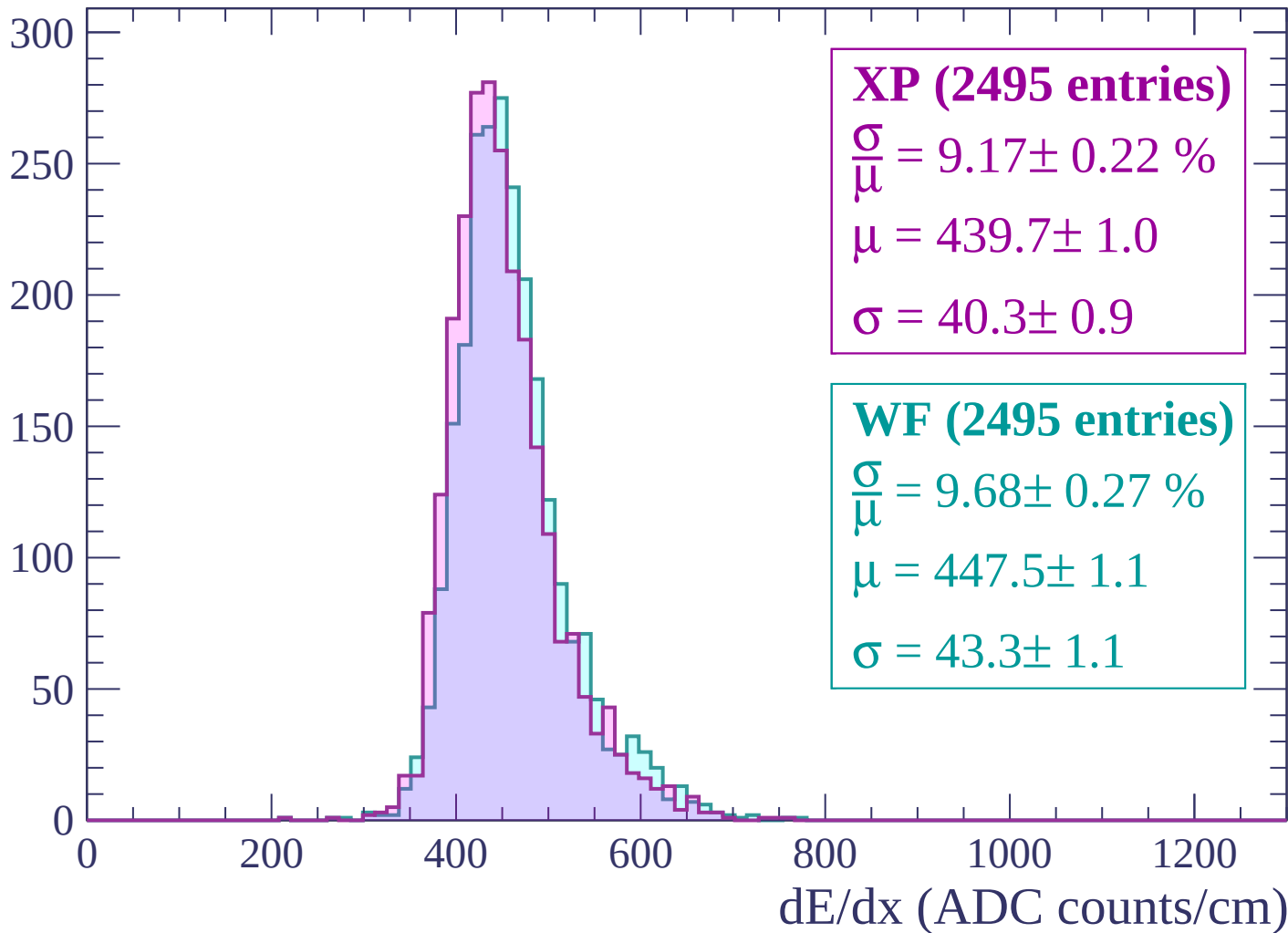
Energy loss | $-920 < p < -880$

Count



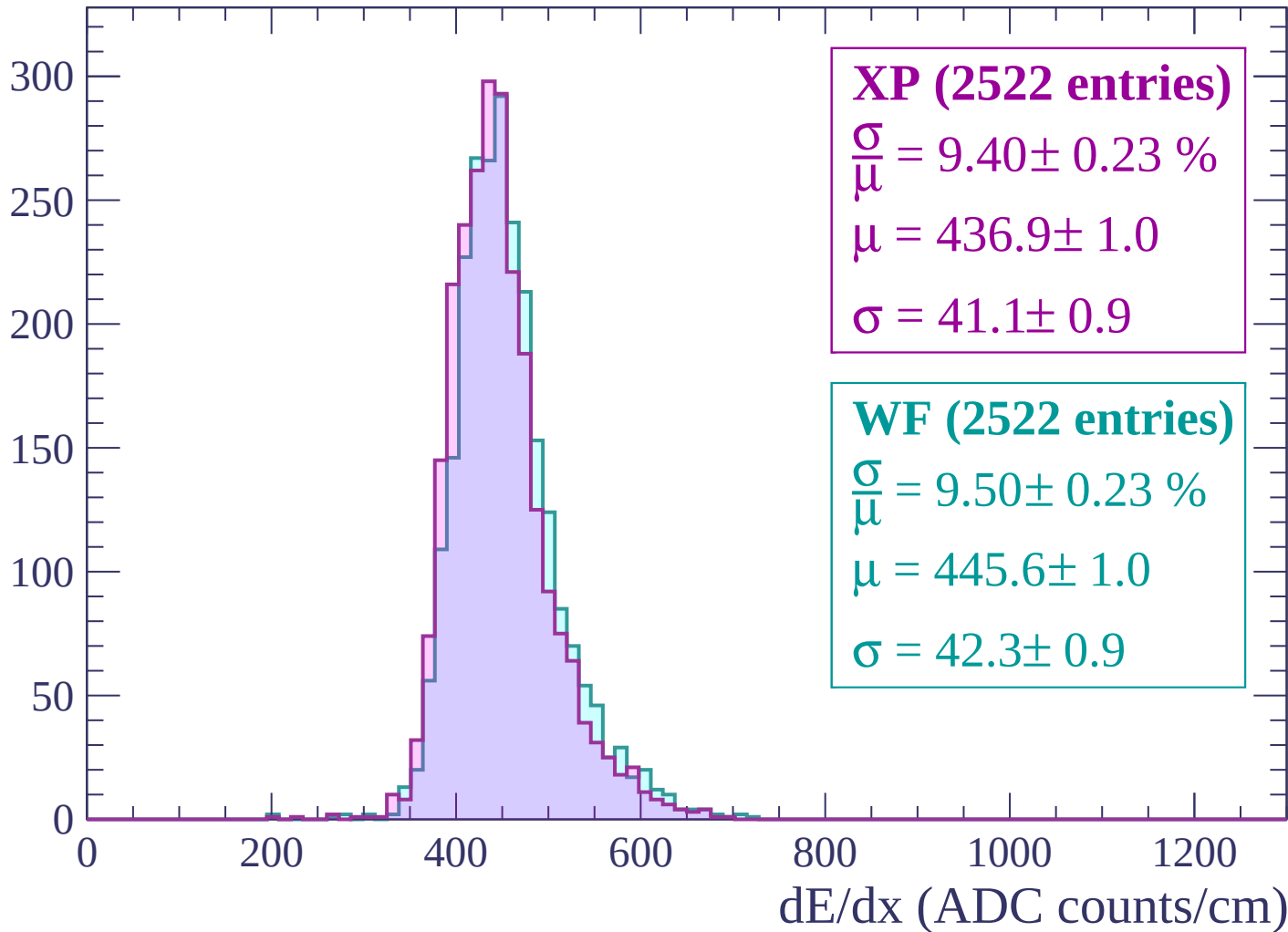
Energy loss | $-880 < p < -840$

Count



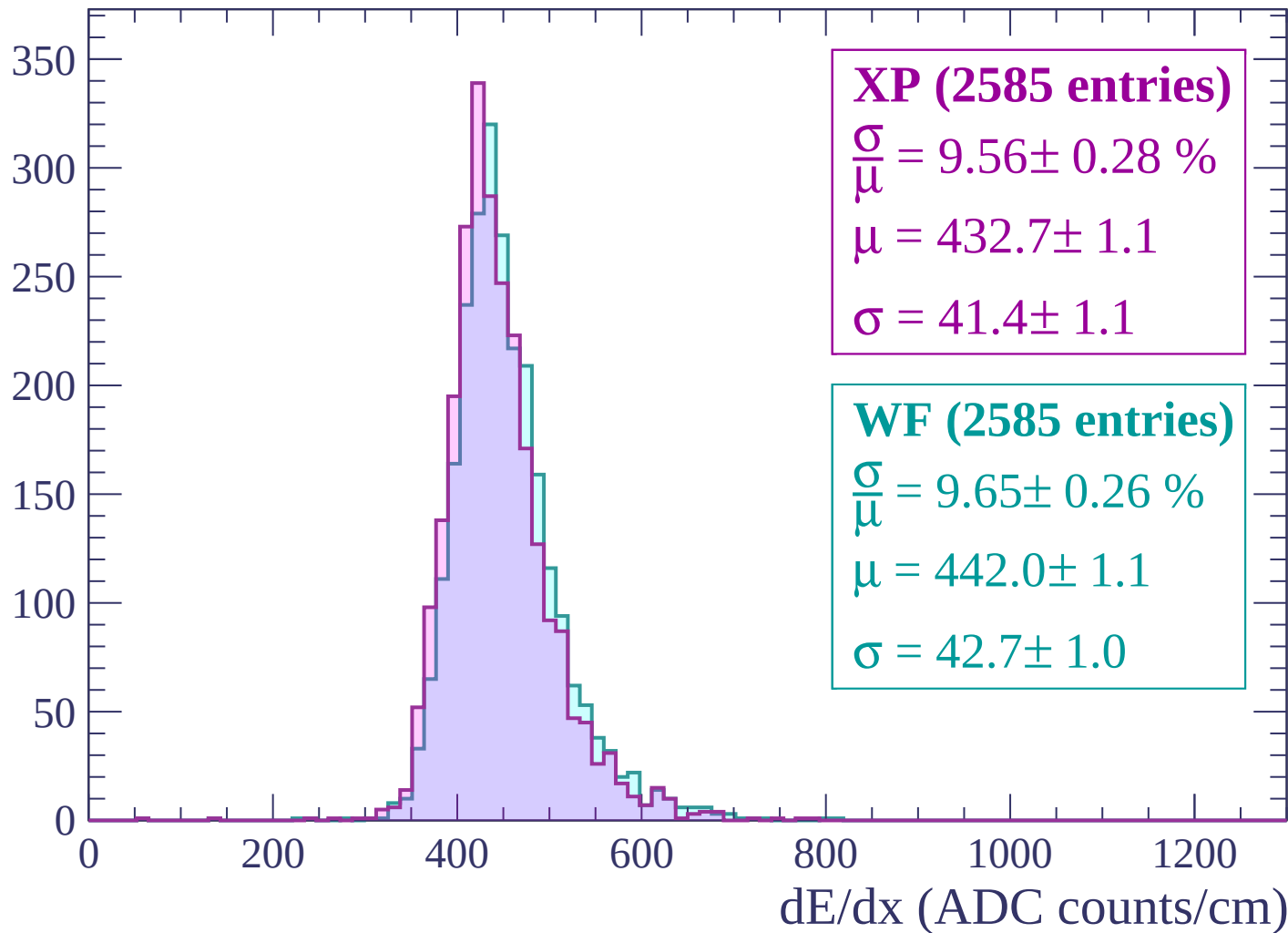
Energy loss | $-840 < p < -800$

Count



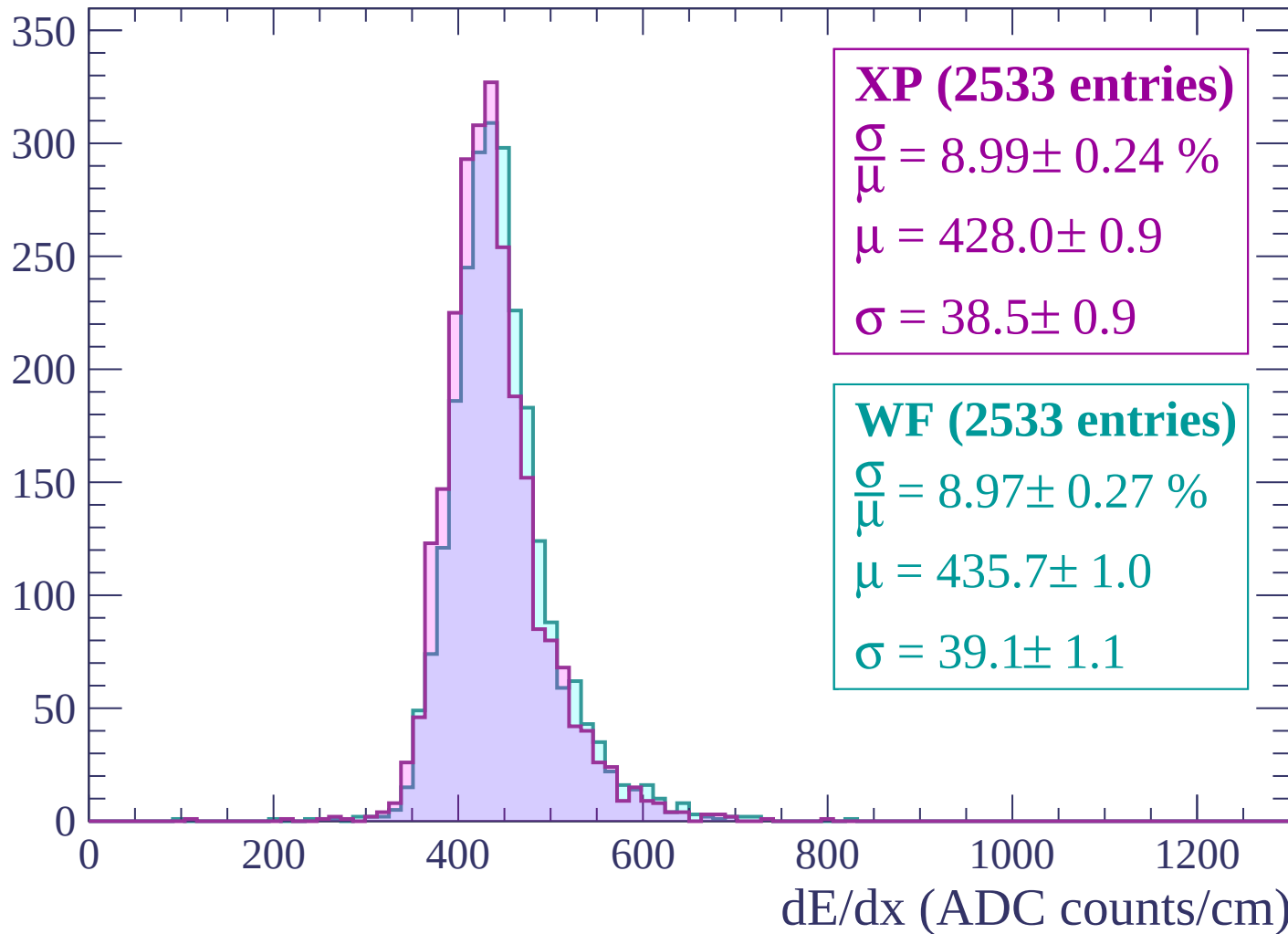
Energy loss | $-800 < p < -760$

Count



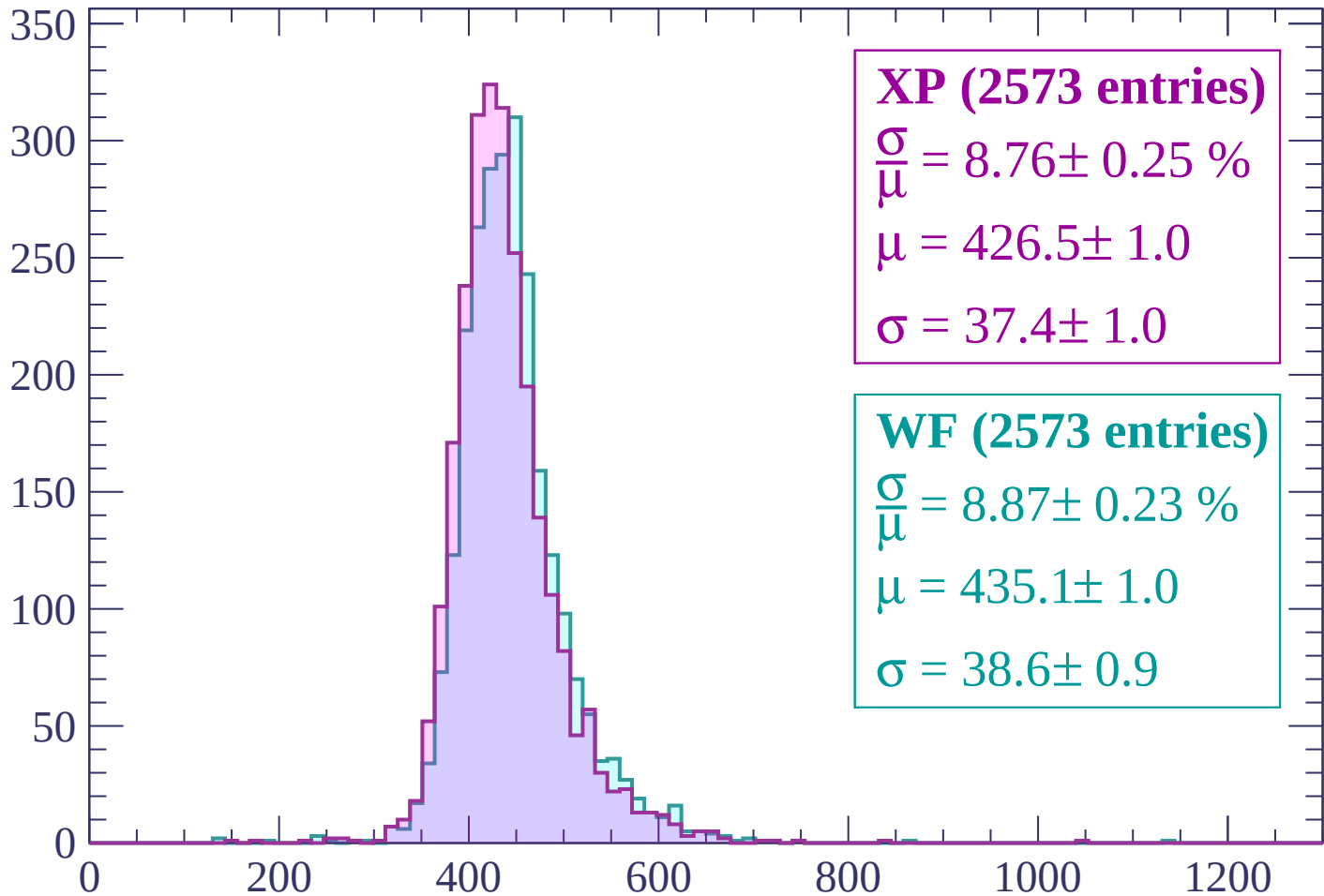
Energy loss | $-760 < p < -720$

Count



Energy loss | $-720 < p < -680$

Count



XP (2573 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.25 \%$$

$$\mu = 426.5 \pm 1.0$$

$$\sigma = 37.4 \pm 1.0$$

WF (2573 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.23 \%$$

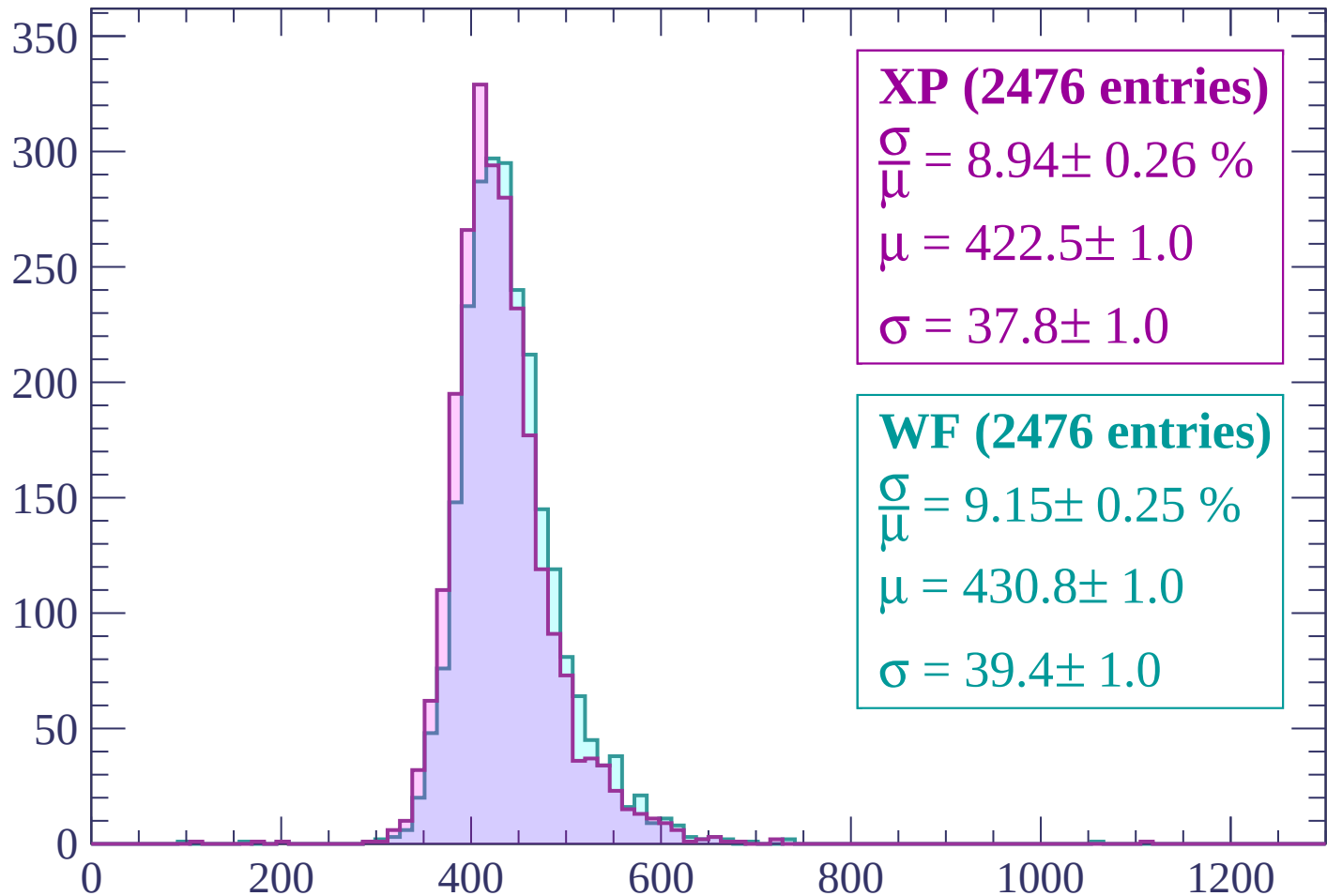
$$\mu = 435.1 \pm 1.0$$

$$\sigma = 38.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP (2476 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.26 \%$$

$$\mu = 422.5 \pm 1.0$$

$$\sigma = 37.8 \pm 1.0$$

WF (2476 entries)

$$\frac{\sigma}{\mu} = 9.15 \pm 0.25 \%$$

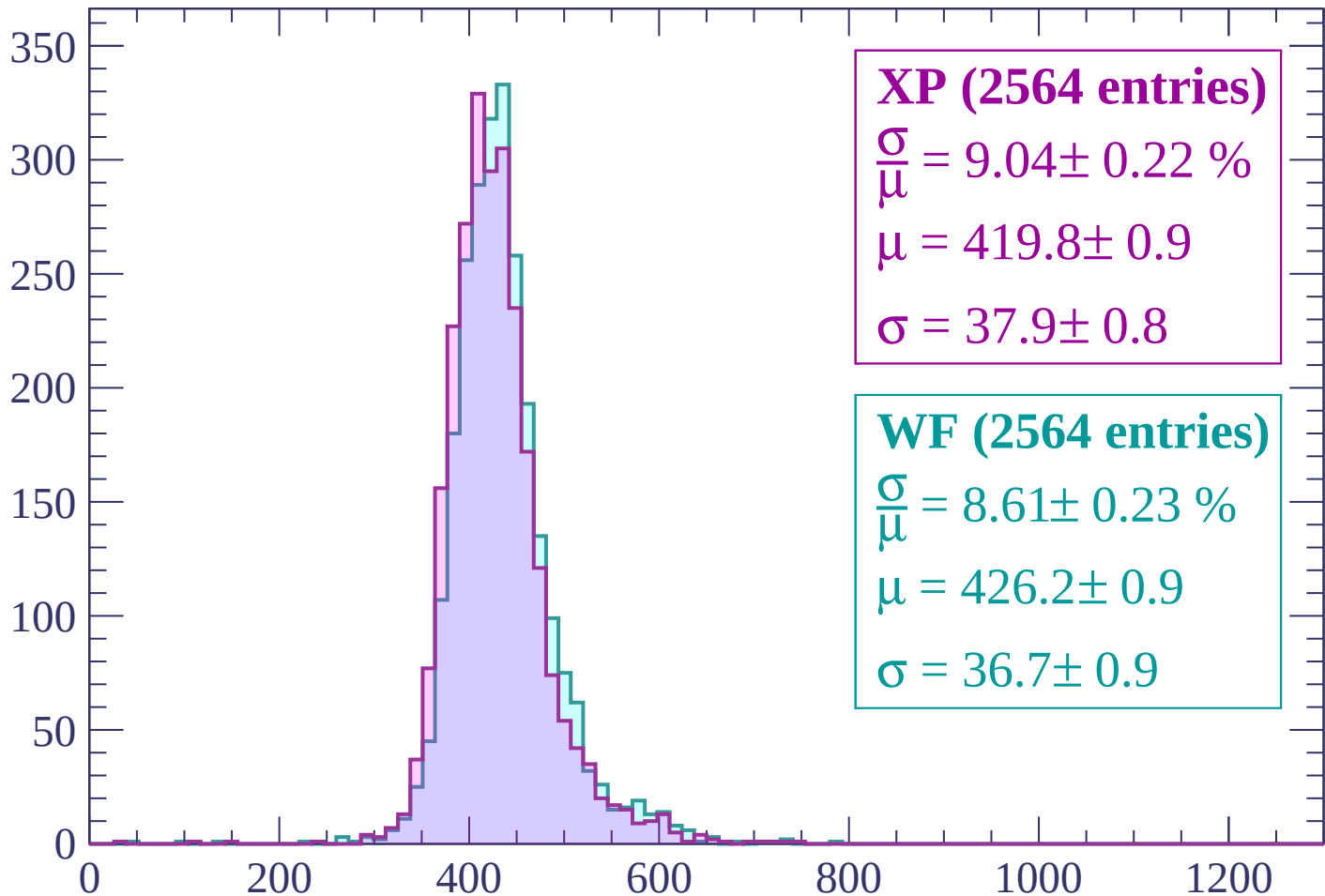
$$\mu = 430.8 \pm 1.0$$

$$\sigma = 39.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP (2564 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.22 \%$$

$$\mu = 419.8 \pm 0.9$$

$$\sigma = 37.9 \pm 0.8$$

WF (2564 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.23 \%$$

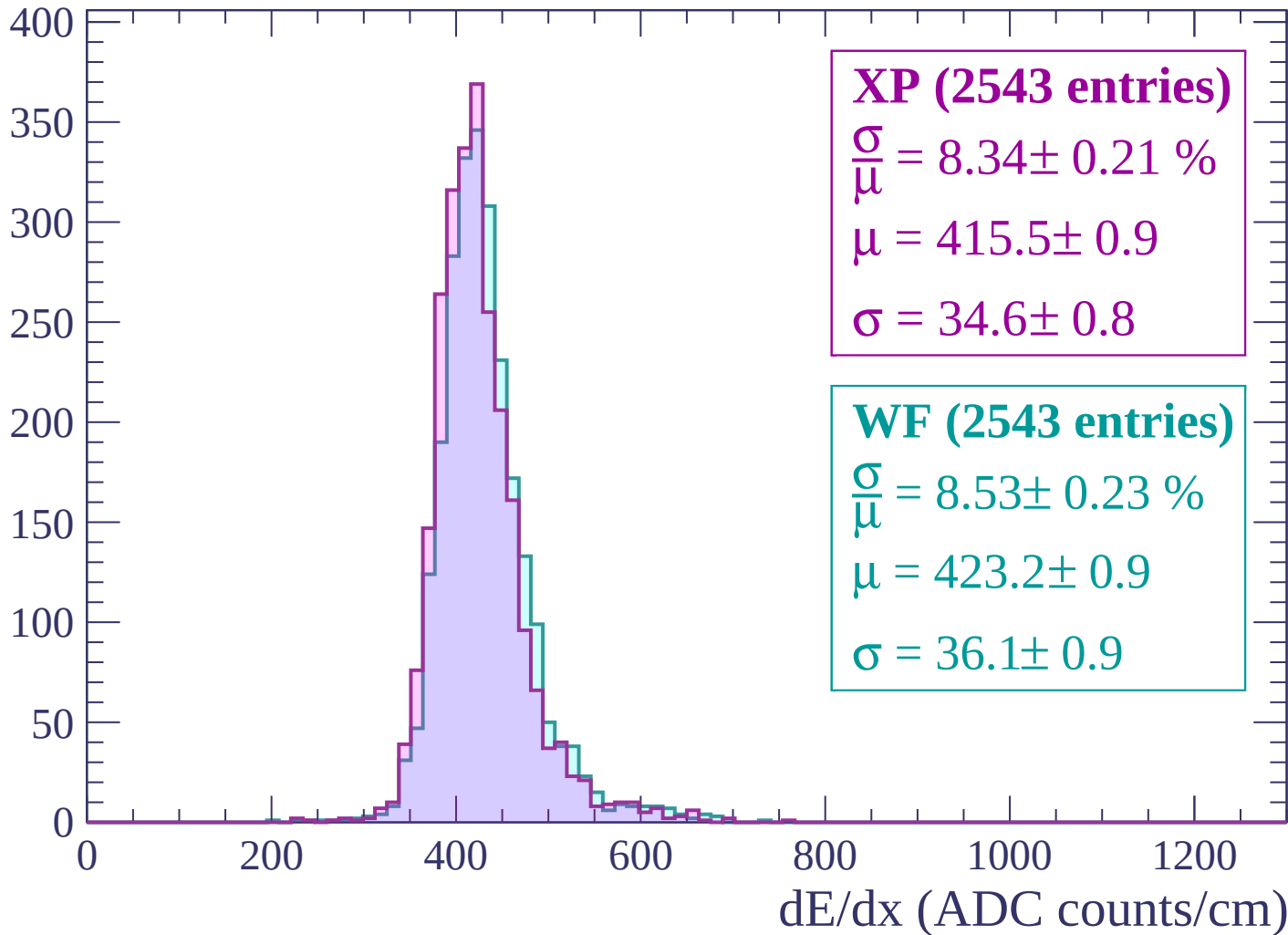
$$\mu = 426.2 \pm 0.9$$

$$\sigma = 36.7 \pm 0.9$$

dE/dx (ADC counts/cm)

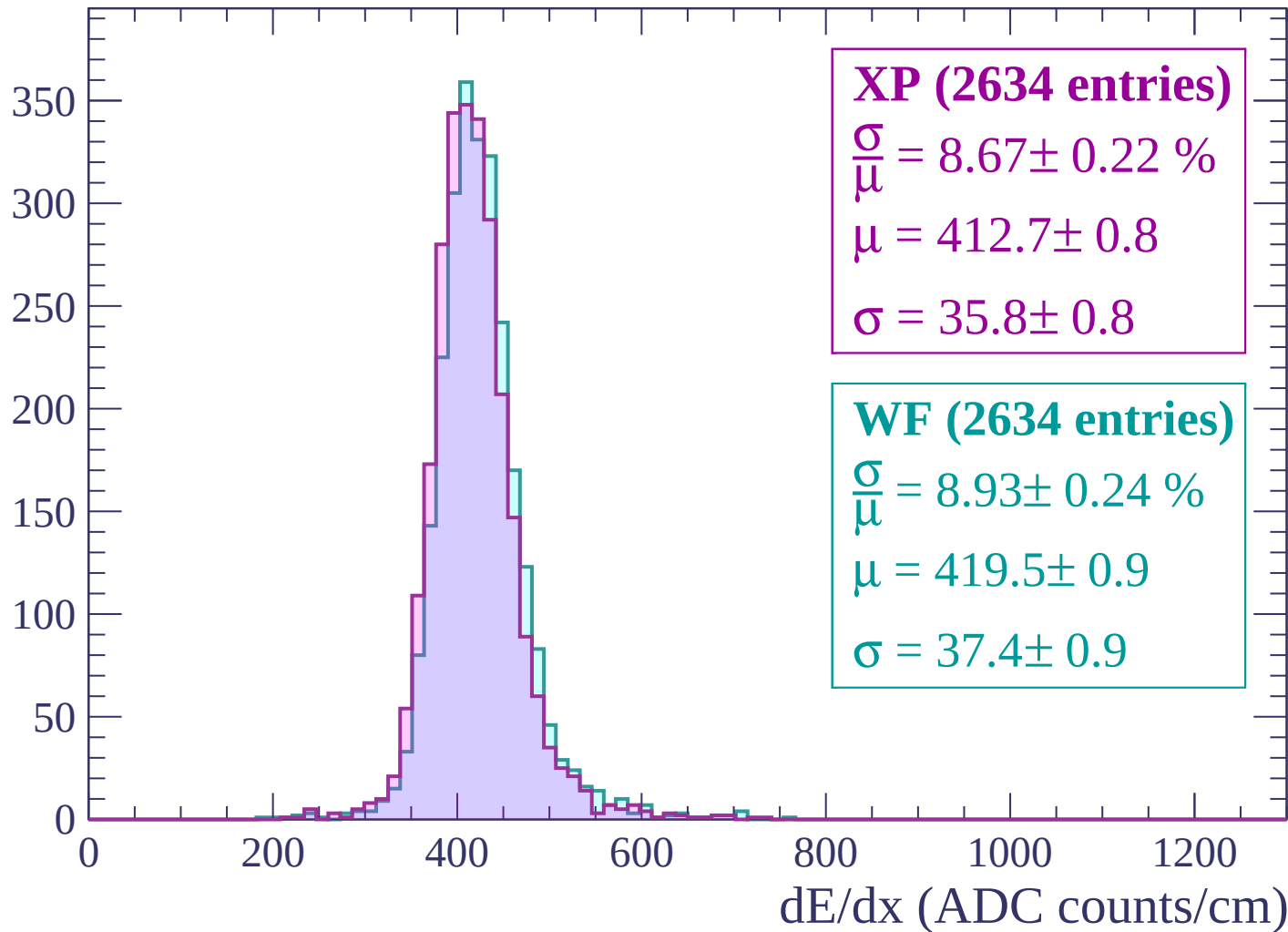
Energy loss | $-600 < p < -560$

Count



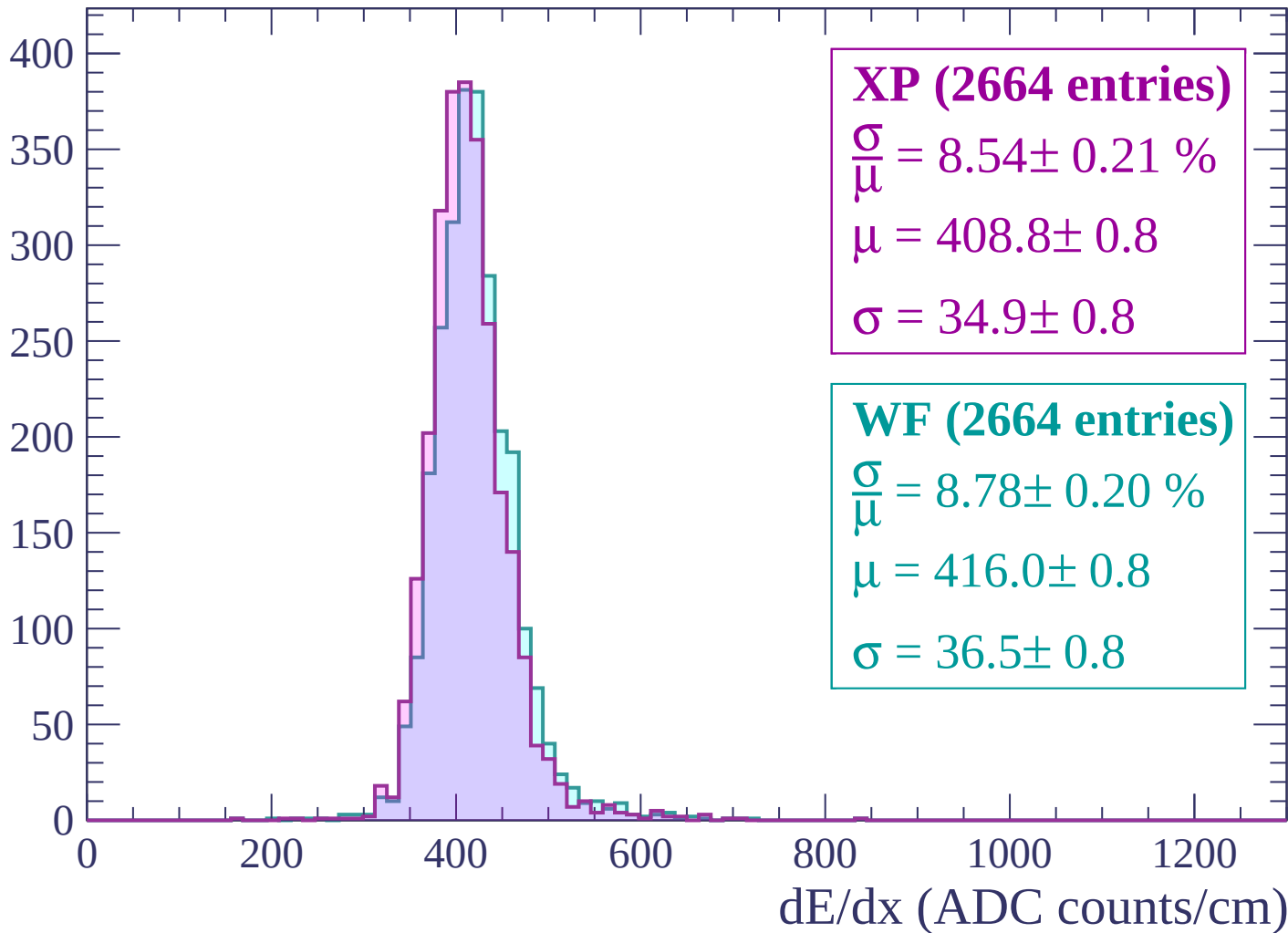
Energy loss | $-560 < p < -520$

Count



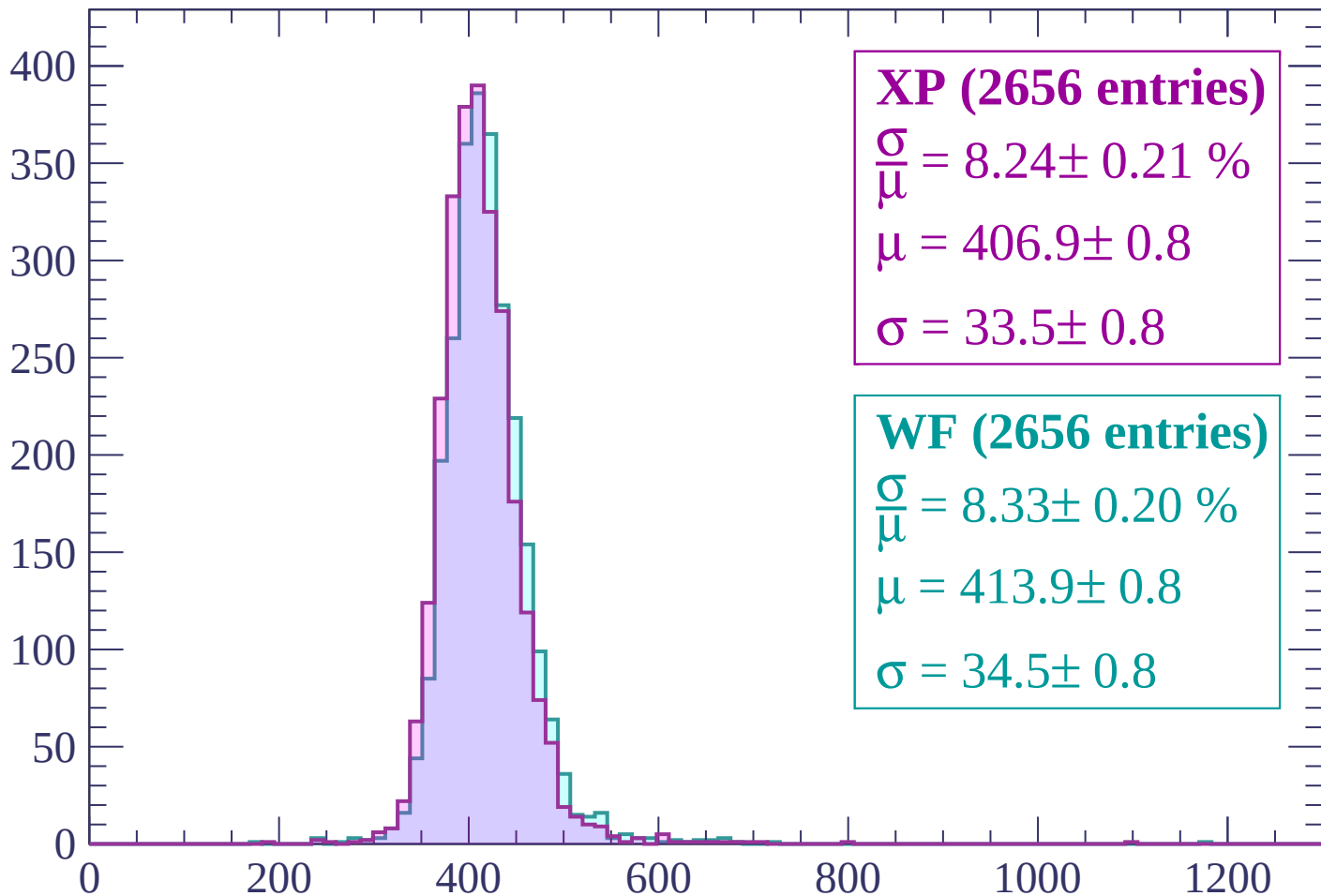
Energy loss | $-520 < p < -480$

Count



Energy loss | $-480 < p < -440$

Count



XP (2656 entries)

$$\frac{\sigma}{\mu} = 8.24 \pm 0.21 \%$$

$$\mu = 406.9 \pm 0.8$$

$$\sigma = 33.5 \pm 0.8$$

WF (2656 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.20 \%$$

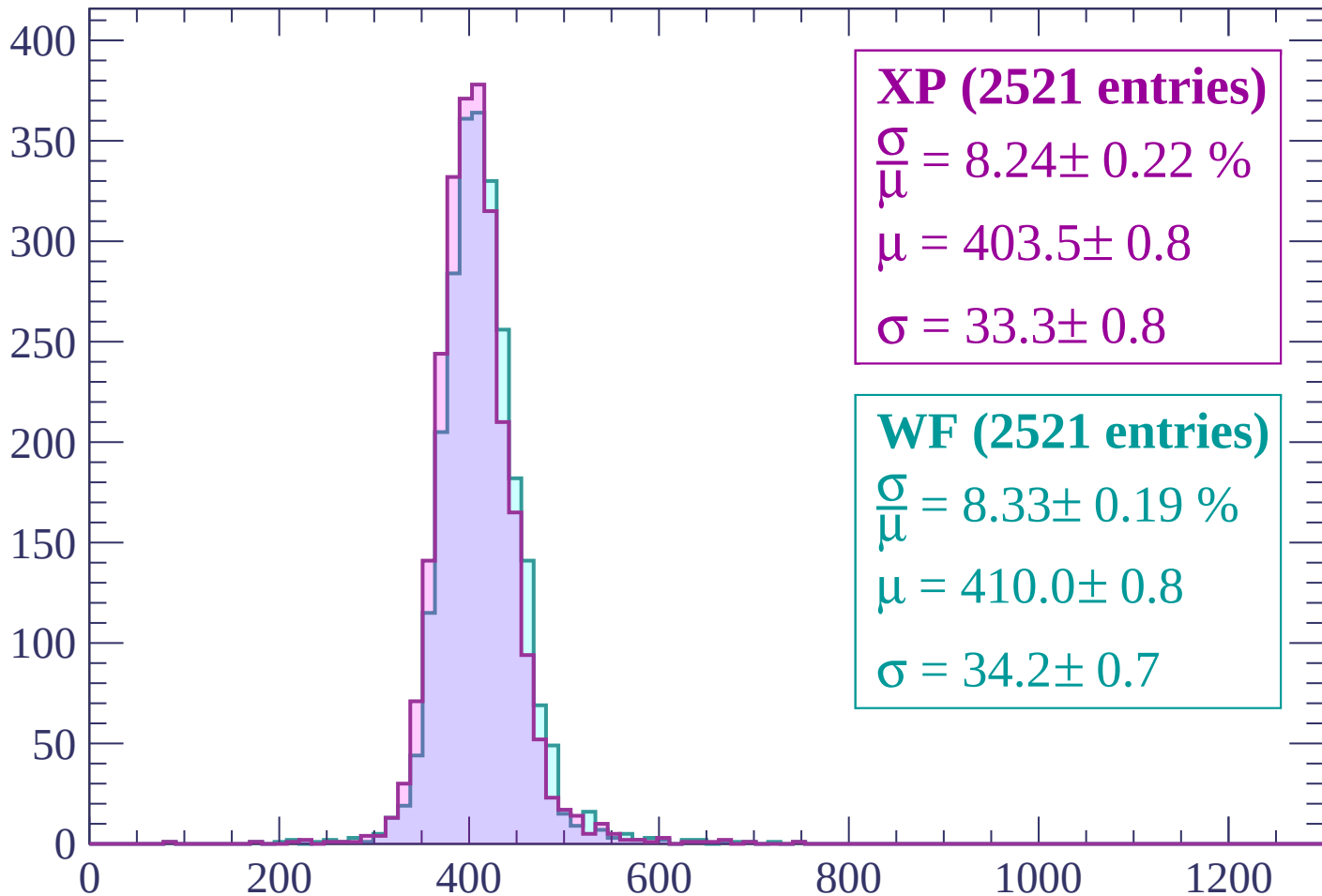
$$\mu = 413.9 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP (2521 entries)

$$\frac{\sigma}{\mu} = 8.24 \pm 0.22 \%$$

$$\mu = 403.5 \pm 0.8$$

$$\sigma = 33.3 \pm 0.8$$

WF (2521 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.19 \%$$

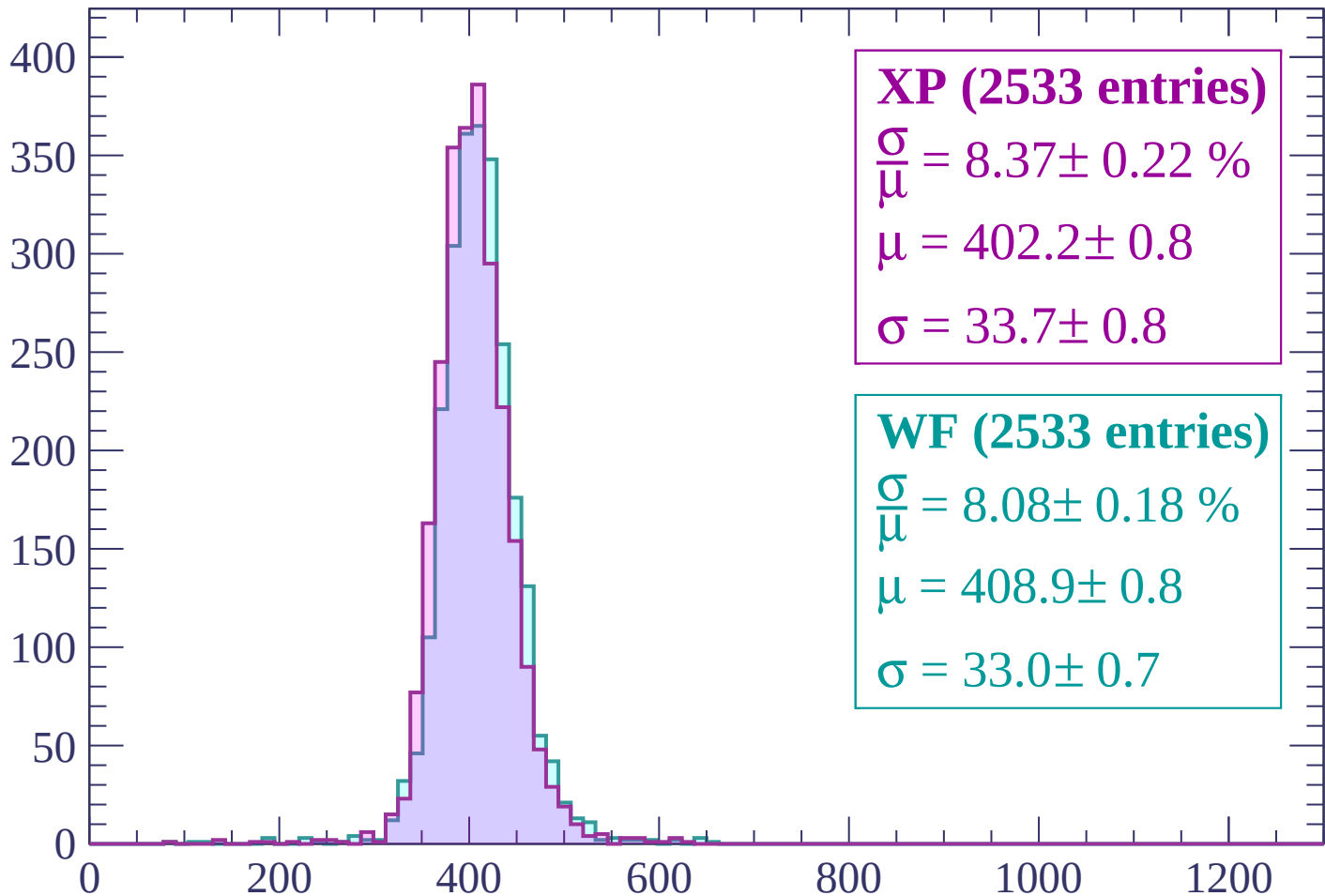
$$\mu = 410.0 \pm 0.8$$

$$\sigma = 34.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP (2533 entries)

$$\frac{\sigma}{\mu} = 8.37 \pm 0.22 \%$$

$$\mu = 402.2 \pm 0.8$$

$$\sigma = 33.7 \pm 0.8$$

WF (2533 entries)

$$\frac{\sigma}{\mu} = 8.08 \pm 0.18 \%$$

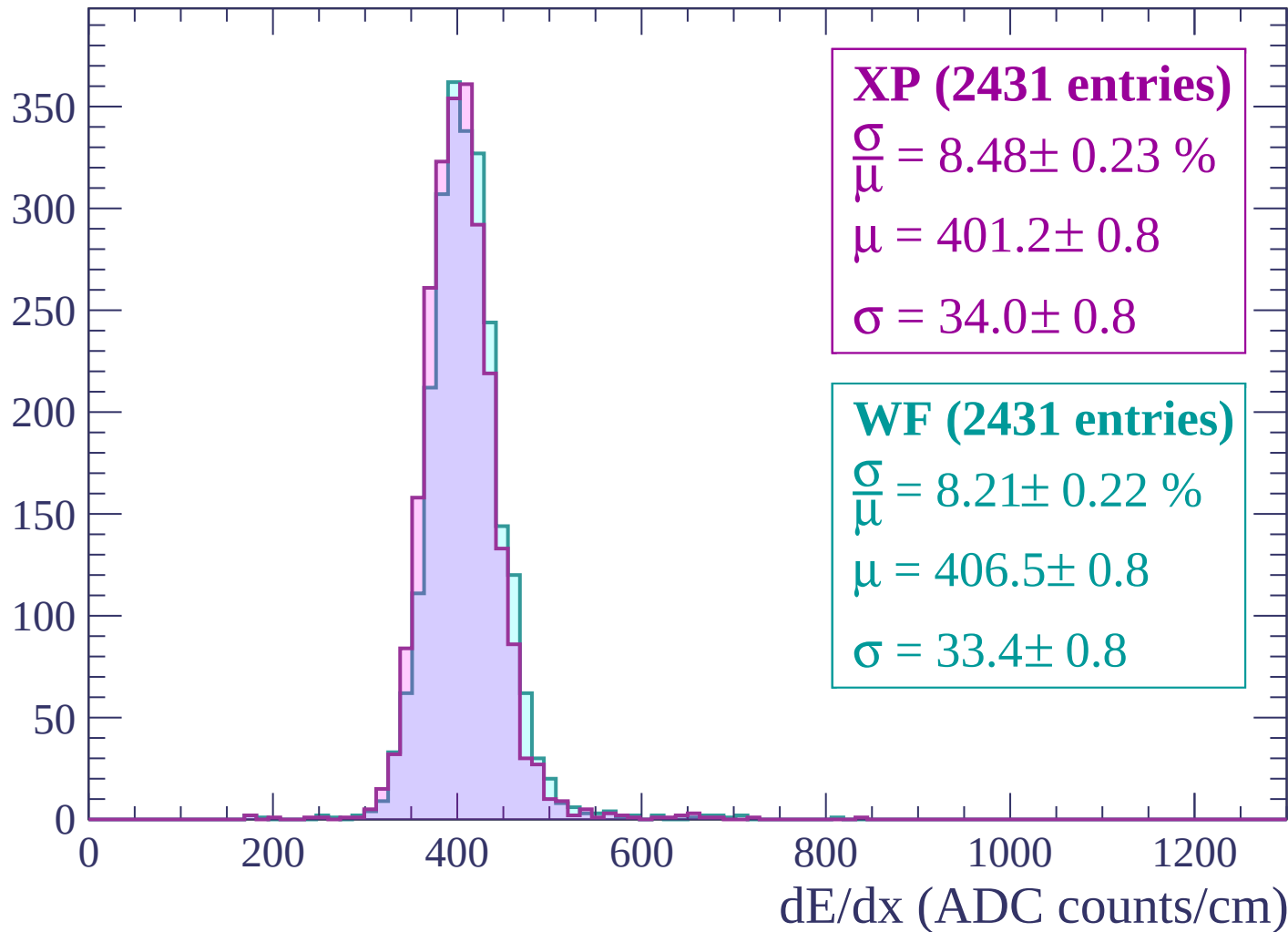
$$\mu = 408.9 \pm 0.8$$

$$\sigma = 33.0 \pm 0.7$$

dE/dx (ADC counts/cm)

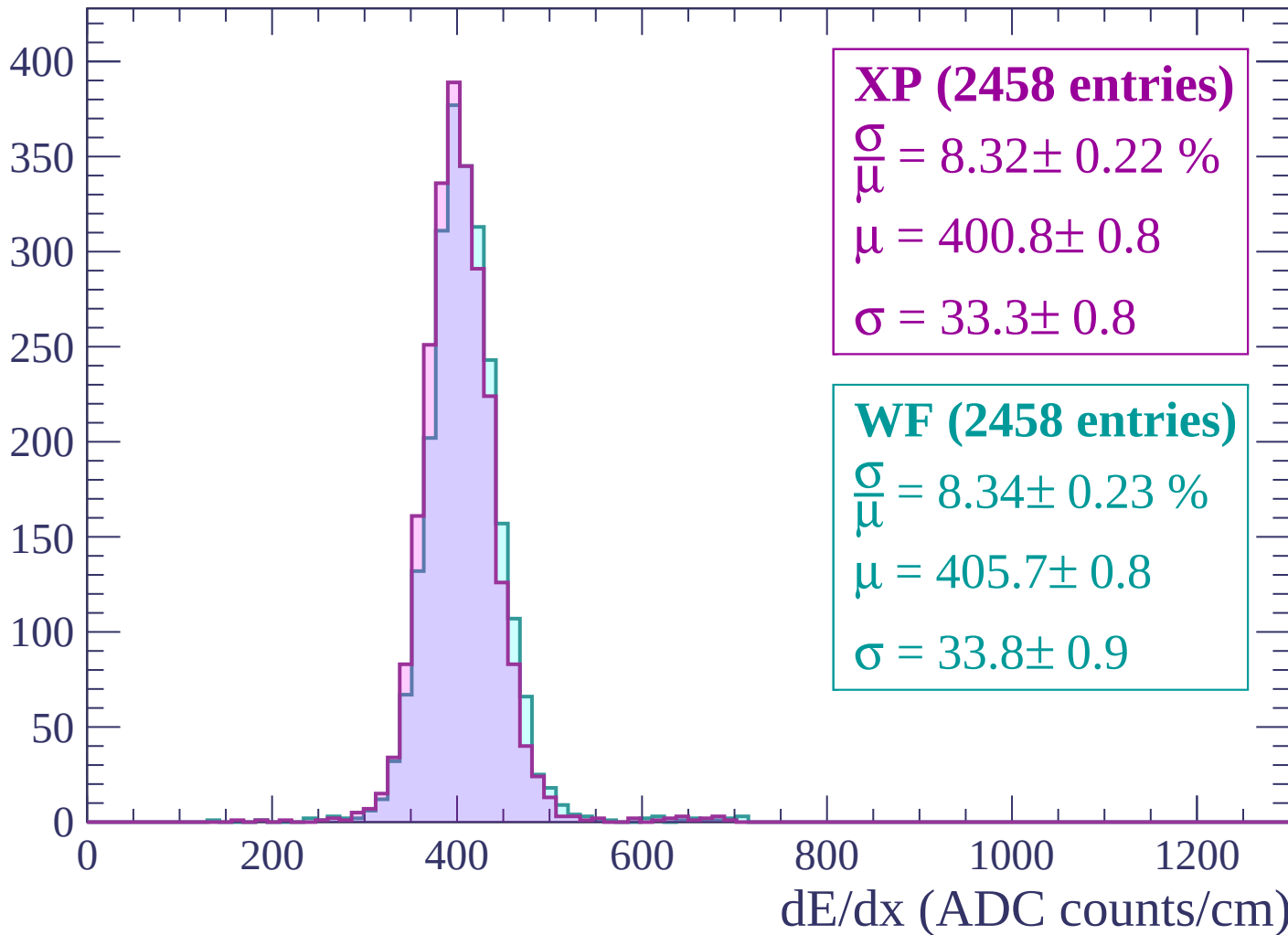
Energy loss | $-360 < p < -320$

Count



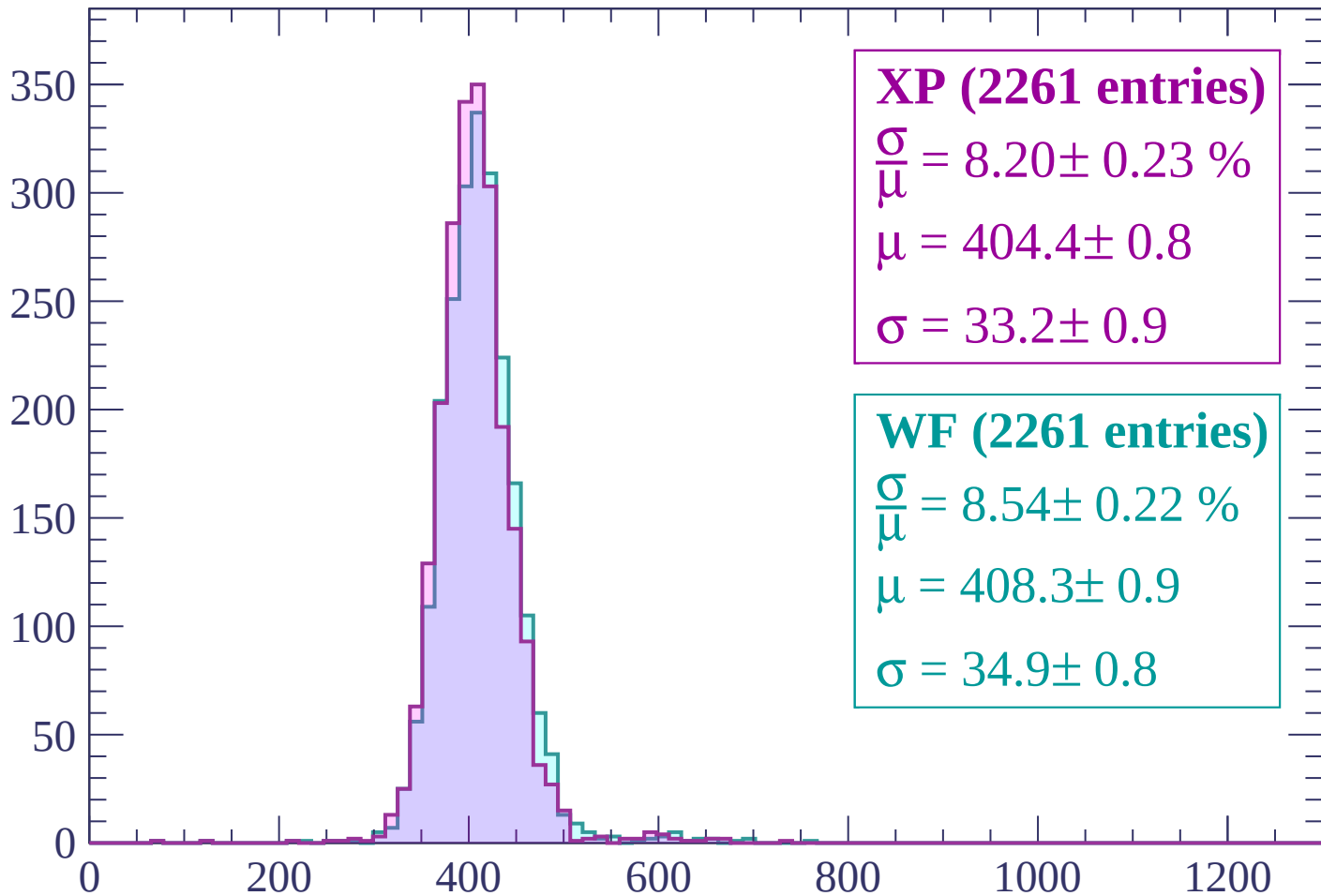
Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count



XP (2261 entries)

$\frac{\sigma}{\mu} = 8.20 \pm 0.23 \%$

$\mu = 404.4 \pm 0.8$

$\sigma = 33.2 \pm 0.9$

WF (2261 entries)

$\frac{\sigma}{\mu} = 8.54 \pm 0.22 \%$

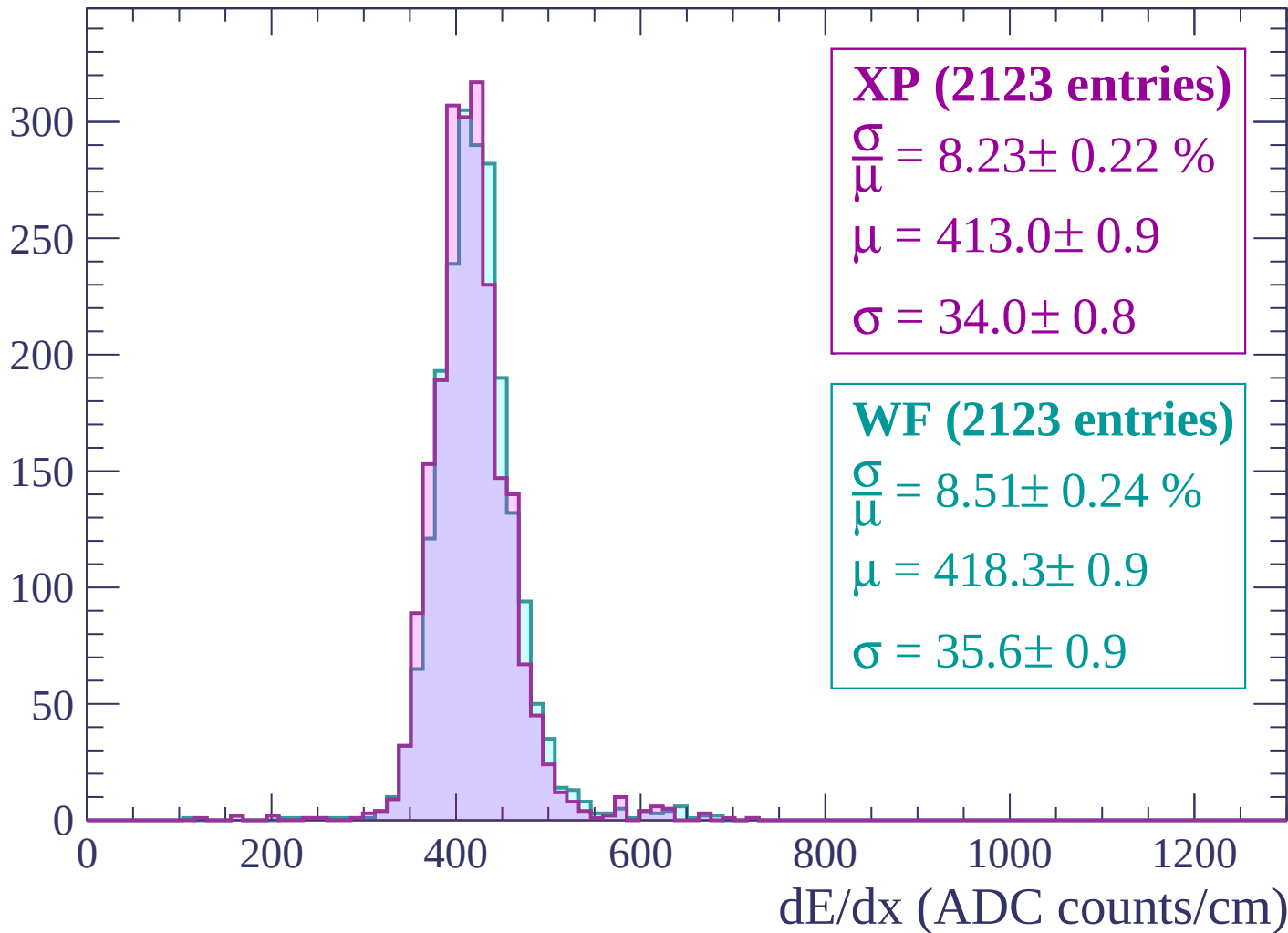
$\mu = 408.3 \pm 0.9$

$\sigma = 34.9 \pm 0.8$

dE/dx (ADC counts/cm)

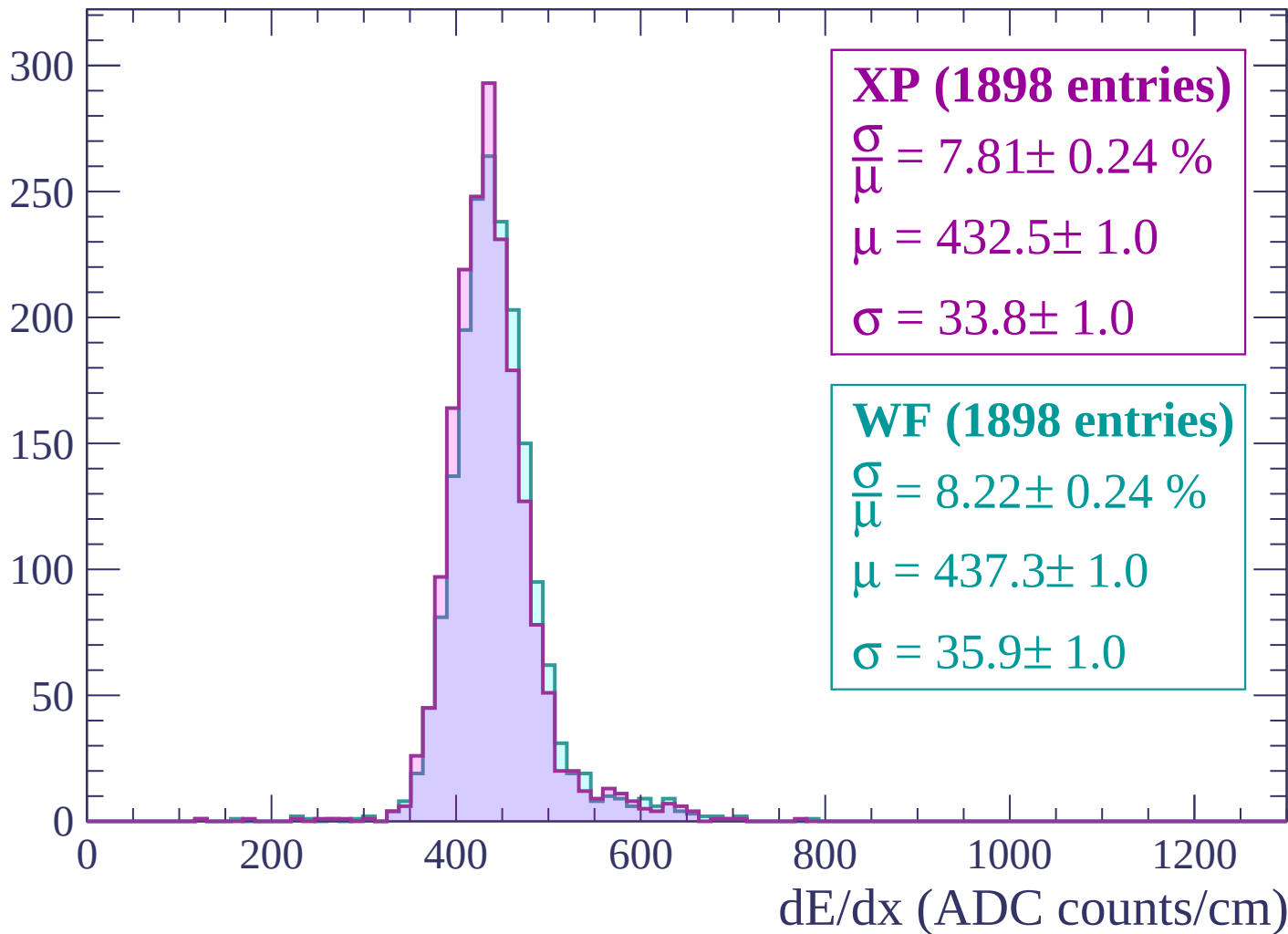
Energy loss | $-240 < p < -200$

Count



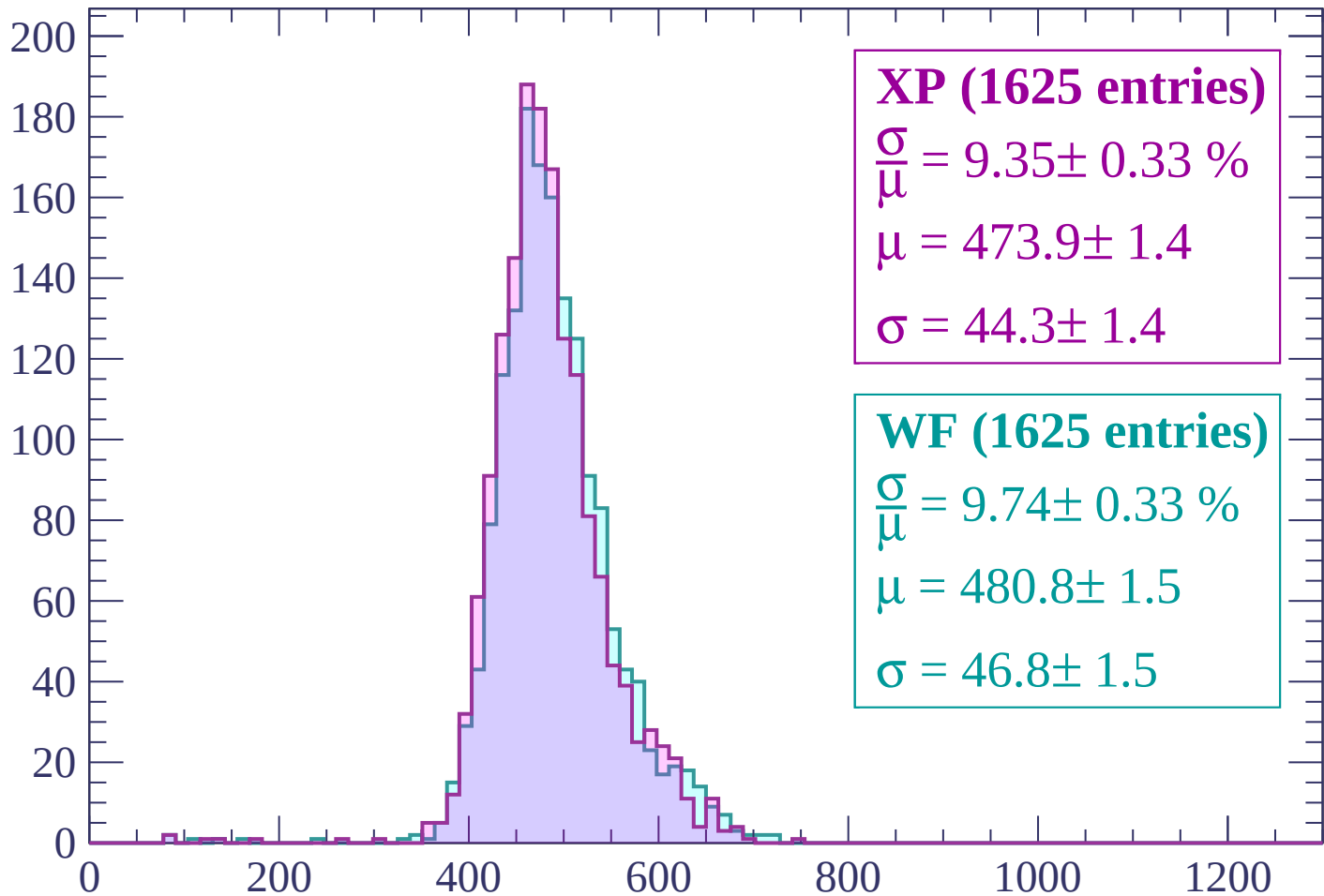
Energy loss | $-200 < p < -160$

Count



Energy loss | -160 < p < -120

Count



XP (1625 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.33 \%$$

$$\mu = 473.9 \pm 1.4$$

$$\sigma = 44.3 \pm 1.4$$

WF (1625 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.33 \%$$

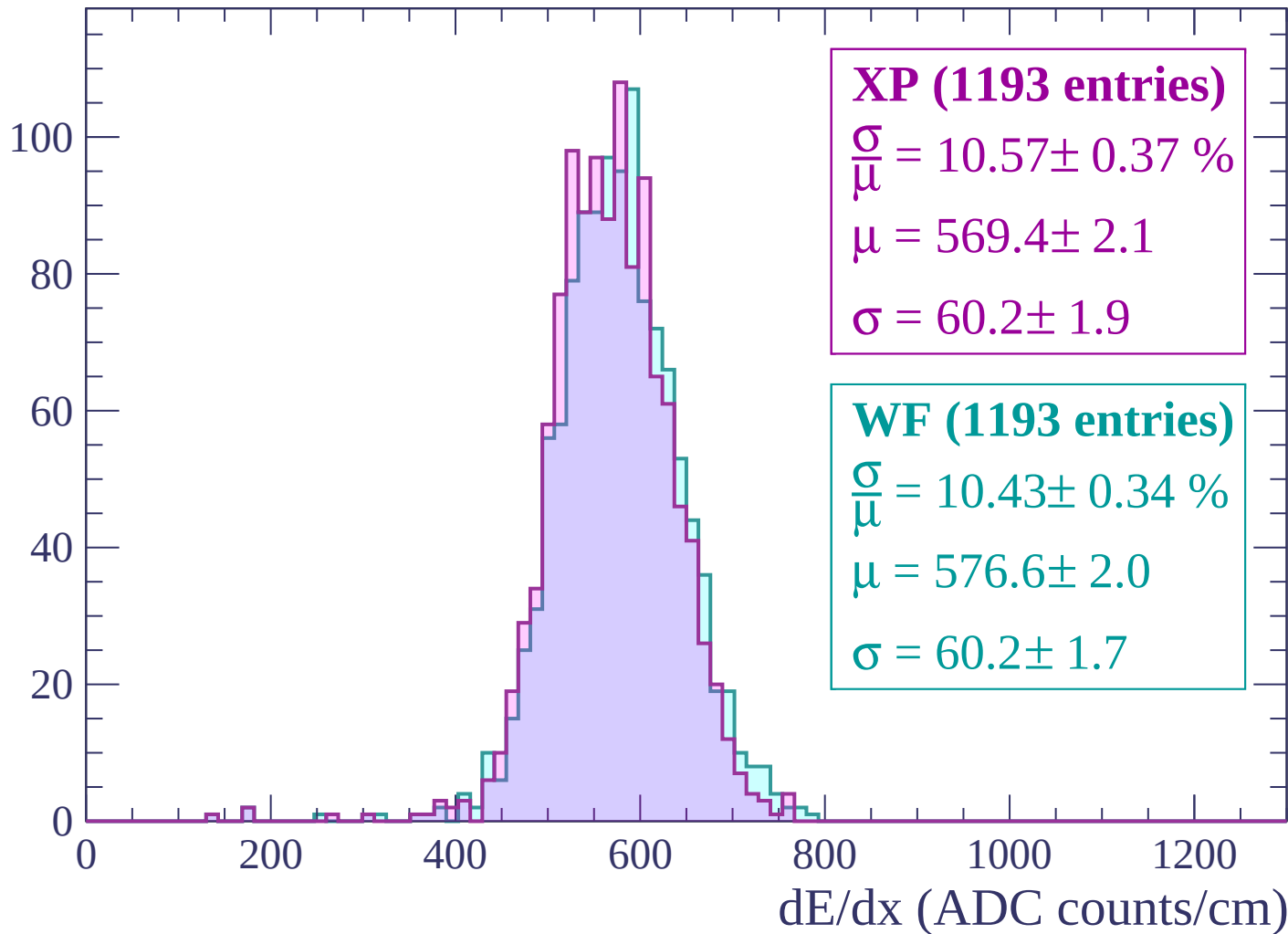
$$\mu = 480.8 \pm 1.5$$

$$\sigma = 46.8 \pm 1.5$$

dE/dx (ADC counts/cm)

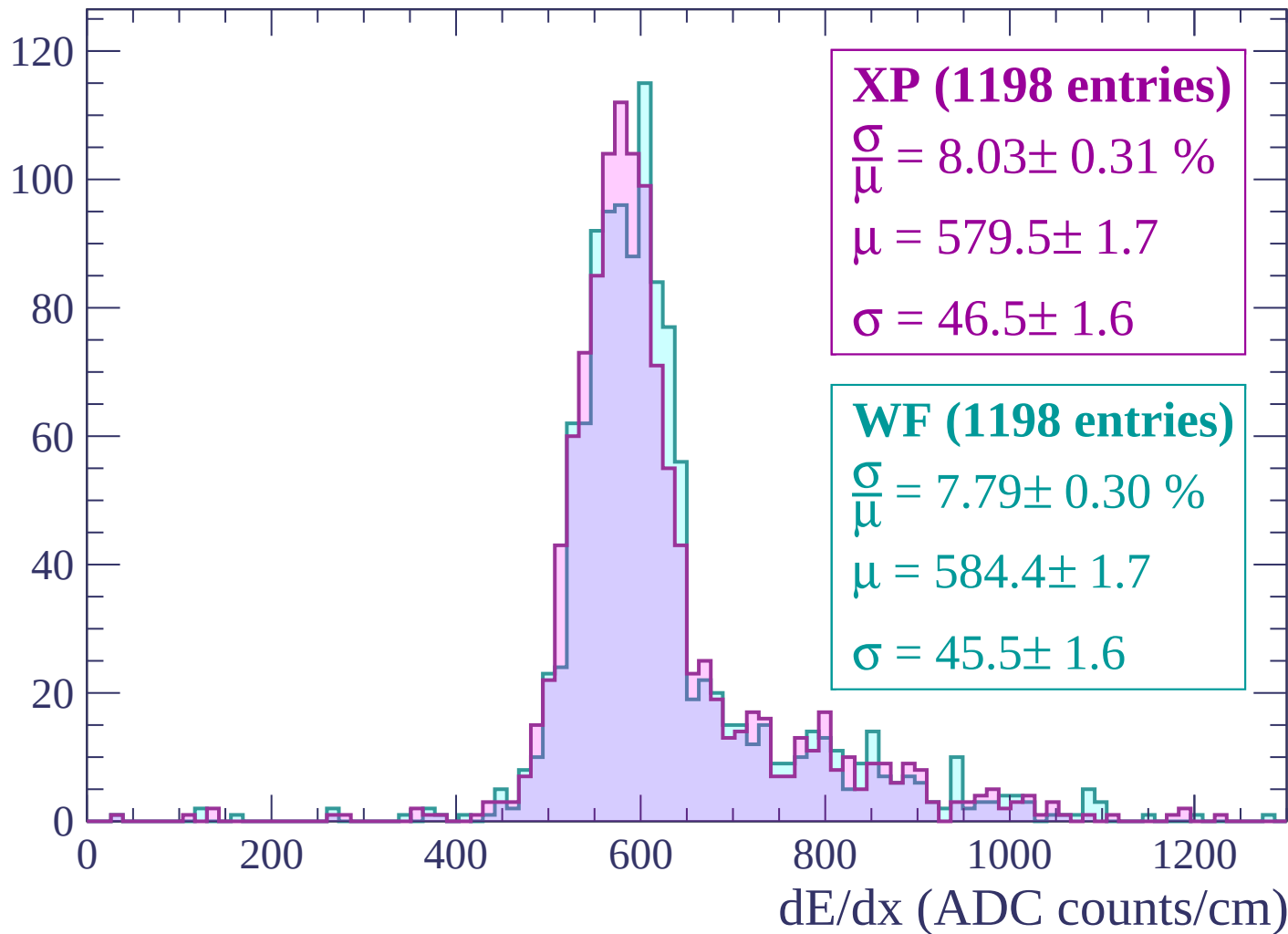
Energy loss | $-120 < p < -80$

Count



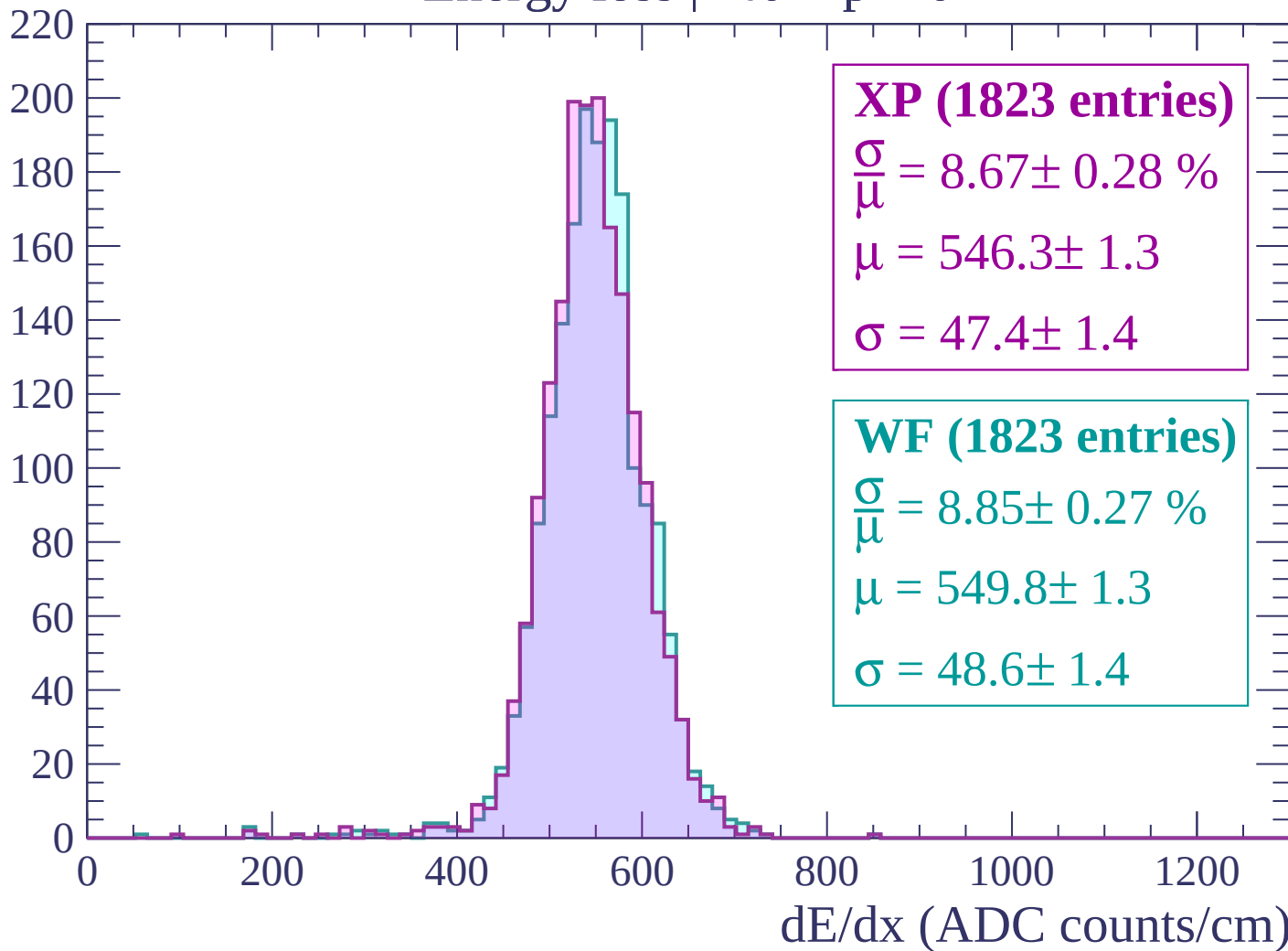
Energy loss | $-80 < p < -40$

Count



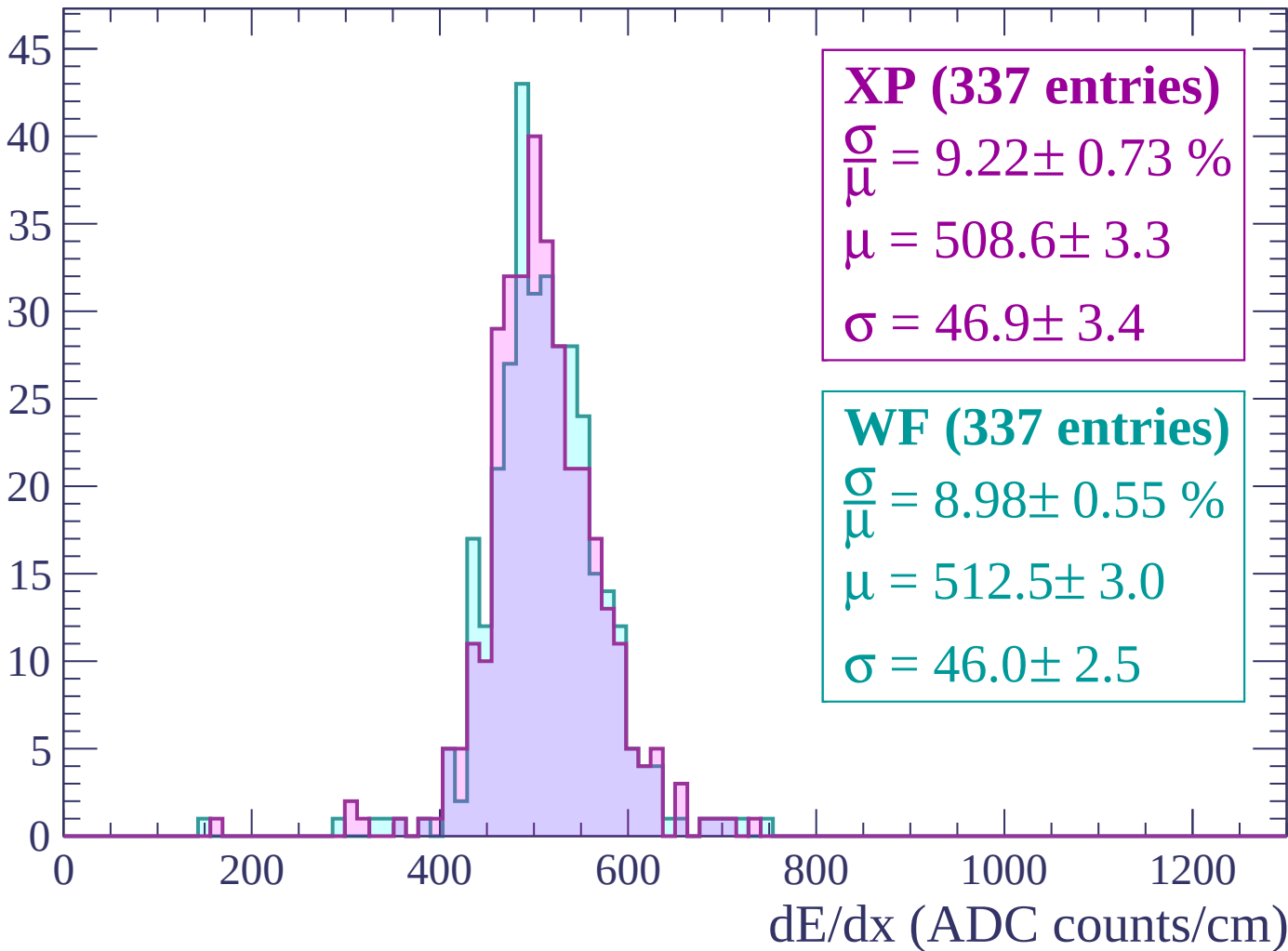
Energy loss | $-40 < p < 0$

Count



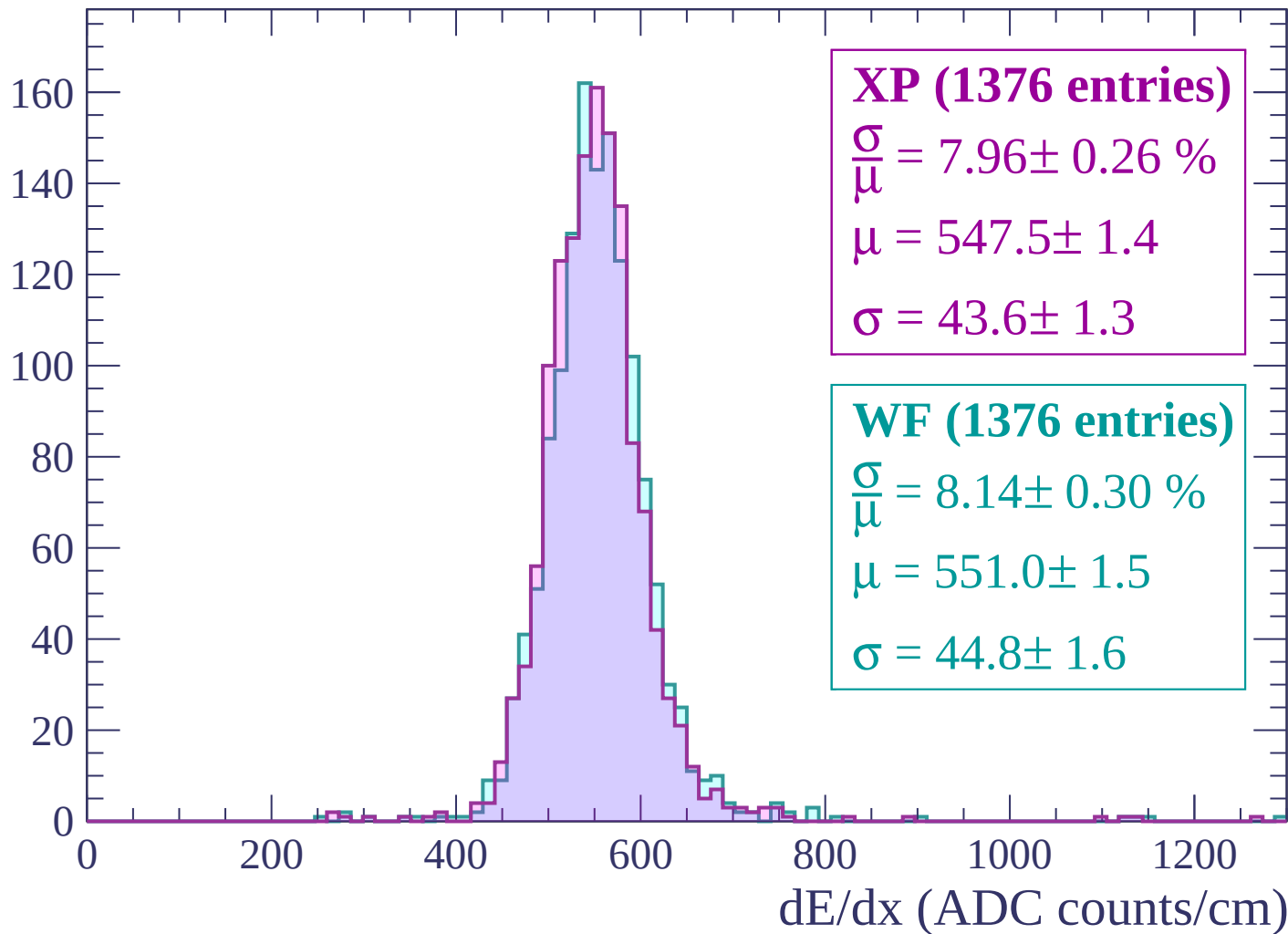
Energy loss | $0 < p < 40$

Count



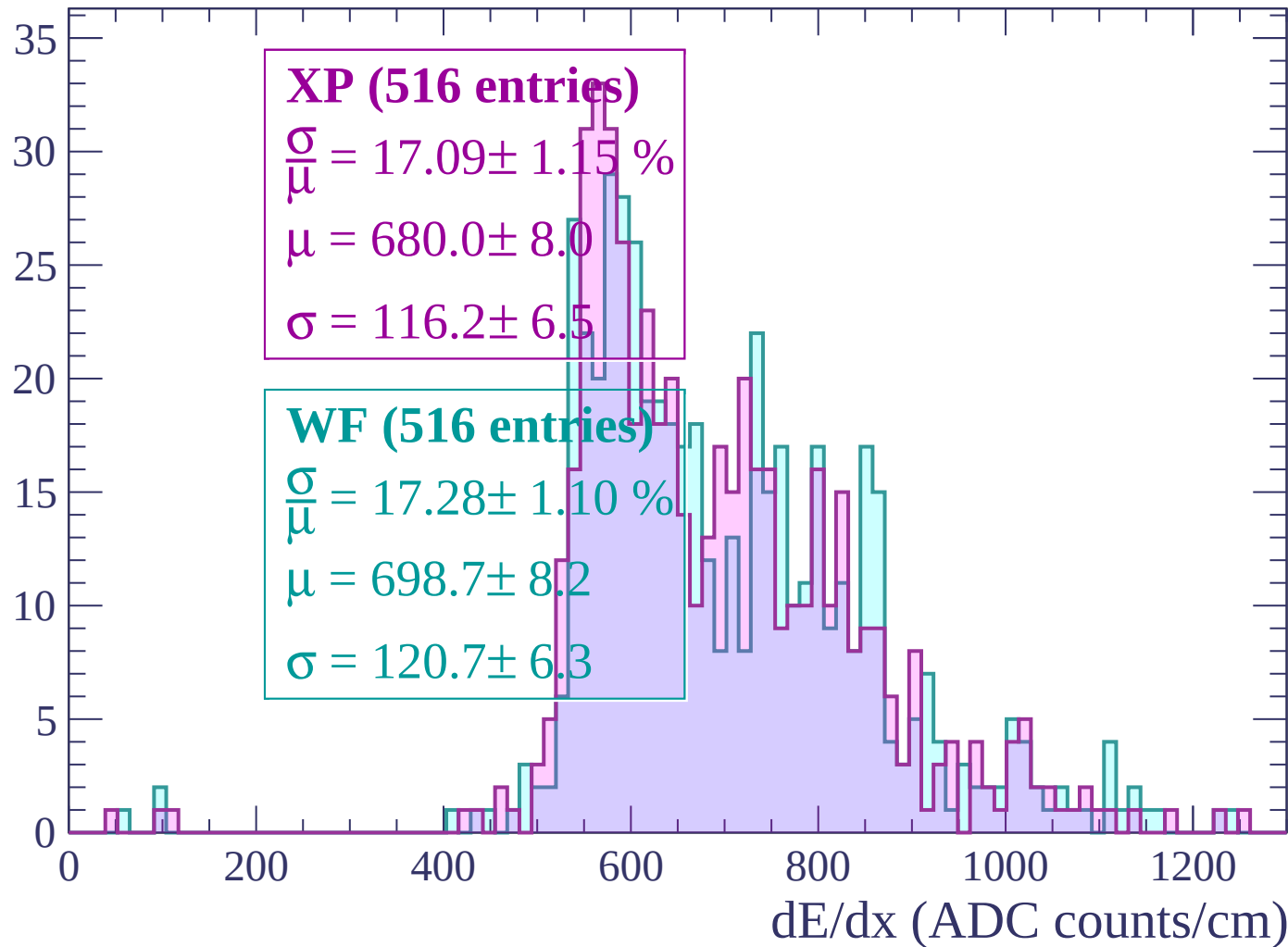
Energy loss | $40 < p < 80$

Count



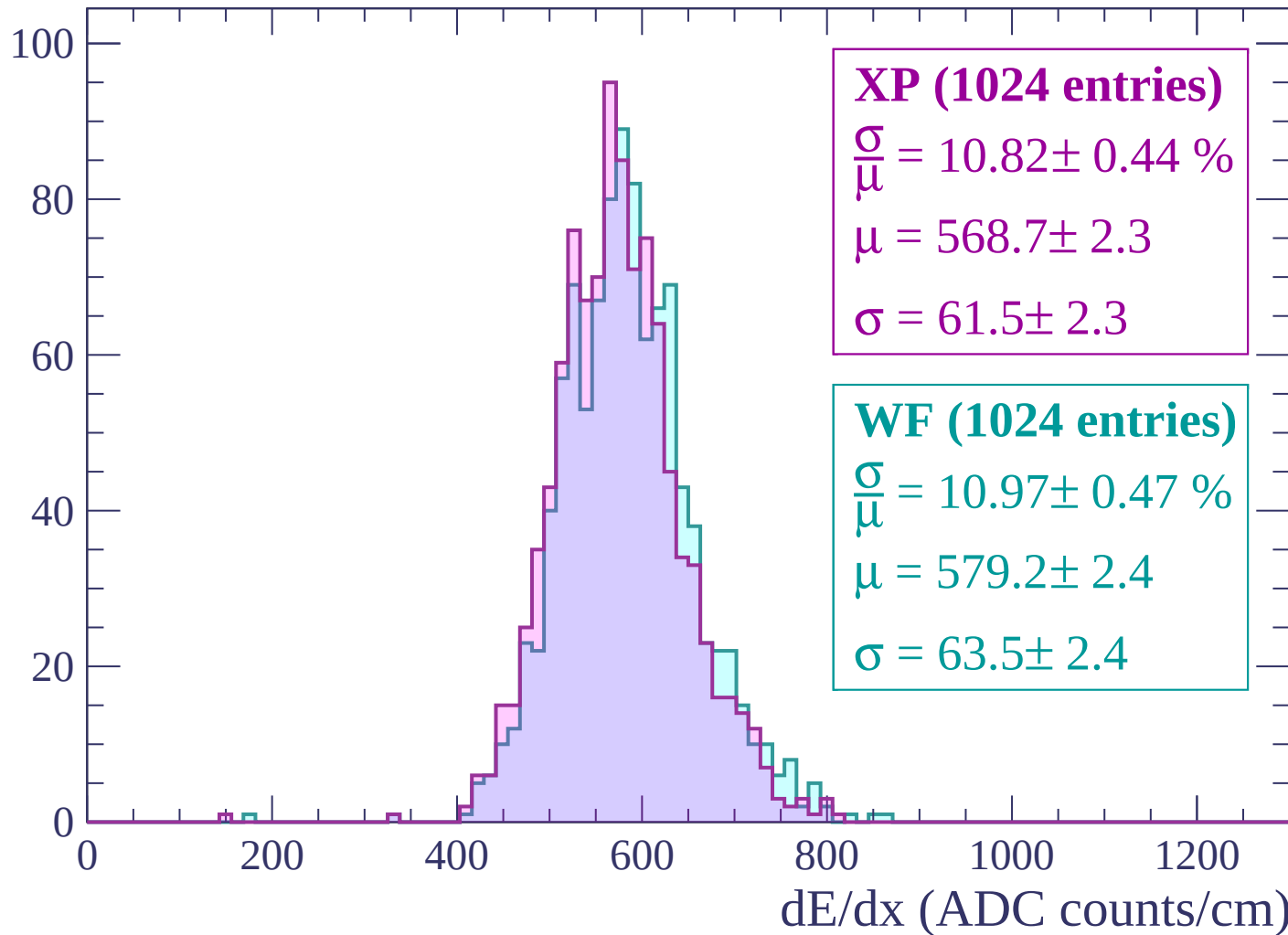
Energy loss | $80 < p < 120$

Count



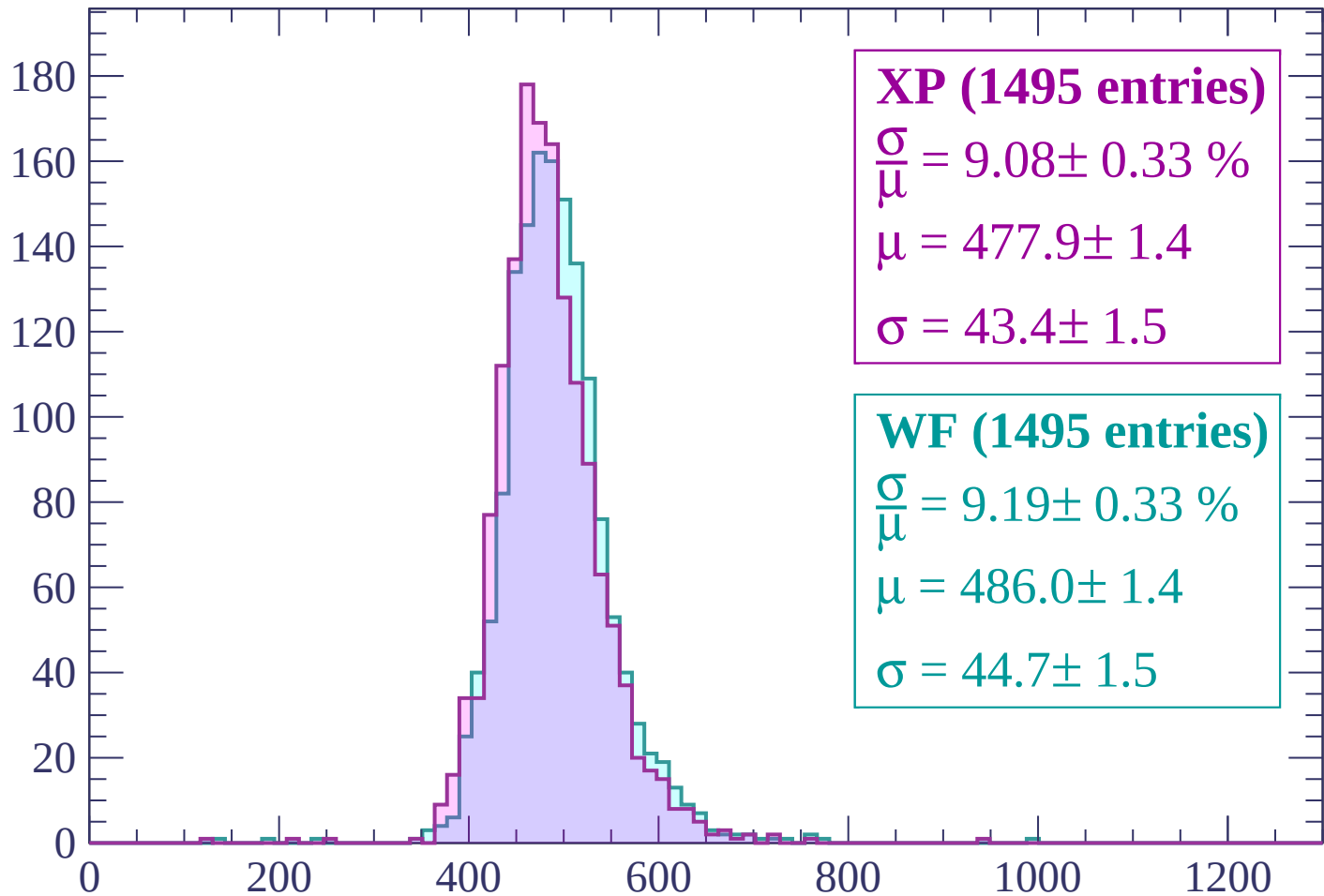
Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count



XP (1495 entries)

$$\frac{\sigma}{\mu} = 9.08 \pm 0.33 \%$$

$$\mu = 477.9 \pm 1.4$$

$$\sigma = 43.4 \pm 1.5$$

WF (1495 entries)

$$\frac{\sigma}{\mu} = 9.19 \pm 0.33 \%$$

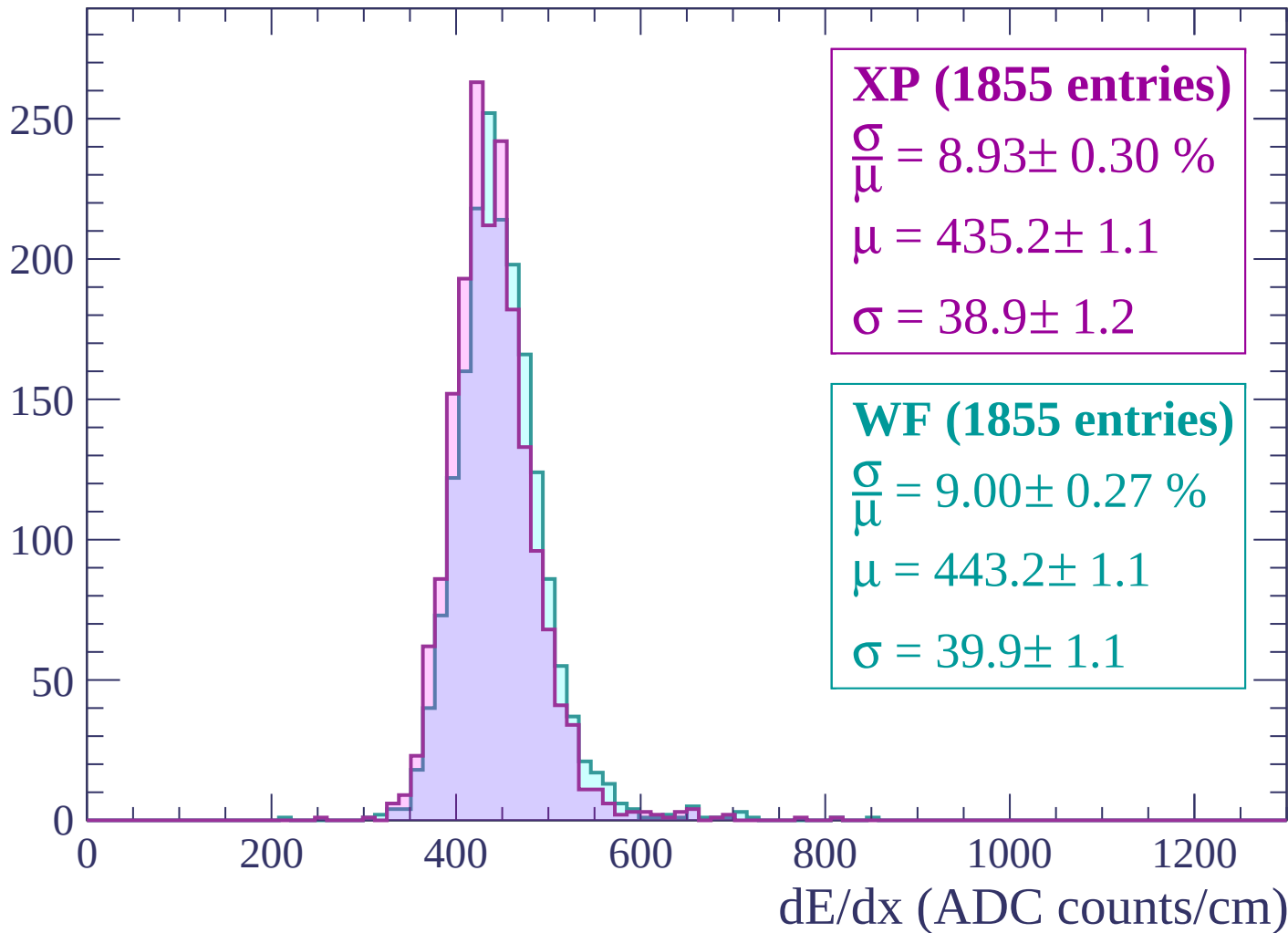
$$\mu = 486.0 \pm 1.4$$

$$\sigma = 44.7 \pm 1.5$$

dE/dx (ADC counts/cm)

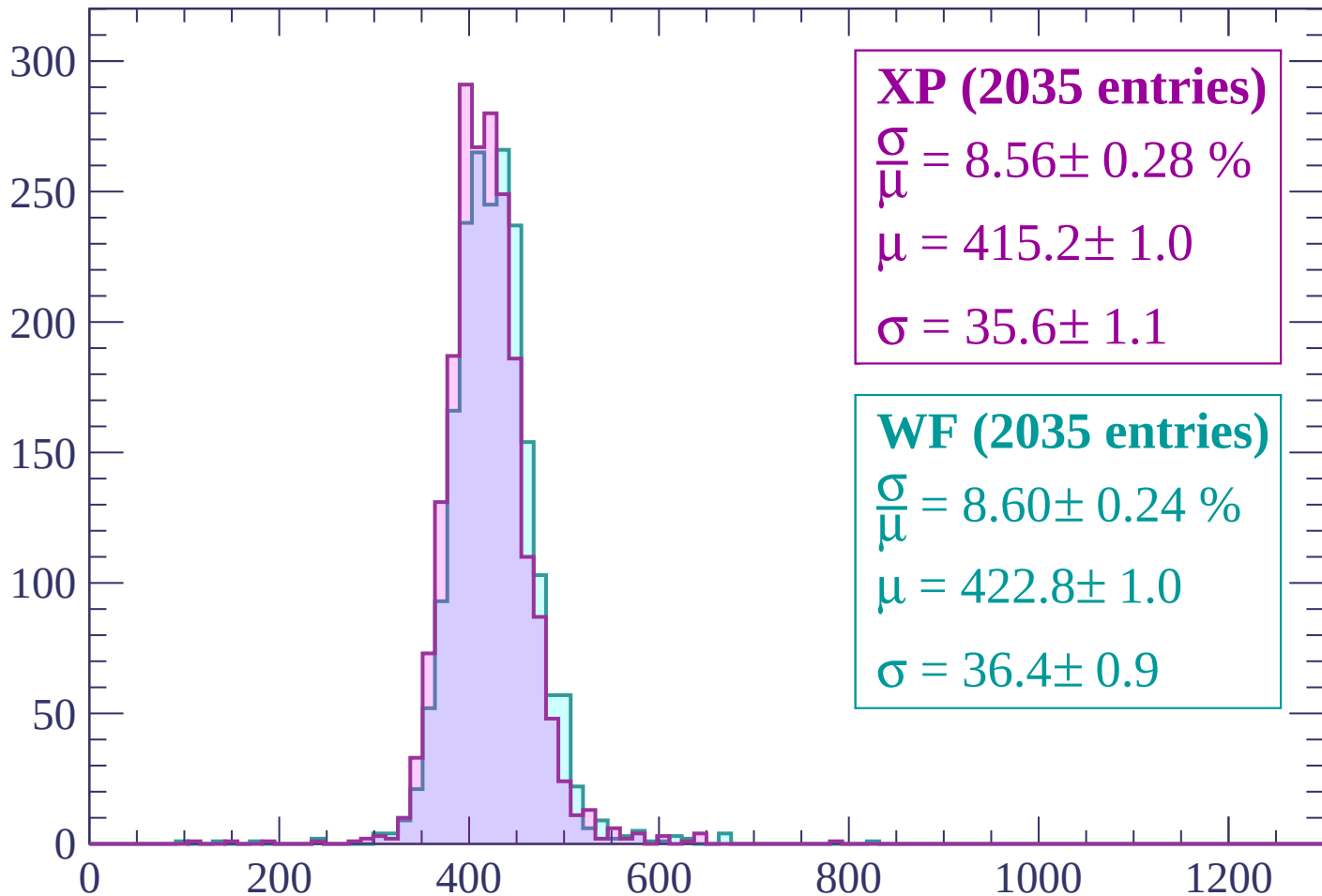
Energy loss | $200 < p < 240$

Count



Energy loss | $240 < p < 280$

Count



XP (2035 entries)

$$\frac{\sigma}{\mu} = 8.56 \pm 0.28 \%$$

$$\mu = 415.2 \pm 1.0$$

$$\sigma = 35.6 \pm 1.1$$

WF (2035 entries)

$$\frac{\sigma}{\mu} = 8.60 \pm 0.24 \%$$

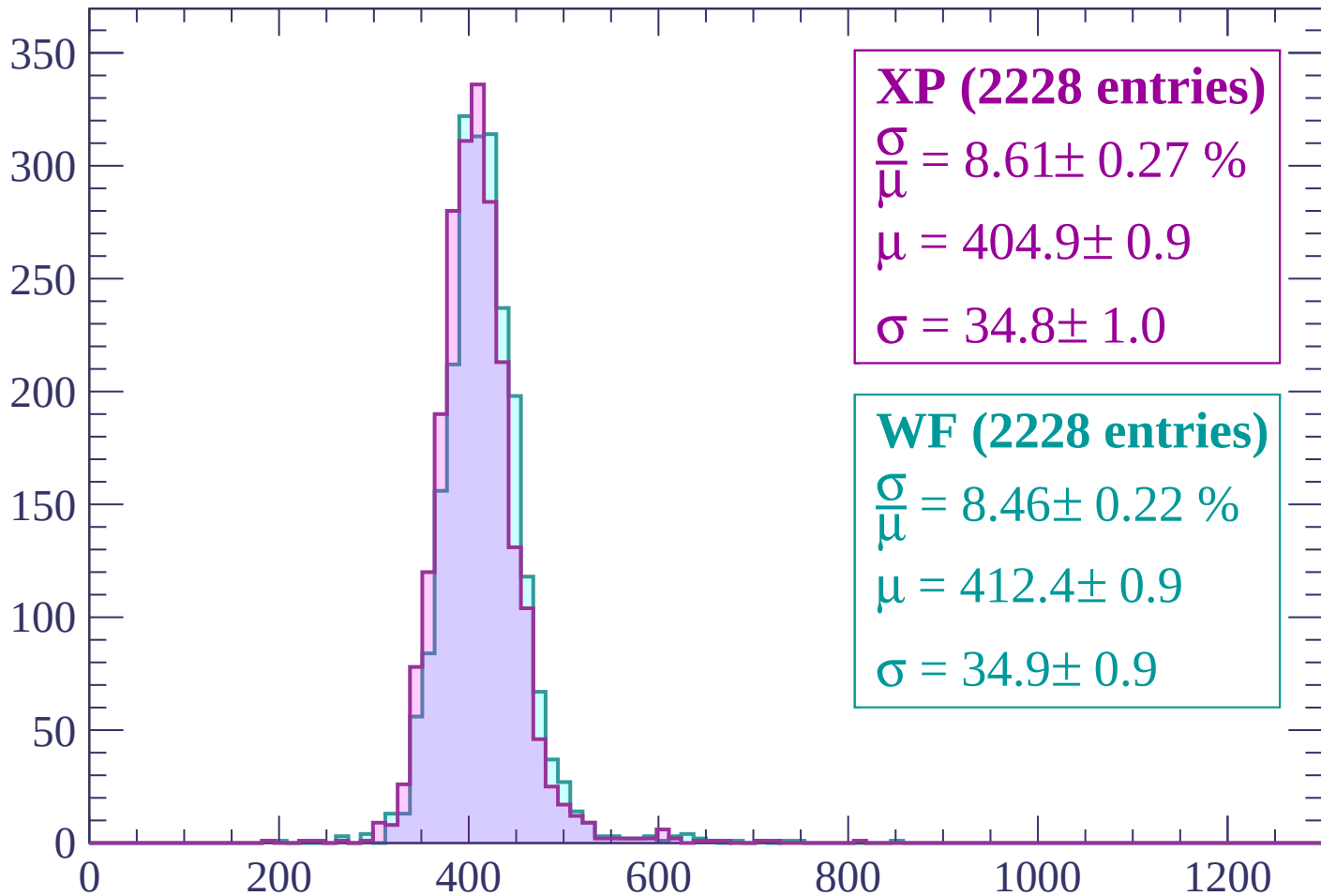
$$\mu = 422.8 \pm 1.0$$

$$\sigma = 36.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP (2228 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.27 \%$$

$$\mu = 404.9 \pm 0.9$$

$$\sigma = 34.8 \pm 1.0$$

WF (2228 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.22 \%$$

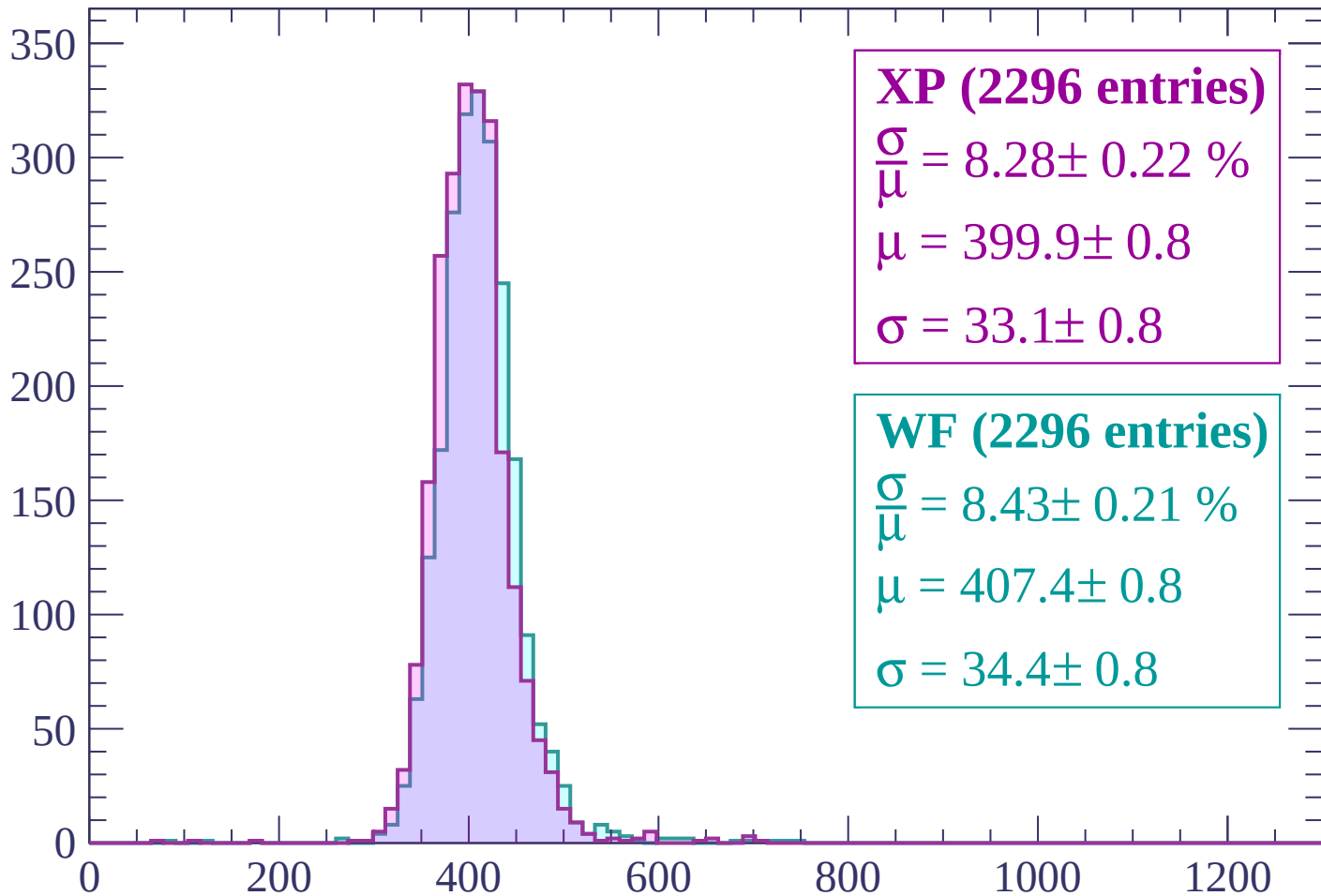
$$\mu = 412.4 \pm 0.9$$

$$\sigma = 34.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP (2296 entries)

$$\frac{\sigma}{\mu} = 8.28 \pm 0.22 \%$$

$$\mu = 399.9 \pm 0.8$$

$$\sigma = 33.1 \pm 0.8$$

WF (2296 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.21 \%$$

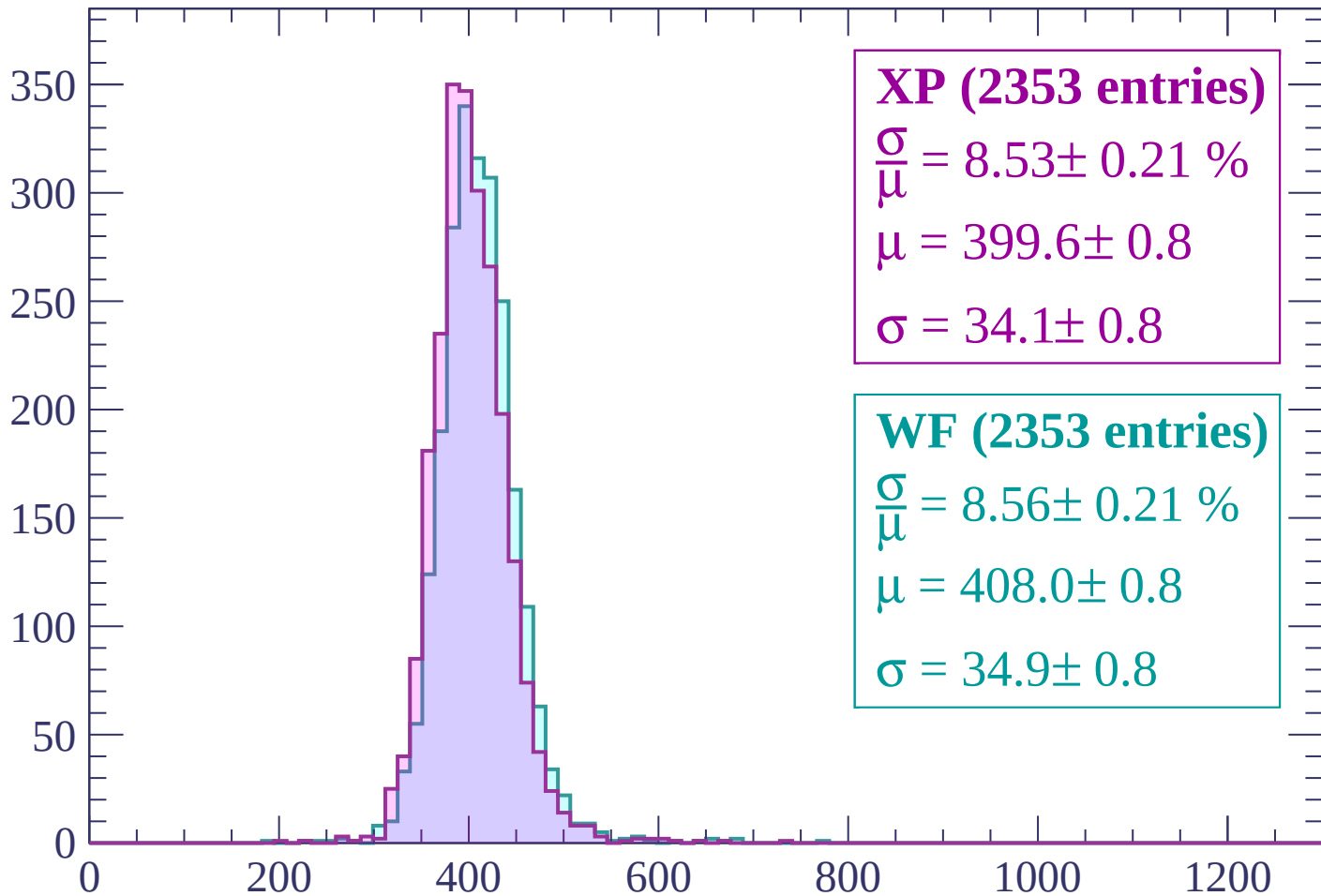
$$\mu = 407.4 \pm 0.8$$

$$\sigma = 34.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP (2353 entries)

$\frac{\sigma}{\mu} = 8.53 \pm 0.21 \%$

$\mu = 399.6 \pm 0.8$

$\sigma = 34.1 \pm 0.8$

WF (2353 entries)

$\frac{\sigma}{\mu} = 8.56 \pm 0.21 \%$

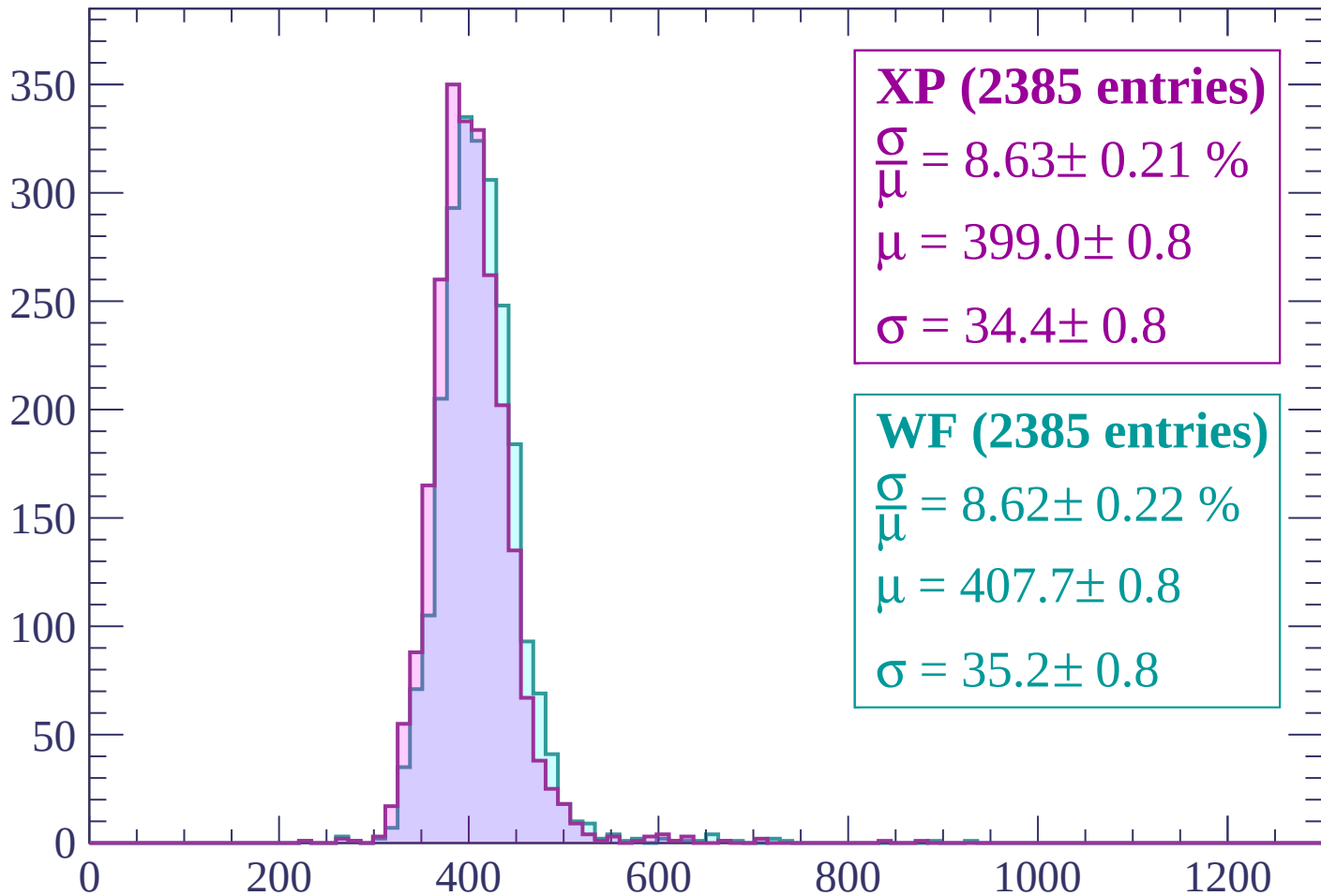
$\mu = 408.0 \pm 0.8$

$\sigma = 34.9 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP (2385 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.21 \%$

$\mu = 399.0 \pm 0.8$

$\sigma = 34.4 \pm 0.8$

WF (2385 entries)

$\frac{\sigma}{\mu} = 8.62 \pm 0.22 \%$

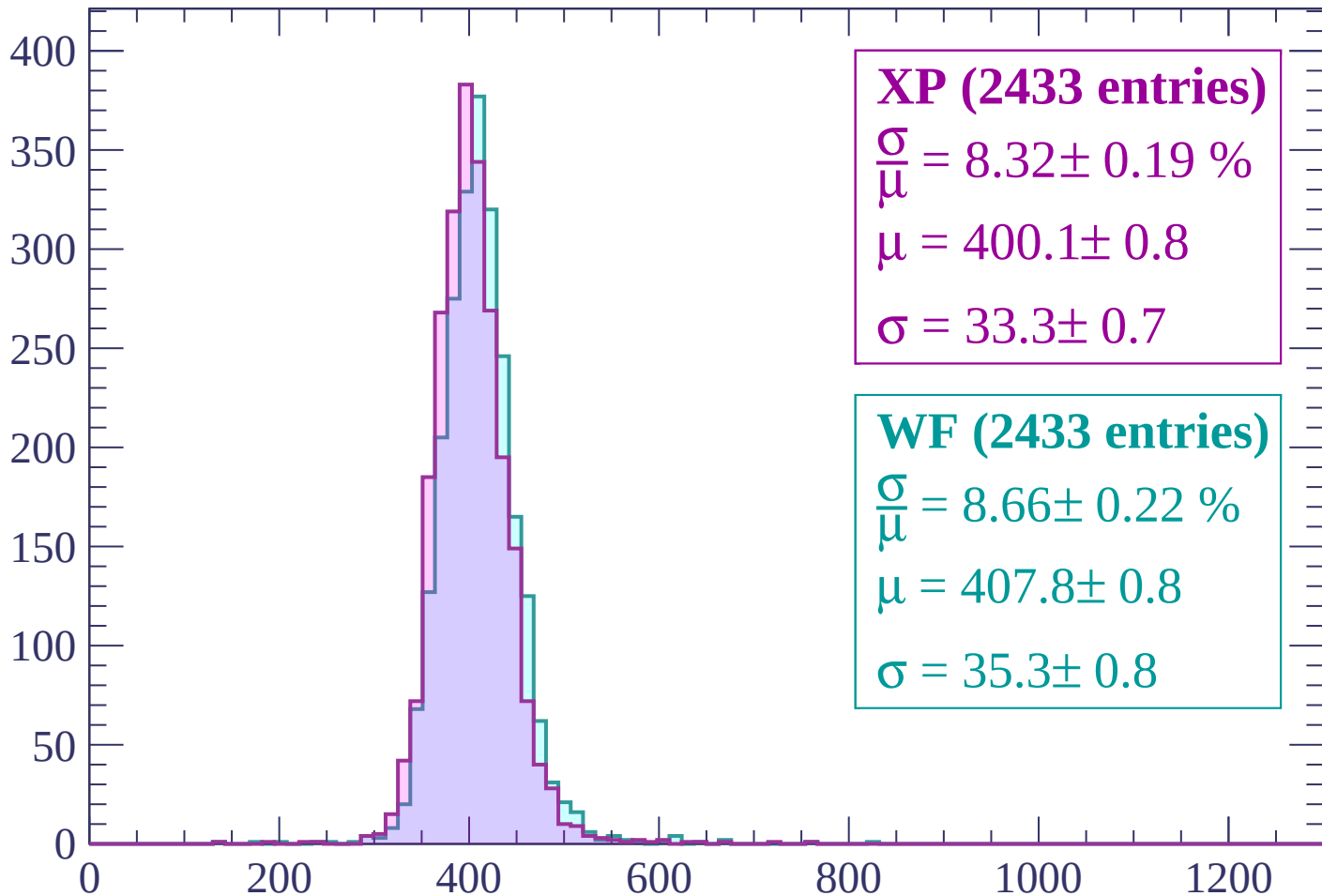
$\mu = 407.7 \pm 0.8$

$\sigma = 35.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP (2433 entries)

$$\frac{\sigma}{\mu} = 8.32 \pm 0.19 \%$$

$$\mu = 400.1 \pm 0.8$$

$$\sigma = 33.3 \pm 0.7$$

WF (2433 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.22 \%$$

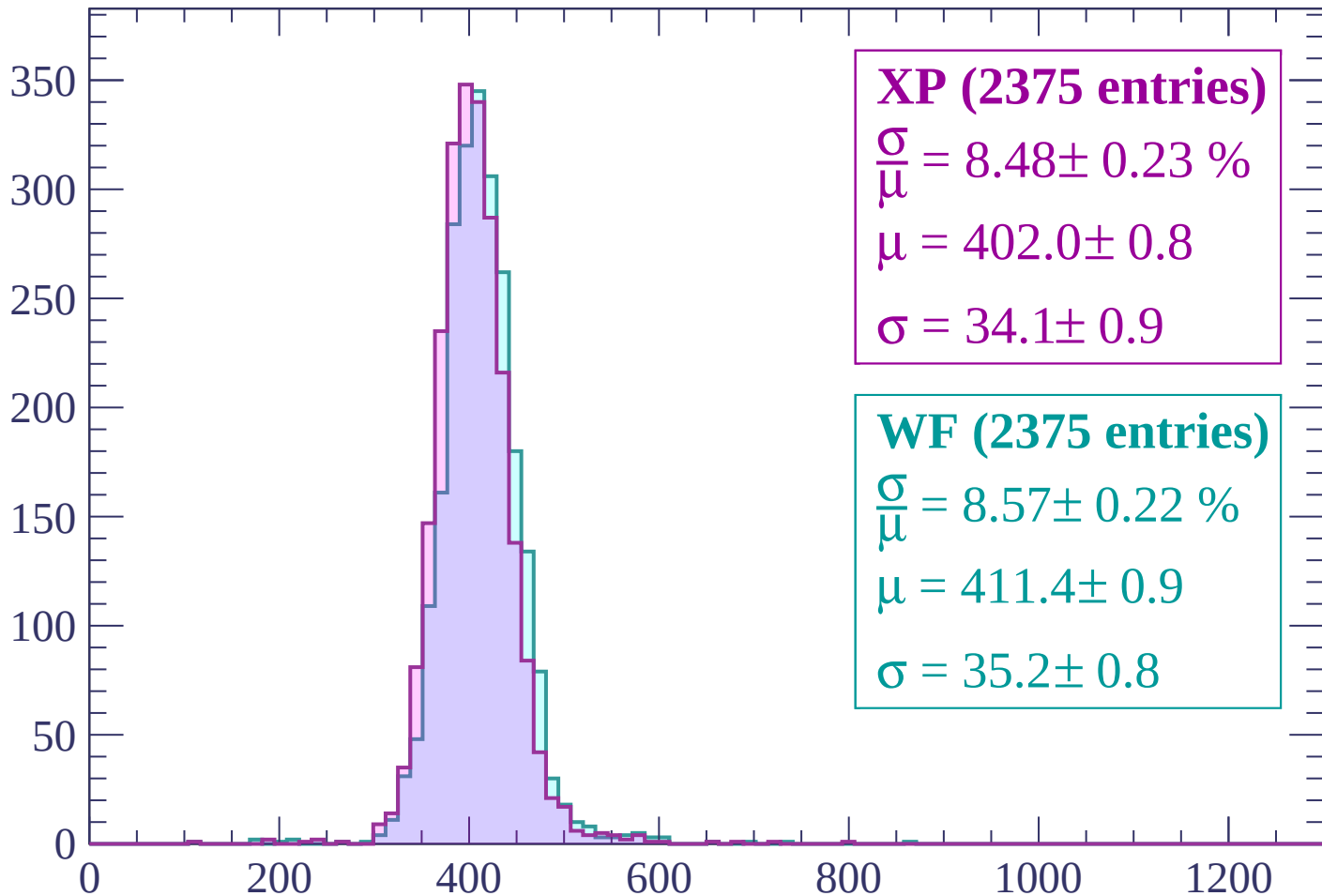
$$\mu = 407.8 \pm 0.8$$

$$\sigma = 35.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP (2375 entries)

$\frac{\sigma}{\mu} = 8.48 \pm 0.23 \%$

$\mu = 402.0 \pm 0.8$

$\sigma = 34.1 \pm 0.9$

WF (2375 entries)

$\frac{\sigma}{\mu} = 8.57 \pm 0.22 \%$

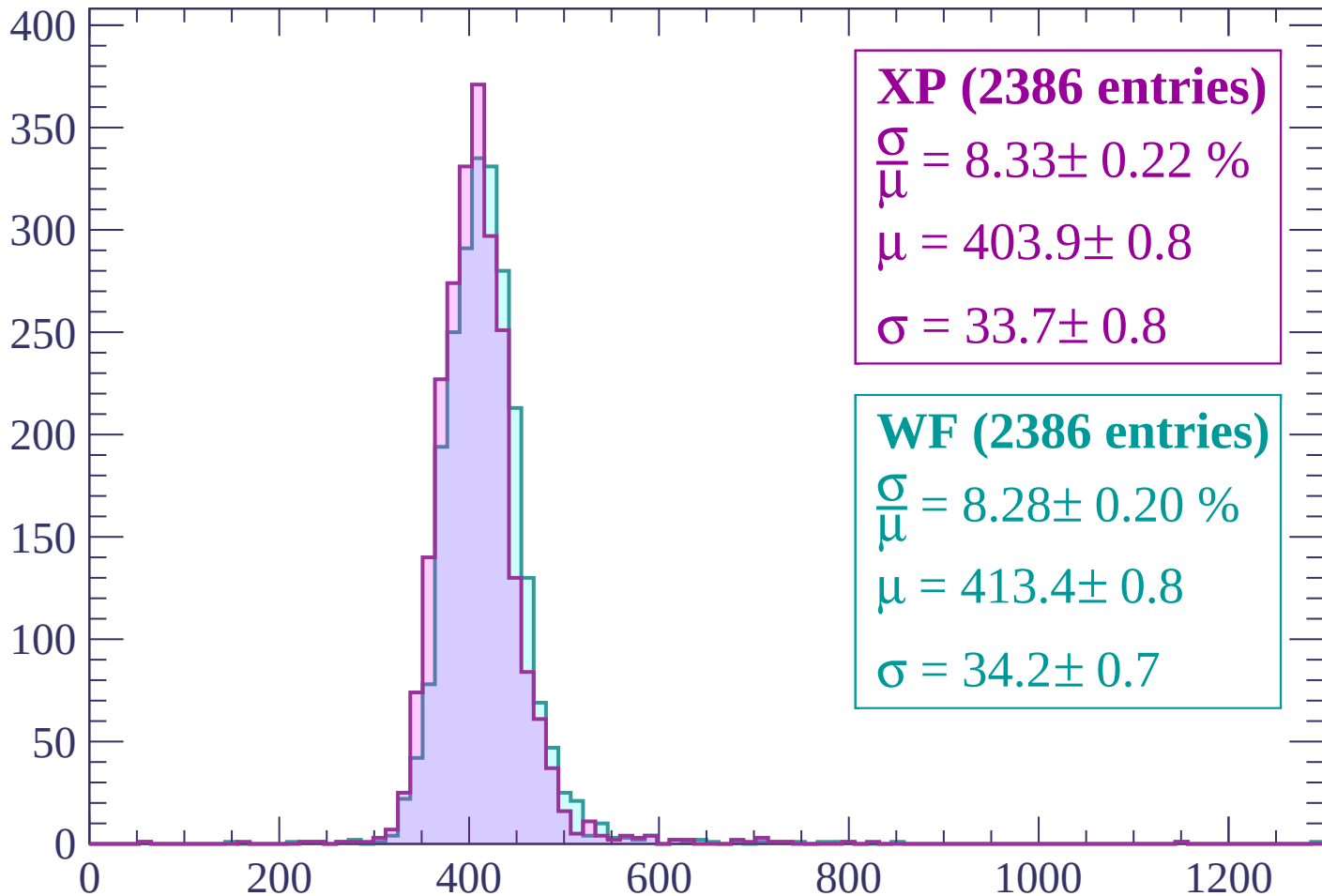
$\mu = 411.4 \pm 0.9$

$\sigma = 35.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count



XP (2386 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.22 \%$$

$$\mu = 403.9 \pm 0.8$$

$$\sigma = 33.7 \pm 0.8$$

WF (2386 entries)

$$\frac{\sigma}{\mu} = 8.28 \pm 0.20 \%$$

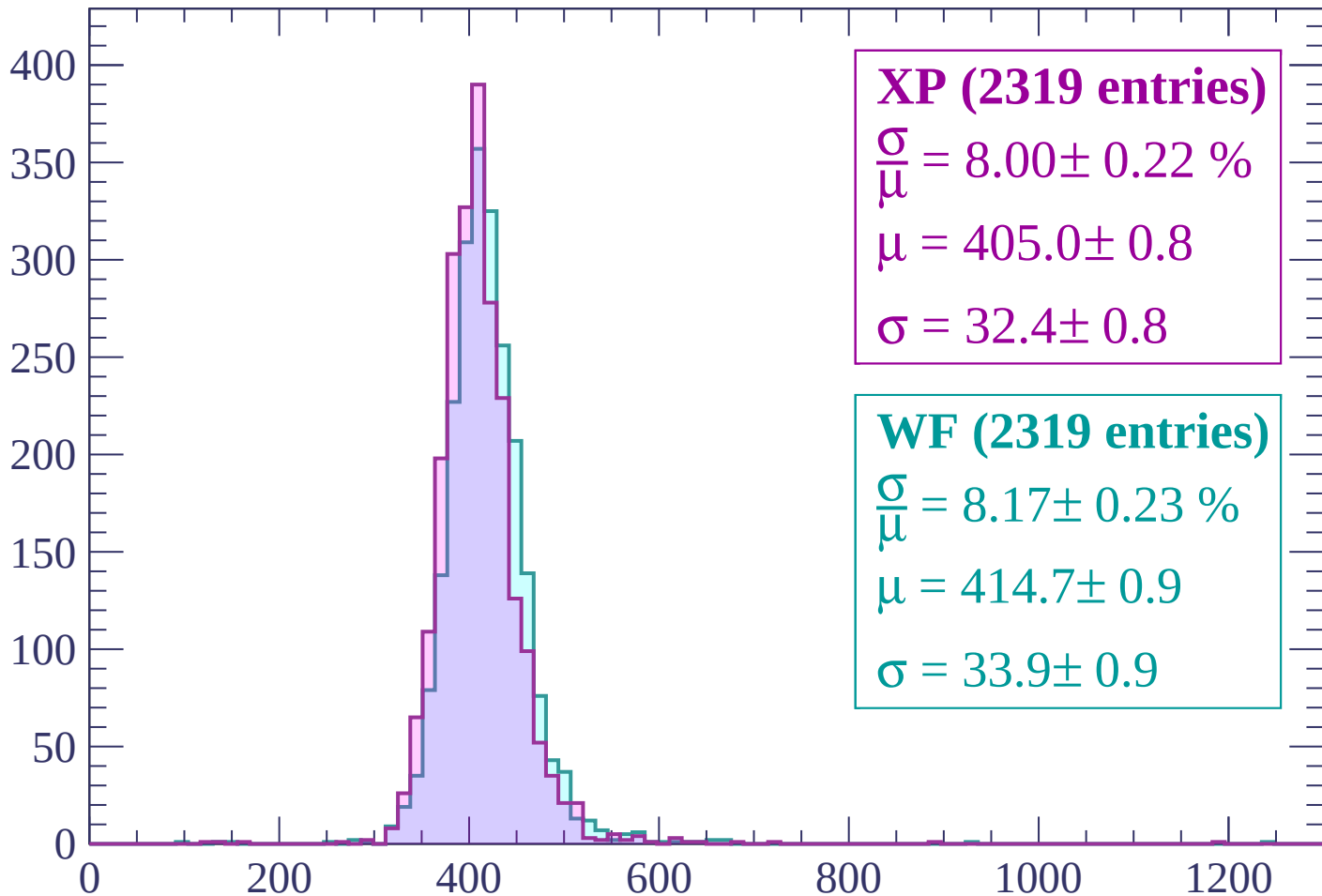
$$\mu = 413.4 \pm 0.8$$

$$\sigma = 34.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP (2319 entries)

$$\frac{\sigma}{\mu} = 8.00 \pm 0.22 \%$$

$$\mu = 405.0 \pm 0.8$$

$$\sigma = 32.4 \pm 0.8$$

WF (2319 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.23 \%$$

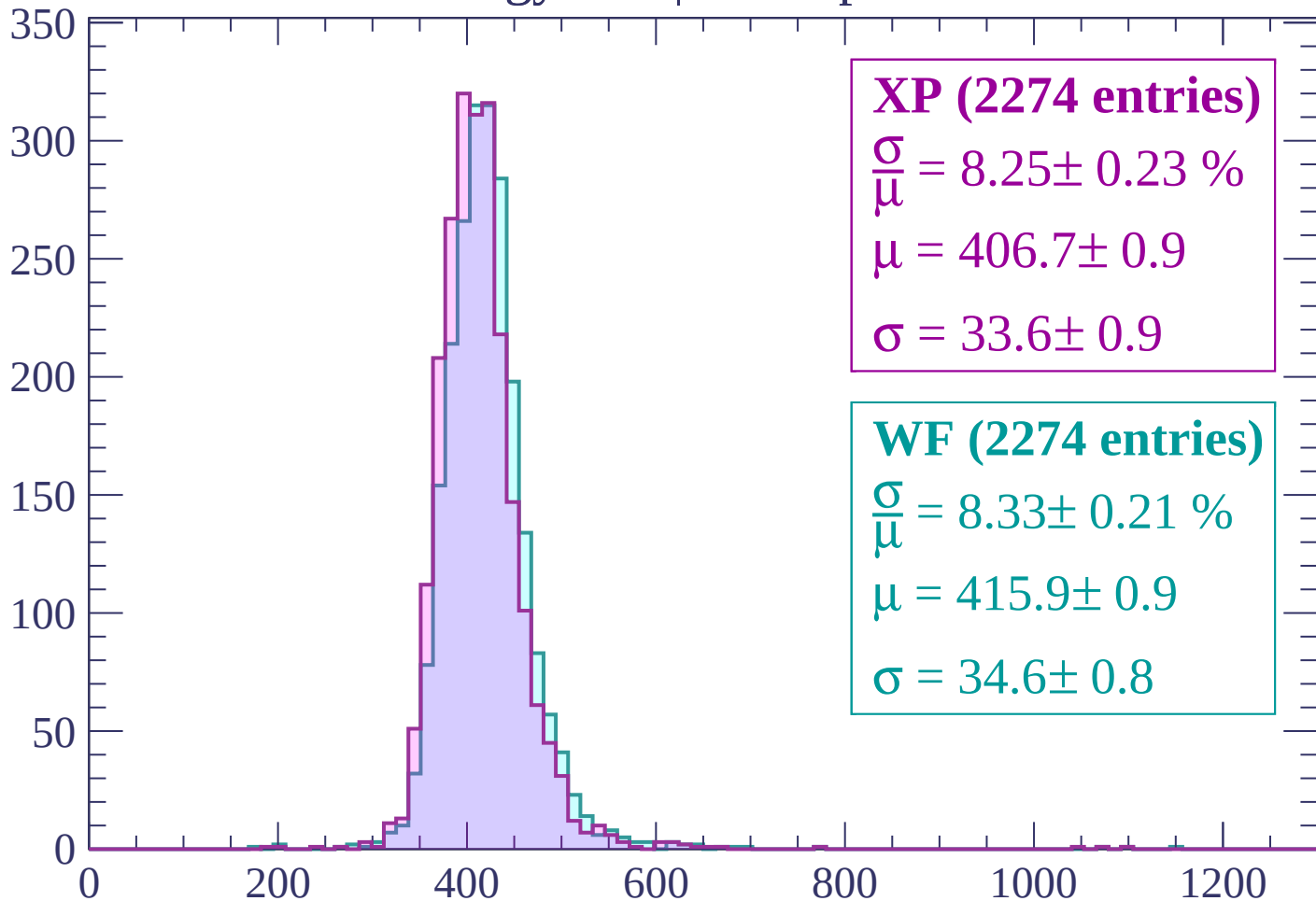
$$\mu = 414.7 \pm 0.9$$

$$\sigma = 33.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP (2274 entries)

$$\frac{\sigma}{\mu} = 8.25 \pm 0.23 \%$$

$$\mu = 406.7 \pm 0.9$$

$$\sigma = 33.6 \pm 0.9$$

WF (2274 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.21 \%$$

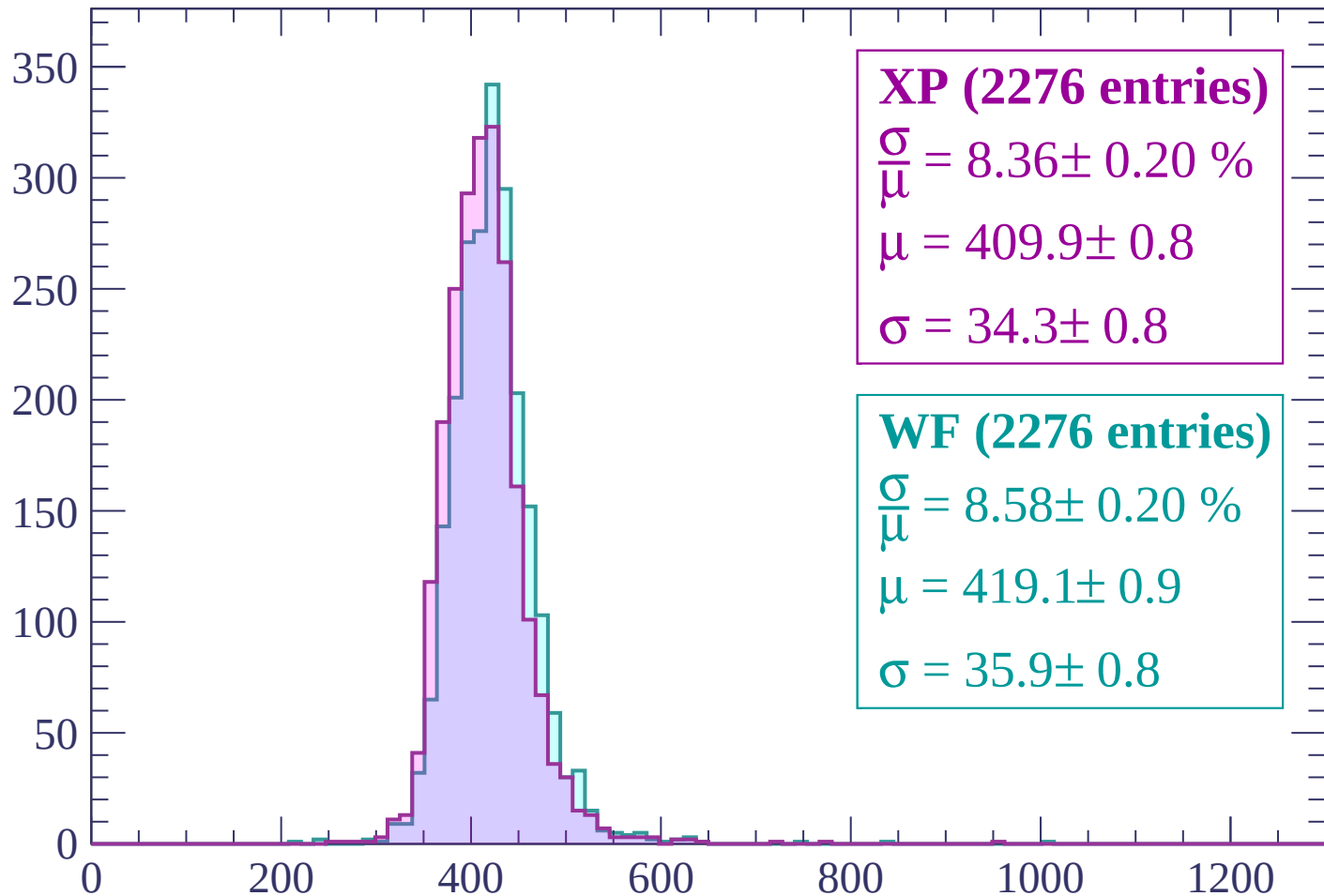
$$\mu = 415.9 \pm 0.9$$

$$\sigma = 34.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP (2276 entries)

$$\frac{\sigma}{\mu} = 8.36 \pm 0.20 \%$$

$$\mu = 409.9 \pm 0.8$$

$$\sigma = 34.3 \pm 0.8$$

WF (2276 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.20 \%$$

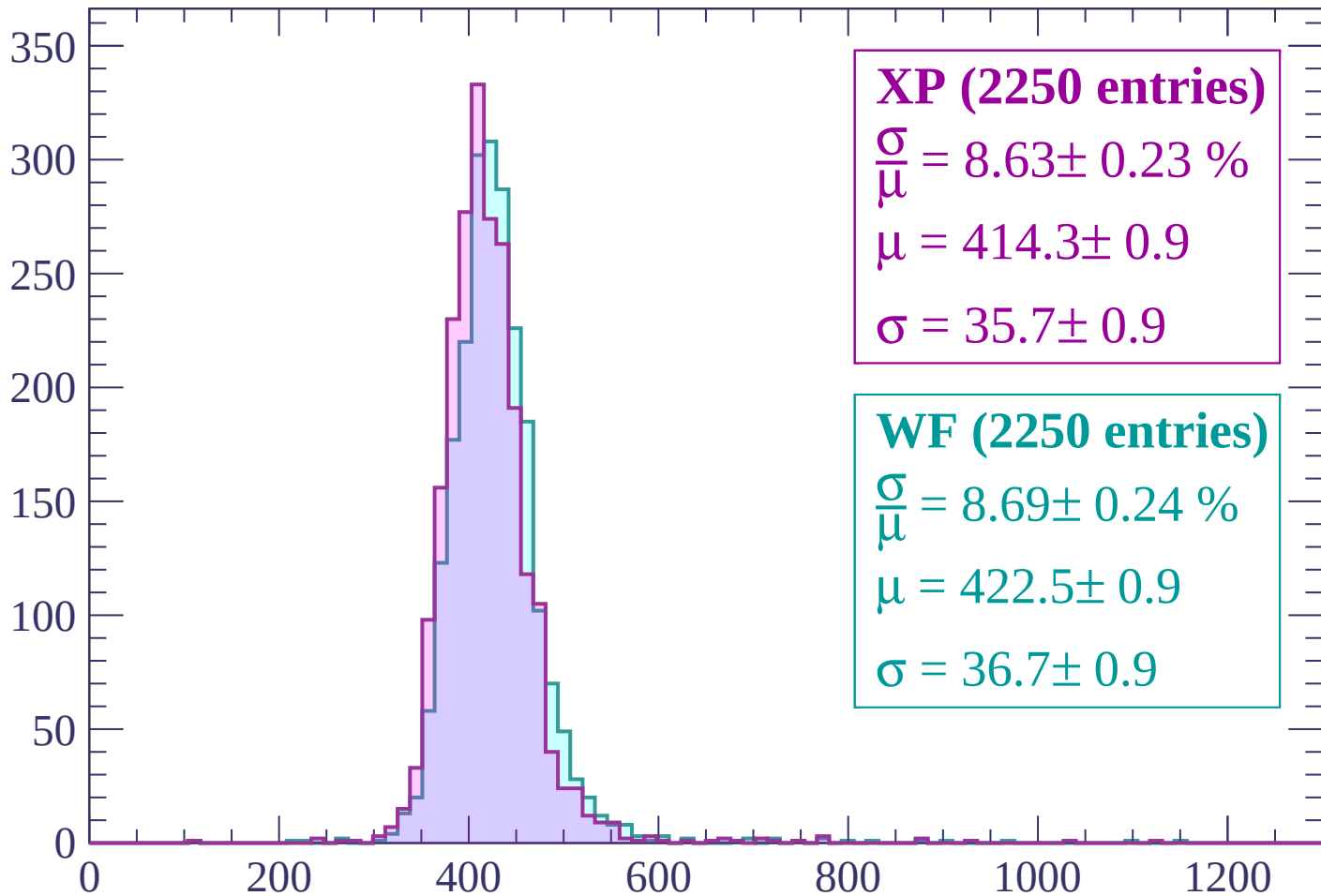
$$\mu = 419.1 \pm 0.9$$

$$\sigma = 35.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP (2250 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.23 \%$$

$$\mu = 414.3 \pm 0.9$$

$$\sigma = 35.7 \pm 0.9$$

WF (2250 entries)

$$\frac{\sigma}{\mu} = 8.69 \pm 0.24 \%$$

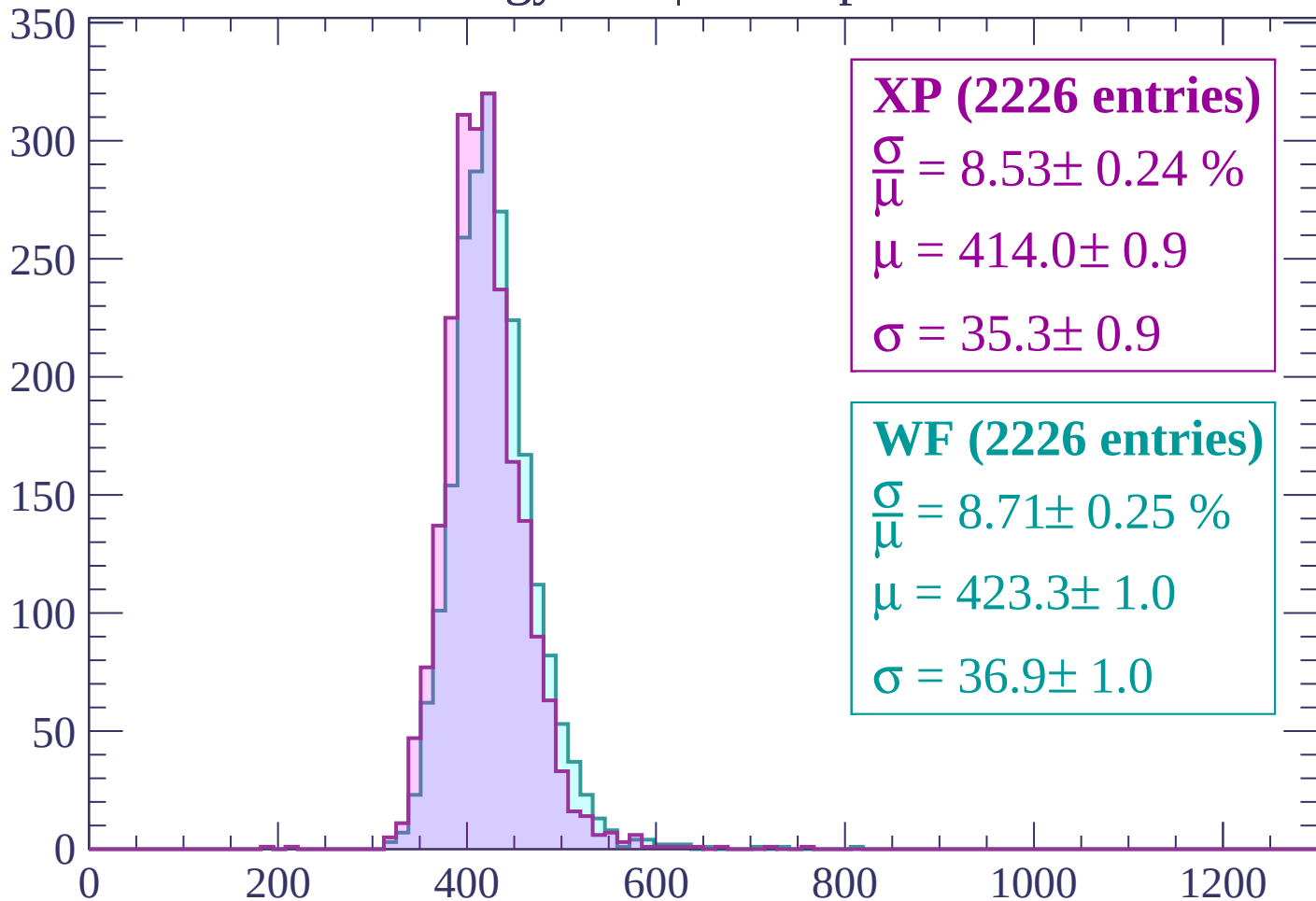
$$\mu = 422.5 \pm 0.9$$

$$\sigma = 36.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP (2226 entries)

$$\frac{\sigma}{\mu} = 8.53 \pm 0.24 \%$$

$$\mu = 414.0 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

WF (2226 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.25 \%$$

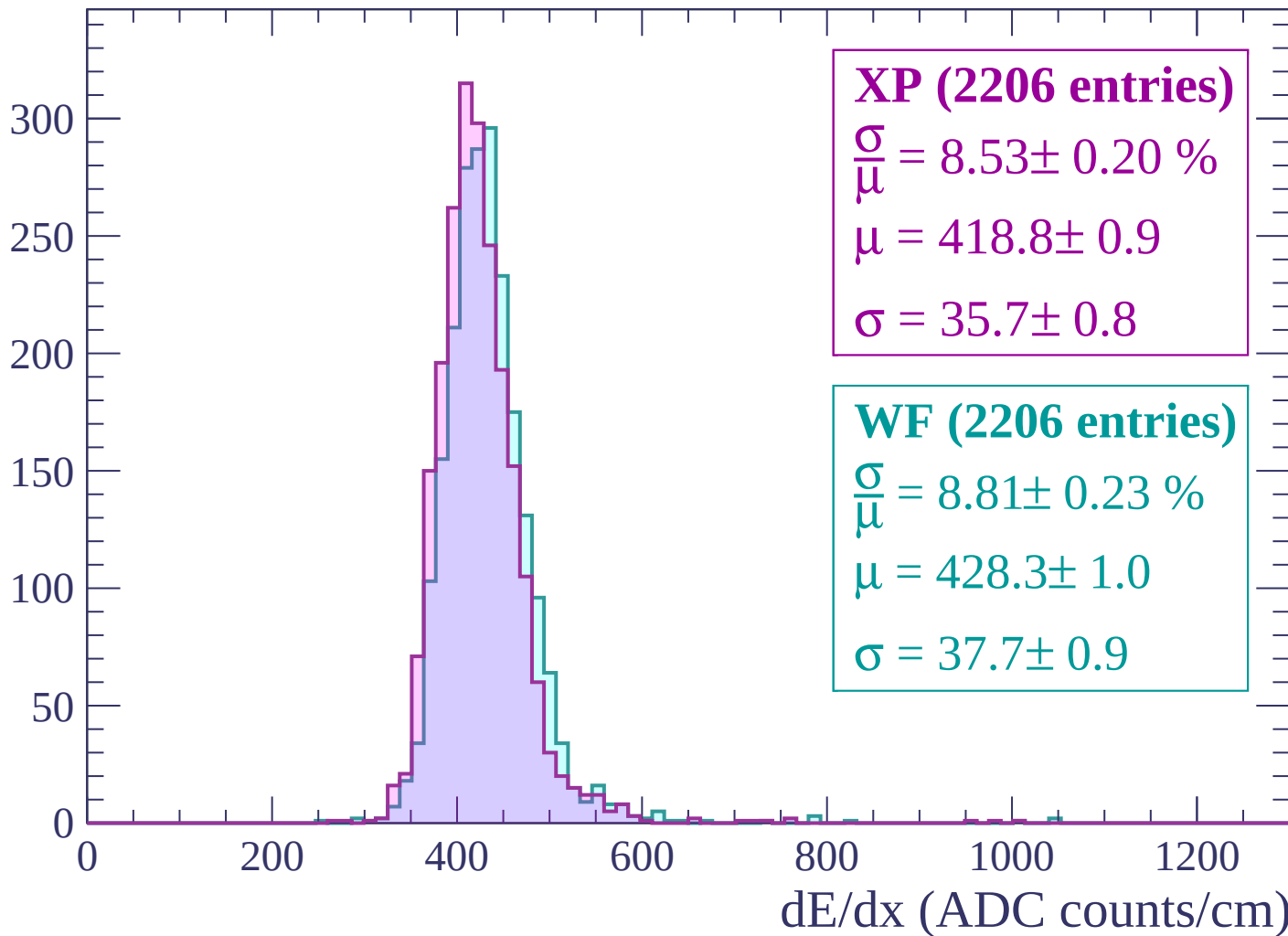
$$\mu = 423.3 \pm 1.0$$

$$\sigma = 36.9 \pm 1.0$$

dE/dx (ADC counts/cm)

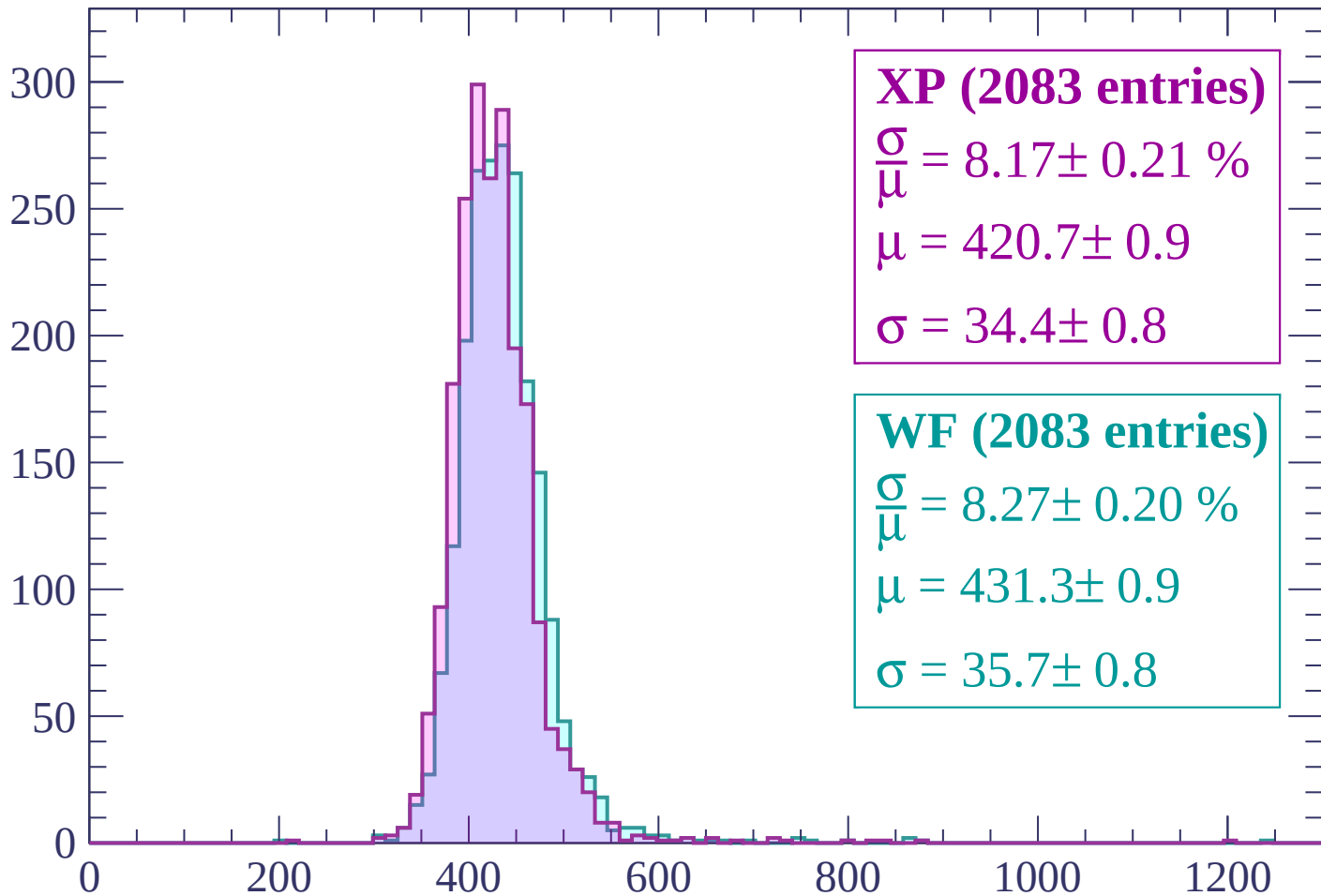
Energy loss | $760 < p < 800$

Count



Energy loss | $800 < p < 840$

Count



XP (2083 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.21 \%$$

$$\mu = 420.7 \pm 0.9$$

$$\sigma = 34.4 \pm 0.8$$

WF (2083 entries)

$$\frac{\sigma}{\mu} = 8.27 \pm 0.20 \%$$

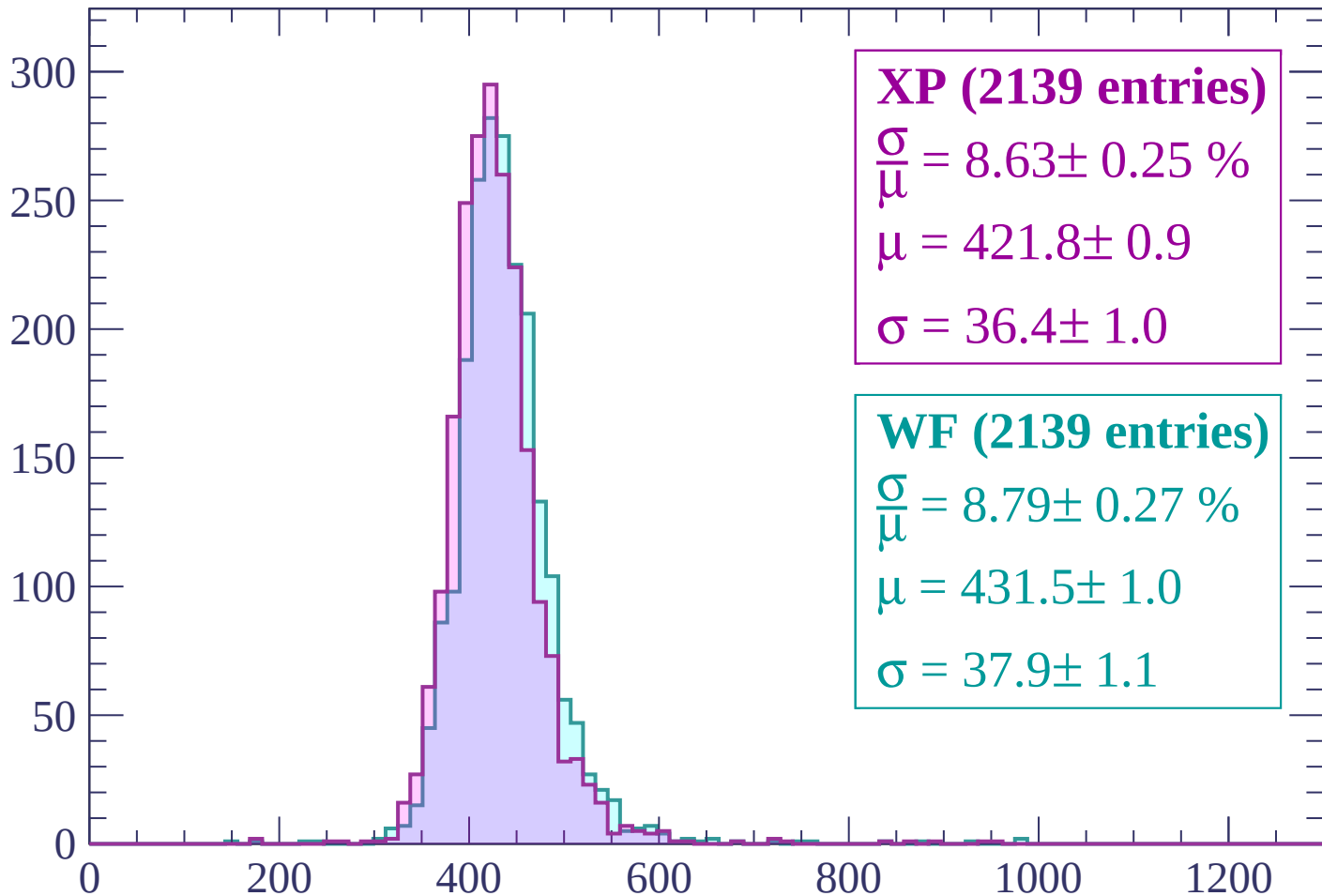
$$\mu = 431.3 \pm 0.9$$

$$\sigma = 35.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count



XP (2139 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.25 \%$

$\mu = 421.8 \pm 0.9$

$\sigma = 36.4 \pm 1.0$

WF (2139 entries)

$\frac{\sigma}{\mu} = 8.79 \pm 0.27 \%$

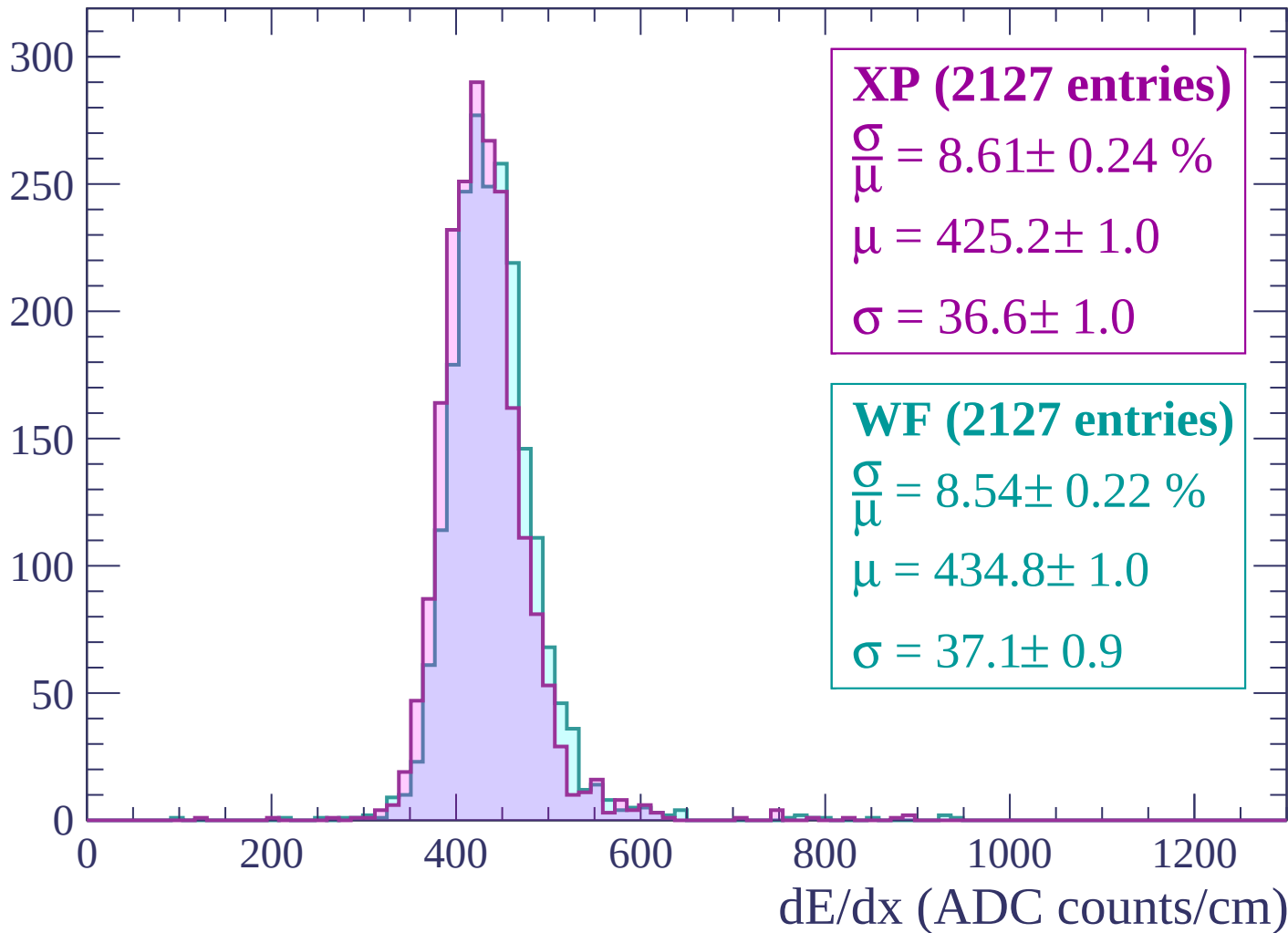
$\mu = 431.5 \pm 1.0$

$\sigma = 37.9 \pm 1.1$

dE/dx (ADC counts/cm)

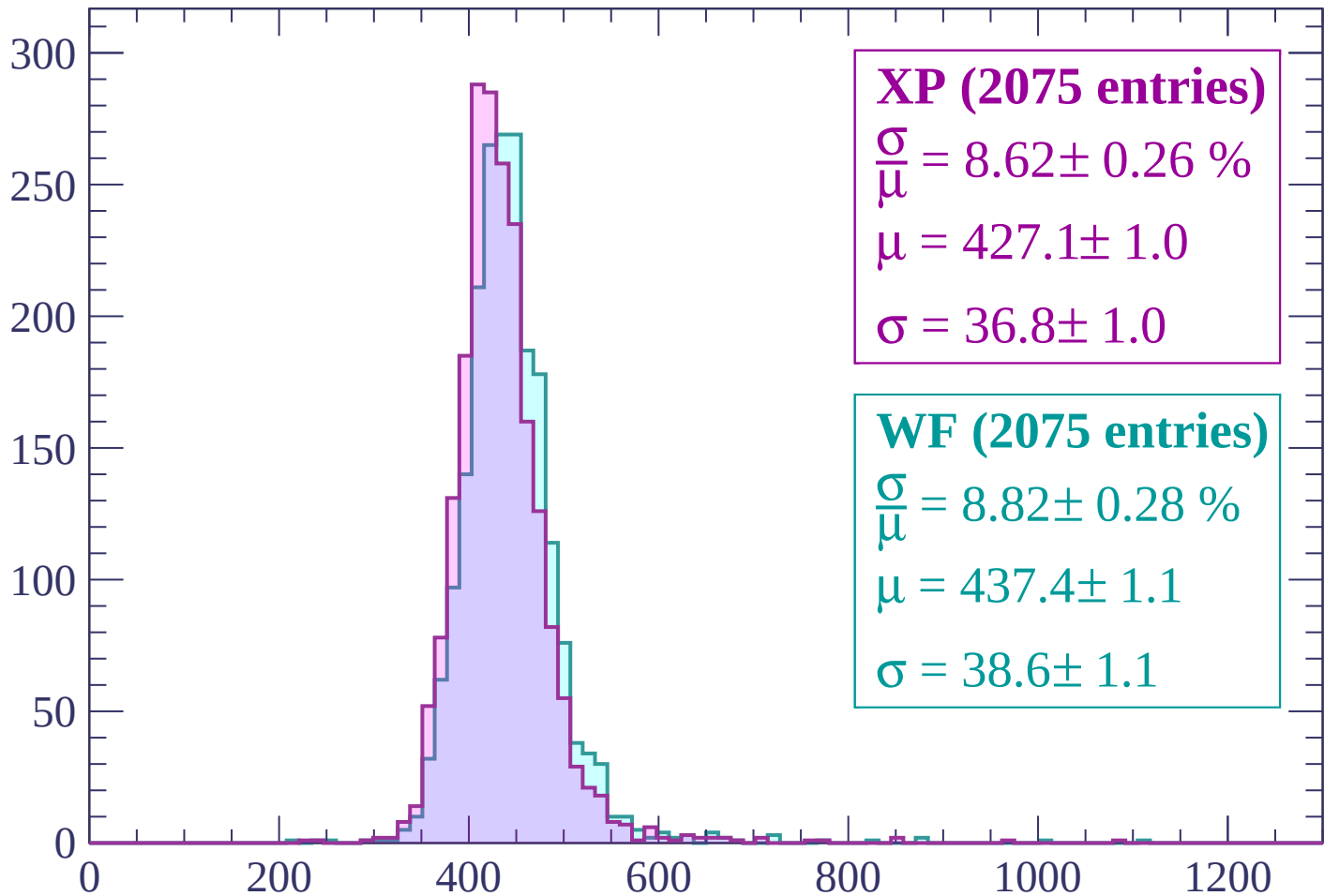
Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count



XP (2075 entries)

$\frac{\sigma}{\mu} = 8.62 \pm 0.26 \%$

$\mu = 427.1 \pm 1.0$

$\sigma = 36.8 \pm 1.0$

WF (2075 entries)

$\frac{\sigma}{\mu} = 8.82 \pm 0.28 \%$

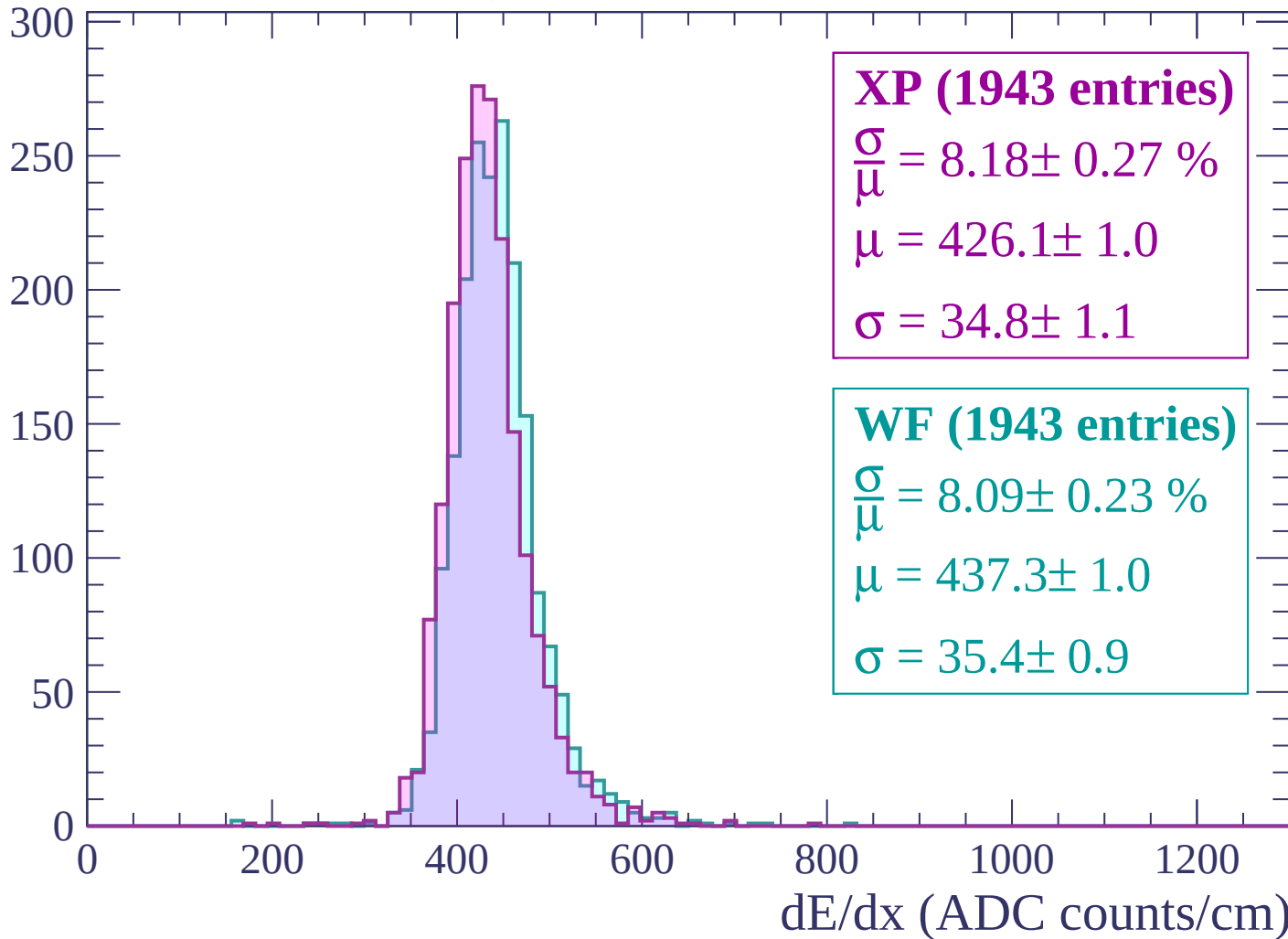
$\mu = 437.4 \pm 1.1$

$\sigma = 38.6 \pm 1.1$

dE/dx (ADC counts/cm)

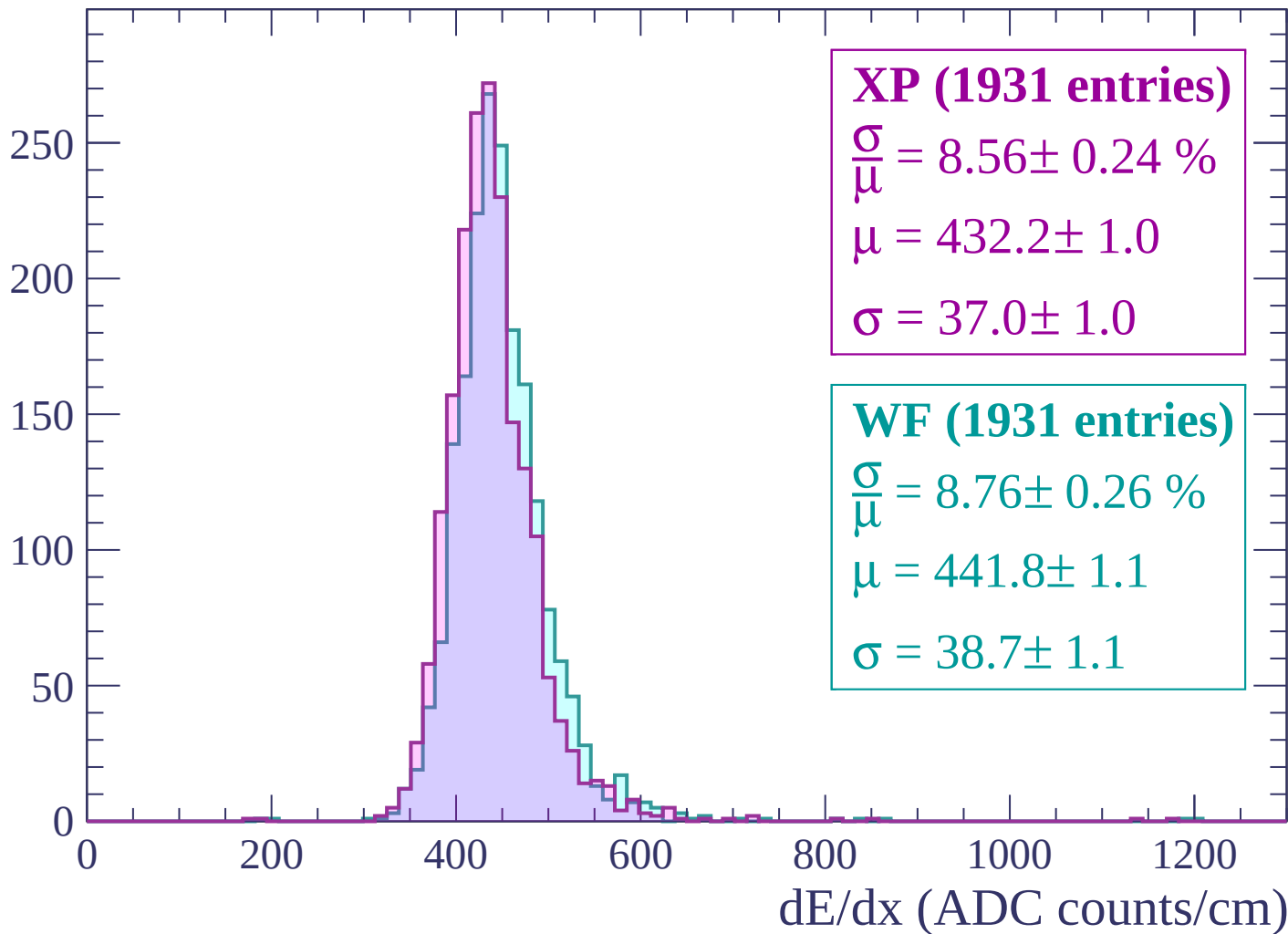
Energy loss | $960 < p < 1000$

Count



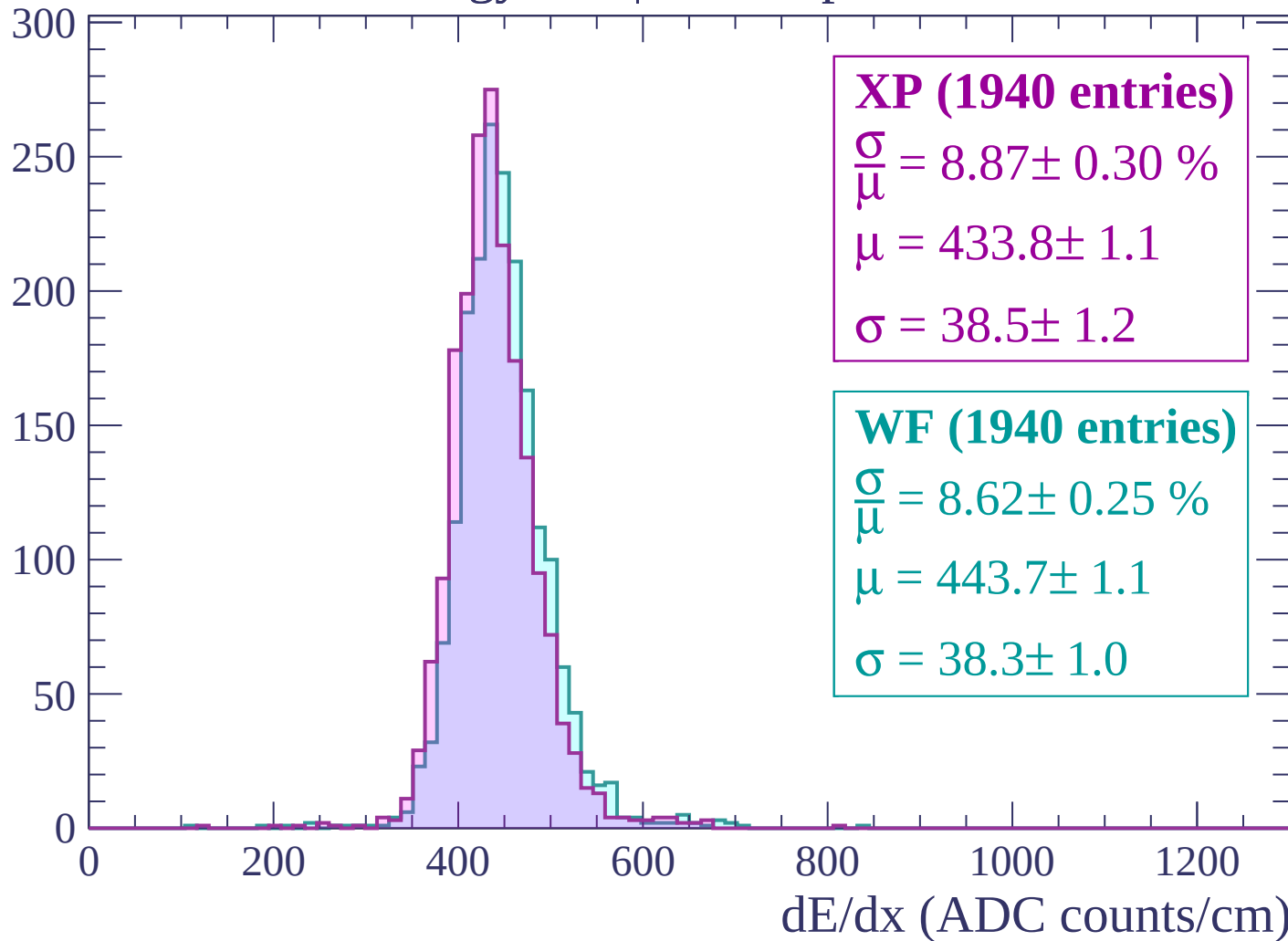
Energy loss | $1000 < p < 1040$

Count



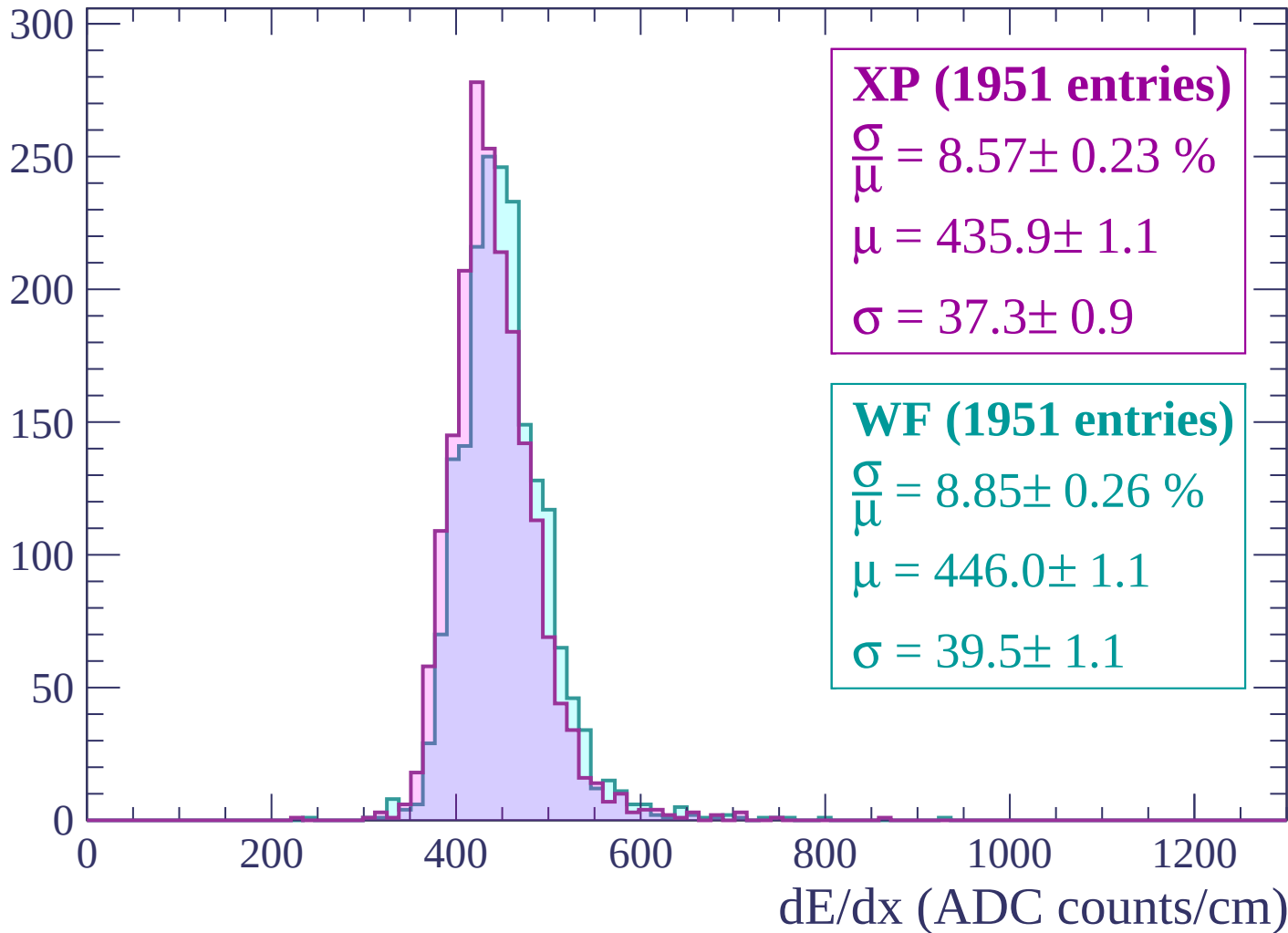
Energy loss | $1040 < p < 1080$

Count



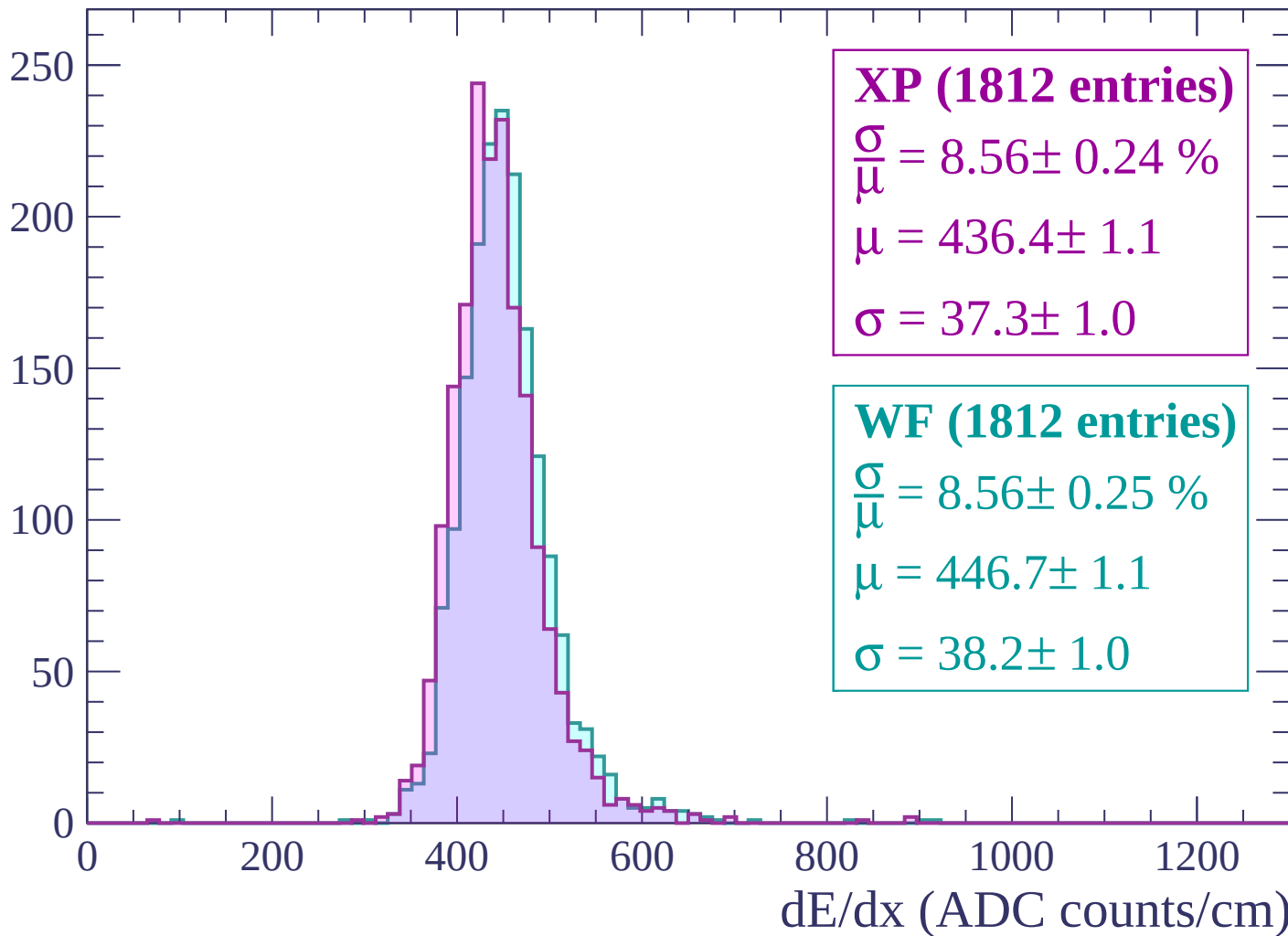
Energy loss | $1080 < p < 1120$

Count



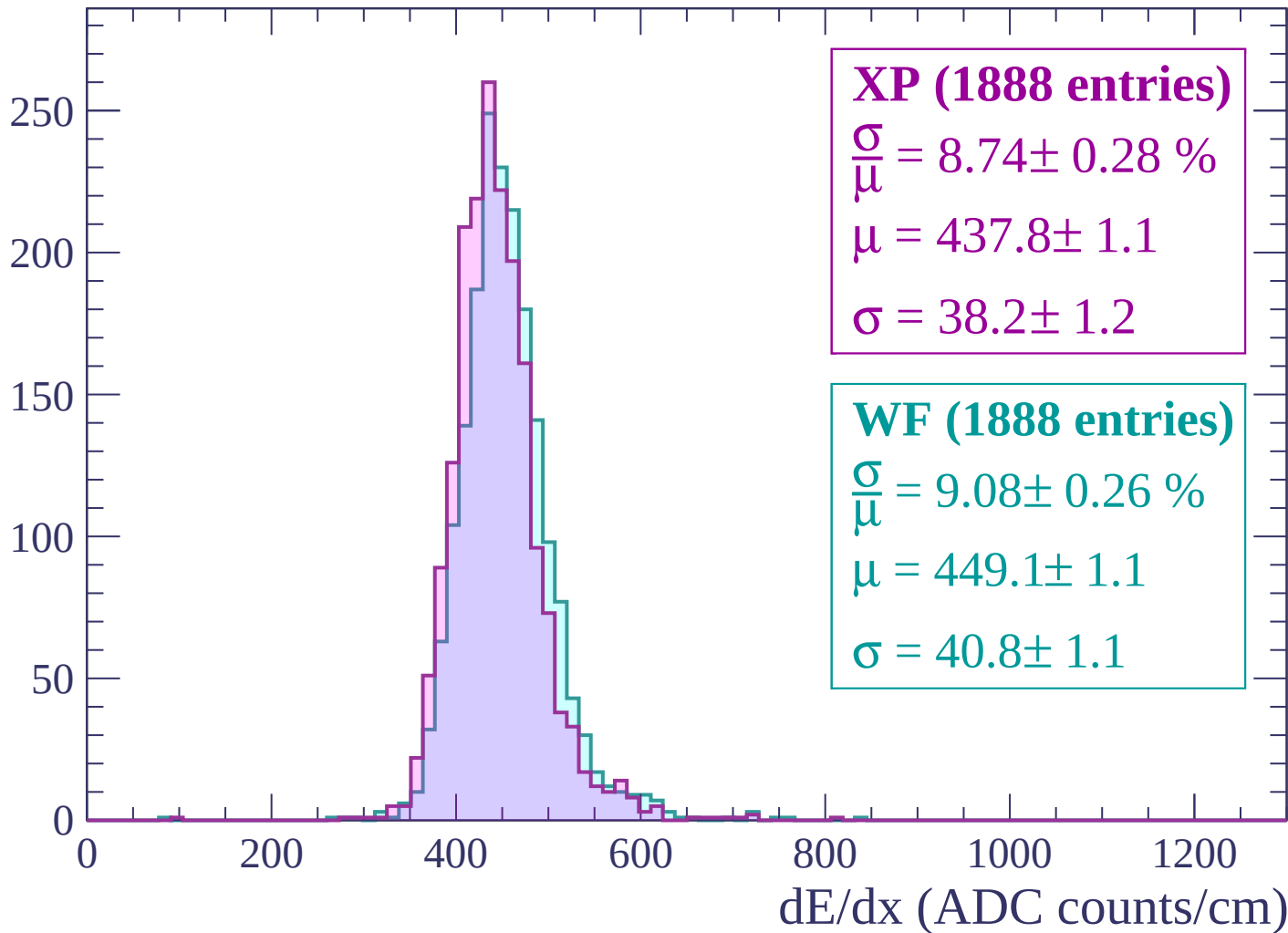
Energy loss | $1120 < p < 1160$

Count



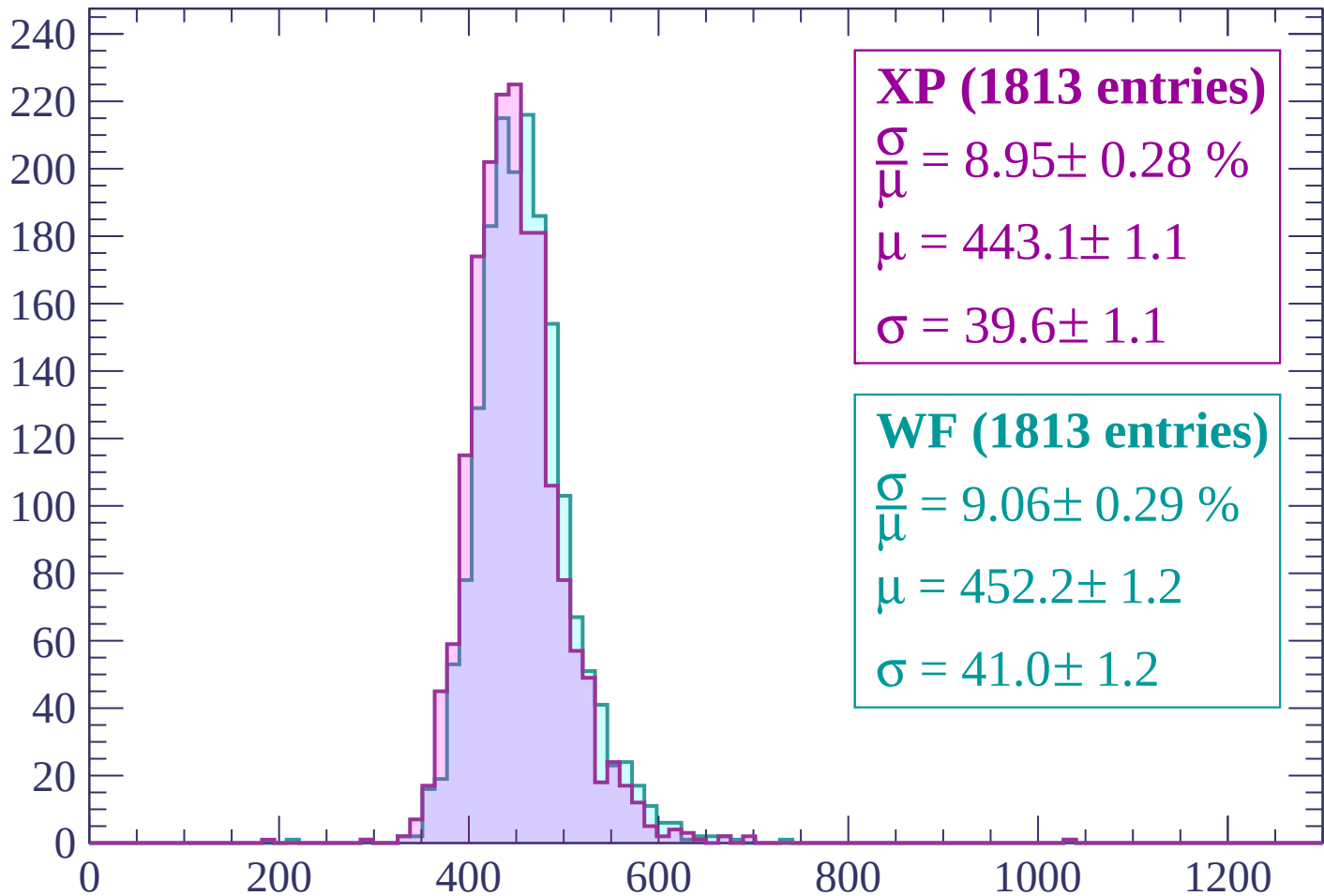
Energy loss | $1160 < p < 1200$

Count



Energy loss | $1200 < p < 1240$

Count



XP (1813 entries)

$\frac{\sigma}{\mu} = 8.95 \pm 0.28 \%$

$\mu = 443.1 \pm 1.1$

$\sigma = 39.6 \pm 1.1$

WF (1813 entries)

$\frac{\sigma}{\mu} = 9.06 \pm 0.29 \%$

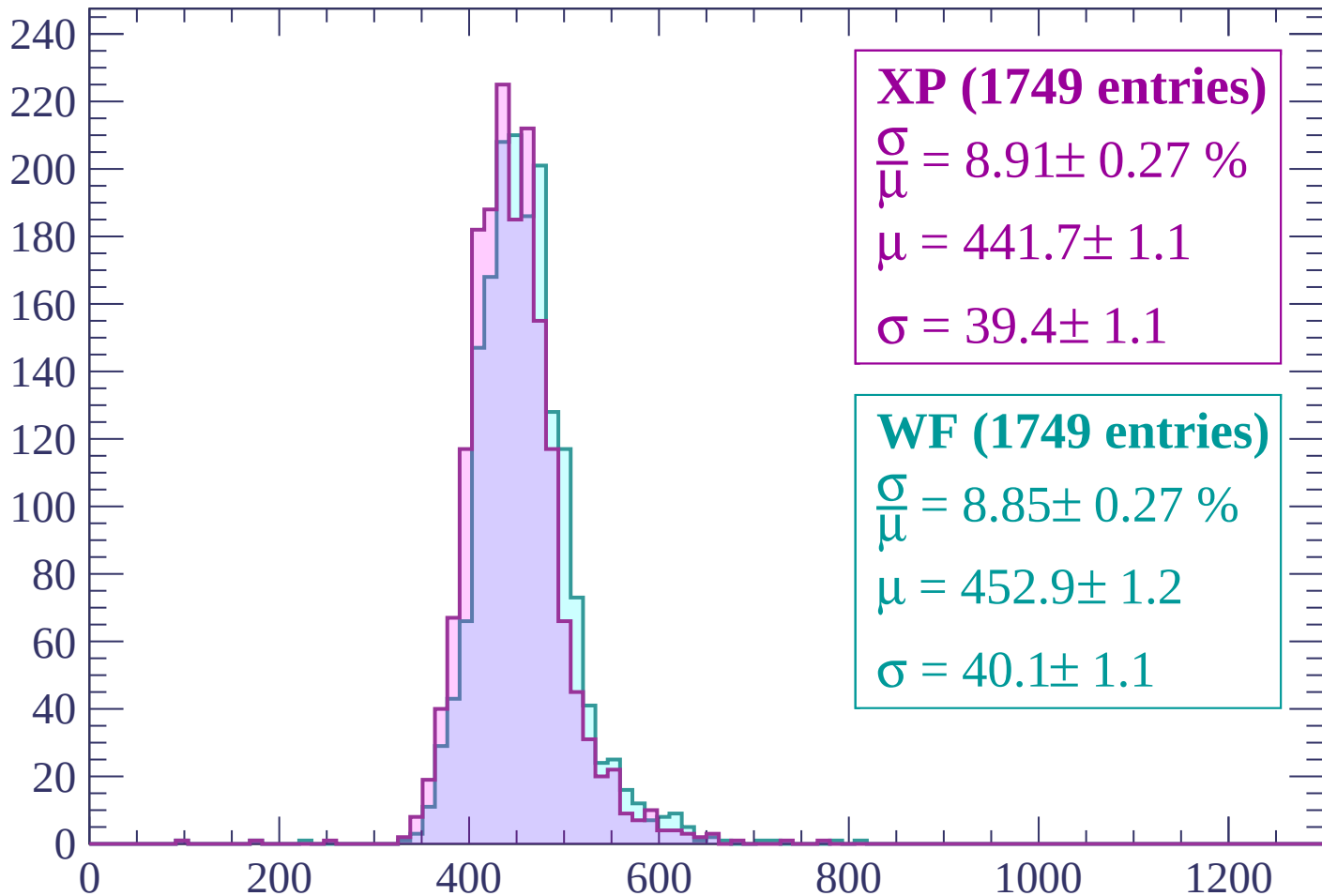
$\mu = 452.2 \pm 1.2$

$\sigma = 41.0 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP (1749 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.27 \%$$

$$\mu = 441.7 \pm 1.1$$

$$\sigma = 39.4 \pm 1.1$$

WF (1749 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.27 \%$$

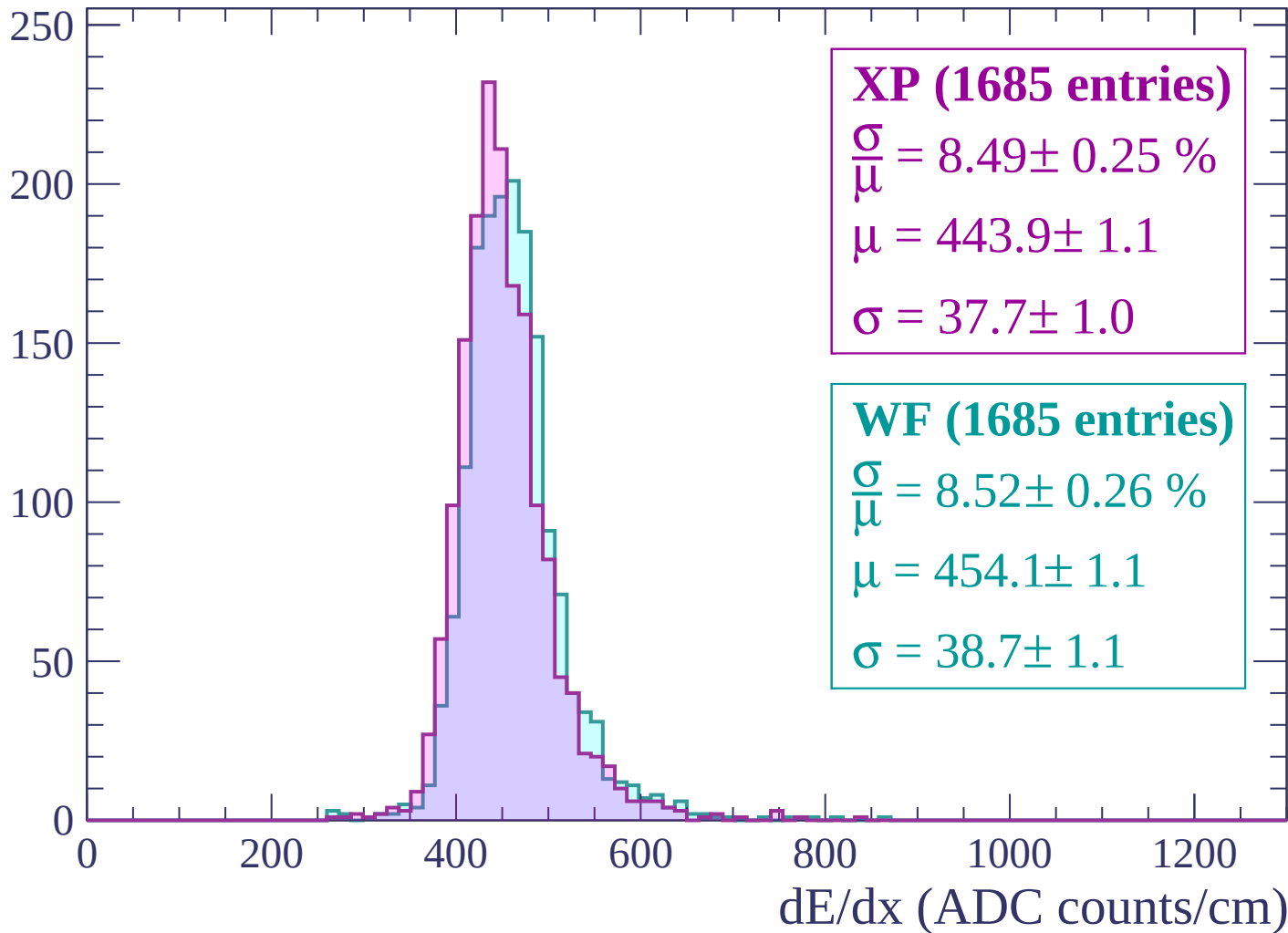
$$\mu = 452.9 \pm 1.2$$

$$\sigma = 40.1 \pm 1.1$$

dE/dx (ADC counts/cm)

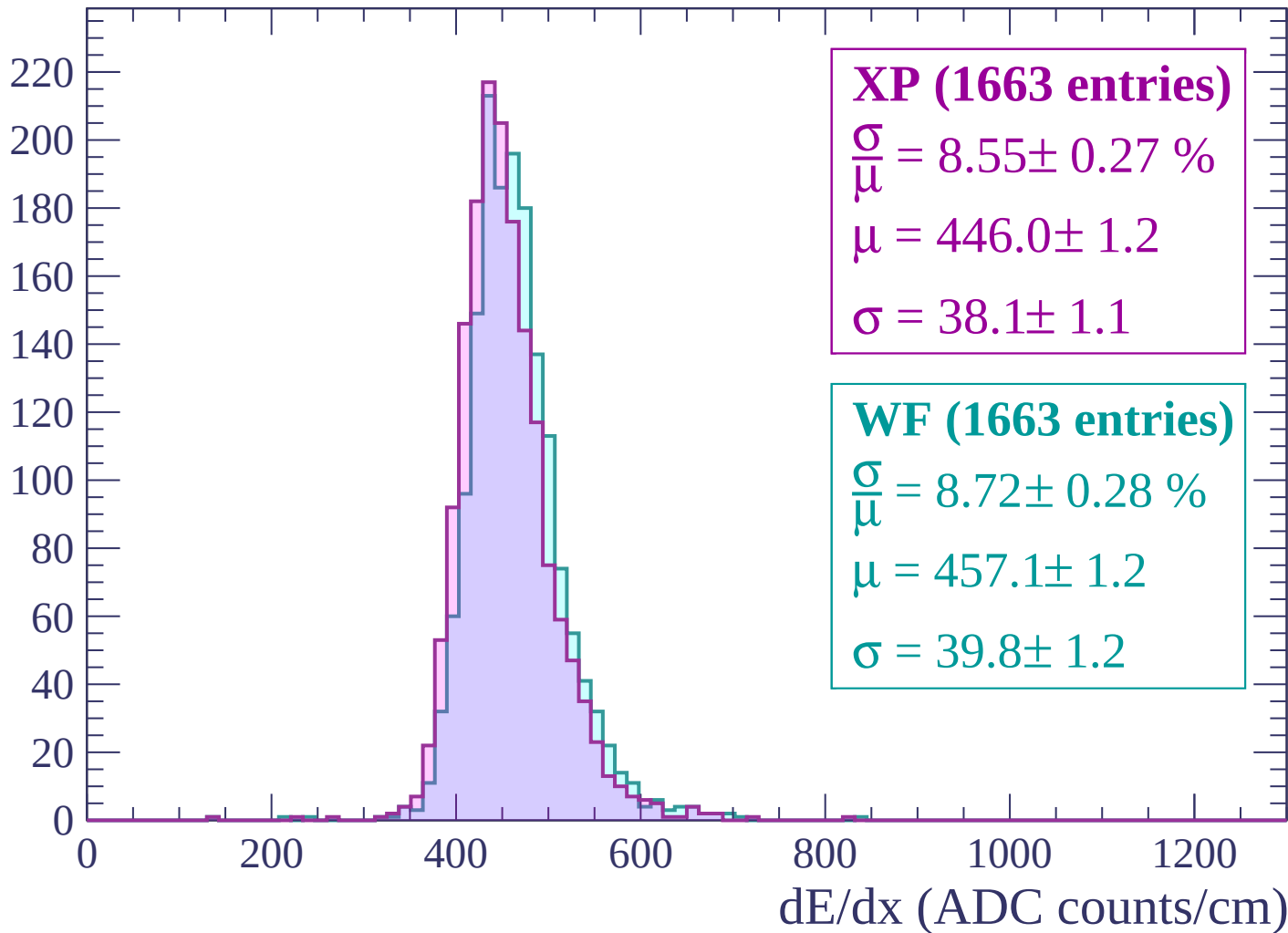
Energy loss | $1280 < p < 1320$

Count



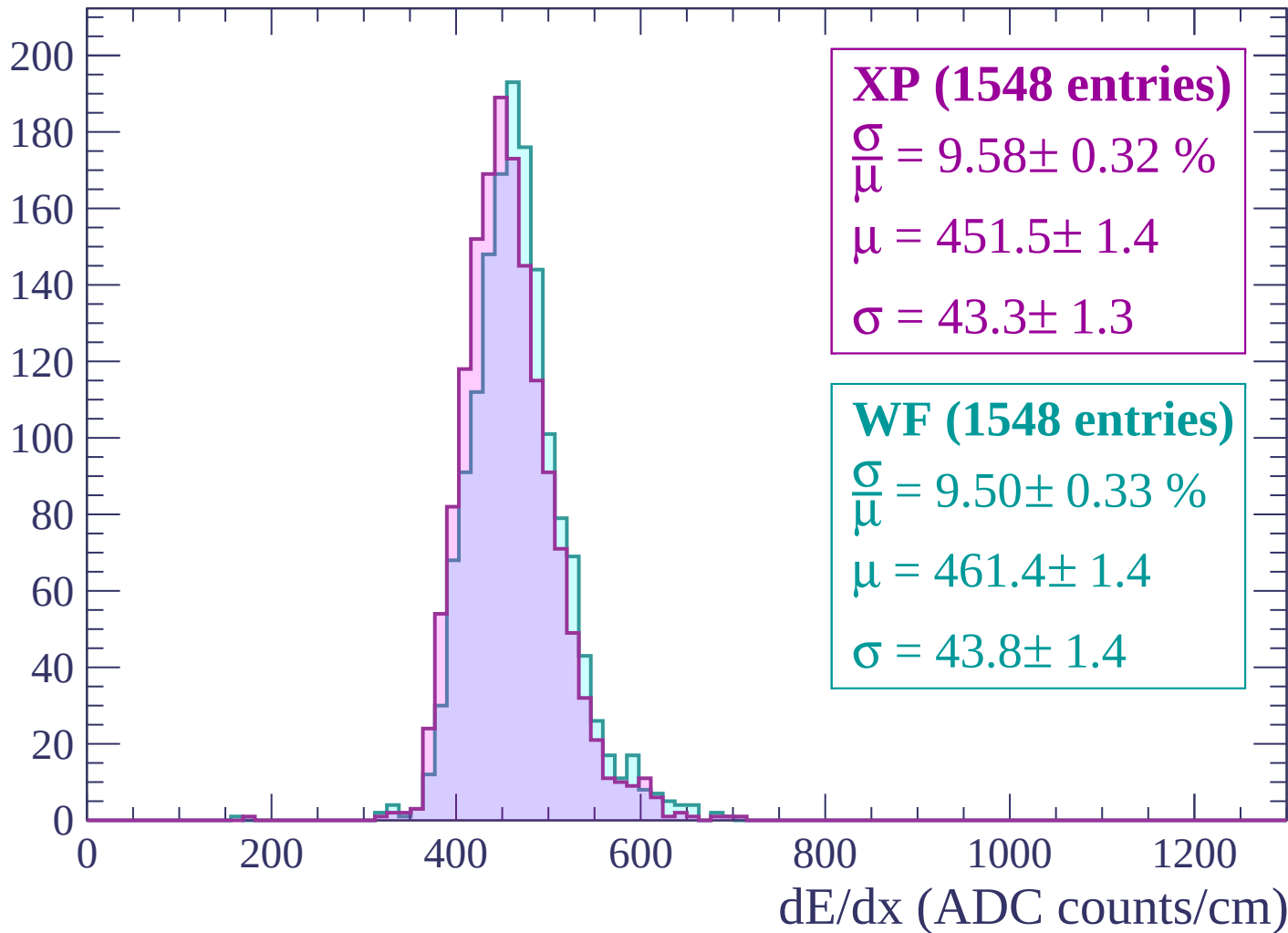
Energy loss | $1320 < p < 1360$

Count



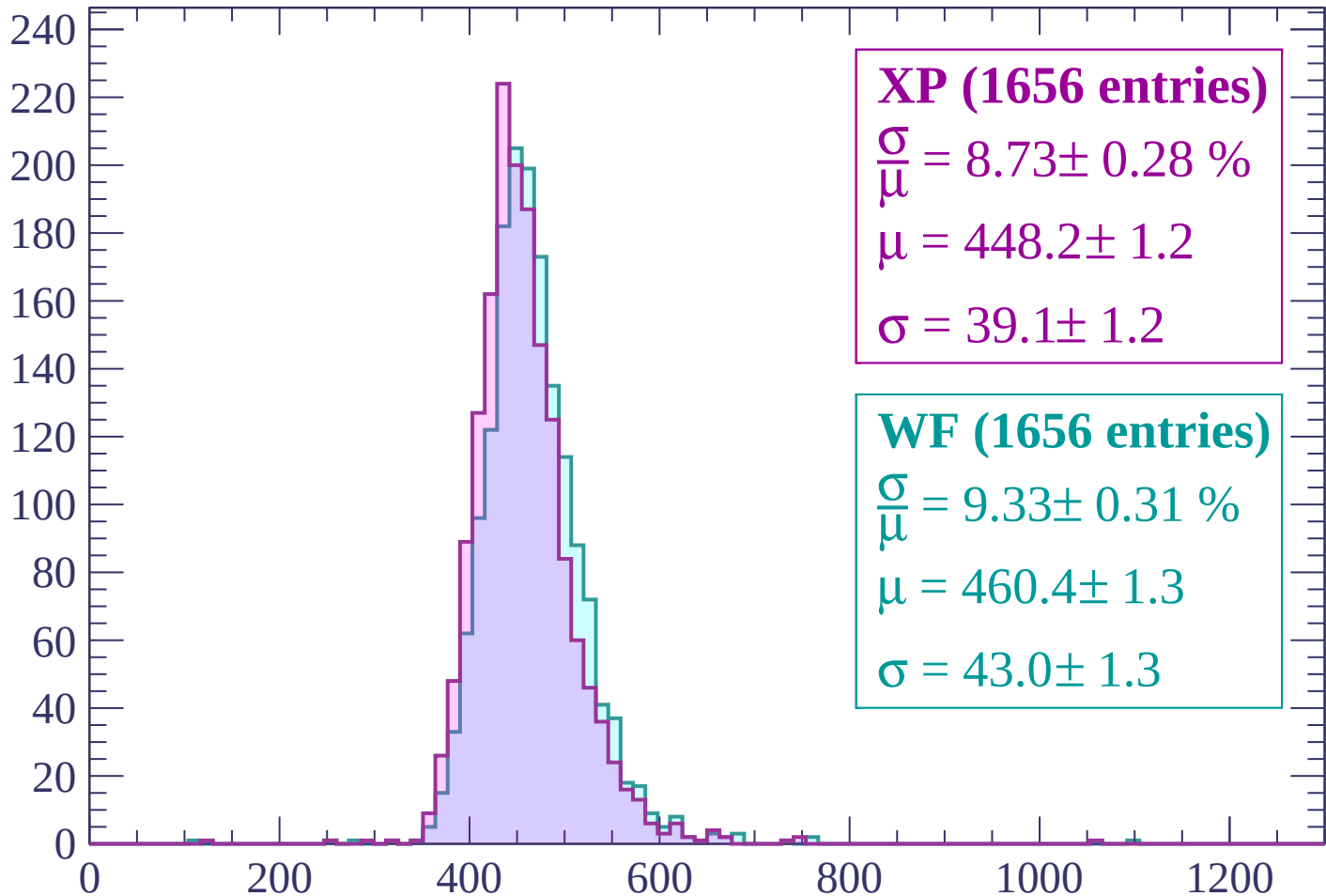
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

Count



XP (1656 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.28 \%$$

$$\mu = 448.2 \pm 1.2$$

$$\sigma = 39.1 \pm 1.2$$

WF (1656 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.31 \%$$

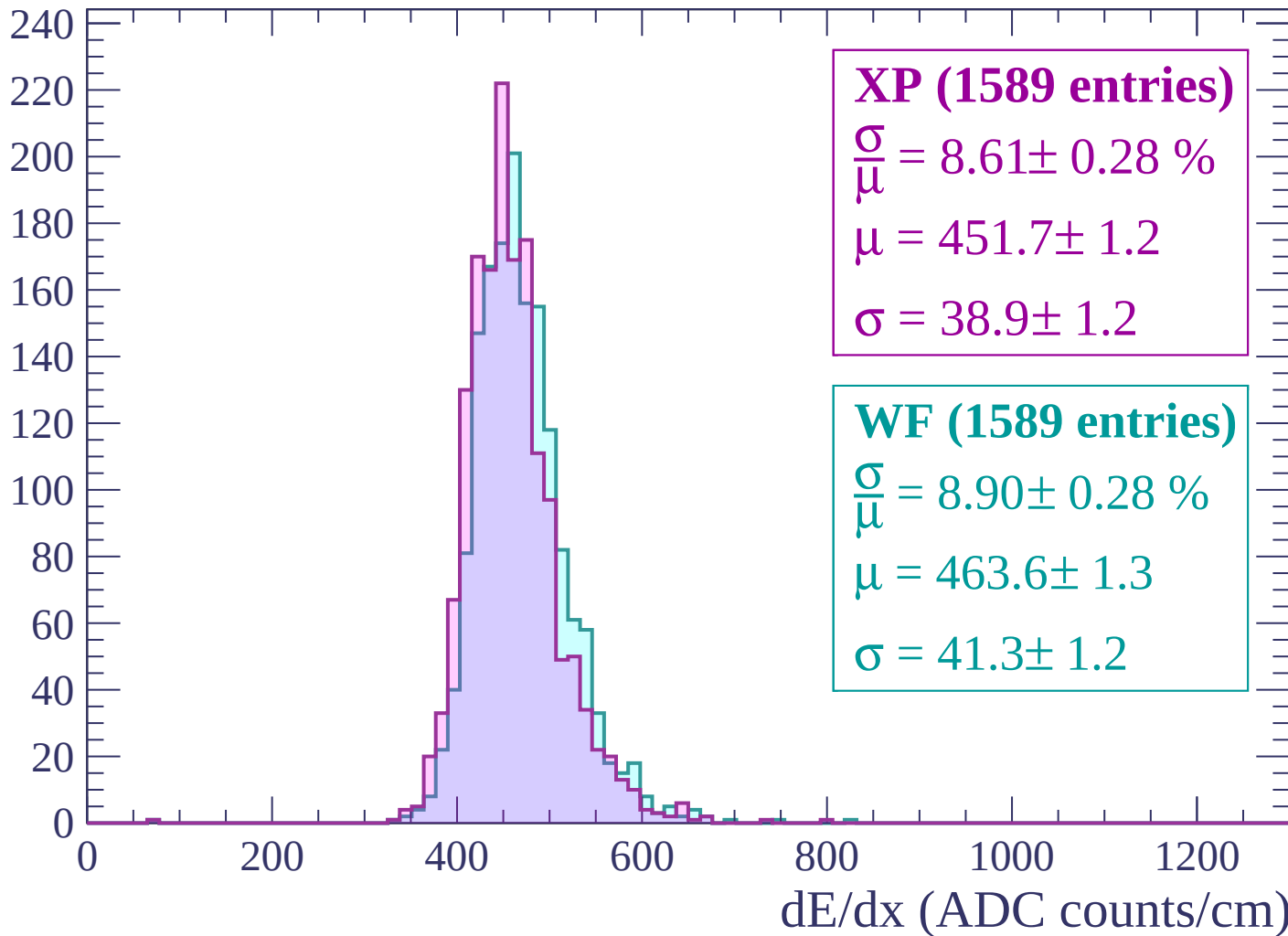
$$\mu = 460.4 \pm 1.3$$

$$\sigma = 43.0 \pm 1.3$$

dE/dx (ADC counts/cm)

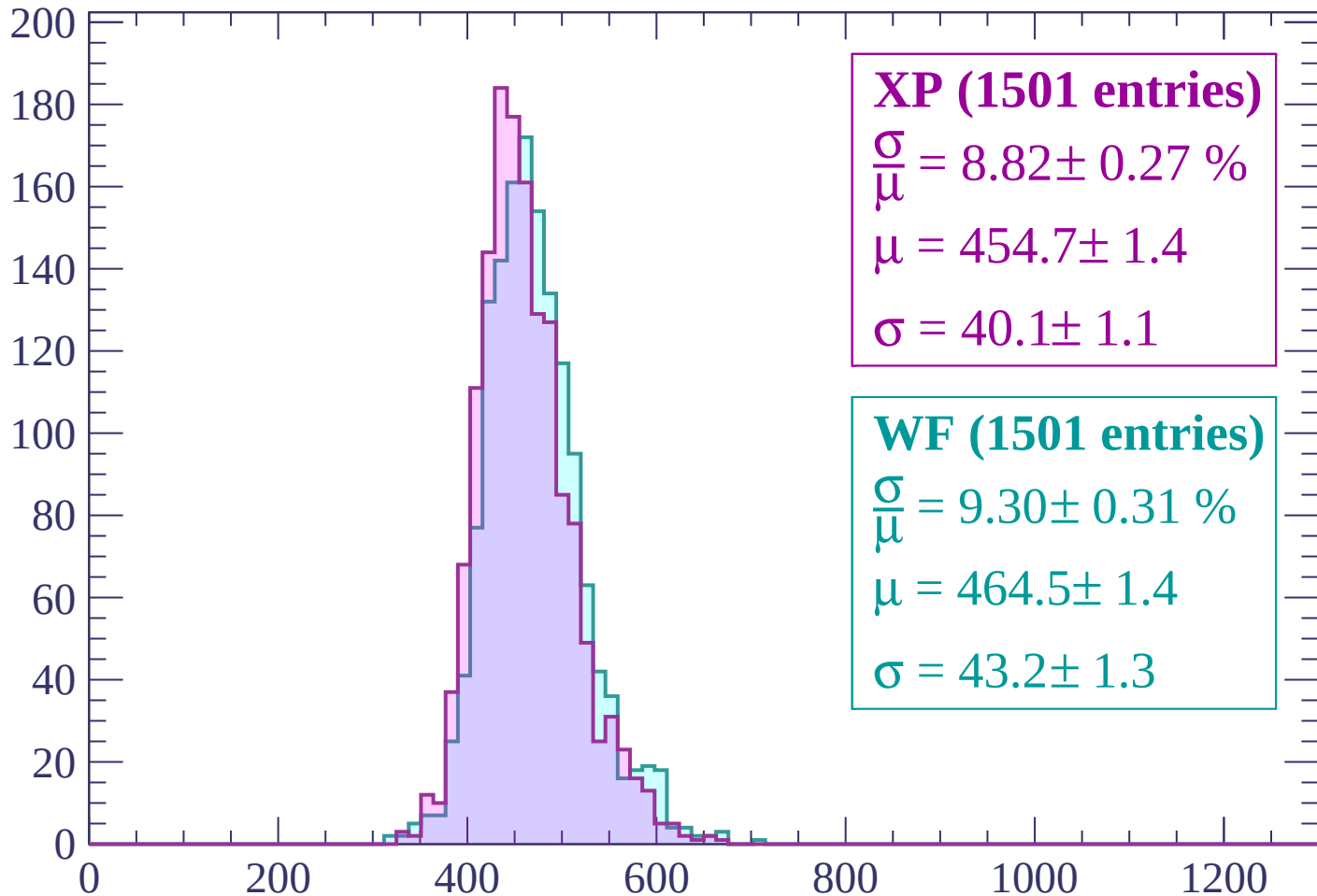
Energy loss | $1440 < p < 1480$

Count



Energy loss | $1480 < p < 1520$

Count



XP (1501 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.27 \%$$

$$\mu = 454.7 \pm 1.4$$

$$\sigma = 40.1 \pm 1.1$$

WF (1501 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.31 \%$$

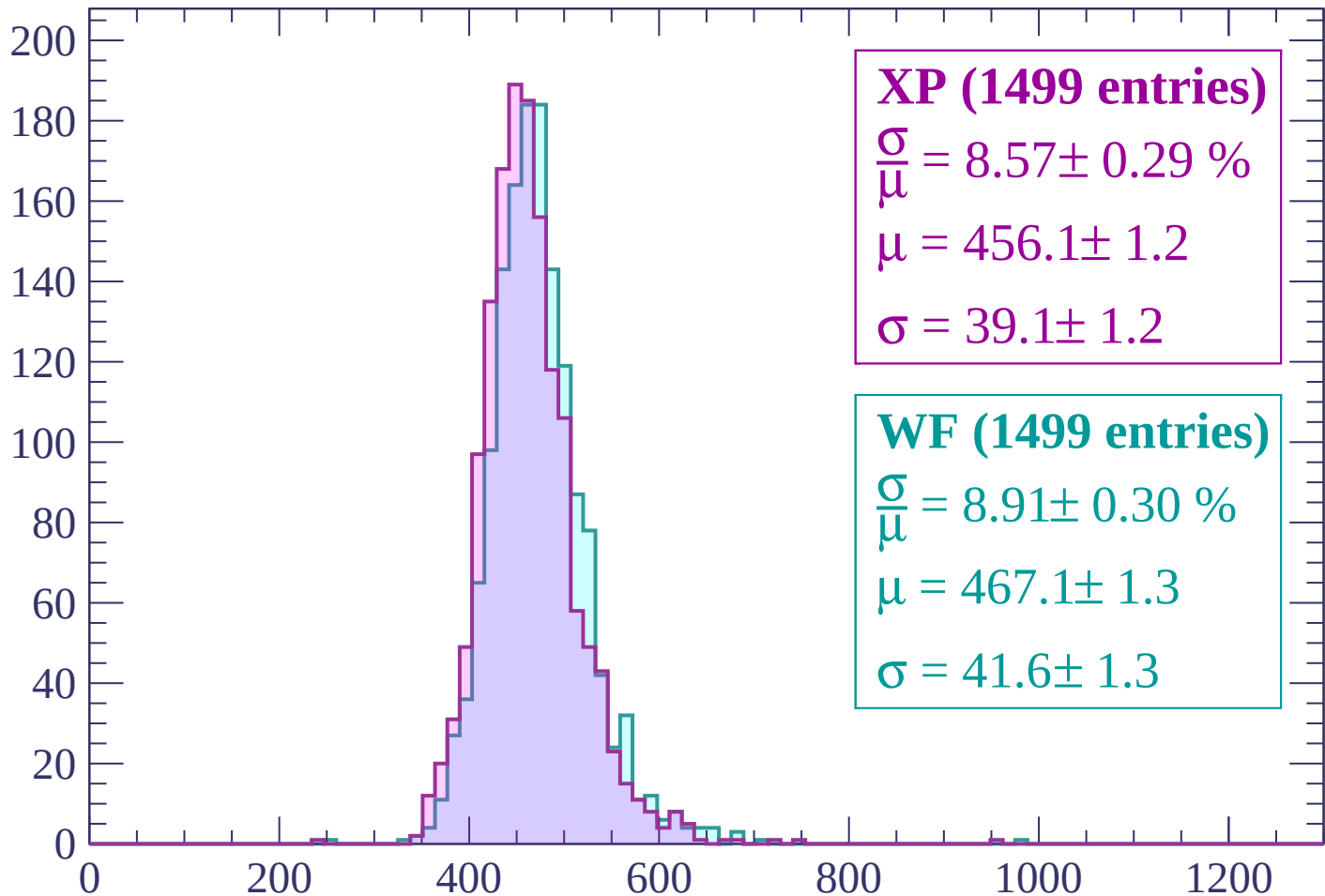
$$\mu = 464.5 \pm 1.4$$

$$\sigma = 43.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP (1499 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.29 \%$$

$$\mu = 456.1 \pm 1.2$$

$$\sigma = 39.1 \pm 1.2$$

WF (1499 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.30 \%$$

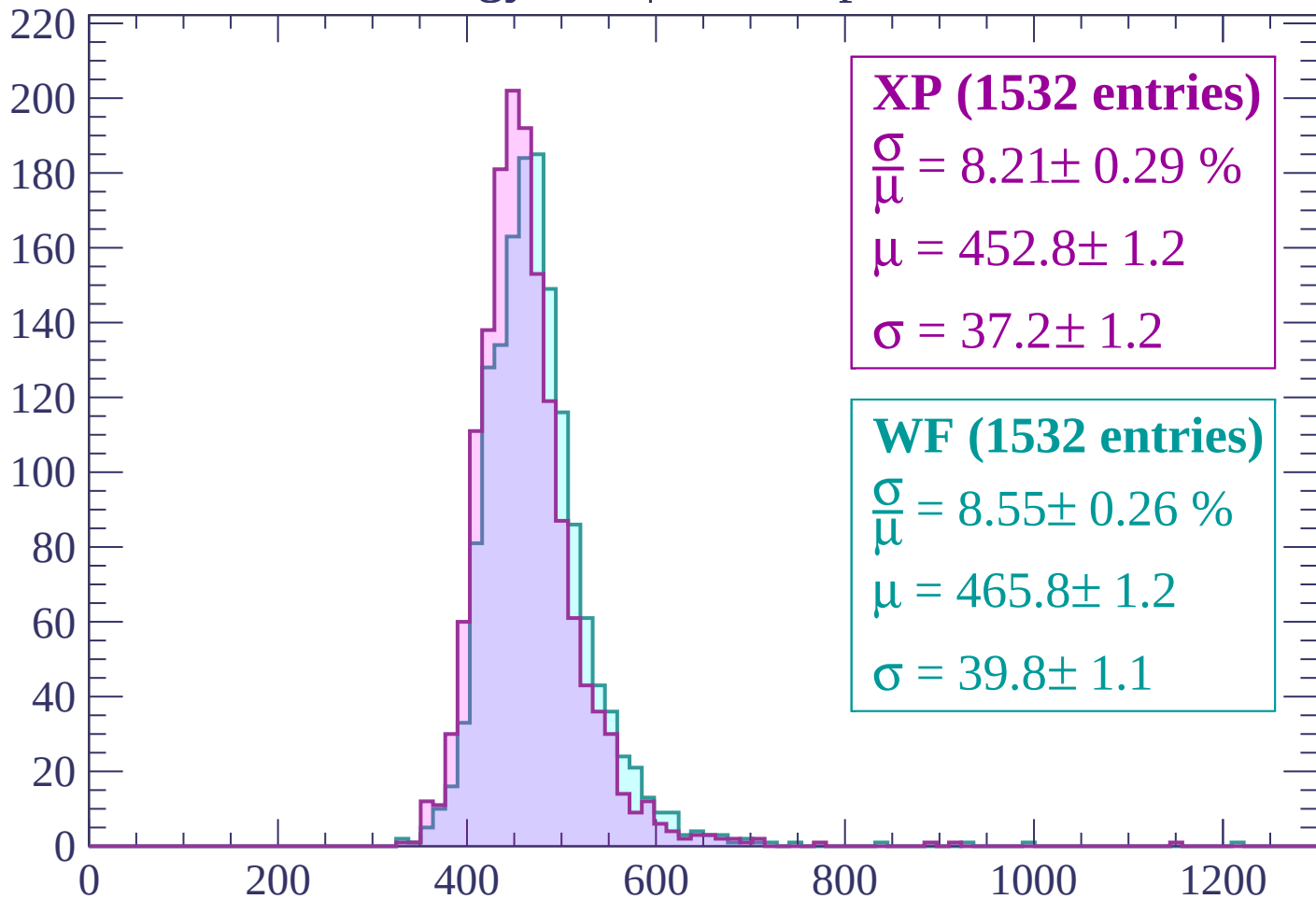
$$\mu = 467.1 \pm 1.3$$

$$\sigma = 41.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP (1532 entries)

$$\frac{\sigma}{\mu} = 8.21 \pm 0.29 \%$$

$$\mu = 452.8 \pm 1.2$$

$$\sigma = 37.2 \pm 1.2$$

WF (1532 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.26 \%$$

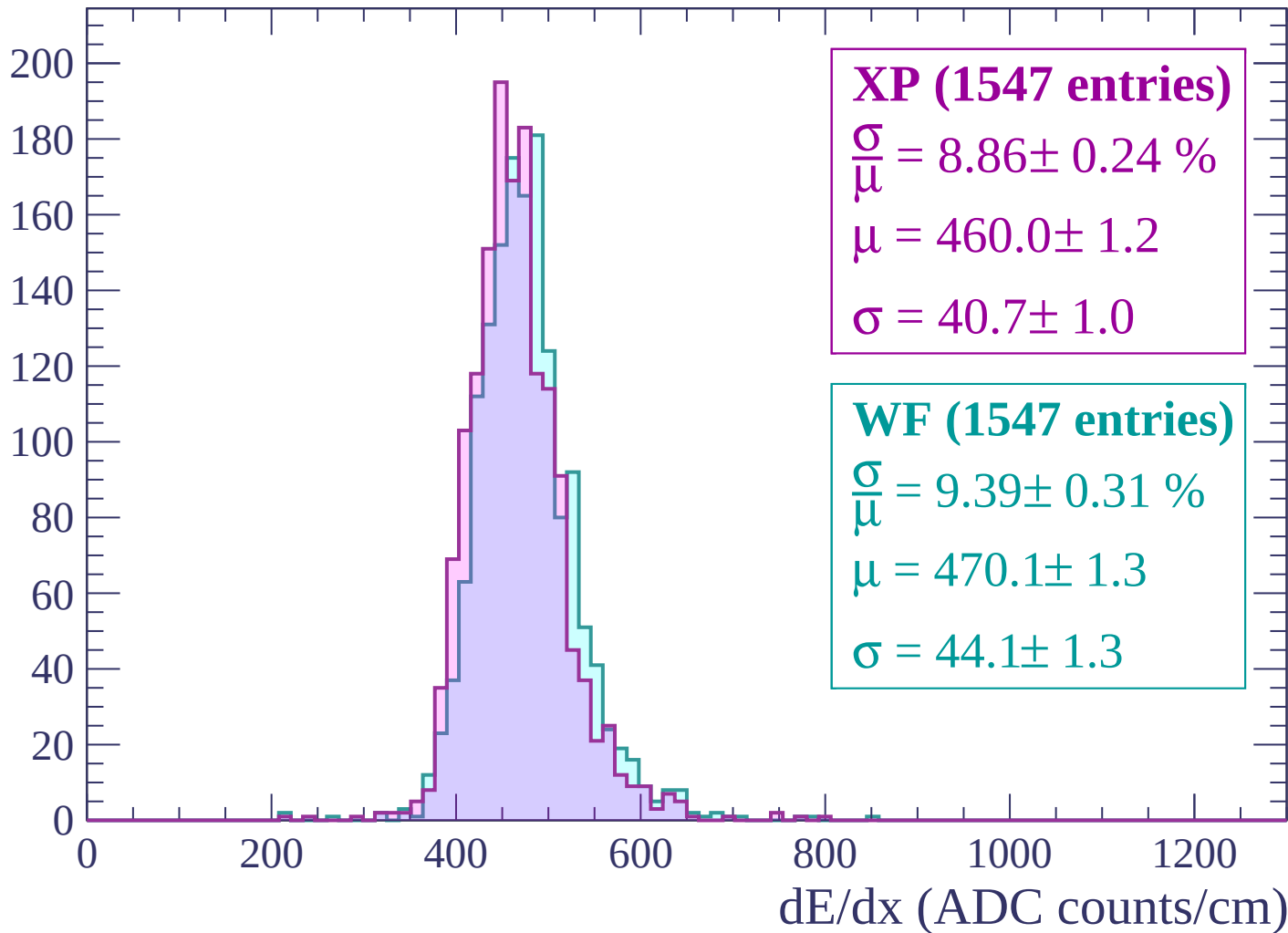
$$\mu = 465.8 \pm 1.2$$

$$\sigma = 39.8 \pm 1.1$$

dE/dx (ADC counts/cm)

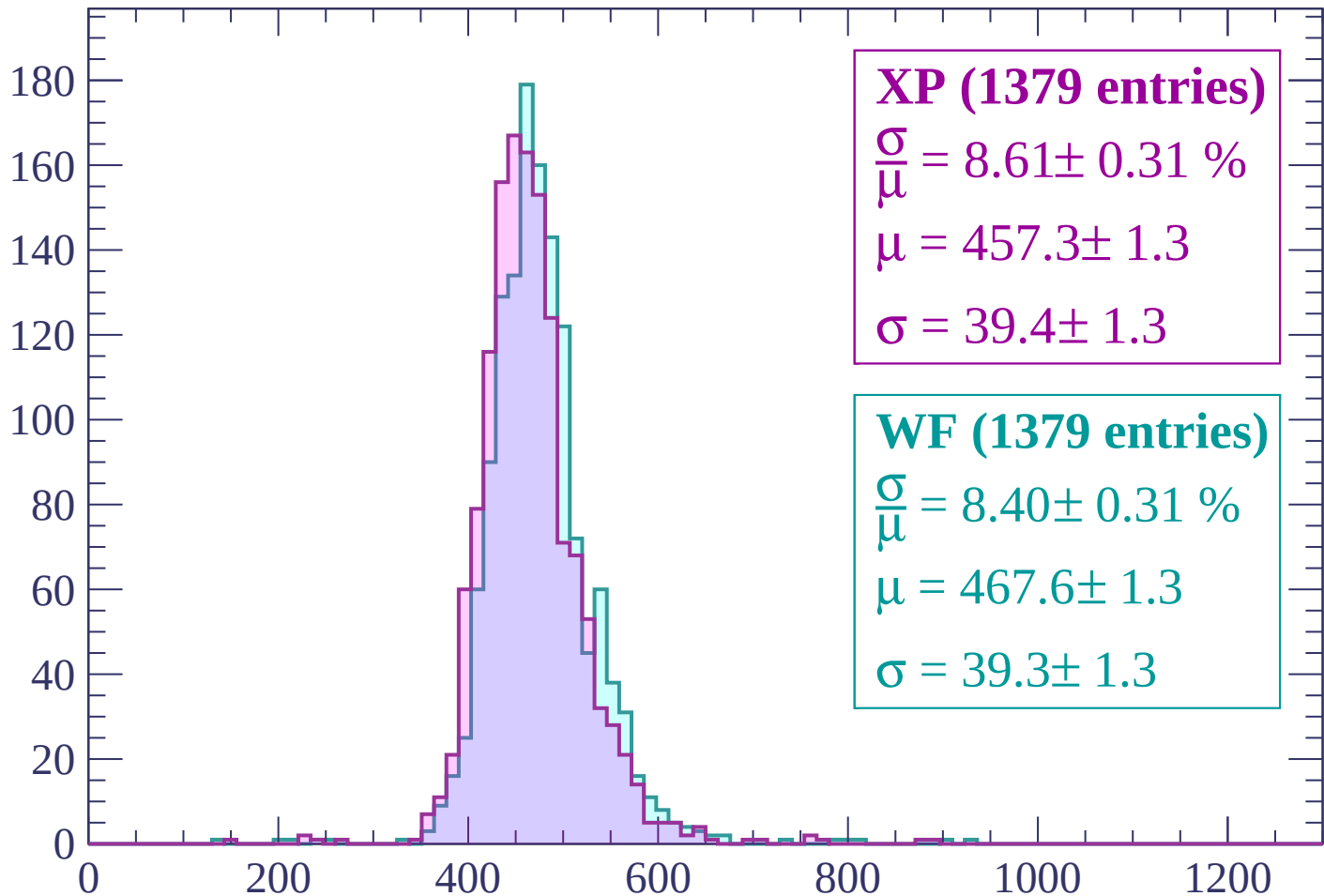
Energy loss | $1600 < p < 1640$

Count



Energy loss | $1640 < p < 1680$

Count



XP (1379 entries)

$$\frac{\sigma}{\mu} = 8.61 \pm 0.31 \%$$

$$\mu = 457.3 \pm 1.3$$

$$\sigma = 39.4 \pm 1.3$$

WF (1379 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.31 \%$$

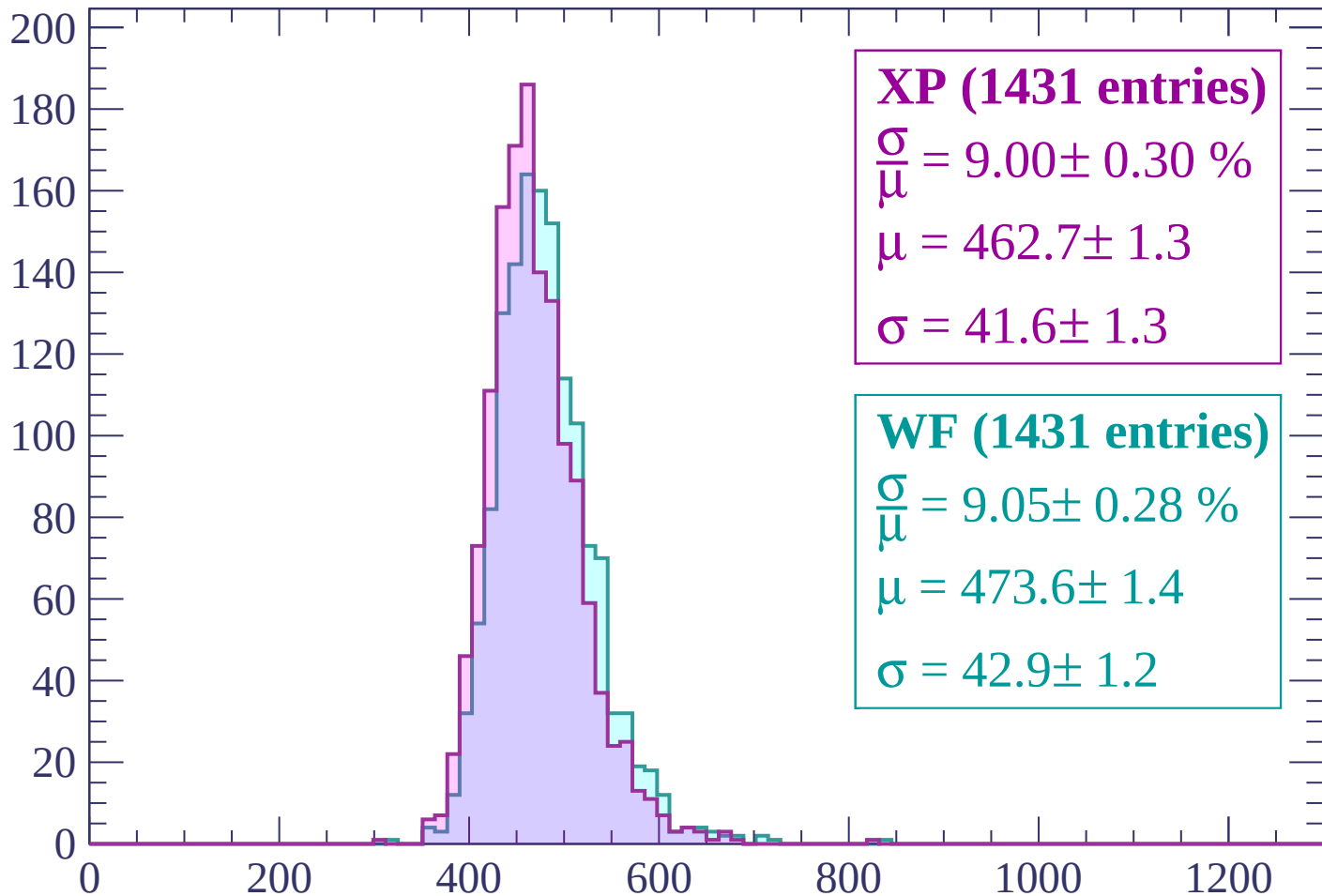
$$\mu = 467.6 \pm 1.3$$

$$\sigma = 39.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP (1431 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.30 \%$

$\mu = 462.7 \pm 1.3$

$\sigma = 41.6 \pm 1.3$

WF (1431 entries)

$\frac{\sigma}{\mu} = 9.05 \pm 0.28 \%$

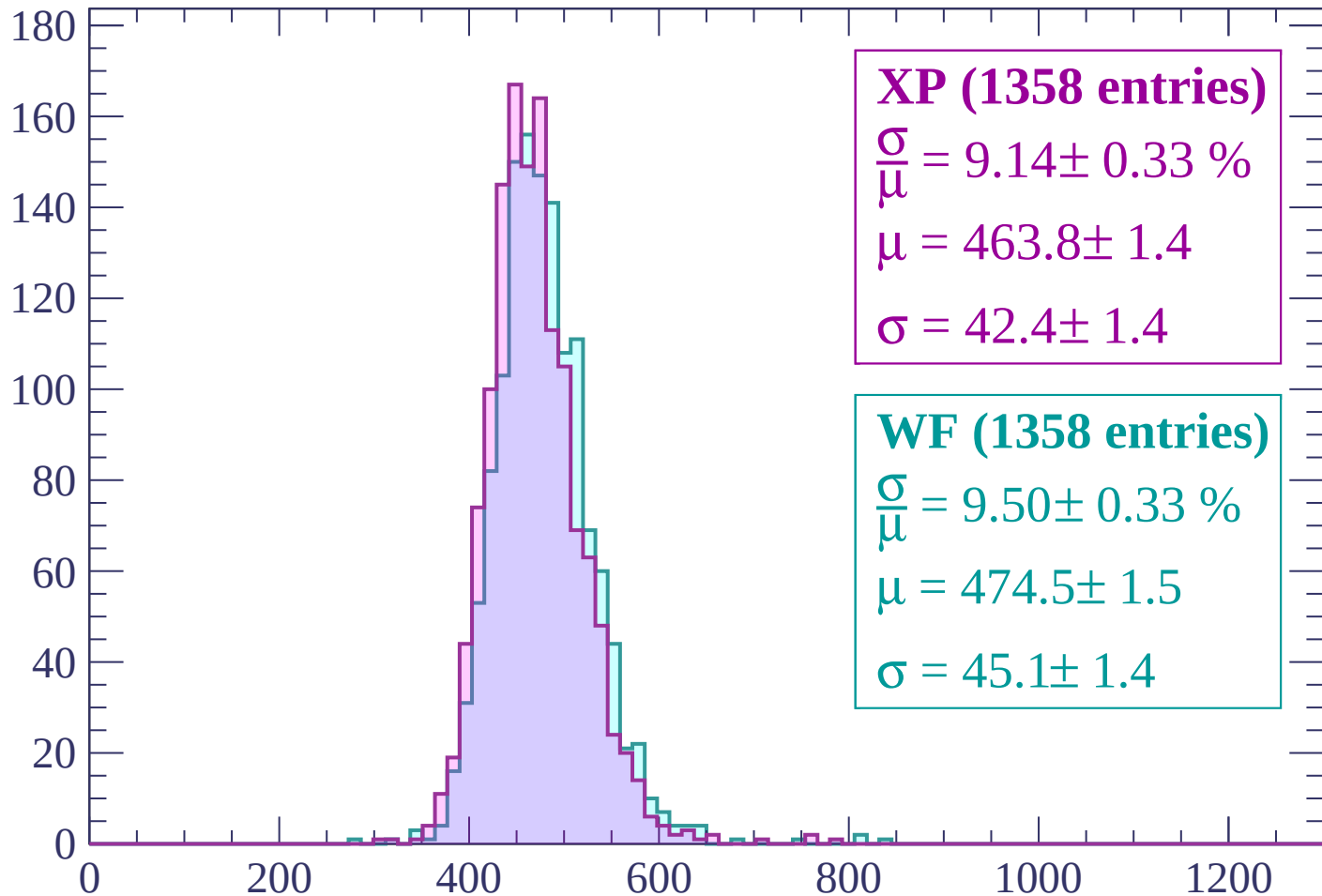
$\mu = 473.6 \pm 1.4$

$\sigma = 42.9 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP (1358 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.33 \%$$

$$\mu = 463.8 \pm 1.4$$

$$\sigma = 42.4 \pm 1.4$$

WF (1358 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.33 \%$$

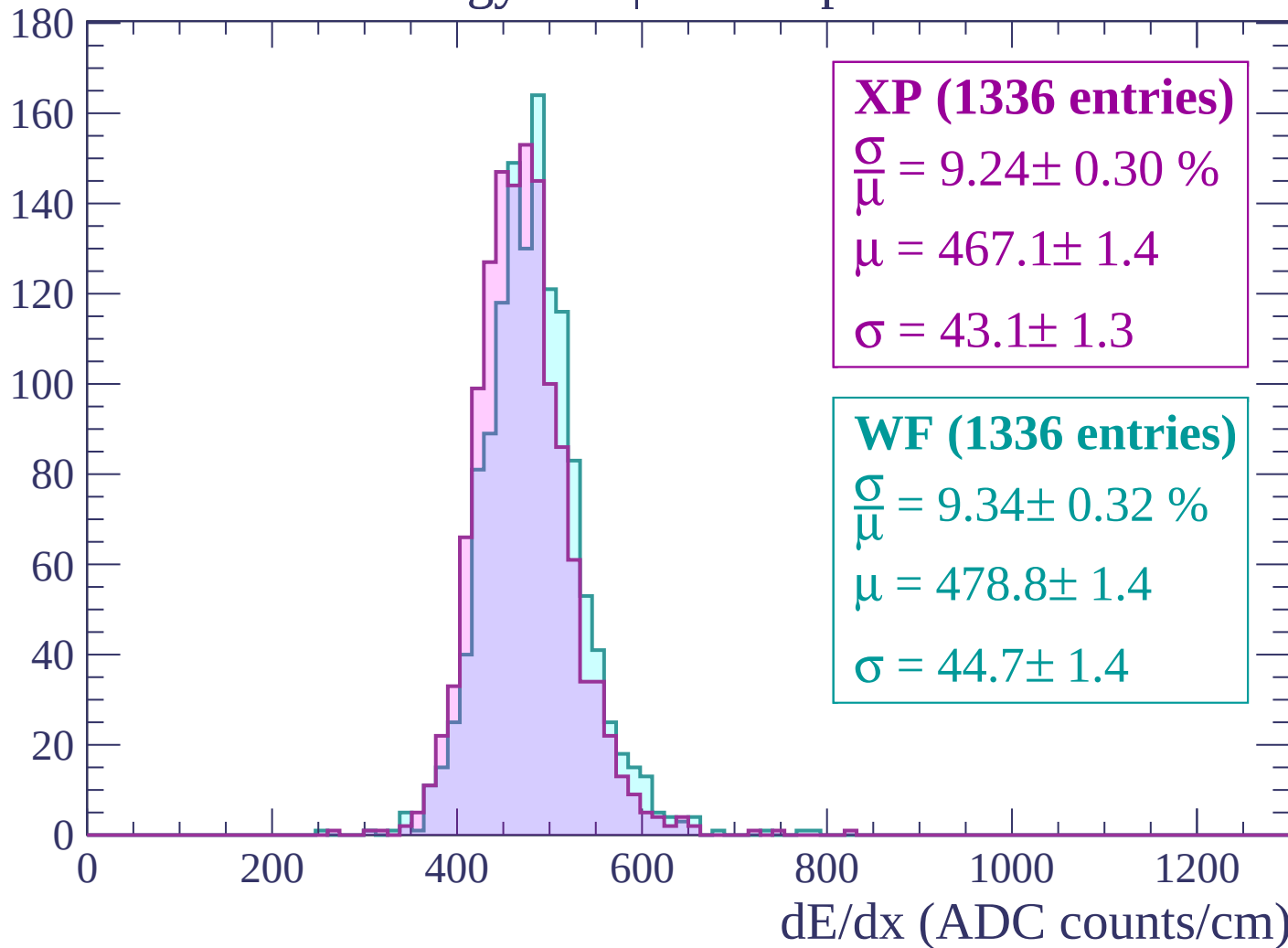
$$\mu = 474.5 \pm 1.5$$

$$\sigma = 45.1 \pm 1.4$$

dE/dx (ADC counts/cm)

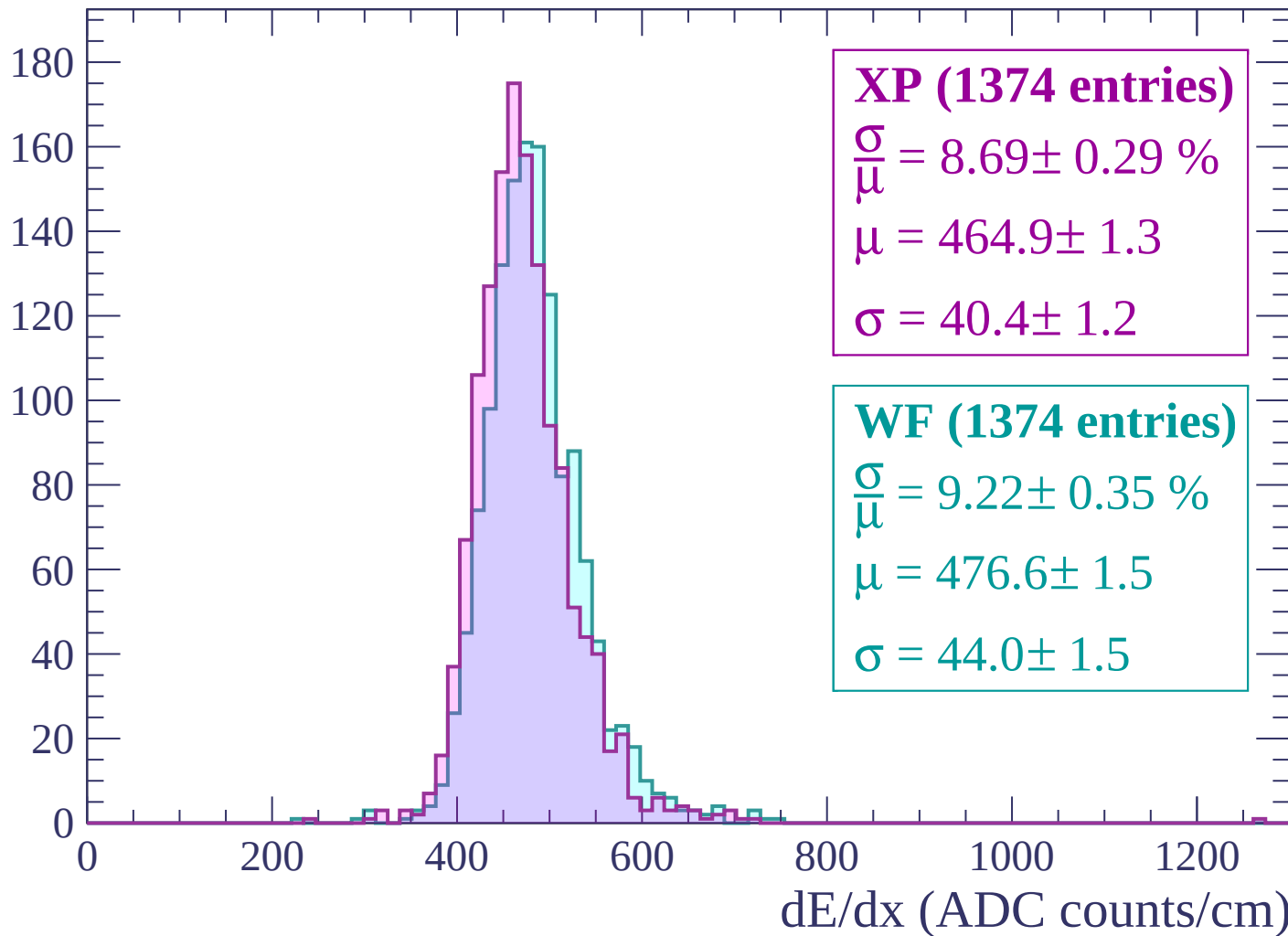
Energy loss | $1760 < p < 1800$

Count



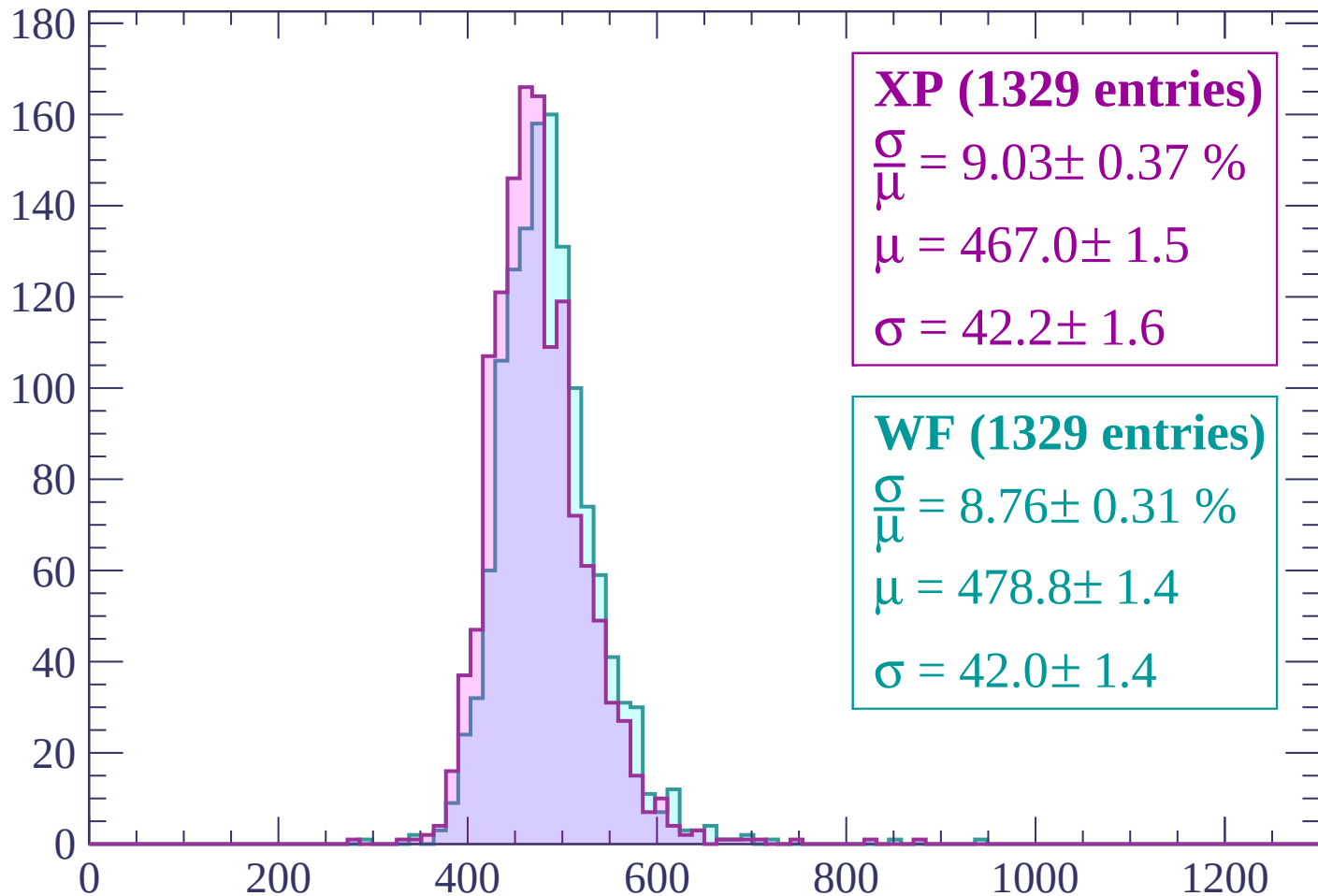
Energy loss | $1800 < p < 1840$

Count



Energy loss | $1840 < p < 1880$

Count



XP (1329 entries)

$$\frac{\sigma}{\mu} = 9.03 \pm 0.37 \%$$

$$\mu = 467.0 \pm 1.5$$

$$\sigma = 42.2 \pm 1.6$$

WF (1329 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.31 \%$$

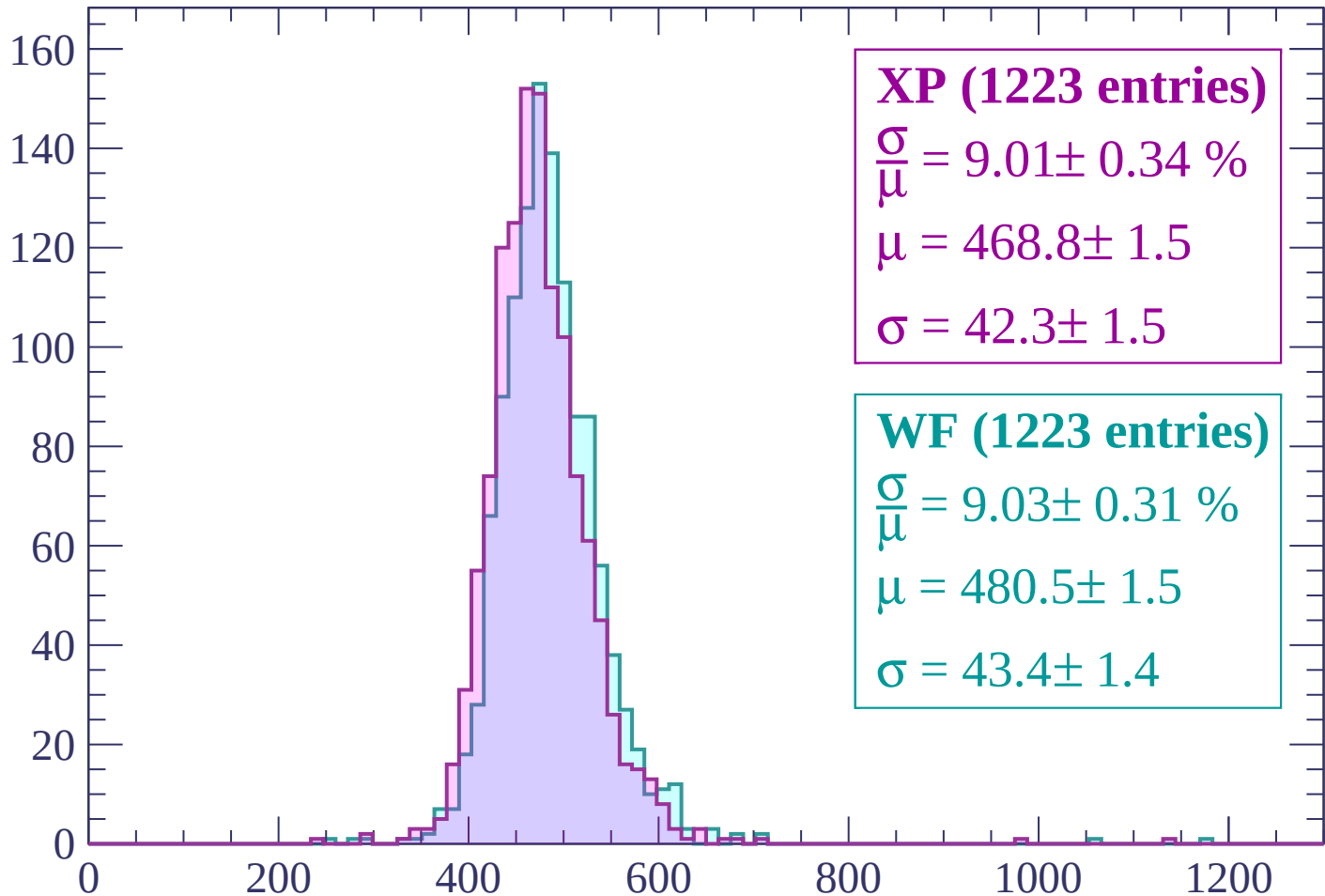
$$\mu = 478.8 \pm 1.4$$

$$\sigma = 42.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP (1223 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.34 \%$$

$$\mu = 468.8 \pm 1.5$$

$$\sigma = 42.3 \pm 1.5$$

WF (1223 entries)

$$\frac{\sigma}{\mu} = 9.03 \pm 0.31 \%$$

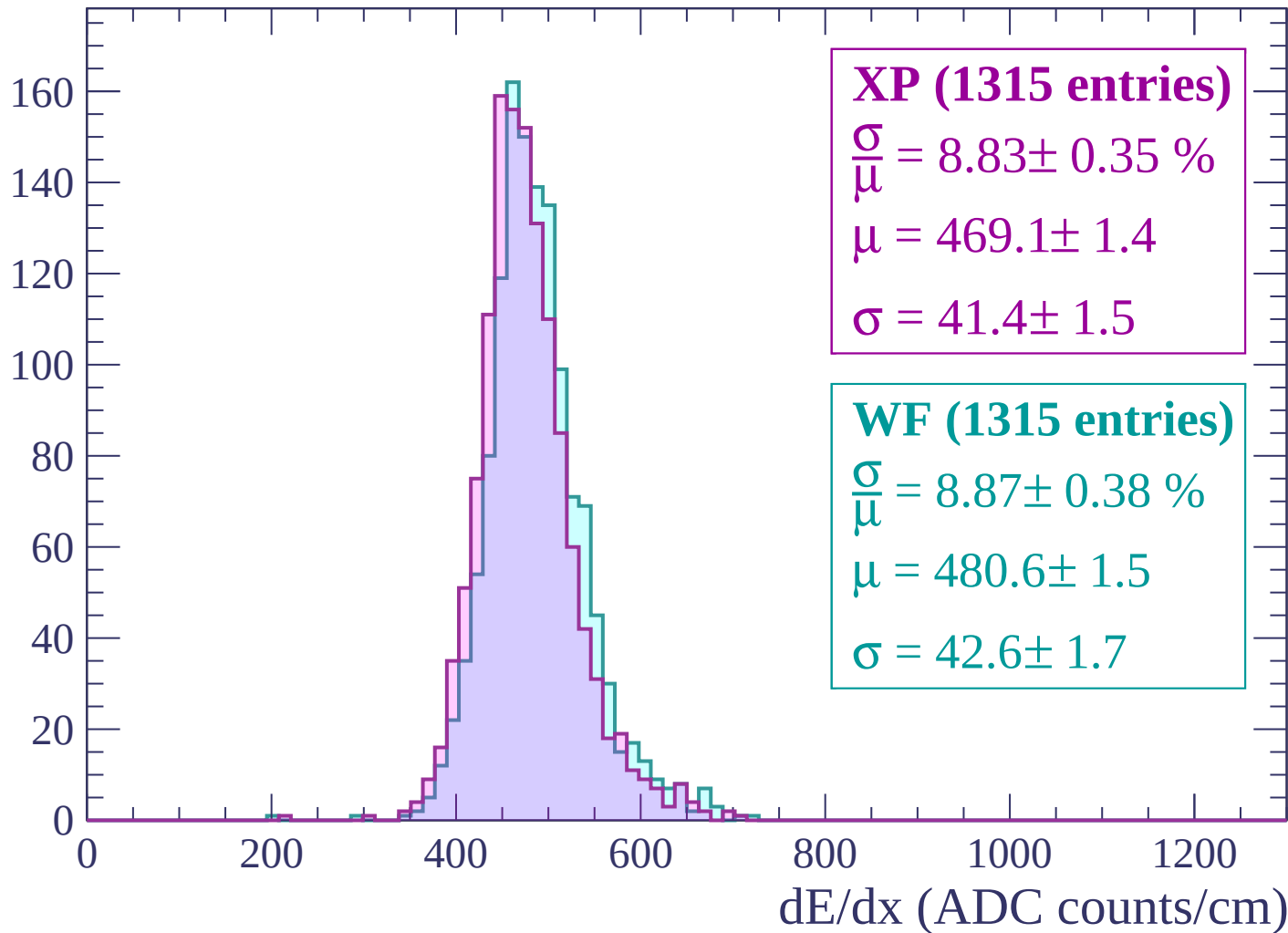
$$\mu = 480.5 \pm 1.5$$

$$\sigma = 43.4 \pm 1.4$$

dE/dx (ADC counts/cm)

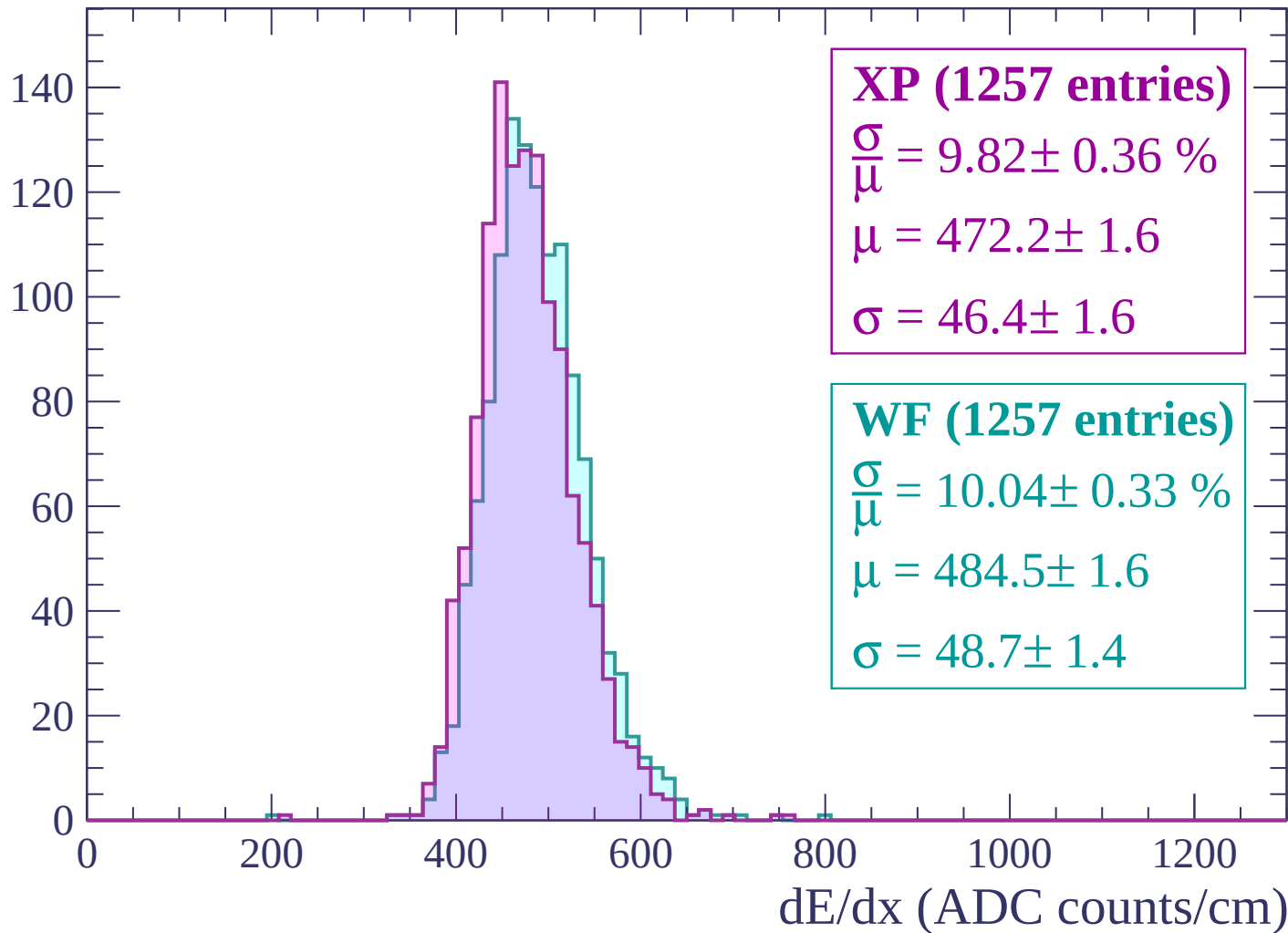
Energy loss | $1920 < p < 1960$

Count



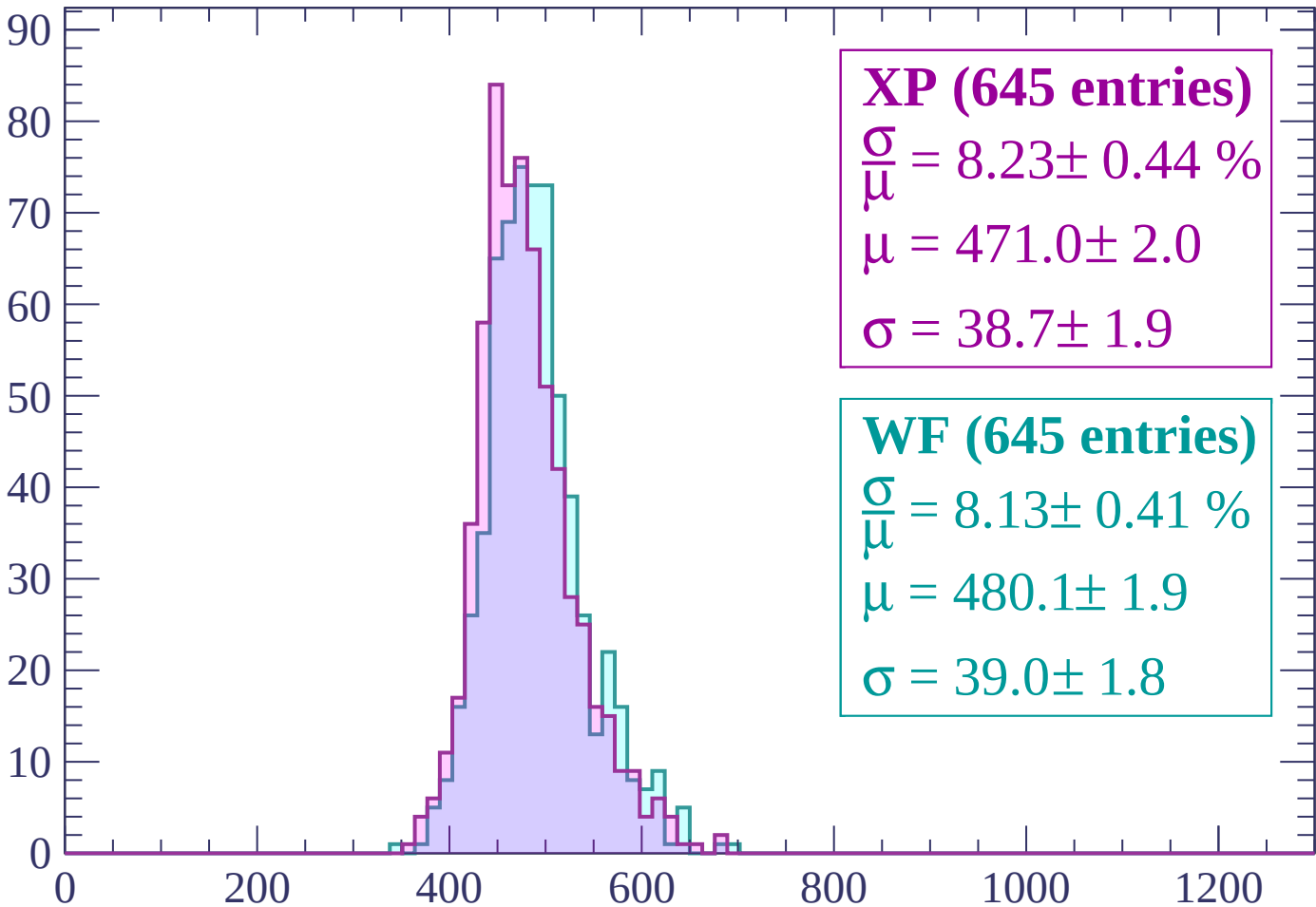
Energy loss | $1960 < p < 2000$

Count



Energy loss | $2000 < p < 2040$

Count



XP (645 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.44 \%$$

$$\mu = 471.0 \pm 2.0$$

$$\sigma = 38.7 \pm 1.9$$

WF (645 entries)

$$\frac{\sigma}{\mu} = 8.13 \pm 0.41 \%$$

$$\mu = 480.1 \pm 1.9$$

$$\sigma = 39.0 \pm 1.8$$

dE/dx (ADC counts/cm)

